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# Real Analysis: Foundations

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Sergei Ovchinnikov

# Real Analysis: Foundations

Springer

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*To my family*

# Preface

This book is an introductory text on elementary Real Analysis, focusing on the modern approach to the foundations of the subject.

All definitions and propositions of Real Analysis are formulated in terms of the field of real numbers **R**. However, most of them do not use all the properties of the field **R**, but only operations and relations defining it as an ordered field. Thus, one can attempt to "develop" Analysis over a general ordered field. Yet, there is a serious problem with this idea—the proofs of all main theorems of Real Analysis depend on the completeness property of the field **R**. In fact, there is no way to bypass this problem because, during the past two decades, it has been recognized that many theorems of Real Analysis are intrinsically equivalent to the completeness property of the underlying ordered field. This brings the foundations of Real Analysis into the paradigm of the "reverse mathematics" (Stillwell 2019), where axioms are deduced from theorems. From this perspective, the foundations of Real Analysis are multifaceted, consisting of completeness axioms for the field **R** and propositions equivalent to them. The main emphasis of the book is to illustrate this unique feature of Real Analysis.

The number systems that are fundamental in mathematics include natural numbers, integers, and rational and real numbers. In order to keep the book self-contained, an overview of natural numbers and integers is included in Appendix A.

The subject of the first chapter is the field of rational numbers. The first two sections of this chapter are devoted to a rigorous development of the arithmetic of rational numbers. The notion of an ordered field is introduced in Section 1.3, where it is shown that the set of rational numbers **Q** can be endowed with the structure of an ordered field and, as such, is contained in every ordered field. There are deficiencies of the number system **Q** that motivate the need for a larger ordered field. Algebraic and geometric examples are found in Section 1.4. In Section 1.5, an important "analytical" concept of convergence of a sequence in an ordered field appears for the first time in the book. Cauchy sequences (also known as fundamental sequences) with terms in an ordered field are important tools in the development of real numbers and also more broadly in analysis. These sequences are introduced in

the same section, where, in particular, an example of a divergent Cauchy sequence in **Q** is given, exhibiting another limitation of the field of rational numbers.

The real numbers are introduced in Chapter 2 via Cantor's method of equivalent Cauchy sequences of rational numbers. Another construction—by Dedekind cuts—is outlined in Appendix B. The chapter opens with two properties of ordered fields, the Dedekind Completeness Property and the Cauchy Completeness Property. The central theorem of Section 2.1 establishes an important relation between these properties: an ordered field is Dedekind complete if and only if it is Cauchy complete and Archimedean. In the book, an ordered field is called "complete" if it is Dedekind complete. A rather lengthy Section 2.2 is devoted to the Cauchy completion of an ordered field **F**, that is, the construction of a minimal ordered field containing the given field in which every Cauchy sequence converges. It is proved that such a field exist, is unique and the given field is a dense subfield of it. The field **R** of real numbers is introduced in Section 2.3 as the Cauchy completion of the field **Q** of rational numbers. It is shown that **R** is the unique Dedekind complete field. Basic properties of the field **R** are established in the last section of Chapter 2, where it is proved that these properties are equivalent to the completeness property. The names of some of these properties—Bounded Monotone Convergence Property, Bolzano–Weierstrass Property, and Nested Intervals Property—are known to every student of Real Analysis.

Continuous functions on an ordered field and their properties are the main topics of Chapter 3. The chapter opens with Section 3.1, where basic topological structures on an ordered field are introduced. Open and closed subsets are defined and the usual properties of these subsets are established. It is proved that every nonempty open subset of the real field **R** is an at most countable union of pairwise disjoint intervals and that this property characterizes the field **R**. Then, the connectedness property is introduced and shown to characterize the field **R** in the class of ordered fields. Next, the compact sets are introduced, and the Heine–Borel theorem of Real Analysis is proved. It is shown that this theorem characterizes the field **R**—any incomplete ordered field contains a non-compact set that is closed and bounded. Section 3.2 focuses on three fundamental properties of continuous functions on closed bounded intervals: the Intermediate Value, Extreme Value, and Fixed Point properties. It is proved that these properties are satisfied only in the field of real numbers **R**. In other words, each of these properties is equivalent to the completeness property of an ordered field. Uniformly continuous functions are discussed in Section 3.3.

The theory of the derivative for functions on ordered fields is given in Chapter 4. The chapter starts with Section 4.1 covering the limits of functions. The derivative is defined in Section 4.2, where its basic properties are also established. In Section 4.3, the Mean Value, Zero Derivative, Monotonicity, and Darbouxies property of differentiable functions are each shown to be equivalent to the completeness property of the underlying field. The chapter ends with a section on convex functions.

The theories of two integrals—Riemann and Darboux—are treated in Chapter 5. The concept of a partition of an interval is instrumental in the definitions of the integrals. This notion and relevant properties of ordered fields are covered in Section 5.1. The Riemann integral and Riemann integrability are defined in

Section 5.2. It is proved in this section that the Riemann integrability of continuous functions is equivalent to the completeness property. Some standard properties of the Riemann integral are established in Section 5.3 for a general ordered field. Step functions are introduced in Section 5.4, where it is shown that these functions are Riemann integrable and their Riemann integrals are evaluated. These functions are then used in the definitions of the Darboux integral and Darboux integrability in Section 5.5. The connections between Darboux integrability and existence of the Darboux integral are established in the same section. Properties of Darboux integrable functions are exhibited in the last section of the chapter.

The last chapter of the book—Chapter 6—is devoted to infinite series with terms in an ordered field and covers standard topics adopted from Real Analysis. Basic definitions are found in Section 6.1, where it is shown, in particular, that convergent series with terms in "large" ordered fields terminate with zeros. In the rest of the chapter, it is shown that the validity of standard propositions about series depend strongly on properties of the underlying ordered field. A theorem of Real Analysis claims that a series with real non-negative terms converges if and only if the series of partial sums is bounded. It is proved in Section 6.2 that this proposition is equivalent to the completeness property of the field. Convergence of an alternating series is established in Section 6.3 for series with terms in a Cauchy complete ordered field. Relations between convergence and absolute convergence of a series are studied in Section 6.4. It is shown there that an Archimedean ordered field is complete if and only if absolute convergence of series implies their convergence. The Ratio Test is a powerful tool in Real Analysis. It is shown in Section 6.5 that the assertion of the Ratio Test theorem is equivalent to the completeness property of the underlying field.

There are three appendices in the book. Theories of number systems of natural numbers and integers are outlined in Appendix A. I include these topics to complement the theories of rational and real number systems presented in the book. Together, the four number systems form the foundation of Real Analysis. The construction of real numbers by means of Dedekind cuts is an alternative to Cantor's method used in Chapter 2. This construction is outlined in Appendix B, where it is also traced back to Euclid's *Elements*. Some theorems in the book use special properties of ordered fields. It was not my goal to cover the theory of ordered fields in the book. These algebraic structures are introduced in the book on the "need only" basis. In order to help the reader to see the "big picture," I include in Appendix C an overview of ordered fields.

The intended audience of this book includes upper-undergraduate and lower-graduate students interested in the foundations of Real Analysis. Although the proofs in the book are rather elementary, some mathematical maturity is required. In writing this book, I used original publications found only in journals and on the Internet. There are many propositions in Real Analysis that are equivalent to the completeness property. The paper (Deveau and Tiesmann 2013/2014) contains a long list of these propositions together with pertinent references. To keep the size of the book within reasonable limits and proofs accessible to the target audience,

I included in this text only selected results that are relevant to some "traditional" topics of elementary Real Analysis.

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Berkeley, CA, USA Sergei Ovchinnikov
October 2020

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# Chapter 1
# Rational Numbers




In this chapter, we describe the system of rational numbers in the framework of ordered fields. All necessary facts about systems of natural numbers and integers that we use in this chapter are found in Appendix A.

Rational numbers and algebraic operations on them are defined in Sections 1.1 and 1.2.

Section 1.3 is central in the chapter. It focuses on properties of an ordered field **F** and its subfield of rational numbers **Q**. The symbols **F** and **Q** are used throughout the book exclusively for ordered fields and the field of rational numbers.

The number system **Q** is the departure point for many developments in mathematics. However, it has limitations, some of which will be demonstrated in the book. Two of these "deficiencies" of **Q** are discussed in Section 1.4. Another one is found in Section 1.5. These features of the field **Q** indicate the need for a larger field. This issue will be resolved in Chapter 2, where the field **R** of real numbers is introduced.

In Section 1.5, we make the first step towards developing analysis over an ordered field **F** by introducing the notion of a convergent sequence in **F**. We also introduce Cauchy sequences, also known as fundamental sequences. Cauchy sequences are instrumental in constructing real numbers in Chapter 2.

## 1.1 Definitions

In elementary mathematics, a *fraction* is an expression in the form $\frac{m}{n}$ (or $m/n$), where $m$ and $n$ are integers, and $n \neq 0$. Two fractions $\frac{m}{n}$ and $\frac{p}{q}$ are said to be "equivalent" if they have the "same value":

$$\frac{m}{n} = \frac{p}{q},$$

or, which is the same, $mq = pn$.

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Bearing in mind these informal elementary notions, we introduce a formal definition. Recall that $\mathbf{Z}$ denotes the ring of integers (cf. Section A.2).

**Definition 1.1.** A *fraction* is an element of the set $\mathbf{Z} \times (\mathbf{Z} \setminus \{0\})$, that is, a pair of integers $(m, n)$ with $n \neq 0$. The integers $m$ and $n$ are called the *numerator* and the *denominator* of the fraction $(m, n)$, respectively. Two fractions $(m, n)$ and $(p, q)$ are said to be *equivalent* if $mq = pn$. In this case, we write $(m, n) \sim (p, q)$.

The term "equivalent" in Definition 1.1 is justified by the following theorem.

**Theorem 1.1.** *The binary relation* $\sim$ *on* $\mathbf{Z} \times (\mathbf{Z} \setminus \{0\})$ *defined by*

$$(m, n) \sim (p, q) \qquad \textit{if and only if} \qquad mq = pn,$$

*is an equivalence relation.*

*Proof.* We verify the reflexivity, symmetry, and transitivity properties of the relation $\sim$ :

(Reflexivity) $(m, n) \sim (m, n)$ because $mn = mn$.

(Symmetry) If $mq = pn$, then $pn = mq$. Hence,

$$(m, n) \sim (p, q) \qquad \text{implies} \qquad (p, q) \sim (m, n).$$

(Transitivity) Suppose that $(m, n) \sim (p, q)$ and $(p, q) \sim (r, s)$, that is, $mq = pn$ and $ps = rq$. Then

$$(mq)(ps) = (pn)(rq),$$

which implies $ms = rn$. Hence, $(m, n) \sim (r, s)$. □

**Definition 1.2.** A *rational number* is an equivalence class of the relation $\sim$ on the set of fractions, $\mathbf{Z} \times (\mathbf{Z} \setminus \{0\})$. The equivalence class of the fraction $(m, n)$ is denoted by $[m, n]$. The set of rational numbers is denoted by $\mathbf{Q}$.

It is clear that $(mp, np) \sim (m, n)$ for every $p \neq 0$ (cf. Exercise 1.1). In particular, $[m, n] = [-m, -n]$. Therefore, any fraction with a negative denominator is equivalent to a fraction with a positive denominator. In what follows, we sometimes assume that $n$ is a positive integer in $[m, n]$.

## 1.2 Operations on Rational Numbers

In elementary algebra, the sum of two fractions is defined by

$$\frac{m}{n} + \frac{p}{q} = \frac{mq + np}{nq}.$$

We mirror this identity in the following definition.

**Definition 1.3.** The *sum* of two rational numbers $[m, n]$ and $[p, q]$ is given by

$$[m, n] + [p, q] = [mq + np, nq].$$

The following theorem justifies this definition.

**Theorem 1.2.** *If* $(m, n) \sim (m', n')$ *and* $(p, q) \sim (p', q')$*, then*

$$[m, n] + [p, q] = [m', n'] + [p', q'].$$

*In words: The sum of two rational numbers does not depend on the choice of the fractions representing these numbers.*

*Proof.* Suppose that $(m, n) \sim (m', n')$ and $(p, q) \sim (p', q')$, so $mn' = m'n$ and $pq' = p'q$. We have

$$\begin{aligned}
(mq + np)n'q' &= (mq)(n'q') + (np)(n'q') \\
&= (mn')(qq') + (nn')(pq') \\
&= (m'n)(qq') + (nn')(p'q) \qquad \text{(since } mn' = m'n,\ pq' = p'q\text{)} \\
&= (m'q')(nq) + (nq)(n'p') \\
&= (m'q' + n'p')nq.
\end{aligned}$$

Hence, $[mq + np, nq] = [m'q' + n'p', n'q']$, which is the desired result. □

The next definition mirrors the product of fractions in arithmetic.

**Definition 1.4.** The *product* of rational numbers $[m, n]$ and $[p, q]$ is given by

$$[m, n] \cdot [p, q] = [mp, nq].$$

Note that sometimes the symbol $\cdot$ is dropped in products of fractions. The following theorem shows that the product in Definition 1.4 is well defined.

**Theorem 1.3.** *If* $(m, n) \sim (m', n')$ *and* $(p, q) \sim (p', q')$*, then*

$$[m, n][p, q] = [m', n'][p', q'].$$

*Proof.* We need to show that

$$[mp, nq] = [m'p', n'q'].$$

Since $mn' = m'n$ and $pq' = p'q$, we have

$$(mp)(n'q') = (mn')(pq') = (m'n)(p'q) = (m'p')(nq).$$

The result follows. □

It is convenient to use the traditional "fractional notation" for rational numbers. We denote by $\frac{m}{n}$ (sometimes by $m/n$), the equivalence class of the fraction $(m, n)$. In this notation,

$$\frac{m}{n} + \frac{p}{q} = \frac{mq + np}{nq} \qquad \text{and} \qquad \frac{m}{n} \cdot \frac{p}{q} = \frac{mp}{nq}. \tag{1.1}$$

We also have the *Cancellation Property*:

$$\frac{mp}{np} = \frac{m}{n}, \qquad \text{for } p \neq 0, \tag{1.2}$$

because, clearly, $mpn = mnp$.

The function $\varphi : \mathbf{Z} \to \mathbf{Q}$ is defined by

$$\varphi : n \longmapsto \frac{n}{1}. \tag{1.3}$$

It is easy to verify that this function is an *embedding*, that is, it is one-to-one and preserves the operations of addition and multiplication (cf. Exercise 1.2). In what follows, we identify the ring $\mathbf{Z}$ with $\varphi(\mathbf{Z}) \subseteq \mathbf{Q}$ and denote by $n$ ($n \in \mathbf{Z}$) the rational number $n/1$. In particular,

$$\frac{0}{1} = 0 \qquad \text{and} \qquad \frac{1}{1} = 1. \tag{1.4}$$

It is clear that

$$\frac{m}{n} + \frac{-m}{n} = 0, \qquad \text{for all } m \in \mathbf{Z}, n \in \mathbf{Z} \setminus \{0\}.$$

For this reason, the rational number $\frac{-m}{n}$ is called the *additive inverse* (or the *negative*) of $\frac{m}{n}$ and is denoted by $-\frac{m}{n}$.

We use a similar argument for the operation of multiplication. Clearly,

$$\frac{m}{n} \cdot \frac{n}{m} = 1, \qquad \text{for all } m, n \in \mathbf{Z} \setminus \{0\}.$$

We define

$$\left(\frac{m}{n}\right)^{-1} = \frac{n}{m}, \qquad \text{for all } m, n \in \mathbf{Z} \setminus \{0\}$$

and call this rational number the *multiplicative inverse* of $\frac{m}{n}$.

## 1.3 Q as an Ordered Field

The set of rational numbers **Q** together with the operations of addition and multiplication on it enjoys many algebraic properties that we expect of a number system. Among these properties we select nine that characterize the algebraic structure of a field.

**Definition 1.5.** Let **F** be a set together with two binary operations on it, denoted by $+$ and $\cdot$, and called *addition* and *multiplication*, respectively. The algebraic structure $\langle \mathbf{F}, +, \cdot \rangle$ is called a *field* if it has the following properties:

**A1 Commutative Property of Addition:**

$$a + b = b + a \qquad \text{for all } a, b \in \mathbf{F}.$$

**A2 Associative Property of Addition:**

$$(a + b) + c = a + (b + c) \qquad \text{for all } a, b, c \in \mathbf{F}.$$

**A3 Additive Identity:**

there exists an element $0 \in \mathbf{F}$ such that $a + 0 = a$ for all $a \in \mathbf{F}$.

**A4 Additive Inverse:**

for each $a \in \mathbf{F}$, there is an element $-a \in \mathbf{F}$ such that $a + (-a) = 0$.

**M1 Commutative Property of Multiplication:**

$$a \cdot b = b \cdot a \qquad \text{for all } a, b \in \mathbf{F}.$$

**M2 Associative Property of Multiplication:**

$$(a \cdot b) \cdot c = a \cdot (b \cdot c) \qquad \text{for all } a, b, c \in \mathbf{F}.$$

**M3 Multiplicative Identity:**

there exists an element $1 \in \mathbf{F}$ such that $1 \neq 0$ and $a \cdot 1 = a$ for all $a \in \mathbf{F}$.

**M4 Multiplicative Inverse:**

$$\text{for each } a \neq 0 \text{ in } \mathbf{F}, \text{ there is an element } a^{-1} \in \mathbf{F} \text{ such that } a \cdot a^{-1} = 1.$$

**D Distributive Property:**

$$a \cdot (b + c) = a \cdot b + a \cdot c \qquad \text{for all } a, b, c \in \mathbf{F}.$$

The operations of *subtraction* and *division* in a field $\mathbf{F}$ are defined as follows: For $a, b \in \mathbf{F}$, the *difference* $a - b$ is given by

$$a - b = a + (-b).$$

For $a, b \in \mathbf{F}$ with $b \neq 0$, the *quotient* $\dfrac{a}{b}$ is given by

$$\frac{a}{b} = a \cdot b^{-1}.$$

The quotient of $a$ and $b$ is also denoted by $a/b$ or by $a \div b$. Note that both operations are well defined because the elements $-b$ and $b^{-1}$ are unique (cf. Exercise 1.4).

There are many properties of fields that are known from elementary algebra. Some of these properties are found in Exercises 1.4–1.6. The reader is encouraged to verify these properties.

The set $\mathbf{Q}$ equipped with the operations of addition and multiplication is a prototypical example of a field.

**Theorem 1.4.** *The set* $\mathbf{Q}$ *endowed with the binary operations of addition and multiplication defined by* (1.1) *is a field.*

*Proof.* In all rational numbers below the numerators are integers and the denominators are nonzero integers. For the properties of integers used in the proof the reader is referred to Appendix A.

We verify the nine properties listed in Definition 1.5.

**A1**

$$\frac{m}{n} + \frac{p}{q} = \frac{mq + np}{nq} = \frac{pn + qm}{qn} = \frac{p}{q} + \frac{m}{n}.$$

**A2**

$$\left(\frac{m}{n} + \frac{p}{q}\right) + \frac{r}{s} = \frac{mq + np}{nq} + \frac{r}{s} = \frac{mqs + nps + nqr}{nqs}$$
$$= \frac{m}{n} + \frac{ps + qr}{qs} = \frac{m}{n} + \left(\frac{p}{q} + \frac{r}{s}\right).$$

**A3** Cf. (1.4).

$$\frac{m}{n} + 0 = \frac{m}{n} + \frac{0}{1} = \frac{m \cdot 1 + n \cdot 0}{n \cdot 1} = \frac{m}{n}.$$

**A4**

$$\frac{m}{n} + \left(-\frac{m}{n}\right) = \frac{m}{n} + \frac{-m}{n} = \frac{mn + n(-m)}{n^2} = 0.$$

**M1**

$$\frac{m}{n} \cdot \frac{p}{q} = \frac{mp}{nq} = \frac{pm}{qn} = \frac{p}{q} \cdot \frac{m}{n}.$$

**M2**

$$\left(\frac{m}{n} \cdot \frac{p}{q}\right) \cdot \frac{r}{s} = \frac{mp}{nq} \cdot \frac{r}{s} = \frac{mpr}{nqs} = \frac{m}{n} \cdot \frac{pr}{qs} = \frac{m}{n} \cdot \left(\frac{p}{q} \cdot \frac{r}{s}\right).$$

**M3** Cf. (1.4).

$$\frac{m}{n} \cdot 1 = \frac{m}{n} \cdot \frac{1}{1} = \frac{m \cdot 1}{n \cdot 1} = \frac{m}{n}.$$

**M4** Clearly, $[mn, mn] = [1, 1]$. Hence,

$$\frac{m}{n} \cdot \left(\frac{m}{n}\right)^{-1} = \frac{m}{n} \cdot \frac{n}{m} = \frac{mn}{nm} = \frac{1}{1} = 1.$$

**D** By using the cancellation property (cf. (1.2)) in the third step, we have

$$\frac{m}{n} \cdot \left(\frac{p}{q} + \frac{r}{s}\right) = \frac{m}{n} \cdot \frac{ps + qr}{qs} = \frac{mps + mqr}{nqs} = \frac{mpns + nqmr}{(nq)(ns)}$$
$$= \frac{mp}{nq} + \frac{mr}{ns} = \frac{m}{n} \cdot \frac{p}{q} + \frac{m}{n} \cdot \frac{r}{s}.$$

It follows that the algebraic structure $\langle \mathbf{Q}, +, \cdot \rangle$ is a field. □

Because $1 \neq 0$ in a field, there is no one-element field. Here is an example of a field with two elements.

*Example 1.1.* The set $\mathbf{F} = \{0, 1\}$ with the operations $+$ and $\cdot$ defined by

| $+$ | 0 | 1 |
|---|---|---|
| 0 | 0 | 1 |
| 1 | 1 | 0 |

and

| $\cdot$ | 0 | 1 |
|---|---|---|
| 0 | 0 | 0 |
| 1 | 0 | 1 |

is a field (cf. Exercise 1.3). This field is denoted by $\mathbf{Z}_2$ (or by $GF(2)$).

The next example shows that there are fields whose elements do not even resemble "usual" numbers.

*Example 1.2.* Let $\mathbf{Z}[x]$ be the polynomial ring over $\mathbf{Z}$ in the indeterminate $x$ (cf. Example A.1). Define a binary relation $\sim$ on the set $\mathbf{Z}[x] \times (\mathbf{Z}[x] \setminus \{0\})$ by

$$(f, g) \sim (h, k) \qquad \text{if} \qquad fk = hg$$

(cf. Definition 1.1). This relation is an equivalence relation (cf. Exercise 1.7). The equivalence classes of the relation $\sim$ are called *rational functions* over $\mathbf{Z}$ in the indeterminate $x$. As in the case of rational numbers, we denote the equivalence class of the pair $(f, g)$ by $\dfrac{f}{g}$ and write

$$\frac{f}{g} = \frac{a_n x^n + \cdots + a_1 x + a_0}{b_m x^m + \cdots + b_1 x + b_0}$$

if the polynomials $f$ and $g$ are given explicitly. Note that $(f, g) \sim (-f, -g)$. Therefore, we may assume that $g$ in $\dfrac{f}{g}$ is a positive polynomial, that is, $b_m > 0$ (cf. Example A.1). Rational functions 0 and 1 are simply 0 and 1 polynomials. Operations of addition and multiplication are defined by

$$\frac{f}{g} + \frac{h}{k} = \frac{fk + gh}{gk} \quad \text{and} \quad \frac{f}{g} \cdot \frac{h}{k} = \frac{fh}{gk},$$

respectively. The set of all rational functions over $\mathbf{Z}$ in the indeterminate $x$ is denoted by $\mathbf{Z}(x)$. This set with the operations of addition and multiplication defined above is a field (cf. Exercise 1.8).

At the end of Section 1 we noted that "any fraction with a negative denominator is equivalent to a fraction with a positive denominator". Below, we follow this convention.

In elementary algebra, we compare two fractions with positive denominators as follows

$$\frac{m}{n} < \frac{p}{q} \qquad \text{if} \qquad mq < pn.$$

This condition is the basis of the formal definition.

**Definition 1.6.** Let $a = \dfrac{m}{n}$ and $b = \dfrac{p}{q}$ be rational numbers with positive denominators. We put

$$a < b \qquad \text{if} \qquad mq < pn.$$

The following theorem asserts that the definition of the relation $<$ on the set $\mathbf{Q}$ does not depend on the choice of representatives $(m, n)$ and $(p, q)$ of the classes $a$ and $b$ in Definition 1.6.

**Theorem 1.5.** *Let $(m, n)$, $(p, q)$, $(m', n')$, and $(p', q')$ be fractions with positive denominators such that $(m, n) \sim (m', n')$ and $(p, q) \sim (p', q')$. Then*

$$mq < pn \qquad \text{if and only if} \qquad m'q' < p'n'.$$

*Proof.* By symmetry, it suffices to prove that $mq < pn$ implies $m'q' < p'n'$.

Since $(m, n) \sim (m', n')$ and $(p, q) \sim (p', q')$, we have

$$mn' = m'n \qquad \text{and} \qquad pq' = p'q.$$

Multiplying both sides of the inequality $mq < pn$ by the positive integer $n'q'$ (cf. Theorem A.16(ii)), we obtain

$$mqn'q' < pnn'q'.$$

Since $mn' = m'n$ and $pq' = p'q$, we have

$$(m'q')(qn) < (p'n')(qn).$$

Hence, since $qn > 0$, we obtain

$$m'q' < p'n',$$

again by Theorem A.16(ii). □

The binary relations $>$, $\leq$, and $\geq$ on the set $\mathbf{Q}$ are defined by

$$\begin{aligned} a > b \quad &\text{if} \quad b < a, \\ a \leq b \quad &\text{if} \quad a < b \text{ or } a = b, \\ a \geq b \quad &\text{if} \quad a > b \text{ or } a = b, \end{aligned}$$

respectively.

As in the case of the order relation $<$ on the set of integers $\mathbf{Z}$ (cf. Section A.2), Trichotomy and Transitivity properties hold for the order relation $<$ on the set $\mathbf{Q}$.

**Theorem 1.6.**

(a) (Trichotomy Property.) *For any rational numbers $a$, $b$ exactly one of the following holds:*

$$a < b, \qquad a = b, \qquad b < a.$$

(b) (Transitivity Property.) *For any rational numbers $a$, $b$, $c$, if $a < b$ and $b < c$, then $a < c$.*

*Proof.*

(a) Let $a = \dfrac{m}{n}$ and $b = \dfrac{p}{q}$ be rational numbers with $n > 0$, $q > 0$. By Theorem A.14(i), exactly one of the following holds:

$$mq < np, \qquad mq = np, \qquad np < mq.$$

The result follows.

(b) Let $a = \dfrac{m}{n}$, $b = \dfrac{p}{q}$, and $c = \dfrac{r}{s}$ be rational numbers with positive denominators. Suppose that $a < b$ and $b < c$, that is,

$$mq < pn \qquad \text{and} \qquad ps < rq.$$

Since as $s > 0$ and $n > 0$, we have (cf. Theorem A.16(ii))

$$mqs < pns \qquad \text{and} \qquad nps < nrq,$$

that is, $mqs < nrq$ (cf. Theorem A.14(ii)). Hence, $ms < rq$, because $q > 0$. It follows that $a < c$.

□

According to Theorem 1.6, the binary relation $<$ is a linear order on $\mathbf{Q}$ (cf. Definition A.7).

By the Trichotomy Property, for any rational number $a$, there are three mutually exclusive possibilities:

$$a < 0, \qquad a = 0, \qquad a > 0.$$

A rational number $a$ is said to be *negative* if $a < 0$ and *positive* if $a > 0$.

The next theorem shows that addition and multiplication by positive numbers preserve order on $\mathbf{Q}$ (cf. Theorem A.16(ii)).

**Theorem 1.7.**

(a) *$a < b$ if and only if $a + c < b + c$, for all $a, b, c \in \mathbf{Q}$.*
(b) *$a < b$ if and only if $a \cdot c < b \cdot c$, for all $a, b \in \mathbf{Q}$ and $c > 0$ in $\mathbf{Q}$.*

*Proof.* Let $a = \dfrac{m}{n}$, $b = \dfrac{p}{q}$, and $c = \dfrac{r}{s}$ be rational numbers with positive denominators.

(a) The following is a set of equivalent inequalities:

$$\frac{m}{n} + \frac{r}{s} < \frac{p}{q} + \frac{r}{s},$$
$$\frac{ms + rn}{ns} < \frac{ps + qr}{qs},$$
$$msqs + rnqs < psns + qrns$$

By Theorem A.16(i), the last inequality is equivalent to $mq < pn$, which in turn is equivalent to

$$\frac{m}{n} < \frac{p}{q}.$$

(b) Note that $r > 0$ since $c > 0$. The inequality

$$\frac{m}{n} \cdot \frac{r}{s} < \frac{p}{q} \cdot \frac{r}{s}$$

is equivalent to

$$mrqs < prns,$$

which is equivalent to $mq < pn$ (by Theorem A.16(ii)). The latter inequality is equivalent to $\frac{m}{n} < \frac{p}{q}$. □

Theorem 1.7 suggests the following definition.

**Definition 1.7.** A field **F** is said to be an *ordered field* if there is a linear order $<$ on the set **F** such that

(a) $a < b$ if and only if $a + c < b + c$, for all $a, b, c \in \mathbf{F}$, and
(b) $a < b$ if and only if $a \cdot c < b \cdot c$, for all $a, b \in \mathbf{F}$ and $c > 0$ in **F**.

An element $a$ of the field **F** is said to be *positive* if $a > 0$, and *negative* if $a < 0$. The binary relations $>$, $\leq$, and $\geq$ on the set **F** are defined by

$$a > b \quad \text{if} \quad b < a,$$
$$a \leq b \quad \text{if} \quad a < b \text{ or } a = b,$$
$$a \geq b \quad \text{if} \quad a > b \text{ or } a = b,$$

respectively.

By Theorem 1.7, the field **Q** equipped with the linear order $<$ is an ordered field.

A distinguished feature of the field $\mathbf{Q}$ is that it is contained (as a subfield) in every ordered field, as the following theorem asserts. Note that, as before, we consider $\mathbf{Z}$ as a subset of $\mathbf{Q}$.

**Theorem 1.8.** *Let* $\mathbf{F}$ *be an ordered field. There is an embedding* $\tau$ *of the field* $\mathbf{Q}$ *into the field* $\mathbf{F}$*, that is, there is a one-to-one mapping* $\tau : \mathbf{Q} \to \mathbf{F}$ *that preserves the operations of addition and multiplication and the order relation on the set* $\mathbf{Q}$.

*Proof.* First, we define $\tau$ on the set $\mathbf{Z} \subseteq \mathbf{Q}$ recursively by

$$\tau(0) = 0 \qquad \text{and} \qquad \tau(n+1) = \tau(n) + 1 \qquad \text{for } n \in \mathbf{Z}.$$

(Note that $1 \in \mathbf{Z}$ on the left-hand side of the above equation, whereas $1 \in \mathbf{F}$ on the right-hand side.)

Since

$$\tau(n+1) - \tau(n) = 1 > 0$$

(cf. Exercise 1.9), we have $\tau(n) < \tau(n+1)$ for all $n \in \mathbf{Z}$. By induction (cf. Exercise 1.12), $\tau(m) < \tau(n)$ in $\mathbf{F}$ if $m < n$ in $\mathbf{Z}$. Hence, $\tau$ preserves the order relation on $\mathbf{Z}$ and, therefore, is a one-to-one mapping.

We prove that $\tau(m+n) = \tau(m) + \tau(n)$ for $n \geq 0$ by induction. The case $n = 0$ is trivial. Suppose that $\tau(m+n) = \tau(m) + \tau(n)$ holds for some $n > 0$. Then

$$\tau(m+n+1) = \tau(m+n) + 1 = \tau(m) + \tau(n) + 1 = \tau(m) + \tau(n+1),$$

which proves the case $n \geq 0$. For $n < 0$, we have

$$\tau(m) = \tau(m+n+(-n)) = \tau(m+n) + \tau(-n).$$

Note that $\tau(-n) = -\tau(n)$ for all $n \in \mathbf{Z}$ (cf. Exercise 1.10). Therefore,

$$\tau(m+n) = \tau(m) - \tau(-n) = \tau(m) + \tau(n).$$

Hence $\tau$ preserves the operation of addition on $\mathbf{Z}$.

We use induction again to show that $\tau(mn) = \tau(m)\tau(n)$ for $n \geq 0$. The case $n = 0$ is trivial. Suppose that $\tau(mn) = \tau(m)\tau(n)$ for some $n > 0$. We have

$$\begin{aligned} \tau(m(n+1)) &= \tau(mn+m) = \tau(mn) + \tau(m) = \tau(m)\tau(n) + \tau(m) \\ &= \tau(m)(\tau(n) + 1) = \tau(m)\tau(n+1), \end{aligned}$$

which proves the case $n \geq 0$. For $n < 0$, we have (cf. Exercise 1.10)

$$\tau(mn) = -\tau(m(-n)) = -\tau(m)\tau(-n) = \tau(m)\tau(n)$$

We proved that $\tau$ preserves the operation of multiplication on $\mathbf{Z}$. In summary, $\tau$ is an embedding of $\mathbf{Z}$ into $\mathbf{F}$.

Now, we extend the mapping $\tau$ from $\mathbf{Z}$ to $\mathbf{Q}$. Namely, we define

$$\tau\Big(\frac{m}{n}\Big) = \frac{\tau(m)}{\tau(n)}, \qquad \text{for all } m \in \mathbf{Z},\, n \in \mathbf{Z} \setminus \{0\}.$$

This function is well defined. Indeed, suppose that fractions $(m, n)$ and $(p, q)$ are equivalent, that is, $mq = pn$. Then $\tau(m)\tau(q) = \tau(p)\tau(n)$. Therefore (cf. Exercise 1.6 (g)),

$$\tau\Big(\frac{m}{n}\Big) = \frac{\tau(m)}{\tau(n)} = \frac{\tau(p)}{\tau(q)} = \tau\Big(\frac{p}{q}\Big).$$

Moreover, for every $n \in \mathbf{N}$,

$$\tau\Big(\frac{n}{1}\Big) = \frac{\tau(n)}{\tau(1)} = \tau(n).$$

Hence, $\tau$ is indeed an extension of the function defined on $\mathbf{Z}$ in the first part of the proof.

Suppose that $\dfrac{m}{n} < \dfrac{p}{q}$ in $\mathbf{Q}$. We may assume that $n, q > 0$. By the definition of the order $<$ on $\mathbf{Q}$, we have $mq < pn$, so $\tau(m)\tau(q) < \tau(p)\tau(n)$. Since $\tau(n) > 0$ and $\tau(q) > 0$, we obtain

$$\tau\Big(\frac{m}{n}\Big) = \frac{\tau(m)}{\tau(n)} < \frac{\tau(p)}{\tau(q)} = \tau\Big(\frac{p}{q}\Big)$$

(cf. Exercise 1.11). It follows that $\tau$ preserves order on $\mathbf{Q}$ and therefore is one-to-one.

The mapping $\tau : \mathbf{Q} \to \mathbf{F}$ preserves the addition operation on $\mathbf{Q}$. Indeed,

$$\begin{aligned}\tau\Big(\frac{m}{n} + \frac{p}{q}\Big) = \tau\Big(\frac{mq + pn}{nq}\Big) &= \frac{\tau(m)\tau(q) + \tau(p)\tau(n)}{\tau(n)\tau(q)} \\ &= \frac{\tau(m)}{\tau(n)} + \frac{\tau(p)}{\tau(q)} = \tau\Big(\frac{m}{n}\Big) + \tau\Big(\frac{p}{q}\Big),\end{aligned}$$

for all $m, n, p, q \in \mathbf{Z}$ with $n, q \neq 0$.

It also preserves the operation of multiplication on $\mathbf{Q}$, because

$$\tau\Big(\frac{m}{n} \cdot \frac{p}{q}\Big) = \tau\Big(\frac{mp}{nq}\Big) = \frac{\tau(mp)}{\tau(nq)} = \frac{\tau(m)\tau(p)}{\tau(n)\tau(q)} = \frac{\tau(m)}{\tau(n)} \cdot \frac{\tau(p)}{\tau(q)} = \tau\Big(\frac{m}{n}\Big)\tau\Big(\frac{p}{q}\Big)$$

(cf. Exercise 1.6 (d)). □

Theorem 1.8 shows that we can identify rational numbers with their images under the mapping $\tau$ in the ordered field $\mathbf{F}$. Recall that the mappings

$$\mathbf{N} \xrightarrow{\psi} \mathbf{Z} \xrightarrow{\phi} \mathbf{Q}$$

are embeddings (cf. (A.4) and (1.3)). We combine these embeddings with $\tau$ in the diagram.

$$\mathbf{N} \xrightarrow{\psi} \mathbf{Z} \xrightarrow{\phi} \mathbf{Q} \xrightarrow{\tau} \mathbf{F}$$

In what follows, the sets $\mathbf{N}$, $\mathbf{Z}$, and $\mathbf{Q}$ are considered as subsets of every ordered field. According to this convention, natural numbers are integers, integers are rational numbers, and rational numbers are elements of each ordered field.

Another distinguished property of the field $\mathbf{Q}$ is the Archimedean Property.

**Definition 1.8.** An ordered field $\mathbf{F}$ has the *Archimedean Property* if for any two positive elements $x$ and $y$ of $\mathbf{F}$ there is a positive integer $n$ such that $nx > y$. In this case, $\mathbf{F}$ is said to be an *Archimedean field*.

**Theorem 1.9.** *The field* $\mathbf{Q}$ *is an Archimedean field.*

*Proof.* Let $x = \dfrac{p}{q}$ and $y = \dfrac{r}{s}$ be positive elements of $\mathbf{Q}$. We may assume that $p$, $q$, $r$, and $s$ are positive integers. Let $n = q(r+1)$. Then

$$nx = \frac{q(r+1)}{1} \cdot \frac{p}{q} = \frac{(r+1)p}{1} > \frac{r}{s} = y,$$

because $(r+1)ps > r$. □

Not every ordered field is Archimedean, as the following example demonstrates.

*Example 1.3.* Let $\mathbf{Z}(x)$ be the field of rational functions from Example 1.2. For positive polynomials $g$ and $k$ (cf. Example A.1), we define

$$\frac{f}{g} < \frac{h}{k} \qquad \text{if} \qquad fk < gh.$$

Thus, $<$ is a binary relation on $\mathbf{Z}(x)$. It can be easily verified (cf. Exercise 1.14) that $<$ is a linear order on $\mathbf{Z}(x)$ with properties (a) and (b) of Definition 1.7. Hence, $\mathbf{Z}(x)$ is an ordered field. For polynomials 1 and $x$, we have $n1 < x$ for every $n \in \mathbf{N}$. (Formally, $0x + n < 1x + 0$ for every $n \in \mathbf{N}$ because $0 < 1$.) Therefore, the ordered field $\mathbf{Z}(x)$ is not Archimedean.

## 1.4 Limitations of Q

A linear equation $ax = b$ over **Q**, where $a \neq 0$, has a unique solution $x = b/a$. However, as the next theorem shows, there are simple quadratic equations over **Q** that do not have solutions.

**Theorem 1.10.** *The equation $x^2 = 2$ does not have a solution in* **Q**.

*Proof.* Suppose to the contrary that there is $r \in \mathbf{Q}$ such that $r^2 = 2$. Recall that we may assume that $b$ is a positive integer in any fraction $(a, b) \in r$. Let

$$A = \{b \in \mathbf{N} : (a, b) \in r \text{ for some } a \in \mathbf{Z}.\}$$

Because $r^2 = 2$, we must have $a^2 = 2b^2$ for every $(a, b) \in r$. (Note that $(a^2, b^2) \sim (2, 1)$ if and only if $a^2 = 2b^2$.) Hence,

$$A = \{b \in \mathbf{N} : a^2 = 2b^2 \text{ for some } a \in \mathbf{Z}\}.$$

The set $A$ is not empty because we assumed that $r^2 = 2$. By the Well-Ordering Principle (cf. Theorem A.6), the set $A$ contains a least element, say $n$. Then, for some $m$, we have

$$m^2 = 2n^2.$$

Since $m^2 = 2n^2 > n^2$, we have $m - n > 0$. Also, since $m^2 = 2n^2 < 4n^2$, we have $2n - m > 0$. We obtain the desired contradiction from

$$(2n - m)^2 = 4n^2 - 4mn + m^2 = 2m^2 - 4mn + 2n^2 = 2(m - n)^2,$$

because $0 < m - n < n$. □

The results of Theorem 1.10 and Exercise 1.15 suggest that algebraic equations of higher degrees over **Q** may have no solutions in **Q**.

Another deficiency of the field **Q** is demonstrated by an ancient theorem in geometry asserting that the diagonal of a square is incommensurable with its side. Recall that "two (straight) segments are called commensurable if they have a common measure, and incommensurable if such a common measure does not exist" (Kiselev 2006, Section 3.1). Because the diagonal divides the square into two isosceles right triangles, we establish the desired result in the following form.

**Theorem 1.11.** *The hypotenuse and leg of an isosceles right triangle are incommensurable.*

*Proof.* We proceed by contradiction. Suppose the hypotenuse and leg of an isosceles right triangle are commensurable, that is, they have a common measure. Let $ABC$ be an isosceles right triangle with the right angle at the vertex $B$ (cf. Figure 1.1) and

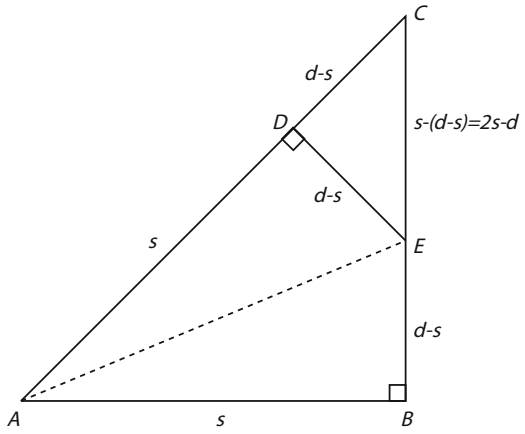



**Fig. 1.1** Proof of Theorem 1.11.

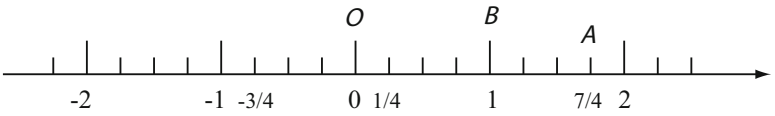



**Fig. 1.2** The number line.

$$|AB| = |BC| = s \qquad \text{and} \qquad |AC| = d,$$

where $s$ and $d$ are positive integers. We mark on the hypotenuse $AC$ the segment $AD$ congruent to the leg $AB$ and draw the line $DE$ through $D$ perpendicular to $AC$. The triangle $CDE$ is clearly isosceles, so $|DE| = d - s$. It is also clear that the triangles $AED$ and $ABE$ are congruent. Hence, $|BE| = d - s$, implying $|EC| = 2s - d$. The isosceles right triangle $ECD$ has sides of integral length that are strictly smaller than those of $ABC$ ($d - s < s$ by the triangle inequality). A applying the same construction to the triangle $ECD$, we obtain an even smaller right triangle with sides of integral length. This construction can be repeated indefinitely, contradicting the Well-Ordering Principle (cf. Theorem A.6). □

Rational numbers are represented in the usual way by points on a number line (cf. Figure 1.2). For instance, the segment $OA$ is commensurable with the unit segment $OB$, so its length is $7/4$.

The drawing in Figure 1.3 reproduces the right triangle $ABC$ from Figure 1.1 with $s = |AB| = 1$ and the line segment $DE$ extended to intersect the number line at point $F$. Because $|AD| = |AB| = 1$, the right triangle $AFD$ is congruent to $ABC$. As we have seen above, the hypotenuse $AF$ is incommensurable with the side $AB$. Therefore, the point $F$ is not representable by a rational number. This example shows that there are "gaps" in the number line representing the rational numbers. Intuitively (think about the Pythagorean Theorem), the point $F$ should represent $\sqrt{2}$, which then should be treated as the "length" of the hypotenuse.

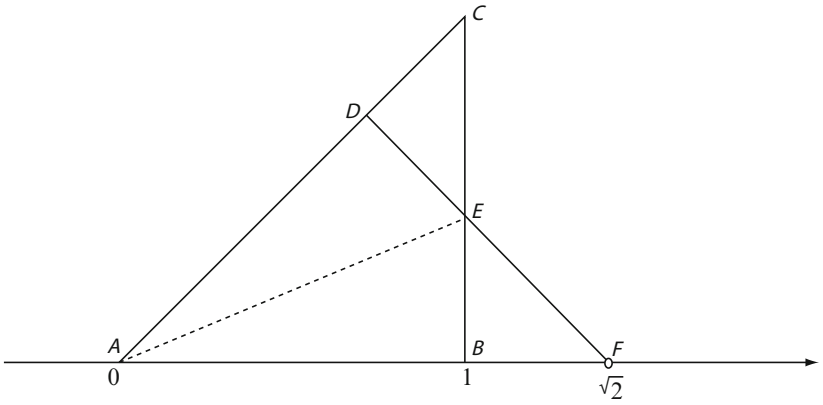



**Fig. 1.3** Plot of $\sqrt{2}$.

## 1.5 Convergence in an Ordered Field

**Definition 1.9.** For every $x \in \mathbf{F}$, the element $|x|$ of $\mathbf{F}$ defined by

$$|x| = \begin{cases} x, & \text{if } x \geq 0, \\ -x, & \text{if } x < 0. \end{cases}$$

is called the *absolute value* of the element $x$.

The following properties of the absolute value characterize an ordered field as a "metric space" with the "distance function" $d(x, y) = |x - y|$:

**Theorem 1.12.** *For all $x, y, z \in \mathbf{F}$,*

$|x - y| = 0$ *if and only if* $x = y$;
$|x - y| = |y - x|$ *symmetry;*
$|x - z| \leq |x - y| + |y - z|$ *triangle inequality;*

The proof of Theorem 1.12 is left as an exercise (cf. Exercise 1.16). Several other properties of the absolute value function $x \mapsto |x|, x \in \mathbf{F}$, are found in Exercise 1.17.

The notion of a sequence is ubiquitous in analysis. We proceed with formal definitions.

**Definition 1.10.** A *sequence* in $\mathbf{F}$ is a function $h : \mathbf{N} \to \mathbf{F}$. If $h(n) = a_n, n \in \mathbf{N}$, then we write the function $h$ as $(a_1, a_2, \ldots)$ or as $(a_n)$. (Note that $n$ in the symbol $(a_n)$ is a "dummy" variable.) Let $a$ be an element of $\mathbf{F}$. The sequence $(a_n)$ where $a_n = a$ for all $n \in \mathbf{N}$ is called a *constant sequence* in $\mathbf{F}$. We denote this sequence by $(a)$.

**Definition 1.11.** A sequence $(a_n)$ in $\mathbf{F}$ is said to be *bounded above* if there is an element $M \in \mathbf{F}$ such that

$$a_n < M, \qquad \text{for all } n \in \mathbf{N}.$$

A sequence $(a_n)$ in $\mathbf{F}$ is said to be *bounded below* if there is an element $M' \in \mathbf{F}$ such that

$$a_n > M', \qquad \text{for all } n \in \mathbf{N}.$$

A sequence is said to be *bounded* if it is bounded above and below.

**Definition 1.12.** Let $(a_n)$ be a sequence in $\mathbf{F}$. An element $a \in \mathbf{F}$ is said to be a *limit* of the sequence $(a_n)$ if, for every positive $\varepsilon \in \mathbf{F}$, there exists $N \in \mathbf{N}$ such that

$$|a_n - a| < \varepsilon, \qquad \text{for all } n > N,$$

or, equivalently,

$$a - \varepsilon < a_n < a + \varepsilon, \qquad \text{for all } n > N.$$

If $a$ is a limit of the sequence $(a_n)$, we say that $(a_n)$ *converges* to $a$ and write

$$a_n \longrightarrow a.$$

If a sequence has a limit, it is said to be *convergent*. Otherwise, it is said to be *divergent*.

**Theorem 1.13.** *A sequence has at most one limit.*

*Proof.* Suppose to the contrary that $(a_n)$ is a convergent sequence such that $a_n \to a$ and $a_n \to b$ for $a \neq b$. Since $a_n \to a$, there is $N' \in \mathbf{N}$ such that

$$|a_n - a| < \frac{|a-b|}{2}, \qquad \text{for all } n > N'.$$

Similarly, $a_n \to b$ means that there is $N'' \in \mathbf{N}$ such that

$$|a_n - b| < \frac{|a-b|}{2}, \qquad \text{for all } n > N''.$$

Let $N$ be the maximum of the numbers $N'$ and $N''$. Then, by the triangle inequality,

$$\begin{aligned} |a-b| = |(a-a_n) + (a_n - b)| &\leq |a_n - a| + |a_n - b| \\ &< \frac{|a-b|}{2} + \frac{|a-b|}{2} = |a-b|, \end{aligned}$$

for $n > N$. This contradiction proves the assertion of the theorem. □

If a sequence $(a_n)$ converges to $a$, that is, $a_n \to a$, we also write

$$\lim a_n = a.$$

The next theorems establish two properties of convergent sequences in **F** that are well known in analysis.

**Theorem 1.14. (The Squeeze Theorem)** *Let $(a_n)$, $(b_n)$, and $(c_n)$ be sequences in* **F** *such that*

$$a_n \le c_n \le b_n, \qquad \text{for all } n \in \mathbf{N}.$$

*If $a_n \to c$ and $b_n \to c$, then $c_n \to c$.*

*Proof.* Let $\varepsilon$ be a positive element of **F**. Because $a_n \to c$ and $b_n \to c$, there is $N \in \mathbf{N}$ (cf. the proof of Theorem 1.13) such that

$$c - \varepsilon < a_n < c + \varepsilon \quad \text{and} \quad c - \varepsilon < b_n < c + \varepsilon, \qquad \text{for all } n > N.$$

Hence,

$$c - \varepsilon < a_n \le c_n \le b_n < c + \varepsilon, \qquad \text{for all } n > N.$$

It follows that the sequence $(c_n)$ converges to $c$. □

**Theorem 1.15. (The Comparison Principle)** *Let $(a_n)$ and $(b_n)$ be sequences in* **F** *such that*

$$a_n \le b_n, \qquad \text{for all } n \in \mathbf{N}.$$

*If $a_n \to a$ and $b_n \to b$, then $a \le b$.*

*Proof.* We proceed by contradiction. Suppose that $a > b$ and let $\varepsilon = (a - b)/2$. Because $a_n \to a$ and $b_n \to b$, there is $N \in \mathbf{N}$ such that

$$a - \varepsilon < a_n < a + \varepsilon \quad \text{and} \quad b - \varepsilon < b_n < b + \varepsilon, \qquad \text{for all } n > N.$$

Consequently,

$$b_{N+1} < b + \varepsilon = a - \varepsilon < a_{N+1},$$

which contradicts our assumption that $a_n \le b_n$ for all $n \in \mathbf{N}$. □

We now proceed to introduce an important class of sequences in **F** that will be used in the construction of real numbers in Chapter 2.

**Definition 1.13.** A sequence $(a_n)$ in $\mathbf{F}$ is said to be a *Cauchy sequence* (or a *fundamental sequence*) if, for every positive $\varepsilon \in \mathbf{F}$, there is $N \in \mathbf{N}$ such that

$$|a_m - a_n| < \varepsilon, \qquad \text{for all } m, n > N.$$

**Theorem 1.16.** *Every convergent sequence is a Cauchy sequence.*

*Proof.* Suppose $a_n \to a$. Then, given $\varepsilon > 0$, there is $N \in \mathbf{N}$ such that

$$|a_n - a| < \frac{\varepsilon}{2}, \qquad \text{for all } n > N.$$

By the triangle inequality,

$$|a_m - a_n| = |(a_m - a) + (a - a_n)| \le |a_m - a| + |a_n - a| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon,$$

for all $m, n > N$. Hence, $(a_n)$ is a Cauchy sequence. □

Note that the converse of Theorem 1.16 does not hold, for instance, in the field $\mathbf{Q}$ (cf. Example 1.4 below).

Basic general properties of Cauchy sequences are subjects of the next two theorems.

**Theorem 1.17.** *Every Cauchy sequence in* $\mathbf{F}$ *is bounded.*

*Proof.* Let $(a_n)$ be a Cauchy sequence in $F$. For $\varepsilon = 1$ there is $N \in \mathbf{N}$ such that

$$|a_m - a_n| < 1, \qquad \text{for all } m, n > N - 1.$$

Let $a$ be the maximum element of the finite set $\{|a_1|, \ldots, |a_N|\}$ (cf. Exercise 1.25). Then

$$|a_n| \le a, \qquad \text{for all } n \le N,$$

and, by the triangle inequality,

$$|a_n| = |a_N + (a_n - a_N)| \le |a_N| + |a_n - a_N| < a + 1, \qquad \text{for all } n > N.$$

Hence, $|a_n| < a + 1$ for all $n \in \mathbf{N}$, so the sequence $(a_n)$ is bounded. □

By Theorems 1.16 and 1.17, we have the following corollary.

**Corollary 1.1.** *Every convergent sequence is bounded.*

**Theorem 1.18.** *If* $(a_n)$ *and* $(b_n)$ *are Cauchy sequences in* $\mathbf{F}$*, then* $(a_n + b_n)$ *and* $(a_n b_n)$ *are also Cauchy sequences in* $\mathbf{F}$.

*Proof.* Let $\varepsilon$ be a positive element of $\mathbf{F}$.

Since $(a_n)$ and $(b_n)$ are Cauchy sequences, there are $N'$ and $N''$ in $\mathbf{N}$ such that

$$|a_m - a_n| < \frac{\varepsilon}{2}, \qquad \text{for all } n, m > N',$$

and

$$|b_m - b_n| < \frac{\varepsilon}{2}, \qquad \text{for all } n, m > N''.$$

Let $N$ be the maximum of $\{N', N''\}$. Then

$$\begin{aligned} |(a_m + b_m) - (a_n + b_n)| &= |(a_m - a_n) + (b_m - b_n)| \\ &\leq |a_m - a_n| + |b_m - b_n| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon, \end{aligned}$$

for all $m, n > N$. Hence, $(a_n + b_n)$ is a Cauchy sequence.

By Theorem 1.17, there are positive elements $A$ and $B$ of $\mathbf{F}$ such that

$$|a_n| < A \quad \text{and} \quad |b_n| < B, \qquad \text{for all } n \in \mathbf{N}.$$

Since $(a_n)$ and $(b_n)$ are Cauchy sequences, there are natural numbers $N'$ and $N''$ such that

$$|a_m - a_n| < \frac{\varepsilon}{2B}, \qquad \text{for all } m, n > N',$$

and

$$|b_m - b_n| < \frac{\varepsilon}{2A}, \qquad \text{for all } m, n > N''.$$

Let $N$ be the maximum of $\{N', N''\}$. Then

$$\begin{aligned} |a_m b_m - a_n b_n| &= |(a_m b_m - a_m b_n) + (a_m b_n - a_n b_n)| \\ &= |a_m(b_m - b_n) + (a_m - a_n)b_n| \\ &\leq |a_m||b_m - b_n)| + |a_m - a_n||b_n| \\ &< A \cdot \frac{\varepsilon}{2A} + \frac{\varepsilon}{2B} \cdot B = \varepsilon, \end{aligned}$$

for all $m, n > N$. Therefore, $(a_n b_n)$ is a Cauchy sequence. □

We will need the following rather special property of bounded sequences in the next chapter.

**Theorem 1.19.** *If $(a_n)$ is a bounded sequence and a sequence $(b_n)$ converges to zero, then the sequence $(a_n b_n)$ converges to zero.*

*Proof.* Let $\varepsilon$ be a positive element of $\mathbf{F}$. Since $(a_n)$ is a bounded sequence, there is a positive element $M \in \mathbf{F}$ such that $|a_n| < M$ for all $n \in \mathbf{N}$. As $\lim b_n = 0$, there is $N \in \mathbf{N}$ such that

$$|b_n| < \frac{\varepsilon}{M}, \qquad \text{for all } n > N.$$

Therefore,

$$|a_n b_n| < M \cdot \frac{\varepsilon}{M} = \varepsilon, \qquad \text{for all } n > N.$$

The result follows from the definition of a limit. □

The next theorem establishes two standard algebraic properties of convergent sequences (cf. Theorem 1.18).

**Theorem 1.20.** *Let $(a_n)$ and $(b_n)$ be convergent sequences in $\mathbf{F}$ and let*

$$\lim a_n = a, \quad \lim b_n = b.$$

*Then the sequences $(a_n + b_n)$ and $(a_n b_n)$ are convergent and*

$$\lim(a_n + b_n) = a + b, \quad \lim(a_n b_n) = ab.$$

*Proof.* Let $\varepsilon$ be a positive element of $\mathbf{F}$.

Since $a_n \to a$ and $b_n \to b$, there are $N'$ and $N''$ in $\mathbf{N}$ such that

$$|a_n - a| < \frac{\varepsilon}{2}, \qquad \text{for all } n > N',$$

and

$$|b_n - b| < \frac{\varepsilon}{2}, \qquad \text{for all } n > N''.$$

Let $N$ be the maximum of $\{N', N''\}$. Then

$$\begin{aligned} |(a_n + b_n) - (a + b)| &= |(a_n - a) + (b_n - b)| \\ &\le |a_n - a| + |b_n - b| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon, \end{aligned}$$

for all $n > N$. Hence, $(a_n + b_n) \to (a + b)$.

By Theorems 1.16 and 1.17, there is a positive element $A \in \mathbf{F}$ such that

$$|a_n| < A \qquad \text{for all } n \in \mathbf{N}.$$

Let $B$ be the maximum of $\{A, |b|\}$. It is clear that $B \geq A > 0$. Since as $a_n \to a$ and $b_n \to b$, there are natural numbers $N'$ and $N''$ such that

$$|a_n - a| < \frac{\varepsilon}{2B}, \qquad \text{for all } n > N',$$

and

$$|b_n - b| < \frac{\varepsilon}{2B}, \qquad \text{for all } n > N''.$$

Let $N$ be the maximum of $\{N', N''\}$. Then

$$\begin{aligned} |a_n b_n - ab| &= |(a_n b_n - a_n b) + (a_n b - ab)| \\ &= |a_n(b_n - b) + (a_n - a)b| \\ &\leq |a_n||b_n - b)| + |a_n - a||b| \\ &< B \cdot \frac{\varepsilon}{2B} + \frac{\varepsilon}{2B} \cdot B = \varepsilon, \end{aligned}$$

for all $n > N$. Therefore, $(a_n b_n) \to (ab)$. □

**Definition 1.14.** A sequence $(a_n)$ in $\mathbf{F}$ is said to be *increasing* if $a_n \leq a_{n+1}$ for all $n \in \mathbf{N}$, and *strictly increasing* if $a_n < a_{n+1}$ for all $n \in \mathbf{N}$.

A sequence $(a_n)$ in $\mathbf{F}$ is said to be *decreasing* if $a_n \geq a_{n+1}$ for all $n \in \mathbf{N}$, and *strictly decreasing* if $a_n > a_{n+1}$ for all $n \in \mathbf{N}$.

A sequence in $\mathbf{F}$ is called *monotone* if it is either increasing or decreasing.

**Definition 1.15.** Let $(a_n)$ be a sequence in $\mathbf{F}$ and let $k_1 < k_2 < \cdots < k_n < \cdots$ be a strictly increasing sequence in $\mathbf{N}$. The sequence

$$(a_{k_n}) = (a_{k_1}, a_{k_2}, \ldots)$$

is called a *subsequence* of $(a_n)$. If $k_1 = m, k_2 = m + 1, \ldots$, where $m \in \mathbf{N}$, then the subsequence

$$(a_m, a_{m+1}, \ldots)$$

is called a *tail* (or more precisely the *m-tail*) of the sequence $(a_n)$.

Note that a sequence is a subsequence (and a tail) of itself.

**Theorem 1.21.**

(a) *A sequence in* $\mathbf{F}$ *is convergent if and only if every subsequence of it is convergent.*
(b) *Similarly, a sequence is convergent if and only if every tail of it is convergent.*

*Proof.* Let $(a_n)$ be a sequence in $\mathbf{F}$ and $(a_{k_n})$ a subsequence of $(a_n)$. Clearly, it suffices to show that $a_n \to a$ implies $a_{k_n} \to a$.

Let $\varepsilon > 0$. There is $N \in \mathbf{N}$ such that $|a_n - a| < \varepsilon$ for all $n > N$. Since $(k_n)$ is a strictly increasing sequence of natural numbers, there is $N' \in \mathbf{N}$ such that $k_n > N$ for all $n > N'$. It follows that $|a_{k_n} - a| < \varepsilon$ for all $n > N'$. Hence, $a_{k_n} \to a$.

The proof of part (b) is left as an exercise (cf. Exercise 1.26). □

The next two theorems will be used later in the book.

**Theorem 1.22.** *Every sequence in* $\mathbf{F}$ *has a monotone subsequence.*

*Proof.* Let $(a_n)$ be a sequence in $\mathbf{F}$ and set

$$S = \{n \in \mathbf{N} : a_m > a_n \text{ for all } m > n\}.$$

Consider the following two mutually exclusive cases:

1. The set $S$ is infinite. Let $k_1 < k_2 < \cdots < k_n < \cdots$ be the elements of the countable set $S$ enumerated in the increasing order (cf. Exercise 1.27). Then $(a_{k_n})$ is an increasing subsequence of $(a_n)$.
2. The set $S$ is finite (or empty). Then there is a natural number $k_1$ such that $k_1 > n$ for all $n \in S$. Because $k_1 \notin S$, there is $k_2 > k_1$ such that $a_{k_2} \le a_{k_1}$. Since $k_2 \notin S$ there is $k_3 > k_2$ such that $a_{k_3} \le a_{k_2}$. By repeating this process, we obtain a decreasing subsequence $(a_{k_n})$ of $(a_n)$.

□

**Theorem 1.23.** *If a Cauchy sequence has a convergent subsequence, then the Cauchy sequence converges to the same limit.*

*Proof.* Let $(a_{k_n})$ be a convergent subsequence of a Cauchy sequence $(a_n)$ and let $a = \lim a_{k_n}$. Then, for $\varepsilon > 0$ in $\mathbf{F}$, there is $N_1 \in \mathbf{N}$ such that

$$|a_{k_n} - a| < \frac{\varepsilon}{2}, \qquad \text{for all } n > N_1.$$

Since $(a_n)$ is a Cauchy sequence, there is $N_2 \in \mathbf{N}$ such that

$$|a_m - a_n| < \frac{\varepsilon}{2}, \qquad \text{for all } m, n > N_2.$$

Because $(k_n)$ is a strictly increasing sequence of natural numbers, there is $N_3 \in \mathbf{N}$ such that

$$k_n > N_2, \qquad \text{for all } n > N_3.$$

Let $N = \max\{N_1, N_2, N_3\}$. By the triangle inequality,

$$|a_n - a| = |(a_n - a_{k_n}) + (a_{k_n} - a)| \le |a_n - a_{k_n}| + |a_{k_n} - a| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon,$$

for all $n > N$. The result follows. □

The next example shows that the converse of Theorem 1.16 does not hold in the field of rational numbers $\mathbf{Q}$.

*Example 1.4.* We define sequences $(a_n)$ and $(b_n)$ in $\mathbf{Q}$ recursively as follows.

First, we set $a_1 = 1$ and $b_1 = 2$. Note that

$$a_1^2 < 2 \leq b_1^2.$$

Suppose that, for some $n \in \mathbf{N}$, $a_n$ and $b_n$ are defined and satisfy

$$a_n^2 < 2 \leq b_n^2.$$

To define $a_{n+1}$ and $b_{n+1}$, we consider two mutually exclusive cases:

(1) $\left(\dfrac{a_n + b_n}{2}\right)^2 < 2$. We set $a_{n+1} = \dfrac{a_n + b_n}{2}$ and $b_{n+1} = b_n$.

(2) $\left(\dfrac{a_n + b_n}{2}\right)^2 \geq 2$. We set $a_{n+1} = a_n$ and $b_{n+1} = \dfrac{a_n + b_n}{2}$.

Note that in both cases

$$a_{n+1}^2 < 2 \leq b_{n+1}^2.$$

It is a straightforward algebraic exercise to verify that:

$$a_n \leq a_{n+1} < b_{n+1} \leq b_n, \qquad \text{for all } n \in \mathbf{N}. \tag{1.5}$$

From this, it easily follows that $a_m \leq a_n$, $b_m \geq b_n$ if $m \leq n$, and $a_m < b_n$ for all $m, n \in \mathbf{N}$. Furthermore,

$$b_{n+1} - a_{n+1} = \frac{1}{2}(b_n - a_n) = \cdots = \frac{1}{2^n}(b_1 - a_1) = \frac{1}{2^n}, \tag{1.6}$$

which can be formally proved by induction.

For $m \geq n > 1$, we have

$$0 \leq a_m - a_n < b_n - a_n = \frac{1}{2^{n-1}} < \frac{1}{n-1} \longrightarrow 0$$

(cf. Exercise 1.21 (a)). Hence, the sequence $(a_n)$ is Cauchy by the Squeeze Theorem (cf. Theorem 1.14).

The sequence $(a_n^2)$ converges to 2. Indeed, we have

$$\begin{aligned} 0 < |2 - a_n^2| = 2 - a_n^2 &\leq b_n^2 - a_n^2 = (b_n - a_n)(b_n + a_n) \\ &\leq \frac{1}{2^{n-1}}(b_1 + 2) = \frac{4}{2^{n-1}} < \frac{4}{n-1} \longrightarrow 0. \end{aligned} \tag{1.7}$$

Therefore, again by the Squeeze Theorem, $\lim a_n^2 = 2$.

Suppose that the sequence $(a_n)$ converges in $\mathbf{Q}$ and $\lim a_n = r$. Then we must have $\lim a_n^2 = r^2$. Hence, $r^2 = 2$, contradicting Theorem 1.10. Therefore, $(a_n)$ is a Cauchy sequence that is not convergent in the ordered field $\mathbf{Q}$.

We need the following definition to give an interpretation of Example 1.4.

**Definition 1.16.** Let $a < b$ be elements of an ordered field $\mathbf{F}$. A *closed bounded interval* in $\mathbf{F}$ is a set of the form

$$[a, b] = \{x \in \mathbf{F} : a \leq x \leq b\}.$$

A family of intervals $\{[a_n, b_n] : n \in \mathbf{N}\}$ is said to be *nested* if

$$[a_1, b_1] \supseteq [a_2, b_2] \supseteq \cdots \supseteq [a_n, b_n] \supseteq \cdots,$$

or, equivalently, if

$$a_m \leq a_n < b_n \leq b_m, \qquad \text{for all } m \leq n.$$

Clearly, any closed bounded interval is a nonempty subset of $\mathbf{F}$. Moreover, any closed bounded interval in $\mathbf{F}$ contains at least countably many elements (cf. Exercise 1.28).

*Example 1.5.* Let $(a_n)$ and $(b_n)$ be sequences in the field $\mathbf{Q}$ from Example 1.4. By (1.5), $[a_n, b_n]$ is a closed bounded interval in $\mathbf{Q}$ and

$$[a_{n+1}, b_{n+1}] \subseteq [a_n, b_n], \qquad \text{for all } n \in \mathbf{N}.$$

Thus, $\{[a_n, b_n] : n \in \mathbf{N}\}$ is a nested family of closed bounded intervals in the field $\mathbf{Q}$. Let $S = \bigcap_{n\in\mathbf{N}} [a_n, b_n]$. By (1.6) $(b_n - a_n) \to 0$, so $S$ cannot contain more than one element. Suppose that $S = \{r\}$ for some $r \in \mathbf{Q}$. It is not difficult to show that then we must have $r = \lim a_n$, contradicting the conclusion of Example 1.4. Therefore,

$$\bigcap_{n\in\mathbf{N}} [a_n, b_n] = \varnothing.$$

## Notes

Equivalence relations are abundant in many mathematical constructions. For instance, we define integers as equivalence classes of pairs of natural numbers (cf. Definition A.5) and rational numbers as classes of equivalent fractions

(cf. Definition 1.2). Real numbers are defined in Chapter 2 as equivalence classes of Cauchy sequences in **Q**. An example from a different branch of mathematics is the definition of a real projective space (cf. Gowers 2008, III.72).

Fields (including ordered fields) are instances of rings and integral domains (cf. Section A.3). These algebraic structures are among central concepts of abstract algebra. Only very basic definitions and properties of fields are introduced in this chapter.

The Archimedean Property of an ordered field (cf. Definition 1.8) stems from the foundations of geometry, where one of the axioms of geometry bears the same name (Hilbert Axiom V, 1950, p. 15; Efimov Axiom IV.1, 1980, p. 69; and Kiselev 2006, p. 119). "Its attribution to Archimedes is purely accidental: in the introduction to his "Quadrature of the Parabola", Archimedes emphasizes that this axiom was used by his predecessors ..." (Bourbaki 1966, p. 408). See also comments on Definition 4 from Euclid's *Elements*, book V, in Section B.2.

The notation $\lim_{n\to\infty} a_n$ is used in many texts on Real Analysis for the limit of a convergent sequence $(a_n)$. We use the more compact notation $\lim a_n$ because it is unambiguous throughout this book.

The French mathematician Augustin-Louis Cauchy (1789–1857) (the last name pronounced *koh-shee*) was an early pioneer of analysis. His analysis textbook, *Cours d'Analyse de l'École Royale Polytechnique* (1821), was very popular in the 19th century and is still a good read.

In Real Analysis, the converse of Theorem 1.16 holds. Namely, every Cauchy sequence in **R** converges (Theorem 2.3 in Chapter 2). Thus Example 1.4 is another instance of deficiency of the number system **Q** (cf. Section 1.4).

Notably, Definition 1.12 (of the limit) does not use all properties of the field **F**. Here is an example of convergence in the ring **Z** of integers.

Let $(a_n)$ be a sequence of integers and $a$ an integer. Suppose that for a positive integer $\varepsilon$, there is $N \in \mathbf{N}$ such that

$$|a_n - a| < \varepsilon, \qquad \text{for all } n > N.$$

For $\varepsilon = 1$, it follows that $a_n = a$ for all $n > N$, that is, the sequence $(a_n)$ is "eventually constant". Clearly, every eventually constant sequence converges. Thus eventually constant sequences are the only convergent sequences in **Z**. Surprisingly, there are ordered fields with the same property (cf. Lemma 4.5).

In some Real Analysis texts, the interval $[a, b]$ is defined for arbitrary elements $a$ and $b$ of the ordered field **F**. Then the empty set and the singletons are intervals. They are called *degenerate* intervals. We assume that $a < b$ in Definitions 1.16 and 3.1 (in Chapter 3), thereby ensuring that all intervals under consideration in this book are non-degenerate.

## Exercises

**1.1.** Show that $(mp, np) \sim (m, n)$ for any $p \neq 0$ (cf. Definition 1.1).

**1.2.** Prove that the function $\varphi$ defined by (1.3) is one-to-one and

$$\varphi(m+n) = \varphi(m) + \varphi(n), \qquad \varphi(m \cdot n) = \varphi(m) \cdot \varphi(n)$$

for all $m, n \in \mathbf{Z}$.

**1.3.** Show that the algebraic structure from Example 1.1 is a field.

**1.4.** Show that in a field $\mathbf{F}$ the additive and multiplicative inverses are unique. That is, prove that for every $a \in \mathbf{F}$ there is a unique element $-a \in \mathbf{F}$ such that $a+(-a) = 0$ and for every nonzero element $a \in \mathbf{F}$ there is a unique element $a^{-1} \in \mathbf{F}$ such that $a \cdot a^{-1} = 1$.

**1.5.** Show that $a \cdot 0 = 0$ for every element $a$ of a field $\mathbf{F}$. (Hint: $0 + 0 = 0$.)

**1.6.** Let $\mathbf{F}$ be a field and let $a$, $b$, $c$, and $d$ be elements of $\mathbf{F}$. Then

*(a)* $1^{-1} = 1$.
*(b)* If $a \neq 0$, $b \neq 0$, then $(a \cdot b)^{-1} = a^{-1} \cdot b^{-1}$ or, equivalently,

$$\frac{1}{a \cdot b} = \frac{1}{a} \cdot \frac{1}{b}.$$

*(c)* If $b \neq 0$, then $c \cdot b = a$ if and only if $c = \dfrac{a}{b}$.
*(d)* $\dfrac{a}{b} \cdot \dfrac{c}{d} = \dfrac{a \cdot c}{b \cdot d}$, provided that $b \cdot d \neq 0$.
*(e)* If $b \cdot c \neq 0$, then $\dfrac{a}{b} = \dfrac{a \cdot c}{b \cdot c}$.
*(f)* $\dfrac{a}{b} + \dfrac{c}{d} = \dfrac{a \cdot d + c \cdot b}{b \cdot d}$, provided that $b \cdot d \neq 0$.
*(g)* If $b \cdot d \neq 0$, then $\dfrac{a}{b} = \dfrac{c}{d}$ if and only if $a \cdot d = c \cdot b$.
*(h)* $\left(\dfrac{a}{b}\right)^{-1} = \dfrac{b}{a}$, provided that $a \cdot b \neq 0$.
*(i)* If $b \neq 0$, $c \neq 0$, and $d \neq 0$, then $\dfrac{a}{b} \div \dfrac{c}{d} = \dfrac{a \cdot b}{b \cdot c}$.

**1.7.** Show that the relation $\sim$ introduced in Example 1.2 is an equivalence relation on $\mathbf{Z}[x] \times (\mathbf{Z}[x] \setminus \{0\})$.

**1.8.** Prove that the set $\mathbf{Z}(x)$ of rational functions in the indeterminate $x$ (cf. Example 1.2) has the algebraic structure of a field.

**1.9.** Show that $0 < 1$ in an ordered field $\mathbf{F}$.

**1.10.** Prove that $\tau(-n) = -\tau(n)$ for all $n \in \mathbf{Z}$, where $\tau$ is the function introduced in the first half of the proof of Theorem 1.8.

**1.11.** Let $a$, $b$, $c$, and $d$ be elements of an ordered field $\mathbf{F}$ such that $b > 0$, $d > 0$. Show that

$$\frac{a}{b} < \frac{c}{d} \qquad \text{if and only if} \quad ad < cb.$$

**1.12.** Let $\{a_1, \ldots, a_n\}$ $(n > 2)$ be a subset of an ordered field such that $a_k < a_{k+1}$ for all $1 \leq k \leq n-1$. Show that $a_1 < a_n$.

**1.13.** Let $\mathbf{F}$ be an Archimedean field. Prove that for every $x \in \mathbf{F}$ there is $m \in \mathbf{Z}$ such that

$$m - 1 \leq x < m.$$

(Hint: suppose first that $x > 0$.)

**1.14.** Prove that $\mathbf{Z}(x)$ is an ordered field (cf. Example 1.2).

**1.15.** Prove that the equation $x^2 = D$, where the positive integer $D$ is not a perfect square, has no solutions in $\mathbf{Q}$. (Hint: There is an integer $t$ such that $t^2 < D < (t+1)^2$, Dedekind 1863.)

**1.16.** Prove Theorem 1.12.

**1.17.** Let $\mathbf{F}$ be an ordered field. Show that for all $a, b \in \mathbf{F}$,

(a) $|a| \geq 0$.
(b) $|a| = 0$ if and only if $a = 0$.
(c) $|-a| = |a|$.
(d) $|a \cdot b| = |a| \cdot |b|$.
(e) $-|a| \leq a \leq |a|$.
(f) $|a| \leq b$ if and only if $-b \leq a \leq b$.
(g) $|a|^2 = a^2$.
(h) $|a - b| \geq \big||a| - |b|\big|$.
(i) $|a + b| = |a| + |b|$ if and only if $ab \geq 0$.

**1.18.** Let $a$, $b$, and $c$ be elements of an ordered field such that $a \leq c$. Prove that

$$a \leq b \leq c \quad \text{if and only if} \quad |a - b| + |b - c| = |a - c|.$$

**1.19.** If $a \leq x \leq b$ and $a \leq y \leq b$ in an ordered field, show that

$$|x - y| \leq b - a.$$

**1.20.** Prove that a sequence $(a_n)$ in an ordered field $\mathbf{F}$ does not converge to $a \in \mathbf{F}$ if and only if there exists $\varepsilon > 0$ in $\mathbf{F}$ such that for every $n \in \mathbf{N}$,

$$|a_k| > \varepsilon, \qquad \text{for some } k > n.$$

**1.21.** Prove that

(a) $\lim(1/n) = 0$,
(b) $\lim(1/n^3) = 0$,
(c) $\lim \dfrac{2+n-n^2}{4+5n+3n^2} = -\dfrac{1}{3}$,

in the field $\mathbf{Q}$.

**1.22.** Show that if $(a_n)$ is Cauchy sequence, then so is the sequence $(|a_n|)$.

**1.23.** Let $(a_n)$ be a convergent sequence in $\mathbf{F}$ and $k \in \mathbf{F}$. Prove that

$$\lim(ka_n) = k \lim a_n.$$

**1.24.** Let $p$ be a positive rational number and $(a_n)$ be a sequence of positive rational numbers such that $a_n^2 \to p$. Show that $(a_n)$ is a Cauchy sequence.

**1.25.** Let $A$ be a finite set of elements of an ordered field. Prove that there are $a, b \in A$ such that $a \leq c \leq b$ for all $c \in A$. The elements $a$ and $b$ are called the *minimum* (denoted by $\min A$) and *maximum* (denoted by $\max A$) elements of $A$, respectively.

**1.26.** Show that if a sequence $(a_n)$ converges to $a \in \mathbf{F}$, so does every tail of $(a_n)$.

**1.27.** Let $S$ be an infinite subset of the set of natural numbers $\mathbf{N}$. Show that there if a mapping $f : \mathbf{N} \to S$ such that $f(n) < f(n+1)$ for all $n \in \mathbf{N}$.

**1.28.** Let $[a, b]$ be a closed interval in an ordered field $\mathbf{F}$. Show that for every natural number $n > 1$,

$$a < \frac{(n-1)\,a + b}{n} < b.$$

# Chapter 2
# Real Numbers




The limitations of the field **Q** that we discussed in the previous chapter are due to the fact that in **Q** there are not enough elements to provide for solutions of simple quadratic equations (cf. Theorem 1.10 and Exercise 1.15), to measure the length of a hypotenuse of a right triangle (cf. Theorem 1.11), and to find a limit of an intuitively convergent Cauchy sequence (cf. Example 1.4).

The first section (Section 2.1) of this chapter introduces "completeness" properties of an ordered field that remedy the aforementioned limitations of the field **Q**. The Archimedean Property and its relation to the completeness properties are also discussed in Section 2.1.

In Section 2.2, we present the construction of the Cauchy completion of an ordered field. This construction is used in Section 2.3 to define the field of real numbers **R** and prove that this field exists and is unique (up to isomorphism).

Fundamental properties of the field **R** are established in Section 2.4, where it is also shown that these properties are equivalent to the (Dedekind) completeness property of an ordered field.

## 2.1 Completeness Properties of Ordered Fields

The first "completeness" property of an ordered field utilizes the notion of the "supremum" of a subset of the field.

**Definition 2.1.** A nonempty subset $E$ of an ordered field $\mathbf{F}$ is said to be *bounded above* if there is an element $b \in \mathbf{F}$, called an *upper bound* of $E$, such that $x \leq b$ for all $x \in E$.

An element $b \in \mathbf{F}$ is called a *supremum* (a *least upper bound*) of a nonempty set $E \subseteq \mathbf{F}$ if

(a) $b$ is an upper bound of $E$, and

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(b) $b \le b'$ for all upper bounds $b'$ of $E$.

Similarly, a nonempty set $E \subseteq \mathbf{F}$ is *bounded below* if there is $c \in \mathbf{F}$ such that $c \le x$ for all $x \in E$. An element $c \in \mathbf{F}$ is called an *infimum* (a *greatest lower bound*) of a nonempty set $E$ if

(c) $c$ is a lower bound of $E$, and
(d) $c \ge c'$ for all lower bounds $c'$ of $E$.

Note that a nonempty subset of $\mathbf{F}$ has at most one supremum and at most one infimum (cf. Exercise 2.1). We denote by $\sup E$ and $\inf E$ the supremum and infimum of a subset $E$, respectively, provided they exist.

*Example 2.1.* For $c \in \mathbf{F}$, let

$$A = \{x \in \mathbf{F} : x < c\}, \quad B = \{x \in \mathbf{F} : x \ge c\}$$

and

$$A' = \{x \in \mathbf{F} : x \le c\}, \quad B' = \{x \in \mathbf{F} : x > c\}.$$

It is not difficult to see that

$$\sup A = \inf B = \sup A' = \inf B' = c.$$

We show only that $\sup A = c$. The other cases are treated similarly. It is clear that every element of $B$ (including $c$) is an upper bound of $A$. For every $x \in A$, we have $(x + c)/2 \in A$ because

$$x < \frac{x + c}{2} < c.$$

It follows that $A$ does not contain its own upper bounds, that is, $\sup A = c$.

*Example 2.2.* Let

$$C = \{x \in \mathbf{Q} : x^2 < 2\}.$$

It is clear that no non-positive rational number is a supremum of $C$. Let $x$ be a positive rational number in $C$, so $2 - x^2 > 0$. Since $\lim(1/n) = 0$ (cf. Exercise 2.5), there is $n \in \mathbf{N}$ such that

$$2x\frac{1}{n} + \frac{1}{n^2} < 2 - x^2,$$

or, equivalently,

$$\left(x + \frac{1}{n}\right)^2 < 2.$$

Hence, $x + 1/n \in C$, implying that $x$ is not a supremum of $C$.

By Theorem 1.10, $x^2 - 2 \neq 0$. Let $x$ be a positive rational number such that $x^2 - 2 > 0$, so $x \notin C$. As in the previous paragraph, there is $n \in \mathbf{N}$ such that

$$2x\frac{1}{n} - \frac{1}{n^2} < x^2 - 2,$$

or, equivalently,

$$\left(x - \frac{1}{n}\right)^2 > 2.$$

Therefore, $x - 1/n \notin C$. Since $x - 1/n < x$, $x$ is not the supremum of $C$.

We conclude that the set $C$ does not have a supremum in $\mathbf{Q}$.

The following property of supremum is instrumental in applications.

**Theorem 2.1.** *An upper bound $b$ of a nonempty set $E \subseteq \mathbf{F}$ is the supremum of $E$ if and only if for each positive $\varepsilon \in \mathbf{F}$ there exists $x \in E$ such that $x > b - \varepsilon$.*

*Proof.* (Necessity.) Suppose to the contrary that $x \leq b - \varepsilon$ for every $x \in E$, where $b = \sup E$ and $\varepsilon > 0$. We obtain a contradiction by noting that $b - \varepsilon < b$.

(Sufficiency.) Suppose that there is an upper bound $b'$ such that $b' < b$. For $\varepsilon = b - b'$ there is $x \in E$ such that $x > b - \varepsilon = b'$, a contradiction. □

**Definition 2.2 (Dedekind Completeness Property).** An ordered field is said to be *Dedekind complete* if every bounded above subset of the field has a supremum.

Equivalently, an ordered field is *Dedekind complete* if every bounded below subset of the field has an infimum (cf. Exercise 2.2).

Example 2.2 shows that the ordered field of rational numbers $\mathbf{Q}$ is not Dedekind complete.

Every Dedekind complete ordered field is Archimedean (cf. Definition 1.8), as the following theorem asserts.

**Theorem 2.2.** *If an ordered field satisfies the Dedekind Completeness Property, then it is Archimedean.*

*Proof.* We proceed by contradiction. Suppose that the Dedekind complete ordered field $\mathbf{F}$ is not Archimedean, that is, there are positive elements $a, b \in \mathbf{F}$ such that $nb \leq a$ for all $n \in \mathbf{N}$. Hence the set $\{nb\}_{n\in\mathbf{N}}$ is bounded. By the Dedekind Completeness Property, there is $c = \sup\{nb\}_{n\in\mathbf{N}}$ in $\mathbf{F}$. Note that $nb < c$ for all $n$, because $(nb)$ is a strictly increasing sequence. By Theorem 2.1 (with $\varepsilon = b$), there is a natural number $n$ such that

$$c - b < nb < (n + 1)b < c.$$

Therefore (cf. Exercise 2.3),

$$b = (n+1)b - b < c - (c-b) = b,$$

a contradiction. □

Example 1.4 suggests another concept of completeness.

**Definition 2.3 (Cauchy Completeness Property).** An ordered field is said to be *Cauchy complete* if every Cauchy sequence in the field converges.

**Theorem 2.3.** *Every Dedekind complete ordered field is Cauchy complete.*

*Proof.* Let $(a_n)$ be a Cauchy sequence in $\mathbf{F}$. By Theorem 1.22, $(a_n)$ contains a monotone subsequence $(a_{k_n})$. Without loss of generality, we may assume that $(a_{k_n})$ is increasing. It is clear that this subsequence is Cauchy. By Theorem 1.17, it is bounded. We define

$$A = \{x_{k_n} : n \in \mathbf{N}\},$$

the range of the subsequence $(a_{k_n})$.

Let $c = \sup A$. Then, for $\varepsilon > 0$ in $\mathbf{F}$, there is $N \in \mathbf{N}$ such that

$$c - \varepsilon < a_{k_N} \leq c$$

(cf. Theorem 2.1). Since $(a_{k_n})$ is an increasing sequence bounded by $c$, we have

$$|a_{k_n} - c| < \varepsilon, \qquad \text{for all } n \geq N,$$

that is, the sequence $(a_{k_n})$ converges to $c$. By Theorem 1.23, the sequence $(a_n)$ converges to $c$. □

The converse of Theorem 2.3 is not true—there exist non-Archimedean Cauchy complete ordered fields (cf. Example 2.3 at the end of the next section). It follows that the Cauchy completeness property is weaker than the Dedekind completeness property. However, the following holds.

**Theorem 2.4.** *Every Cauchy complete and Archimedean ordered field is Dedekind complete.*

The proof of this theorem is broken into several lemmas. First, for a given nonempty bounded above subset $E$ of the field $\mathbf{F}$, we define recursively two sequences $(a_n)$ and $(b_n)$ as follows (cf. Example 1.4).

Let $a_1$ be an element of $E$ and $b_1$ be an upper bound of $E$. Clearly, $a_1 \leq b_1$. Suppose that $a_n$ and $b_n$ are defined and satisfy the conditions:

(a) $a_n \leq b_n$,
(b) $b_n$ is an upper bound of $E$.

To define $a_{n+1}$ and $b_{n+1}$ we consider two mutually exclusive cases:

(1) $\dfrac{a_n + b_n}{2}$ is an upper bound of $E$. Then we set:

$$a_{n+1} = a_n, \qquad b_{n+1} = \frac{a_n + b_n}{2}.$$

(2) $\dfrac{a_n + b_n}{2}$ is not an upper bound of $E$. Then we set:

$$a_{n+1} = \frac{a_n + b_n}{2}, \qquad b_{n+1} = b_n.$$

It is readily verified that $a_{n+1}$ and $b_{n+1}$ satisfy conditions (a) and (b), so the sequences $(a_n)$ and $(b_n)$ satisfying conditions (a) and (b) are well defined.

In what follows, we often consider cases (1) and (2) in the definition of sequences $(a_n)$ and $(b_n)$ separately.

Our goal is to show that both sequences $(a_n)$ and $(b_n)$ converge to the same limit, which is the supremum of the set $E$.

**Lemma 2.1.** *For all $n \in \mathbf{N}$,*

$$b_n - a_n \le \frac{1}{n}(b_1 - a_1).$$

*Proof.* We proceed by induction. The case $n = 1$ is trivial. Suppose that

$$b_n - a_n \le \frac{1}{n}(b_1 - a_1), \qquad \text{for some } n \in \mathbf{N}.$$

Case (1): We have

$$b_{n+1} - a_{n+1} = \frac{a_n + b_n}{2} - a_n = \frac{b_n - a_n}{2} \le \frac{1}{2n}(b_1 - a_1) \le \frac{1}{n+1}(b_1 - a_1),$$

by the induction hypothesis and because $n + 1 \le 2n$ for $n \in \mathbf{N}$.

Case (2): We have

$$b_{n+1} - a_{n+1} = b_n - \frac{a_n + b_n}{2} = \frac{b_n - a_n}{2} \le \frac{1}{2n}(b_1 - a_1) \le \frac{1}{n+1}(b_1 - a_1),$$

by the induction hypothesis and because $n + 1 \le 2n$ for $n \in \mathbf{N}$.

The result follows. □

**Lemma 2.2.** *The sequences $(a_n)$ and $(b_n)$ are increasing and decreasing, respectively, that is,*

$$a_n \le a_{n+1} \quad \textit{and} \quad b_n \ge b_{n+1} \qquad \textit{for all } n \in \mathbf{N}.$$

*Hence,*

$$a_n \le a_m \quad \textit{and} \quad b_n \ge b_m \qquad \textit{for natural numbers } n < m.$$

*Proof.* First, we prove that $a_n \le a_{n+1}$ for all $n \in \mathbf{N}$.

Case (1): Clearly, $a_n = a_{n+1} \le a_{n+1}$.

Case (2):

$$a_{n+1} = \frac{a_n + b_n}{2} \ge \frac{a_n + a_n}{2} = a_n,$$

because $a_n \le b_n$.

For the sequence $(b_n)$ we also consider two cases.

Case (1):

$$b_{n+1} = \frac{a_n + b_n}{2} \le \frac{b_n + b_n}{2} = b_n.$$

Case (2): Clearly, $b_{n+1} = b_n \le b_n$.

The second claim of the lemma follows easily by induction from the transitivity property of the order relation $\le$ (cf. Exercise 2.4). □

**Lemma 2.3.** *For all $m, n \in \mathbf{N}$,*

$$a_m \le b_n.$$

*Proof.* By definition, $a_m \le b_m$ for all $m \in \mathbf{N}$. By Lemma 2.2, we have

$$a_m \le b_m \le b_n, \qquad \text{for } n < m,$$

and

$$a_m \le a_n \le b_n, \qquad \text{for } m \le n.$$

Hence the result. □

**Lemma 2.4.** *The sequences $(a_n)$ and $(b_n)$ are Cauchy sequences.*

*Proof.* We may assume that $a_1 < b_1$. Otherwise, the sequences $(a_n)$ and $(b_n)$ are constant and the assertion of the lemma is trivial.

We prove that $(a_n)$ is a Cauchy sequence. The proof for the sequence $(b_n)$ is similar and left to the reader.

By Lemmas 2.3 and 2.1, we have

$$0 \le a_m - a_n \le b_n - a_n \le \frac{1}{n}(b_1 - a_1), \qquad \text{for } m > n.$$

Let $\varepsilon > 0$ be a real number. Since $\lim(1/n) = 0$ (cf. Exercise 2.5), there is a natural number $N$ such that

$$\frac{1}{n} < \frac{\varepsilon}{b_1 - a_1}, \qquad \text{for all } n > N.$$

It follows that, for all $m > n > N$,

$$|a_m - a_n| < \varepsilon.$$

Therefore, $(a_n)$ is a Cauchy sequence. □

By the Cauchy completeness property, the sequences $(a_n)$ and $(b_n)$ are convergent. Let

$$\lim a_n = a \qquad \text{and} \qquad \lim b_n = b.$$

**Lemma 2.5.** *For all $n \in \mathbf{N}$,*

$$a_n \le a \qquad \textit{and} \qquad b_n \ge b.$$

*Proof.* We prove the claim for the sequence $(a_n)$. The proof for $(b_n)$ is similar and left to the reader.

Suppose that there is $N \in \mathbf{N}$ such that $a_N > a$. Let $\varepsilon = a_N - a > 0$. Then, by Lemma 2.2,

$$a_n - a \ge a_N - a = \varepsilon, \qquad \text{for all } n > N,$$

which contradicts our assumption that $a_n \to a$. □

**Lemma 2.6.** $a = b$.

*Proof.* By Lemmas 2.5 and 2.1,

$$b - a \le b_n - a_n \le \frac{1}{n}(b_1 - a_1), \qquad \text{for all } n \in \mathbf{N}.$$

By the Comparison Principle (cf. Theorem 1.15), $b \le a$, because $(1/n) \to 0$ (cf. Exercise 2.5). By the same principle, $a_n \le b_n$ for all $n \in \mathbf{N}$ implies $a \le b$. Hence, $a = b$. □

**Lemma 2.7.** *For each $n \in \mathbf{N}$ there is $x_n \in E$ such that $x_n \ge a_n$.*

*Proof.* The proof is by induction. For $n = 1$, we set $x_1 = a_1 \in E$.

Suppose that $x_n \in E$ and $x_n \ge a_n$ for some $n \in \mathbf{N}$.

Case (1). We set $x_{n+1} = x_n$. Since $a_{n+1} = a_n$, we have, by the induction hypothesis,

$$x_{n+1} = x_n \geq a_n = a_{n+1}.$$

Case (2). Since $(a_n + b_n)/2$ is not an upper bound of $E$, there exists $x_{n+1} \in E$ such that

$$x_{n+1} > \frac{a_n + b_n}{2} = a_{n+1}.$$

The result follows. □

The next lemma completes the proof of Theorem 2.4.

**Lemma 2.8.** $a = \sup E$.

*Proof.* First, we show that $a$ is an upper bound of $E$. Suppose to the contrary that there is $x \in E$ such that $x > a$ and let $\varepsilon = x - a > 0$. Since $(b_n)$ is a decreasing sequence converging to $a$, there is $N \in \mathbf{N}$ such that

$$b_n - a < \varepsilon, \qquad \text{for all } n > N.$$

Therefore,

$$b_{N+1} < \varepsilon + a = x.$$

We obtained a contradiction because all terms of the sequence $(b_n)$ are upper bounds of $E$.

Now suppose that there is an upper bound $c$ of the set $E$ such that $c < a$. Let $\varepsilon = a - c > 0$. Since $(a_n)$ is an increasing sequence converging to $a$, there is $N \in \mathbf{N}$ such that

$$a - a_n < \varepsilon, \qquad \text{for all } n > N.$$

Hence, $a_{N+1} > a - \varepsilon = c$. By Lemma 2.7, there is $x_{N+1} \in E$ such that $x_{N+1} \geq a_{N+1} > c$. This contradicts our assumption that $c$ is an upper bound of $E$.

It follows that $a$ is the supremum of the set $E$. □

Since Dedekind completeness implies both Cauchy completeness and the Archimedean Property, we have the following result:

**Theorem 2.5.** *An ordered field is Dedekind complete if and only it is Cauchy complete and Archimedean.*

In what follows, we will often simply say that an ordered field is *complete* if it is Dedekind complete.

The main goal of this chapter is to show that there exists a unique (up to isomorphism) complete ordered field. The elements of this field are called *real numbers* and the field is traditionally denoted by $\mathbf{R}$. The elements of the set $\mathbf{R} \setminus \mathbf{Q}$ are called *irrational numbers*. To realize this goal we first show in the next section that for every ordered field $\mathbf{F}$ there exists a unique (up to isomorphism) Cauchy complete ordered field $\widetilde{\mathbf{F}}$ such that $\mathbf{F}$ is isomorphic to a dense (cf. Definition 2.6 below) subfield of $\widetilde{\mathbf{F}}$.

## 2.2 Cauchy Completion of an Ordered Field

We say that two Cauchy sequences $(a_n)$ and $(b_n)$ in the field $\mathbf{F}$ are *equivalent*, and write $(a_n) \sim (b_n)$, if

$$\lim(a_n - b_n) = 0. \tag{2.1}$$

In fact, the relation $\sim$ is an equivalence relation on the set of Cauchy sequences in the field $\mathbf{F}$ (cf. Exercise 2.14).

We define the set $\widetilde{\mathbf{F}}$ as the set of equivalence classes of the relation $\sim$ and denote by $[(a_n)]$ the equivalence class of $\sim$ containing $(a_n)$.

For an element $a \in \mathbf{F}$, the constant sequence $(a)$ converges to $a$. The function $\beta : \mathbf{F} \to \widetilde{\mathbf{F}}$ defined by

$$\beta(a) = [(a)], \qquad a \in \mathbf{F} \tag{2.2}$$

is said to be the *canonical embedding* of $\mathbf{F}$ into $\widetilde{\mathbf{F}}$. It can be readily verified that a sequence $(a_n)$ belongs to the class $[(a)]$ of a constant sequence $(a)$ if and only if $a_n \to a$ (cf. Exercise 2.15). Informally, one can say that the element $\beta(a) \in \widetilde{\mathbf{F}}$ is the set of all sequences in $\mathbf{F}$ converging to $a \in \mathbf{F}$.

Our next goal is to define operations $+$ and $\cdot$ and an order relation $<$ on $\widetilde{\mathbf{F}}$ that are consistent with the canonical embedding $\beta$, that is, we want the following properties:

$$\beta(a + b) = \beta(a) + \beta(b), \qquad \beta(a \cdot b) = \beta(a) \cdot \beta(b),$$

and

$$a < b \qquad \text{implies} \qquad \beta(a) < \beta(b),$$

for all $a, b \in \mathbf{F}$.

We recall the following properties of bounded and Cauchy sequences in $\mathbf{F}$ (cf. Theorems 1.19 and 1.20):

(1) The sum and product of Cauchy sequences are also Cauchy sequences.

(2) If $(a_n)$ is a bounded sequence and $\lim b_n = 0$, then $\lim(a_n \cdot b_n) = 0$.

By Property (1), the operations of addition and multiplication on $\mathbf{F}$ can be extended to Cauchy sequences as follows:

$$(a_n) + (b_n) = (a_n + b_n) \qquad \text{and} \qquad (a_n) \cdot (b_n) = (a_n \cdot b_n).$$

The next step is to extend these operations on Cauchy sequences to the set $\widetilde{\mathbf{F}}$ of equivalence classes of the relation $\sim$. For this, we need the following lemma.

**Lemma 2.9.** *Let $(a_n)$, $(b_n)$, $(c_n)$, and $(d_n)$ be Cauchy sequences in $\mathbf{F}$ such that $(a_n) \sim (b_n)$ and $(c_n) \sim (d_n)$. Then*

(a) $(a_n + c_n) \sim (b_n + d_n)$.
(b) $(a_n \cdot c_n) \sim (b_n \cdot d_n)$.

*Proof.*

(a) Since

$$(a_n + c_n) - (b_n + d_n) = (a_n - b_n) + (c_n - d_n)$$

and $\lim(a_n - b_n) = 0$, $\lim(c_n - d_n) = 0$, we have

$$\lim((a_n + c_n) - (b_n + d_n)) = 0.$$

Hence, $(a_n + c_n) \sim (b_n + d_n)$.

(b) By Property (2), we have

$$\lim(a_n \cdot c_n - a_n \cdot d_n) = \lim(a_n \cdot (c_n - d_n)) = 0,$$

because $(a_n)$ is Cauchy (and hence bounded) and $\lim(c_n - d_n) = 0$. Therefore, $(a_n \cdot c_n) \sim (a_n \cdot d_n)$. A similar argument shows that $(a_n \cdot d_n) \sim (b_n \cdot d_n)$. By the transitivity property of $\sim$, we have $(a_n \cdot c_n) \sim (b_n \cdot d_n)$. □

The above lemma justifies the following definition of operations of addition and multiplication on $\widetilde{\mathbf{F}}$.

**Definition 2.4.** For elements $[(a_n)]$ and $[(b_n)]$ of $\widetilde{\mathbf{F}}$, their *sum* and *product* are defined by

$$[(a_n)] + [(b_n)] = [(a_n + b_n)]$$

and

$$[(a_n)] \cdot [(b_n)] = [(a_n \cdot b_n)],$$

respectively.

Note that the same symbols $+$ and $\cdot$ are used on both $\mathbf{F}$ and $\widetilde{\mathbf{F}}$ for the operations of addition and multiplication. These operations on $\widetilde{\mathbf{F}}$ are extensions of their counterparts on $\mathbf{F}$ in the following sense:

For elements $a$ and $b$ of $\mathbf{F}$, we have (cf. (2.2))

$$\beta(a+b) = [(a+b)] = [(a)] + [(b)] = \beta(a) + \beta(b)$$

and

$$\beta(a \cdot b) = [(a \cdot b)] = [(a)] \cdot [(b)] = \beta(a) \cdot \beta(b).$$

Informally, if we identify elements of $\mathbf{F}$ with the corresponding (under the mapping $\beta$) elements of $\widetilde{\mathbf{F}}$, then sums and products of elements of $\mathbf{F}$ are the same as sums and products of the corresponding elements of $\widetilde{\mathbf{F}}$. This observation justifies the name "canonical embedding" for the mapping $\beta$.

Now we show that the set $\widetilde{\mathbf{F}}$ endowed with operations $+$ and $\cdot$ is a field whose zero and identity are $[(0)]$ and $[(1)]$, respectively. For this we verify the defining properties of a field (cf. Definition 1.5). Let $[(a_n)]$, $[(b_n)]$, and $[(c_n)]$ be elements of $\widetilde{\mathbf{F}}$.

(1) The Commutative Properties of Addition and Multiplication:

$$[(a_n)] + [(b_n)] = [(a_n + b_n)] = [(b_n + a_n)] = [(b_n)] + [(a_n)],$$
$$[(a_n)] \cdot [(b_n)] = [(a_n \cdot b_n) = [(b_n \cdot a_n)] = [(b_n)] \cdot [(a_n)].$$

(2) The Associative Property of Addition:

$$\begin{aligned}[(a_n)] + ([(b_n)] + [(c_n)]) &= [(a_n)] + [(b_n + c_n)] = [(a_n + b_n + c_n)] \\ &= [(a_n + b_n)] + [(c_n)] = ([(a_n)] + [(b_n)]) + [(c_n)].\end{aligned}$$

(3) The Associative Property of Multiplication:

$$\begin{aligned}[(a_n)] \cdot ([(b_n)] \cdot [(c_n)]) &= [(a_n)] \cdot [(b_n \cdot c_n)] = [(a_n \cdot b_n \cdot c_n)] \\ &= [(a_n \cdot b_n)] \cdot [(c_n)] = ([(a_n)] \cdot [(b_n)]) \cdot [(c_n)].\end{aligned}$$

(4) Additive and Multiplicative Identities:

$$[(a_n)] + [(0)] = [(a_n + 0)] = [(a_n)],$$
$$[(a_n)] \cdot [(1)] = [(a_n \cdot 1)] = [(a_n)].$$

(5) Additive Inverses:

$$[(a_n)] + [(-a_n)] = [(a_n + (-a_n)] = [(0)].$$

(6) The Distributive Property:

$$\begin{aligned}[(a_n)] \cdot ([(b_n)] + [(c_n)]) &= [(a_n)] \cdot [(b_n + c_n)] = [(a_n \cdot b_n + a_n \cdot c_n)] \\ &= [(a_n \cdot b_n)] + [(a_n \cdot c_n)] = [(a_n)] \cdot [(b_n)] + [(a_n)] \cdot [(c_n)].\end{aligned}$$

To show that every nonzero element of $\widetilde{\mathbf{F}}$ has the multiplicative inverse, we first establish the following property of Cauchy sequences in $\mathbf{F}$.

**Lemma 2.10.** *If $(a_n)$ is Cauchy sequence in $\mathbf{F}$ that does not converge to zero, then there exist a positive element $a$ in $\mathbf{F}$ and a natural number $N$ such that*

$$|a_n| > a, \qquad \text{for all } n > N.$$

*Proof.* Since the sequence $(a_n)$ does not converge to zero, there exists $\varepsilon > 0$ in $\mathbf{F}$ such that for every $n \in \mathbf{N}$,

$$|a_k| > \varepsilon, \qquad \text{for some } k > n.$$

Because $(a_n)$ is Cauchy, there is $N \in \mathbf{N}$ such that

$$|a_p - a_q| < \frac{\varepsilon}{2}, \qquad \text{for all } p, q > N.$$

By the last two displayed inequalities, there exists $k > N$ such that

$$|a_k| > \varepsilon \quad \text{and} \quad |a_k - a_m| < \frac{\varepsilon}{2}, \qquad \text{for all } m > N.$$

By the triangle inequality,

$$|a_k| = |a_m + (a_k - a_m)| \le |a_m| + |a_k - a_m|.$$

Therefore, for all $m > N$,

$$|a_m| \ge |a_k| - |a_k - a_m| > \varepsilon - \frac{\varepsilon}{2} = \frac{\varepsilon}{2}.$$

The assertion of the lemma follows for $a = \varepsilon/2$. □

(7) Multiplicative Inverses:

If $[(a_n)] \neq [(0)]$, the Cauchy sequence $(a_n)$ does not converge to zero. Then, by the previous lemma, there are a positive $a \in \mathbf{F}$ and $N \in \mathbf{N}$ such that $|a_n| > a$ for all $n > N$. We define a sequence $(a'_n)$ in $\mathbf{F}$ by

$$a'_n = \begin{cases} a, & \text{if } n \le N, \\ a_n, & \text{if } n > N, \end{cases} \qquad \text{for } n \in \mathbf{N}.$$

It is clear that $(a'_n)$ is Cauchy and $(a'_n) \sim (a_n)$, so $[(a_n)] = [(a'_n)]$. Let

$$b_n = \frac{1}{a'_n}, \qquad \text{for } n \in \mathbf{N}.$$

Then

$$[(a_n)] \cdot [(b_n)] = [(a'_n)] \cdot [(b_n)] = [(1)] = 1.$$

Similarly, $[(b_n)] \cdot [(a_n)] = 1$.

We proceed by showing that the order on $\mathbf{F}$ induces an order on $\widetilde{\mathbf{F}}$.

**Lemma 2.11.** *Let $(a_n)$ and $(b_n)$ be Cauchy sequences in $\mathbf{F}$ such that there are $d \in \mathbf{F}$, $d > 0$, and $N \in \mathbf{N}$ satisfying the condition:*

$$a_n + d < b_n, \qquad \textit{for all } n > N.$$

*If $(a'_n)$ and $(b'_n)$ are Cauchy sequences which are equivalent to $(a_n)$ and $(b_n)$, respectively, then there are $d' \in \mathbf{F}$, $d' > 0$, and $N' \in \mathbf{N}$ such that*

$$a'_n + d' < b'_n, \qquad \textit{for all } n > N'.$$

*Proof.* Since $(a'_n) \sim (a_n)$ and $(b'_n) \sim (b_n)$, we have

$$\lim(a_n - a'_n) = 0 \quad \text{and} \quad \lim(b_n - b'_n) = 0.$$

Therefore, there exists $N \in \mathbf{N}$ such that

$$|a_n - a'_n| < \frac{d}{4} \quad \text{and} \quad |b_n - b'_n| < \frac{d}{4} \qquad \text{for all } n > N.$$

Hence, for all $n > N$,

$$-\frac{d}{4} < b_n - b'_n < \frac{d}{4}$$

and

$$-\frac{d}{4} < a'_n - a_n < \frac{d}{4}.$$

By adding the displayed inequalities, we obtain

$$-\frac{d}{2} < (b_n - a_n) - (b'_n - a'_n) < \frac{d}{2}, \qquad \text{for all } n > N.$$

Therefore,

$$b'_n - a'_n > (b_n - a_n) - \frac{d}{2} > d - \frac{d}{2} = \frac{d}{2},$$

for all $n > N$. The result follows if we set $d' = \dfrac{d}{2}$ and $N' = N$. □

The result of the previous lemma justifies the following definition.

**Definition 2.5.** Let $[(a_n)]$ and $[(b_n)]$ be elements of $\widetilde{\mathbf{F}}$. We write

$$[(a_n)] < [(b_n)]$$

if and only if there are a positive $d \in \mathbf{F}$ and $N \in \mathbf{N}$ such that

$$a_n + d < b_n, \qquad \text{for all } n > N.$$

Note that we use the same symbol $<$ for the order relation on $\widetilde{\mathbf{F}}$ that we used for the order relation on $\mathbf{F}$. This can be justified as follows. Let $\beta : \mathbf{F} \to \widetilde{\mathbf{F}}$ be the canonical embedding and $a$, $b$ be elements of $\mathbf{F}$. Then

$$a < b \qquad \text{if and only if} \qquad \beta(a) < \beta(b).$$

In other words, if we identify elements of $\mathbf{F}$ with their images under the canonical embedding, then the order relation on $\widetilde{\mathbf{F}}$ can be regarded an extension of the order relation on $\mathbf{F}$. This assertion follows immediately from the definition of the canonical embedding.

In what follows, we show that $<$ is a linear order on $\widetilde{\mathbf{F}}$, making it an ordered field.

**Theorem 2.6.** *The binary relation $<$ is a linear order on $\widetilde{\mathbf{F}}$.*

*Proof.* We need to show that $<$ satisfies the transitivity condition and the Trichotomy Law.

1. *Transitivity.* If $[(a_n)] < [(b_n)]$ and $[(b_n)] < [(c_n)]$, then there are positive elements $d'$, $d''$ of $\mathbf{F}$ and natural numbers $N'$, $N''$ such that

   $$a_n + d' < b_n, \qquad \text{for all } n > N'$$

   and

   $$b_n + d'' < c_n, \qquad \text{for all } n > N''.$$

   For $n > N$, where $N = \max\{N', N''\}$, we have

$$a_n + d' < b_n < c_n - d''.$$

Therefore, for $d = d' + d''$, we have $d > 0$ and

$$a_n + d < c_n \qquad \text{for all } n > N.$$

Hence, $[(a_n)] < [(c_n)]$, so $<$ is a transitive relation on $\mathbf{F}$.

2. *Trichotomy Law*. Suppose that $[(a_n)] \neq [(b_n)]$, so the Cauchy sequence $(b_n - a_n)$ does not converge to zero. By Lemma 2.10, there are a positive $d$ in $\mathbf{F}$ and a natural number $N'$ such that

$$|b_n - a_n| > d, \qquad \text{for all } n > N'.$$

Since $(b_n - a_n)$ is a Cauchy sequence, there is $N'' \in \mathbf{N}$ such that

$$-\frac{d}{2} < (b_n - a_n) - (b_m - a_m) < \frac{d}{2}, \qquad \text{for all } m, n > N''.$$

In the rest of the proof, $N = \max\{N', N''\}$. Since $|b_{N+1} - a_{N+1}| > d$, there are two mutually exclusive cases:

Case 1. $b_{N+1} - a_{N+1} > d$. Then, for all $n > N$,

$$b_n - a_n > (b_{N+1} - a_{N+1}) - \frac{d}{2} > d - \frac{d}{2} = \frac{d}{2},$$

that is, $[(a_n)] < [(b_n)]$.

Case 2. $b_{N+1} - a_{N+1} < -d$. Then, for all $n > N$,

$$b_n - a_n < (b_{N+1} - a_{N+1}) + \frac{d}{2} < -d + \frac{d}{2} = -\frac{d}{2},$$

that is, $[(a_n)] > [(b_n)]$.

Note that $[(a_n)] = [(b_n)]$ means that $(b_n - a_n) \to 0$. Hence, we cannot have simultaneously $[(a_n)] = [(b_n)]$ and $[(a_n)] < [(b_n)]$, or $[(a_n)] = [(b_n)]$ and $[(b_n)] < [(a_n)]$. □

**Theorem 2.7.** *The field $\widetilde{\mathbf{F}}$ endowed with the linear order $<$ is an ordered field.*

*Proof.* We need to verify that:

(a) For all elements $[(a_n)]$, $[(b_n)]$, and $[(c_n)]$ of $\widetilde{\mathbf{F}}$, $[(a_n)] < [(b_n)]$ if and only if $[(a_n)] + [(c_n)] < [(b_n)] + [(c_n)]$

(b) For all elements $[(a_n)]$, $[(b_n)]$, and $[(c_n)] > [(0)]$ of $\widetilde{\mathbf{F}}$, $[(a_n)] < [(b_n)]$ if and only if $[(a_n)] \cdot [(c_n)] < [(b_n)] \cdot [(c_n)]$.

(a) (Necessity.) If $[(a_n)] < [(b_n)]$, then there exists a positive $d$ in $\mathbf{F}$ and $N \in \mathbf{N}$ such that $a_n + d < b_n$ for all $n > N$. Since $\mathbf{F}$ is an ordered field, we have

$a_n + c_n + d < b_n + c_n$ for all $n > N$. As $[(a_n)] + [(c_n)] = [(a_n + c_n)]$ and $[(b_n)] + [(c_n)] = [(b_n + c_n)]$, we obtain

$$[(a_n)] + [(c_n)] < [(b_n)] + [(c_n)].$$

(Sufficiency.) If $[(a_n)] + [(c_n)] < [(b_n)] + [(c_n)]$, then, by the argument in the preceding paragraph,

$$[(a_n)] + [(c_n)] + [(-c_n)] < [(b_n)] + [(c_n)] + [(-c_n)],$$

which implies $[(a_n)] < [(b_n)]$.

(b) Note first that $[(c_n)] > 0$ if and only if there is a positive element $c$ of $\mathbf{F}$ and $N' \in \mathbf{N}$ such that $c_n > c$ for all $n > N'$.

(Necessity.) If $[(a_n)] < [(b_n)]$, then there exists a positive $d$ in $\mathbf{F}$ and $N'' \in \mathbf{N}$ such that $a_n + d < b_n$ for all $n > N''$. Since $\mathbf{F}$ is an ordered field, we have

$$a_n c_n + dc < a_n c_n + d c_n < b_n c_n, \qquad \text{for all } n > \max\{N', N''\}.$$

It follows that $[(a_n)] \cdot [(c_n)] < [(b_n)] \cdot [(c_n)]$.

(Sufficiency.) Suppose that $[(a_n)] \cdot [(c_n)] < [(b_n)] \cdot [(c_n)]$, that is, there exist a positive element $d$ of $\mathbf{F}$ and $N' \in \mathbf{N}$ such that

$$b_n c_n - a_n c_n > d, \qquad \text{for all } n > N'.$$

Since as $[(c_n)] > 0$ and $(c_n)$ is bounded, there are an $N'' \in \mathbf{N}$ and a positive element $C$ of $\mathbf{F}$ such that $0 < c_n < C$ for all $n > N''$. Let $N = \max\{N', N''\}$. Then

$$b_n - a_n > \frac{d}{c_n} > \frac{d}{C}, \qquad \text{for all } n > N.$$

It follows that $[(a_n)] < [(b_n)]$.

□

We have proved that $\widetilde{\mathbf{F}}$ is an ordered field. Now we prove that this field is Cauchy complete.

Recall that the mapping $\beta$ "preserves" operations and order on $\mathbf{F}$, in the following sense:

$$\beta(a + b) = \beta(a) + \beta(b), \qquad \beta(a \cdot b) = \beta(a) \cdot \beta(b),$$

and

$$a < b \qquad \text{if and only if} \qquad \beta(a) < \beta(b),$$

for all $a, b \in \mathbf{F}$. Therefore, $\beta(\mathbf{F})$ is an ordered subfield of the field $\widetilde{\mathbf{F}}$. In what follows, we denote $\widetilde{a} = \beta(a)$ for $a \in \mathbf{F}$.

To establish the Cauchy completeness of $\widetilde{\mathbf{F}}$, we need to show that every Cauchy sequence in $\widetilde{\mathbf{F}}$ is convergent. We prove it first for the sequences in $\beta(\mathbf{F})$.

**Lemma 2.12.** *A sequence $(\widetilde{a}_n)$ is a Cauchy sequence in $\widetilde{\mathbf{F}}$ if and only if $(a_n)$ is a Cauchy sequence in $\mathbf{F}$.*

*Proof.* (Necessity.) Let $\varepsilon$ be a positive element of $\mathbf{F}$. Clearly, $\widetilde{\varepsilon}$ is a positive element of $\widetilde{\mathbf{F}}$. Since $(\widetilde{a}_n)$ is Cauchy, there is $N \in \mathbf{N}$ such that

$$-\widetilde{\varepsilon} < \widetilde{a}_n - \widetilde{a}_m < \widetilde{\varepsilon} \qquad \text{for all } m, n > N,$$

or, equivalently,

$$\beta(-\varepsilon) < \beta(a_n) - \beta(a_m) < \beta(\varepsilon), \qquad \text{for all } m, n > N.$$

By the properties of the canonical embedding $\beta$,

$$-\varepsilon < a_n - a_m < \varepsilon, \qquad \text{for all } m, n > N.$$

Hence, the sequence $(a_n)$ is Cauchy.

(Sufficiency.) Let $(a_n)$ be a Cauchy sequence in $\mathbf{F}$ and $\varepsilon > 0$ in $\widetilde{\mathbf{F}}$. Since $\varepsilon = [(e_n)] > [(0)]$ in $\widetilde{\mathbf{F}}$, there is a $c > 0$ in $\mathbf{F}$ and $N' \in \mathbf{N}$ such that $e_n > 2c$ for all $n > N'$. Then, for $n > N'$, we have $e_n > c + c$ in $\mathbf{F}$ implying that $\varepsilon = [(e_n)] > \widetilde{c}$ in $\widetilde{\mathbf{F}}$.

Since the sequence $(a_n)$ is Cauchy, there is $N \in \mathbf{N}$ such that

$$-c < a_n - a_m < c, \qquad \text{for all } m, n > N.$$

By the properties of the canonical embedding,

$$-\varepsilon < -\widetilde{c} < \widetilde{a}_n - \widetilde{a}_m < \widetilde{c} < \varepsilon, \qquad \text{for all } m, n > N.$$

Hence the sequence $(\widetilde{a}_n)$ is Cauchy. □

**Theorem 2.8.** *Every Cauchy sequence $(\widetilde{a}_n)$ is convergent in $\widetilde{\mathbf{F}}$, with the limit $[(a_n)]$.*

*Proof.* Let $\varepsilon = [(e_n)]$ be a positive element of the field $\widetilde{\mathbf{F}}$. We need to show that there is $N \in \mathbf{N}$ such that

$$|\widetilde{a}_n - [(a_m)]| < \varepsilon, \qquad \text{for all } n > N.$$

(Note that $m$ is a "dummy" variable in the symbol $[(a_m)]$.)

As in the proof of Lemma 2.12, there is $c \in \mathbf{F}$ such that $0 < \widetilde{c} < \varepsilon$.

Since $(a_n)$ is a Cauchy sequence in $\mathbf{F}$ (cf. Lemma 2.12), there is $N \in \mathbf{N}$ such that

$$-\frac{c}{2} < a_n - a_m < \frac{c}{2}, \qquad \text{for all } m, n > N.$$

Then, for a given $n > N$, we have

$$a_n - a_m + \frac{c}{2} < c, \qquad \text{for all } m > N,$$

which implies that

$$\widetilde{a}_n - [(a_m)] < \widetilde{c} < \varepsilon \qquad \text{in } \widetilde{\mathbf{F}}.$$

On the other hand,

$$-c + \frac{c}{2} = -\frac{c}{2} < a_n - a_m, \qquad \text{for all } m > N,$$

which implies

$$-\varepsilon < -\widetilde{c} < \widetilde{a}_n - [(a_m)].$$

Hence, for all $n > N$,

$$|\widetilde{a}_n - [(a_m)]| < \varepsilon,$$

which is desired result. □

**Definition 2.6.** A nonempty subset $E$ of an ordered field $\mathbf{F}$ is said to be *dense* in $\mathbf{F}$ if for every $a \in \mathbf{F}$ and a positive $\varepsilon$ in $\mathbf{F}$ there is $x \in E$ such that

$$|x - a| < \varepsilon.$$

The next theorem is a consequence of Lemma 2.12 and Theorem 2.8 (cf. Exercise 2.17).

**Theorem 2.9.** *The subfield $\beta(\mathbf{F})$ is dense in $\widetilde{\mathbf{F}}$, that is, for every $[(a_n)] \in \widetilde{\mathbf{F}}$ and a positive element $\varepsilon$ of $\widetilde{\mathbf{F}}$, there is an element $\widetilde{a}$ of $\widetilde{\mathbf{F}}$ such that*

$$|[(a_n)] - \widetilde{a}| < \varepsilon.$$

Now we establish Cauchy completeness of the ordered field $\widetilde{\mathbf{F}}$. First, we use the result of Theorem 2.9 to "approximate" terms of a Cauchy sequence in $\widetilde{\mathbf{F}}$ by elements of $\beta(\mathbf{F})$. Then, Theorem 2.8 is used to construct the limit of the original sequence.

**Theorem 2.10.** *Every Cauchy sequence in $\widetilde{\mathbf{F}}$ is convergent.*

*Proof.* Let $(x_n)$ be a Cauchy sequence in $\widetilde{\mathbf{F}}$. We need to show that there exists $a \in \widetilde{\mathbf{F}}$ such that for any positive element $\varepsilon$ of $\widetilde{\mathbf{F}}$ there is a natural number $N$ such that

$$|x_n - a| < \varepsilon, \qquad \text{for all } n > N.$$

By Theorem 2.9, for each $n \in \mathbf{N}$ there is an element $a_n$ of $\mathbf{F}$ such that

$$|x_n - \widetilde{a}_n| < \frac{\varepsilon}{3}.$$

Since $(x_n)$ is a Cauchy sequence, there is $N_1 \in \mathbf{N}$ such that

$$|x_n - x_m| < \frac{\varepsilon}{3}, \qquad \text{for all } m, n > N_1.$$

By the triangle inequality,

$$\begin{aligned} |\widetilde{a}_n - \widetilde{a}_m| &= |(\widetilde{a}_n - x_n) + (x_n - x_m) + (x_m - \widetilde{a}_m)| \\ &\le |\widetilde{a}_n - x_n| + |x_n - x_m| + |x_m - \widetilde{a}_m| < \frac{\varepsilon}{3} + \frac{\varepsilon}{3} + \frac{\varepsilon}{3} = \varepsilon, \end{aligned}$$

for all $m, m > N_1$. Hence, $(\widetilde{a}_n)$ is a Cauchy sequence in $\widetilde{\mathbf{F}}$. By Theorem 2.8, this sequence has a limit $a = [(a_n)]$ in $\widetilde{\mathbf{F}}$.

Now we show that $x_n \to a$ in $\widetilde{\mathbf{F}}$. As $\widetilde{a}_n \to a$, there is $N_2 \in \mathbf{N}$ such that

$$|\widetilde{a}_n - a| < \frac{2\varepsilon}{3}, \qquad \text{for all } n > N_2.$$

Therefore, for $n > \max\{N_1, N_2\}$, we have, by the triangle inequality,

$$|x_n - a| = |(x_n - \widetilde{a}_n) + (\widetilde{a}_n - a)| < \frac{\varepsilon}{3} + \frac{2\varepsilon}{3} = \varepsilon.$$

Hence, $x_n \to a$ in $\widetilde{\mathbf{F}}$. □

To complete our program, we need to establish the uniqueness property of the field $\widetilde{\mathbf{F}}$. We begin with a formal definition of "embedding"—a term that was rather informally used in Chapter 1.

**Definition 2.7.** Let $\mathbf{F}$ and $\mathbf{G}$ be ordered fields. A mapping

$$\varphi : \mathbf{F} \longrightarrow \mathbf{G}$$

is called an *embedding of* $\mathbf{F}$ *into* $\mathbf{G}$ if

$$\varphi(a + b) = \varphi(a) + \varphi(b), \quad \varphi(a \cdot b) = \varphi(a) \cdot \varphi(b),$$

and

$$a < b \qquad \text{if and only if} \qquad \varphi(a) < \varphi(b),$$

for all $a, b \in \mathbf{F}$.

It is not difficult to verify that an embedding $\varphi : \mathbf{F} \to \mathbf{G}$ is a one-to-one mapping such that $\varphi(0) = 0$, $\varphi(1) = 1$, and that $\varphi(\mathbf{F})$ is an ordered subfield of $\mathbf{G}$ (cf. Exercise 2.18).

**Definition 2.8.** An embedding $\varphi : \mathbf{F} \to \mathbf{G}$ is said to be a *dense embedding* if $\varphi(\mathbf{F})$ is a dense ordered subfield of $\mathbf{G}$.

The canonical embedding $\beta$ is an instance of a dense embedding (cf. Theorem 2.9).

**Definition 2.9.** An ordered field $\mathbf{G}$ is said to be a (Cauchy) *completion* of an ordered field $\mathbf{F}$ if (1) $\mathbf{G}$ is Cauchy complete and (2) there is a dense embedding $\mathbf{F} \to \mathbf{G}$.

Theorems 2.9 and 2.10 claim that the field $\widetilde{\mathbf{F}}$ is a completion of the field $\mathbf{F}$. The next step is to show that this completion is unique up to isomorphism (cf. Theorem 2.11 below).

**Definition 2.10.** An embedding $\varphi : \mathbf{F} \to \mathbf{G}$ is said to be an *isomorphism of the ordered field* $\mathbf{F}$ *onto the ordered field* $\mathbf{G}$ if $\varphi$ is a bijection. In this case, we say that $\mathbf{F}$ and $\mathbf{G}$ are *isomorphic*.

To establish the uniqueness property of completion, we need two properties of dense embeddings.

**Lemma 2.13.** *Let $\varphi : \mathbf{F} \to \mathbf{G}$ be a dense embedding. Then*

(a) *$(\varphi(a_n))$ is a Cauchy sequence in $\mathbf{G}$ if and only if $(a_n)$ is Cauchy in $\mathbf{F}$.*
(b) *$\varphi(a_n) \to 0$ in $\mathbf{G}$ if and only if $a_n \to 0$ in $\mathbf{F}$.*

*Proof.*

(a) (Necessity.) Let $(\varphi(a_n))$ be a Cauchy sequence in $\mathbf{G}$. For $\varepsilon > 0$ in $\mathbf{F}$, $\varphi(\varepsilon) > 0$ in $\mathbf{G}$. There is $N \in \mathbf{N}$ such that

$$-\varphi(\varepsilon) < \varphi(a_m) - \varphi(a_n) < \varphi(\varepsilon), \qquad \text{for all } m, n > N.$$

As $\varphi$ is an embedding,

$$-\varepsilon < a_m - a_n < \varepsilon, \qquad \text{for all } m, n > N.$$

Therefore, $(a_n)$ is Cauchy in $\mathbf{F}$.

(Sufficiency.) Let $(a_n)$ be a Cauchy sequence in $\mathbf{F}$ and $\varepsilon$ a positive element of $\mathbf{G}$. By the density property of $\varphi$, there is $\delta > 0$ in $\mathbf{F}$ such that $0 < \varphi(\delta) < \varepsilon$ (cf. Exercise 2.16). Since $(a_n)$ is Cauchy, there is $N \in \mathbf{N}$ such that

$$-\delta < a_m - a_n < \delta, \qquad \text{for all } m, n > N.$$

As $\varphi$ is an embedding,

$$-\varepsilon < \varphi(-\delta) < \varphi(a_m) - \varphi(a_n) < \varphi(\delta) < \varepsilon, \qquad \text{for all } m, n > N.$$

Hence, $(\varphi(a_n))$ is a Cauchy sequence in $\mathbf{G}$.

(b) The proof essentially repeats the steps of part (a) and is left to the reader (cf. Exercise 2.19). □

**Theorem 2.11.** *Let* $\mathbf{F}$ *be an ordered field,* $\mathbf{G}$ *a Cauchy complete ordered field, and* $\varphi : \mathbf{F} \to \mathbf{G}$ *a dense embedding. Then* $\mathbf{G}$ *is isomorphic to* $\widetilde{\mathbf{F}}$.

*Proof.* Let $x \in \widetilde{\mathbf{F}}$. Then $x = [(a_n)]$ for a Cauchy sequence $(a_n)$ in $\mathbf{F}$. By Lemma 2.13 (a), $(\beta(a_n))$ is a Cauchy sequence in $\widetilde{\mathbf{F}}$. By Theorem 2.8, $\lim \beta(a_n) = x$. By Lemma 2.13 (a) again, the sequence $(\varphi(a_n))$ is Cauchy in $\mathbf{G}$. Since $\mathbf{G}$ is complete, this sequence converges to some element $y \in \mathbf{G}$, that is, $\varphi(a_n) \to y$.

Suppose that $x = [(b_n)]$, so the Cauchy sequence $(b_n)$ is equivalent to $(a_n)$, that is, $(a_n - b_n) \to 0$ in $\mathbf{F}$. By Lemma 2.13 (b), $\varphi(a_n - b_n) \to 0$, that is, $\varphi(b_n) \to y$. Thus, the correspondence $x \mapsto y$ is well defined.

We define a mapping $\gamma : \widetilde{\mathbf{F}} \to \mathbf{G}$ by $\gamma(x) = y$ and show that $\gamma$ is an isomorphism of $\widetilde{\mathbf{F}}$ onto $\mathbf{G}$. We do it in two steps.

Step 1. The mapping $\gamma$ is an embedding. Let $x = [(a_n)]$ and $x' = [(a'_n)]$ be elements of $\widetilde{\mathbf{F}}$. Then

$$\begin{aligned}\gamma(x + x') &= \lim \varphi([(a_n + a'_n)]) = \lim \varphi([(a_n)] + [(a'_n)]) \\ &= \lim \varphi([(a_n)]) + \lim \varphi([(a'_n)]) = \gamma(x) + \gamma(x')\end{aligned}$$

and

$$\begin{aligned}\gamma(x \cdot x') &= \lim \varphi([(a_n \cdot a'_n)]) = \lim \varphi([(a_n)] \cdot [(a'_n)]) \\ &= \lim \varphi([(a_n)]) \cdot \lim \varphi([(a'_n)]) = \gamma(x) \cdot \gamma(x').\end{aligned}$$

Suppose that $x > x'$, that is, $[(a_n)] > [(a'_n)]$. Then there exist a natural number $N$ and a positive $d$ in $\mathbf{F}$ such that $a_n > a'_n + d$ for $n > N$ (cf. Definition 2.5). Since $\varphi$ is an embedding, $\varphi(a_n) > \varphi(a'_n) + \varphi(d)$ for $n > N$ with $\varphi(d) > 0$. Hence,

$$\lim \varphi(a_n) \geq \lim \varphi(a'_n) + \varphi(d),$$

which implies $\gamma(x) > \gamma(x')$.

Conversely, suppose that $\gamma(x) > \gamma(x')$, that is, $\lim \varphi(a_n) > \lim \varphi(a'_n)$. Then

$$\lim \varphi(a_n - a'_n) > 0,$$

implying that there are a $c > 0$ in $\mathbf{G}$ and an $N \in \mathbf{N}$ such that $\varphi(a_n - a'_n) > c$ for all $n > N$. By the density property of $\varphi$, there is $d > 0$ in $\mathbf{F}$ such that $0 < \varphi(d) < c$. We have

$$\varphi(a_n - a'_n) > c > \varphi(d), \qquad \text{for all } n > N.$$

As $\varphi$ is an embedding, we obtain

$$a_n - a'_n > d, \qquad \text{for all } n > N.$$

Therefore, $x = [(a_n)] > [(a'_n)] = x'$.

Step 2. The mapping $\gamma$ is onto. Let $y \in \mathbf{G}$. Since $\varphi(\mathbf{F})$ is dense in $\mathbf{G}$, there is a sequence $\varphi(a_n)$ such that $\varphi(a_n)$ converges to $y$. Then $\varphi(a_n)$ is Cauchy in $\mathbf{G}$ and, by Lemma 2.13(ii), $(a_n)$ is Cauchy in $\mathbf{F}$. Let $x = [(a_n)] \in \widetilde{\mathbf{F}}$. Then $\gamma(x) = \lim \varphi(a_n) = y$, so $\gamma$ is onto. □

The following theorem summarizes the preceding results of this section.

**Theorem 2.12.** *For every ordered field* $\mathbf{F}$ *there exists up to isomorphism a unique Cauchy complete field* $\widetilde{\mathbf{F}}$ *such that* $\mathbf{F}$ *is isomorphic to a dense subfield of* $\widetilde{\mathbf{F}}$.

We conclude this section by establishing a relation between the Archimedean properties of the fields $\widetilde{\mathbf{F}}$ and $\mathbf{F}$ and give an example of non-Archimedean, Cauchy complete ordered field.

**Theorem 2.13.** *The Cauchy completion* $\widetilde{\mathbf{F}}$ *of an ordered field* $\mathbf{F}$ *is Archimedean if and only if the field* $\mathbf{F}$ *is Archimedean.*

*Proof.* (Necessity.) Left to the reader (cf. Exercise 2.20).

(Sufficiency.) Suppose that $\mathbf{F}$ is Archimedean. Let $[(a_n)]$ and $[(b_n)]$ be positive elements of $\widetilde{\mathbf{F}}$. We need to show that there is $k \in \mathbf{N}$ such that $k[(a_n)] > [(b_n)]$. Note that $k[(a_n)] = [(k)] \cdot [(a_n)] = [(ka_n)]$.

Since $[(a_n)] > 0$, there are a positive $c \in \mathbf{F}$ and $N' \in \mathbf{N}$ such that $a_n > c$ for all $n > N'$. Furthermore, because $[(b_n)] > 0$ and $(b_n)$ is bounded, there are $N'' \in \mathbf{N}$ and a positive element $C \in \mathbf{F}$ such that $0 < b_n < C$ for all $n > N''$. Let $N = \max\{N', N''\}$. Since $\mathbf{F}$ is Archimedean, there is $k \in \mathbf{N}$ such that

$$k > \frac{C+1}{c}.$$

Then

$$ka_n > kc > C + 1 > b_n + 1, \qquad \text{for all } n > N.$$

It follows that $k[(a_n)] > [(b_n)]$. □

*Example 2.3.* Let $\mathbf{F} = \mathbf{Z}(x)$ be the ordered field of rational functions over $\mathbf{Z}$ in the indeterminate $x$. Since this field is not Archimedean (cf. Example 1.3), its Cauchy completion $\widetilde{\mathbf{F}}$ is not Archimedean either. Hence, $\widetilde{\mathbf{F}}$ is an example of a Cauchy complete field that is not Dedekind complete (cf. Theorem 2.5).

## 2.3 The Field R

In this section we show that there exists a unique (up to isomorphism) Dedekind complete field **R**.

For this we need an important property of Archimedean fields. Recall that we consider the field of rational numbers **Q** as a subfield of any ordered field **F** (cf. Theorem 1.8).

**Theorem 2.14.** *The field* **Q** *is a dense subfield of every Archimedean field.*

*Proof.* We show that for $x < y$ in **F** there is $r \in \mathbf{Q}$ such that $x < r < y$ (cf. Exercise 2.16). Since $y - x > 0$, by the Archimedean property, there is $n \in \mathbf{N}$ such that $n(y - x) > 1$. Hence, $ny > nx + 1$. By Exercise 1.13, there is $m \in \mathbf{Z}$ such that

$$m - 1 \leq nx < m,$$

so $x < \dfrac{m}{n}$. Furthermore,

$$ny > nx + 1 \geq m,$$

so $y > \dfrac{m}{n}$. Hence we can choose $r = \dfrac{m}{n}$ to obtain the desired result. □

The next theorem is the central result of this chapter.

**Theorem 2.15.** *There exists a unique Dedekind complete ordered field.*

*Proof.* (Existence.) The Cauchy completion $\widetilde{\mathbf{Q}}$ of the field **Q** is Cauchy complete (cf. Theorem 2.10) and Archimedean (cf. Theorem 2.13). By Theorem 2.4, the field $\widetilde{\mathbf{Q}}$ is Dedekind complete.

(Uniqueness.) Let **R** be a Dedekind complete ordered field. By Theorems 2.3 and 2.2, **R** is a Cauchy complete Archimedean field. By Theorem 2.14, the field **Q** is dense in **R**. Therefore, **R** is the Cauchy completion of **Q** (cf. Definition 2.9). By Theorem 2.12, **R** is a unique (up to isomorphism) Dedekind complete field. □

To conclude this section, we prove that every Archimedean field is a subfield of **R**. First, we prove a property equivalent to density in Archimedean ordered fields.

**Theorem 2.16.** *A nonempty subset $E$ of an Archimedean ordered field* **F** *is dense in* **F** *if and only if every element of* **F** *is the limit of a sequence of elements of $E$.*

*Proof.* (Necessity.) Let $E$ be a dense subset of **F**. Then, for any $a \in \mathbf{F}$ and $n \in \mathbf{N}$, there is $x_n \in E$ such that

$$a - \frac{1}{n} < x_n < a + \frac{1}{n}$$

(cf. Definition 2.6). Since $1/n \to 0$ (cf. Exercise 2.5),

$$\lim\left(a - \frac{1}{n}\right) = \lim\left(a + \frac{1}{n}\right) = a.$$

By the Squeeze Theorem (cf. Theorem 1.14), $x_n \to a$.

(Sufficiency.) Suppose that every element of **F** is the limit of a sequence of elements of $E$. Then, for $a \in \mathbf{F}$ there is a sequence $(x_n)$ in $E$ such that $x_n \to a$. This means that for $\varepsilon > 0$ there is $N \in \mathbf{N}$ such that

$$|x_n - a| < \varepsilon, \qquad \text{for all } n > N.$$

The result follows (cf. Definition 2.6 with $x = x_{N+1}$). □

**Theorem 2.17.** *Let* **F** *be an Archimedean ordered field. There exists an embedding $\varphi : \mathbf{F} \to \mathbf{R}$. In words: every Archimedean ordered field is a subfield of* **R**.

*Proof.* By Theorem 2.16, for $x \in \mathbf{F}$ there is a sequence $(x_n)$ of rational numbers converging to $x$. We define $\varphi : \mathbf{F} \to \mathbf{R}$ by $\varphi(x) = \lim x_n$ in **R**. By the Cauchy completeness of **R**, the limit $\lim x_n$ exists in **R**. The mapping $\varphi$ is well defined because every two sequences of rational numbers converging to $x$ are equivalent (cf. (2.1)).

To show that $\varphi$ is an embedding (cf. Definition 2.7), we first note that, for $x_n \to x$ and $y_n \to y$, where $(x_n), (y_n) \in \mathbf{Q}$ and $x, y \in \mathbf{F}$,

$$\lim(x_n + y_n) = \lim x_n + \lim y_n = x + y$$

and

$$\lim(x_n \cdot y_n) = \lim x_n \cdot \lim y_n = x \cdot y$$

in **F**. Consequently,

$$\varphi(x + y) = \varphi(x) + \varphi(y) \qquad \text{and} \qquad \varphi(x \cdot y) = \varphi(x) \cdot \varphi(y).$$

Next, let $x < y$ be elements of $\mathbf{F}$ and let $(x_n)$ and $(y_n)$ be sequences in $\mathbf{Q}$ that converge to $x$ and $y$, respectively. For $\varepsilon = (y-x)/3$ there are $N_1, N_2 \in \mathbf{N}$ such that

$$|x - x_n| < \varepsilon, \quad \text{for all } n > N_1, \qquad \text{and} \qquad |y - y_n| < \varepsilon, \quad \text{for all } n > N_2.$$

Hence,

$$x_n < x + \varepsilon \quad \text{and} \quad y_n > y - \varepsilon, \qquad \text{for all } n > N = \max\{N_1, N_2\}.$$

Therefore,

$$y_n - x_n > (y - \varepsilon) - (x + \varepsilon) = \varepsilon > 0, \qquad \text{for all} \qquad n > N,$$

that is, $\varphi(x) < \varphi(y)$ (cf. Definition 2.5). □

Thus we have the following chain of ordered fields:

$$\mathbf{Q} \subseteq \mathbf{F} \subseteq \mathbf{R}, \tag{2.3}$$

where $\mathbf{Q}$ is the field of rational numbers, $\mathbf{F}$ is any Archimedean field, and $\mathbf{R}$ is the field of real numbers.

## 2.4 Properties of the Field R

In Section 2.1, we proved that the Dedekind complete ordered field is equivalently characterized as a Cauchy complete, Archimedean field (cf. Theorem 2.5). In this section, we introduce other properties of ordered fields that are equivalent to the Dedekind Completeness Property (cf. Definition 2.2), that is, properties that characterize the field $\mathbf{R}$ of real numbers.

We begin with the property that is instrumental in Dedekind's construction of real numbers (cf. Appendix B).

**Definition 2.11 (Cut Property).** If $A, B$ are non-empty subsets of an ordered field $\mathbf{F}$ such that $A \cup B = \mathbf{F}$ and if $a < b$ for all $a \in A$ and $b \in B$, then there exists an element $c \in \mathbf{F}$, called a *cut point* for the pair $(A, B)$, such that $a \leq c \leq b$ for all $a \in A$ and $b \in B$.

A pair $(A, B)$ of nonempty subsets of $\mathbf{F}$ such that $A \cup B = \mathbf{F}$ and $a < b$, for all $a \in A$ and $b \in B$ is called a *cut* of $\mathbf{F}$. A cut is called a *gap* if it does not have a cut point.

Note that a cut point (if it exists) is unique (cf. Exercise 2.27).

*Example 2.4.* The pairs $(A, B)$ and $(A', B')$ in Example 2.1 are clearly cuts with the same cut point $c$.

*Example 2.5.* Let $A = \{x \in \mathbf{Q} : x < 0 \text{ or } x^2 < 2\}$ and $B = \mathbf{Q} \setminus A$. Clearly, $(A, B)$ is a cut of $\mathbf{Q}$. By Example 2.2, $(A, B)$ is a gap (cf. Exercise 2.28).

**Theorem 2.18.** *A complete ordered field satisfies the Cut Property.*

*Proof.* Let $(A, B)$ be a cut of the field $\mathbf{R}$. The set $A$ is bounded above because $B \neq \varnothing$. By the completeness of $\mathbf{R}$ (cf. Definition 2.2), there exists $c = \sup A$. Let $a \in A$ and $b \in B$. Clearly, $a \leq c$. Suppose that $b < c$. Since $c = \sup A$, for $\varepsilon = c - b$, there is $a' \in A$ such that $a' > c - (c - b) = b$ (cf. Theorem 2.1), which contradicts our assumption that $(A, B)$ is a cut. Hence, $c \leq b$, that is, $c$ is a cut point. □

**Theorem 2.19.** *If an ordered field satisfies the Cut Property, then it is Cauchy complete.*

*Proof.* Let $(a_n)$ be a Cauchy sequence in the ordered field $\mathbf{F}$. By Theorem 1.22, $(a_n)$ has a monotone subsequence $(a_{k_n})$. Without loss of generality, we may assume that $(a_{k_n})$ is increasing. It is clear that this subsequence is Cauchy. By Theorem 1.17, it is bounded. We define

$$A = \{x \in \mathbf{F} : \text{there is } n \in \mathbf{N} \text{ such that } x \leq a_{k_n}\}$$

and

$$B = \{x \in \mathbf{F} : x > a_{k_n} \text{ for all } n \in \mathbf{N}\}$$

Since $a_{k_1} \in A$, the set $A$ is non-empty. The set $B$ is nonempty because the sequence $(a_{k_n})$ is bounded. Clearly, $A \cup B = \mathbf{F}$ and $a < b$ for all $a \in A$ and $b \in B$. Therefore, $(A, B)$ is a cut of $\mathbf{F}$. Let $c$ be its cut point. Since $c \geq a$ for $a \in A$ and $\{a_{k_n}\}_{n\in\mathbf{N}} \subseteq A$, the element $c$ is an upper bound of the set $\{a_{k_n}\}_{n\in\mathbf{N}}$. Since $B$ is the set of upper bounds of $\{a_{k_n}\}_{n\in\mathbf{N}}$ and $c \leq b$ for all $b \in B$, the element $c$ is the least upper bound of $\{a_{k_n}\}_{n\in\mathbf{N}}$. Hence, for every $\varepsilon > 0$ in $\mathbf{F}$, there is $N \in \mathbf{N}$ such that $c - \varepsilon < a_{k_N} \leq c$. As $(a_{k_n})$ is an increasing sequence bounded by $c$, we have (cf. Theorem 2.1)

$$|a_{k_n} - c| < \varepsilon, \qquad \text{for all } n > N,$$

that is, the sequence $(a_{k_n})$ converges to $c$. By Theorem 1.23, the sequence $(a_n)$ is convergent. Therefore, the field $\mathbf{F}$ is Cauchy complete. □

**Theorem 2.20.** *If an ordered field satisfies the Cut Property, then it is Archimedean.*

*Proof.* The proof is by contradiction (cf. the proof of Theorem 2.2). Suppose that the ordered field $\mathbf{F}$ is not Archimedean, that is, there are positive elements $a, b \in \mathbf{F}$ such that $nb \leq a$ for all $n \in \mathbf{N}$. Therefore, the set $X = \{nb\}_{n\in\mathbf{N}}$ is bounded. Set

$$A = \{x \in \mathbf{F} : \text{there is } n \in \mathbf{N} \text{ such that } x \leq nb\}$$

and

$$B = \{x \in \mathbf{F} : x > nb \text{ for all } n \in \mathbf{N}\}.$$

Since $X \subseteq A$, the set $A$ is not empty. The set $B$ consists of all upper bounds of the bounded set $X$ and therefore is not empty either. It is not difficult to see that $A \cup B = \mathbf{F}$ and $x < y$ for all $x \in A$, $y \in B$. Hence, $(A, B)$ is a cut of $\mathbf{F}$. By the Cut Property, there is $c \in \mathbf{F}$ such that $x \leq c \leq y$ for all $x \in A$ and $y \in B$. Since $x \leq c$ for all $x \in A$, the element $c$ is an upper bound of the set $A$. Since $c \leq y$ for all $y \in B$, $c$ is the least upper bound of $A$. Thus, there is $x \in A$ such that $c - b < x$. Since $c > nb$ for all $n \in \mathbf{N}$ ($(nb)$ is a strictly increasing sequence) and there is $n$ such that $x \leq nb$, we have (for this $n$)

$$c - b < x \leq nb < (n + 1)b < c.$$

Therefore (cf. Exercise 2.3),

$$b = (n + 1)b - nb < c - (c - b) = b,$$

a contradiction. □

By Theorem 2.5, we have the following characterization of the field of real numbers **R**.

**Theorem 2.21.** *An ordered field is complete if and only if it satisfies the Cut Property.*

Thus the completeness property of an ordered field may be equivalently expressed as the absence of gaps.

Although the next property appears to be rather special, it is powerful enough to provide a condition that is equivalent to completeness.

**Definition 2.12 (Bounded Monotone Convergence Property).** Every bounded monotone sequence in **F** converges.

In the arguments below we consider only increasing sequences. The reader is encouraged to supply the missing arguments for decreasing sequences.

**Theorem 2.22.** *Every complete ordered field satisfies the Bounded Monotone Convergence Property.*

*Proof.* Let $E = \{a_n : n \in \mathbf{N}\}$, where $(a_n)$ is an increasing sequence in the field. As this sequence is bounded, $a = \sup E$ exists. By Theorem 2.1, for every $\varepsilon > 0$ there is an $N$ such that $a_N \in (a - \varepsilon, a]$. Hence, $a - \varepsilon < a_n \leq a$ for all $n > N$. It follows that $a_n \to a$. □

**Theorem 2.23.** *If an ordered field satisfies the Bounded Monotone Convergence Property, then it is Cauchy complete.*

*Proof.* Let $(a_n)$ be a Cauchy sequence in the field. By Theorem 1.22, $(a_n)$ has a monotone subsequence, say, $(a_{k_n})$. We assume that $(a_{k_n})$ is increasing. By the Bounded Monotone Convergence Property, $(a_{k_n})$ converges to some element $a$ of the field. By Theorem 1.23, $a_n \to a$ and the result follows. □

**Theorem 2.24.** *If an ordered field satisfies the Bounded Monotone Convergence Property, then it is Archimedean.*

*Proof.* Suppose that the field is not Archimedean, that is, there are positive elements $a, b$ of the field such that $na \leq b$ for all $n \in \mathbf{N}$. Then the sequence $(na)$ is bounded and therefore, by the Bounded Monotone Convergence Property, convergent. We obtain a contradiction by noting that the sequence $(na)$ clearly is not Cauchy. □

By Theorem 2.5, the assertions of the last three theorem can be summarized as follows.

**Theorem 2.25.** *An ordered field is complete if and only if it satisfies the Bounded Monotone Convergence Property.*

We introduce below a property of an ordered field which plays an important role not only in real analysis, but also more broadly in analysis.

**Definition 2.13 (Bolzano–Weierstrass Property).** Every bounded sequence has a convergent subsequence.

**Theorem 2.26.** *Every complete ordered field satisfies the Bolzano–Weierstrass Property.*

*Proof.* Let $(a_n)$ be a bounded sequence in a complete ordered field. By Theorem 1.22, $(a_n)$ has a monotone subsequence $(a_{k_n})$ which is clearly bounded. By Theorem 2.22, the sequence $(a_{k_n})$ converges. □

**Theorem 2.27.** *If an ordered field satisfies the Bolzano–Weierstrass Property, then it is Cauchy complete.*

*Proof.* Let $(a_n)$ be a Cauchy sequence in the field. By the Bolzano–Weierstrass Property, $(a_n)$ has a convergent subsequence. By Theorem 1.23, the sequence $(a_n)$ converges. □

**Theorem 2.28.** *If an ordered field satisfies the Bolzano–Weierstrass Property, then it is Archimedean.*

*Proof.* Suppose to the contrary that the field is not Archimedean. Then there are positive elements $a$ and $b$ of the field such that $na \leq b$ for all $n \in \mathbf{N}$. The sequence $(na)$ is bounded. Hence, by the Bolzano–Weierstrass Property, it has a convergent subsequence $(k_n a)$ for some increasing sequence of natural numbers $k_1 < k_2 < \cdots$. This is the desired contradiction, because clearly the sequence $(k_n a)$, is not Cauchy. □

As before, the last three theorems can be combined into the next statement.

**Theorem 2.29.** *An ordered field is complete if and only if it satisfies the Bolzano–Weierstrass Property.*

We conclude our review of properties of ordered fields that are equivalent to the completeness property with the following definition and theorems.

First we recall (cf. Definition 1.16) that a family $\{[a_n, b_n] : n \in \mathbf{N}\}$ of closed bounded intervals in an ordered field is said to be *nested* if

$$[a_1, b_1] \supseteq [a_2, b_2] \supseteq \cdots \supseteq [a_n, b_n] \supseteq \cdots,$$

or, equivalently,

$$a_m \leq a_n < b_n \leq b_m, \qquad \text{for all } m \leq n. \tag{2.4}$$

**Definition 2.14 (Nested Intervals Property).** If $\{[a_n, b_n] : n \in \mathbf{N}\}$ is a nested family of closed bounded intervals in an ordered field such that $(b_n - a_n) \to 0$, then the intersection of all these intervals, $\bigcap_{n\in\mathbf{N}}[a_n, b_n]$, contains exactly one element.

Example 1.5 shows that the field of rational numbers **Q** does not satisfy the Nested Intervals Property. We will see that this is due to incompleteness of the field **Q**.

**Theorem 2.30.** *Every complete ordered field satisfies the Nested Intervals Property.*

*Proof.* Let $\{[a_n, b_n] : n \in \mathbf{N}\}$ be a nested family of intervals in the field **R**. By (2.4), $a_n < b_1$ for all $n \in \mathbf{N}$. Hence, the set $\{a_n : n \in \mathbf{N}\}$ is bounded above. By the completeness property, this set has the least upper bound

$$a = \sup\{a_n : n \in \mathbf{N}\}.$$

First, we prove that $a \in [a_n, b_n]$ for all $n \in \mathbf{N}$.

By (2.4),

$$a_m \leq a_n < b_n \leq b_m, \qquad \text{for all } m \leq n$$

and

$$a_n \leq a_m < b_m \leq b_n, \qquad \text{for all } m > n.$$

Hence,

$$a_m < b_n, \qquad \text{for all } m, n \in \mathbf{N},$$

that is, every $b_n$ is an upper bound of the set $\{a_k : k \in \mathbf{N}\}$. It follows that $b_n \geq a$ for all $n \in \mathbf{N}$. Since $a$ is the supremum of the set $\{a_k : k \in \mathbf{N}\}$, we have $a_n \leq a$ for all $n \in \mathbf{N}$. Hence, $a_n \leq a \leq b_n$ for all $n \in \mathbf{N}$.

Suppose now that an element $b \in \mathbf{F}$ belongs to $[a_n, b_n]$ for all $n \in \mathbf{N}$. As $a, b \in [a_n, b_n]$, we have

$$0 \leq |b - a| \leq (b_n - a_n) \to 0.$$

By the Squeeze Theorem (Theorem 1.14), $b = a$. Hence, $a$ is the unique element of the intersection $\bigcap_{n\in\mathbf{N}}[a_n, b_n]$. □

**Corollary 2.1.** *A complete ordered field, that is, the field of real numbers* **R**, *is uncountable.*

*Proof.* We proceed by contradiction. Suppose that there is a sequence of real numbers

$$x_1, x_2, \ldots, x_n, \ldots$$

such that every real number is a term of this sequence.

It is clear that there is a closed bounded interval $[a_1, b_1]$ of length

$$b_1 - a_1 = \frac{1}{3}$$

such that $x_1 \notin [a_1, b_1]$. We divide this interval into three closed subintervals of equal length. Again, it is clear that there is at least one of these subintervals, say $[a_2, b_2]$, that does not contain $x_2$ (and does not contain $x_1$ either). By using recursion, we construct a nested family of closed bounded intervals $\{[a_n, b_n] : n \in \mathbf{N}\}$ such that, for all $n \in \mathbf{N}$,

$$b_n - a_n = \frac{1}{3^n},$$

and $[a_n, b_n]$ does not contain the numbers $x_1, \ldots, x_n$. By Theorem 2.30, there is a real number $a$ that belongs to all intervals $[a_n, b_n]$ and hence is not a term of the sequence $(x_n)$. This contradiction proves the corollary. □

Note that the field of rational numbers **Q** is a countable set. One may say that Corollary 2.1 claims that the field of real numbers **R** is much "larger" than its subfield **Q**.

The converse of Theorem 2.30 does not hold. A counterexample is outlined in Example 2.6 below. However, we have the following result.

**Theorem 2.31.** *If an Archimedean field satisfies the Nested Intervals Property, then it is complete.*

*Proof.* Let **F** be an Archimedean field. By Theorem 2.17, $\mathbf{Q} \subseteq \mathbf{F} \subseteq \mathbf{R}$. We show that $\mathbf{F} = \mathbf{R}$. Let $a \in \mathbf{R} \setminus \mathbf{Q}$, that is, $a$ is an irrational number. Since **Q** is dense in **R** (cf. Theorem 2.14), there are rational numbers $a_1$ and $b_1$ such that $a_1 < a < b_1$. (It suffices to choose rational numbers between $a - 1$ and $a$ and between $a$ and $a + 1$, respectively (cf, Exercise 2.16)). We construct a nested family of intervals $\{[a_n, b_n] : n \in \mathbf{N}\}$ as follows.

The interval $[a_1, b_1]$ is already defined. As $a_1$ and $b_1$ are rational numbers, the number $(a_1 + b_1)/2$ is also rational. To define $a_2$ and $b_2$, we consider two mutually exclusive cases

(1) $a_1 < a < \dfrac{a_1 + b_1}{2}$. Then we set:

$$a_2 = a_1, \qquad b_2 = \frac{a_1 + b_1}{2}.$$

(2) $\dfrac{a_1 + b_1}{2} < a < b_1$. Then we set:

$$a_2 = \frac{a_1 + b_1}{2}, \qquad b_2 = b_1.$$

Clearly, $[a_1, b_1] \supseteq [a_2, b_2]$, $a \in [a_2, b_2]$, and $(b_2 - a_2) = \dfrac{1}{2}(b_1 - a_1)$.

By continuing the above process recursively, we obtain a nested family of closed bounded intervals

$$[a_1, b_1] \supseteq [a_2, b_2] \supseteq \cdots \supseteq [a_n, b_n] \supseteq \cdots,$$

such that $a \in [a_n, b_n]$ for all $n \in \mathbf{N}$ and $b_n - a_n = \dfrac{1}{2^{n-1}}(b_1 - a_1)$. By the Archimedean property,

$$0 < b_n - a_n = \frac{1}{2^{n-1}}(b_1 - a_1) < \frac{1}{n}(b_1 - a_1) \longrightarrow 0$$

(cf. Exercise 2.5). By the Nested Intervals Property of **F** (note that $a_n, b_n \in \mathbf{F}$ for all $n \in \mathbf{N}$),

$$a = \bigcap_{n \in \mathbf{N}} [a_n, b_n] \in \mathbf{F}.$$

Since $\mathbf{Q} \subseteq \mathbf{F}$, we conclude that $\mathbf{F} = \mathbf{R}$. □

*Example 2.6.* A counterexample to the converse of Theorem 2.30 is the *field of formal Laurent series with coefficients in* **R** (cf. Appendix C):

$$\mathbf{R}((x)) = \Big\{ \sum_{k=n}^{\infty} a_k x^k : a_k \in \mathbf{R},\ n \in \mathbf{Z} \Big\}.$$

To justify this statement it is necessary to (1) show that $\mathbf{R}((x))$ is indeed an ordered field, (2) show that this field is not Archimedean, and (3) prove that this field satisfies the Nested Intervals Property. Then will imply that $\mathbf{R}((x))$ is not Dedekind complete (cf. Theorem 2.2). (It is, however, Cauchy complete (cf. Theorem C.1 in Appendix C).) The proof is rather tedious. For details, the reader is referred to Efimov (1980, pp. 212–215).

## Notes

There are essentially two ways of constructing the real numbers: (a) Dedekind's method of completion by cuts and (b) Cantor's method of Cauchy sequences. Of these we concentrate on (b) because it makes it easier to define the algebraic operations and also because a similar method of completing a metric space plays an important role in analysis. For instance, the Cauchy completeness of the (unordered) field of complex numbers $\mathbf{C}$ is fundamental in the development of complex analysis. For reader's convenience, Dedekind's method is outlined in Section B.1.

The German mathematician Richard Dedekind (1831–1916) made important contributions to abstract algebra, axiomatic foundations of the natural numbers, algebraic number theory and the definition of the real numbers (cf. Dedekind 1863).

The Cut, Bounded Monotone Convergence, and Bolzano–Weierstrass properties completely characterize the field of real numbers $\mathbf{R}$. For instance, the field $\mathbf{R}$ can be defined as an ordered field satisfying the Bolzano–Weierstrass Property (cf. Theorem 2.29). The Bolzano–Weierstrass Property and its variations are fundamental in other branches of analysis.

Bernard Bolzano (1781–1848) was a Czech mathematician, logician, and philosopher. He is mostly remembered for the Bolzano–Weierstrass theorem, which Karl Weierstrass developed independently and published years after Bolzano's first proof. Alas, Bolzano's works remained unknown to most of the mathematical community until years later.

Karl Weierstrass (1815–1897) was a German mathematician often cited as the "father of modern analysis". There are at least ten things (concepts and theorems) named after Karl Weierstrass in analysis.

The Nested Intervals Property is known in real analysis and topology as Cantor's Intersection Theorem. In foundations of geometry, this property is Cantor's Axiom (Axiom IV.2, Efimov 1980, p. 70).

Georg Cantor (1845–1918) was a German mathematician, best remembered as the founder of set theory. He also made important contributions to number theory, trigonometric series, and the rigorous development of real numbers.

## Exercises

**2.1.** Show that a nonempty subset of an ordered field has at most one supremum and at most one infimum.

**2.2.** Let **F** be a Dedekind complete field (cf. Definition 2.2). Prove that every bounded below subset of **F** has an infimum.

**2.3.** Show that if $p < r < s < q$ in an ordered field, then $s - r < q - p$.

**2.4.** Prove the second claim of Lemma 2.2.

**2.5.** Prove that the Archimedean Property of an ordered field **F** is equivalent to the condition

$$\lim \frac{1}{n} = 0$$

(cf. Exercise 1.21 a).

**2.6.** Let $A$ and $B$ be bounded subsets of $\mathbf{R}$ such that $a \leq b$ for any $a \in A$ and $b \in B$. Show that $\sup A \leq \inf B$.

**2.7.** Let $A$ and $B$ be bounded subsets of $\mathbf{R}$. Show that the set $A \cup B$ is bounded and

$$\sup(A \cup B) = \max\{\sup A, \sup B\}.$$

**2.8.** Let $A$ and $B$ be bounded subsets of $\mathbf{R}$ and let

$$A + B = \{a + b : a \in A,\ b \in B\}.$$

Prove that

$$\sup(A + B) = \sup A + \sup B.$$

**2.9.** For a bounded set $E$ of real numbers and $c \in \mathbf{R}$, we define

$$cE = \{ca : a \in E\}.$$

Prove that

$$\inf(cE) = c \inf E, \quad \sup(cE) = c \sup E, \qquad \text{if } c > 0,$$

and

$$\inf(cE) = c \sup E, \quad \sup(cE) = c \inf E, \qquad \text{if } c < 0,$$

**2.10.** Let $E'$ be a nonempty subset of a bounded set $E$ of real numbers. Show that

$$\inf E \le \inf E' \le \sup E' \le \sup E.$$

**2.11.** Let $E$ be a nonempty subset of $\mathbf{R}$. Prove that $b \in \mathbf{R}$ is the supremum of $E$ if and only if for every natural number $n$

(a) the number $b - \frac{1}{n}$ is not an upper bound of $E$, and

(b) the number $b + \frac{1}{n}$ is an upper bound of $E$.

**2.12.** Prove that $\sqrt{2} + \sqrt{3}$ is an irrational number.

**2.13.** If $a \neq 0$ is rational and $b$ is irrational, prove that

$$a + b, \quad a - b, \quad a/b, \quad b/a$$

are all irrational.

**2.14.** Prove that the relation $\sim$ defined by (2.1) is an equivalence relation.

**2.15.** Prove that a Cauchy sequence $(a_n)$ in $\mathbf{F}$ belongs to the class $[(a)]$ of a constant sequence $(a)$ if and only if $(a_n)$ converges to $a$.

**2.16.** Let $E$ be a dense subset of an ordered field $\mathbf{F}$ (cf. Definition 2.6). Show that for $x < y$ in $\mathbf{F}$ there is $r \in E$ such that $x < r < y$.

**2.17.** Prove Theorem 2.9.

**2.18.** Prove the following properties of an embedding $\varphi : \mathbf{F} \to \mathbf{G}$ (cf. Definition 2.7):

(a) $\varphi$ is a one-to-one mapping.
(b) $\varphi(0) = 0$ and $\varphi(1) = 1$.
(c) $\varphi(\mathbf{F})$ is an ordered subfield of $\mathbf{G}$.

**2.19.** Prove the second claim of Lemma 2.13.

**2.20.** Show that the field $\widetilde{\mathbf{F}}$ is not Archimedean if $\mathbf{F}$ is not Archimedean.

**2.21.** If $\mathbf{F}$ and $\mathbf{G}$ are isomorphic ordered fields and $\mathbf{F}$ is Archimedean, prove that $\mathbf{G}$ is also Archimedean.

**2.22.** Prove that a sequence of rational numbers is Cauchy in $\mathbf{Q}$ if and only if it is Cauchy in $\mathbf{R}$.

**2.23.** Let $(a_n)$ be a sequence of rational numbers and $a \in \mathbf{Q}$. Prove that

$$\lim a_n = a \text{ in } \mathbf{Q} \qquad \text{if and only if} \qquad \lim a_n = a \text{ in } \mathbf{R}.$$

**2.24.** Prove that an ordered field $\mathbf{F}$ is Archimedean if and only if the field $\mathbf{Q}$ is dense in $\mathbf{F}$ (cf. Theorem 2.14).

**2.25.** Let $\mathbf{F} = \{a + b\sqrt{2} : a, b \in \mathbf{Q}\}$ and $\mathbf{G} = \{a + b\sqrt{3} : a, b \in \mathbf{Q}\}$. Prove that $\mathbf{F}$ and $\mathbf{G}$ are non-isomorphic ordered subfields of $\mathbf{R}$.

**2.26.** Prove that the set of irrational numbers is dense in $\mathbf{R}$.

**2.27.** Prove that a cut point (if it exists, cf. Definition 2.11) is unique.

**2.28.** Show that if $c$ is a cut point of a cut $(A, B)$ in $\mathbf{F}$, then

$$c = \sup A = \inf B.$$

**2.29.** Let $(A, B)$ be a cut of $\mathbf{F}$. Prove that if every interval

$$(c - \varepsilon, c + \varepsilon) = \{x \in \mathbf{F} : c - \varepsilon < x < c + \varepsilon\}$$

$(\varepsilon > 0)$ contains points from $A$ and $B$, then $c$ is a cut point for $(A, B)$.

**2.30.** Show that if an ordered field has a gap, then it has infinitely many gaps. (Hint: If $(A, B)$ is a gap in $\mathbf{F}$, then $(A + x, B + x)$ is a gap for every $x \in \mathbf{F}$.)

**2.31.** Show that there is a unique positive real number whose square is 2.

# Chapter 3
# Continuous Functions




This chapter begins (Section 3.1) with an introduction of the concepts of open and closed subsets of an ordered field and the basic properties of these sets. Then, the classes of connected and compact subsets of an ordered field are defined and their properties are investigated. We show that some properties of those special sets are equivalent to the completeness property, and therefore characterize the field of real numbers **R**. In conclusion of this section, we prove the Heine–Borel and Borel–Lebesgue theorems characterizing compact sets of real numbers.

Continuous functions on subsets of an ordered field are the subject of Section 3.2. This section focuses on two properties of continuous functions on closed bounded intervals—the Intermediate Value Property and the Extreme Value Property. We prove that these properties are satisfied only in the field **R**, and hence characterize this field. We conclude this section by proving that the Fixed Point Property for continuous functions is equivalent to the completeness property of an ordered field.

Uniformly continuous functions are briefly discussed in Section 3.3. In particular, we prove that continuous functions on closed intervals in **R** are uniformly continuous on these intervals. We also show that Lipschitz functions are uniformly continuous on their domains.

As before, the symbols **F** and **R** denote a general ordered field and the field of real numbers, respectively. We often refer to elements of **F** as points or numbers.

## 3.1 Subsets of an Ordered Field

**Definition 3.1.** For $a < b$ in **F**, the bounded *open interval* $(a, b)$ and the bounded *closed interval* $[a, b]$ (cf. Definition 1.16) are the sets

$$(a, b) = \{x : a < x < b\} \qquad \text{and} \qquad [a, b] = \{x : a \leq x \leq b\},$$

respectively.

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S. Ovchinnikov, *Real Analysis: Foundations*, Universitext,
https://doi.org/10.1007/978-3-030-64701-8_3

The sets $[a, b) = \{x : a \leq x < b\}$ and $(a, b] = \{x : a < x \leq b\}$ are called bounded *half-open intervals*.

The following sets are *unbounded intervals*:

$$\begin{aligned} (a, +\infty) &= \{x : x > a\}, \quad (-\infty, a) = \{x : x < a\} && \text{(open)}, \\ [a, +\infty) &= \{x : x \geq a\}, \quad (-\infty, a] = \{x : x \leq a\} && \text{(closed)}, \\ (-\infty, +\infty) &= \mathbf{F} && \text{(both)}. \end{aligned}$$

An *interval* is any set of one of the forms listed above. The elements $a$ and $b$ are called the *endpoints* of the respective intervals.

All intervals are nonempty sets (cf. Exercise 1.28). Note that the symbols $-\infty$ and $+\infty$ that appear in the definition of unbounded intervals are not elements of $\mathbf{F}$. We call them *extended numbers* and write $\inf E = -\infty$ if the set $E$ is unbounded below, and $\sup E = \infty$ if $E$ is unbounded above.

Open and closed intervals are instances of more general families of subsets of $\mathbf{F}$.

**Definition 3.2.** Open intervals containing $x \in \mathbf{F}$ are called *neighborhoods* of the point $x$. For $\varepsilon > 0$ in $\mathbf{F}$, the interval $(x - \varepsilon, x + \varepsilon)$ is called the $\varepsilon$-neighborhood of $x$. A subset $G$ of $\mathbf{F}$ is called *open* provided that it contains a neighborhood of each of its points.

All open intervals, including $\mathbf{F}$ itself, are open sets. Since the empty set contains no points, it is vacuously open.

The proof of the following characterization of open sets is left as an exercise (cf. Exercise 3.1).

**Theorem 3.1.** *A subset $G$ of $\mathbf{F}$ is open if and only if for every $x \in G$ there is an $\varepsilon$-neighborhood of $x$ such that $(x - \varepsilon, x + \varepsilon) \subseteq G$.*

According to this theorem, it suffices to use $\varepsilon$-neighborhoods to verify that a set is open.

Two fundamental properties of open sets are stated in the following theorem.

**Theorem 3.2.** *The union of any family of open sets is open. The intersection of any finite family of open sets is open.*

*Proof.* The first claim is obvious. Now let $G = \bigcap_{k=1}^{n} G_k$ be a finite intersection of open sets. If $G = \varnothing$, we are done. Suppose that $G \neq \varnothing$ and let $x \in G$. Then $x \in G_k$ for all $1 \leq k \leq n$. Since the sets $G_k$ are open, there are open intervals $(a_k, b_k)$ such that $x \in (a_k, b_k)$ and $(a_k, b_k) \subseteq G_k$ for $1 \leq k \leq n$. It remains to note that the open interval $(a, b)$ with $a = \max\{a_k : 1 \leq k \leq n\}$ and $b = \min\{b_k : 1 \leq k \leq n\}$ is a subset of $G = \bigcap_{k=1}^{n} G_i$ containing $x$. □

It is clear that every nonempty open set is a union of open intervals. For the field of real numbers $\mathbf{R}$ we have a much stronger result.

**Theorem 3.3.** *Every nonempty open set in* $\mathbf{R}$ *is the union of a finite or countable family of pairwise disjoint open intervals.*

*Proof.* Let $x$ be a point in a nonempty open set $G$. There is an open interval $(y, z)$ such that $x \in (y, z) \subseteq G$. Then, $(y, x) \subseteq G$ and $(x, z) \subseteq G$. We define (possibly extended) numbers $a_x$ and $b_x$ by

$$a_x = \inf\{y : (y, x) \subseteq G\} \quad \text{and} \quad b_x = \sup\{z : (x, z) \subseteq G\}.$$

It is clear that $x \in I_x = (a_x, b_x)$. We claim that $I_x \subseteq G$, but $a_x \notin G$, $b_x \notin G$, and prove this assertion in the next two paragraphs.

First we prove that $I_x \subseteq G$ (cf. Figure 3.1). Let $w \in I_x$ and assume that $w > x$. (The case when $w < x$ is treated similarly.) By the definition of $b_x$, there is $z > w$ such that $(x, z) \subseteq G$. Then $w \in (x, z) \subseteq G$. It follows that $I_x \subseteq G$.

Now suppose that $b_x \in G$. Then there is an interval $(u, v) \subseteq G$ with $u < b_x < v$. Clearly, $(x, b_x) \cap (u, v) \neq \varnothing$. We obtain the desired contradiction (recall that $b_x = \sup\{z : (x, z) \subseteq G\}$) by noting that $(x, v) \subseteq G$ (cf. Figure 3.2). A similar argument shows that $a_x \notin I_x$.

Since each $x \in G$ is an element of $I_x$, and each $I_x$ is contained in $G$, we have $G = \bigcup_{x \in G} I_x$. Suppose that two distinct intervals $I_x$ and $I_y$ intersect. Then one of the endpoints of one of these intervals belongs to the other interval—a contradiction, since endpoints of the intervals do not belong to $G$. Thus $G$ is a union of a family of pairwise disjoint intervals. Since each of these intervals contains a rational number, $G$ is a union of a finite or countable family of pairwise disjoint intervals. □

The next theorem shows that the property of the field $\mathbf{R}$ established in Theorem 3.3 characterizes this field. First, we prove a lemma.

**Lemma 3.1.** *If* $(A, B)$ *is a gap in an ordered field* $\mathbf{F}$*, then the sets* $A$ *and* $B$ *are open.*

*Proof.* Let $(A, B)$ be a gap (cf. Definition 2.11) in $\mathbf{F}$. Because there is no cut point for $(A, B)$, for every $a \in A$ there is an $a' \in A$ such that $a' > a$. Clearly, $2a - a' <$

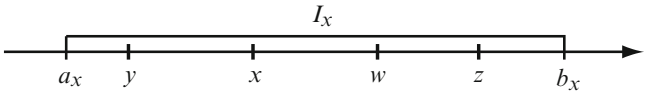



**Fig. 3.1** Proof of Theorem 3.3.

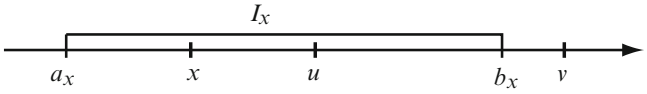



**Fig. 3.2** Proof of Theorem 3.3.

$a < a'$, so $a \in (2a - a', a') \subseteq A$. Hence, $A$ is an open set (cf. Definition 3.2). A similar argument shows that the set $B$ is open. □

**Theorem 3.4.** *An ordered field* $\mathbf{F}$ *is the field of real numbers* $\mathbf{R}$ *if and only if every nonempty open subset of* $\mathbf{F}$ *is an at most countable union of pairwise disjoint open intervals.*

*Proof.* It suffices to show that every incomplete ordered field $\mathbf{F}$ contains a nonempty open subset that is not representable as the disjoint union of a family of open intervals.

Let $(A, B)$ be a gap in the incomplete ordered field $\mathbf{F}$. By Lemma 3.1, the set $A$ is open. As there is no cut point for $(A, B)$, the set $A$ is not an open interval. Suppose that $A$ is a disjoint union of open intervals. Then at least one of these intervals, say $(a, b)$, is bounded. It is easy to see that $a \in A$ and does not belong to any other open interval in the union. This contradiction shows that the open set $A$ is not representable as the disjoint union of a family of open intervals. □

**Definition 3.3.** Let $E$ be a subset of an ordered field $\mathbf{F}$. A point $x \in \mathbf{F}$ is said to be a *limit point* of $E$ if every neighborhood of $x$ contains a point in $E$ distinct from $x$. The set consisting of the points in $E$ and the limit points of $E$ is called the *closure* of $E$ and is denoted by $\overline{E}$.

A point $x \in E$ is said to be an *isolated point* if there is a neighborhood of $x$ that contains no point of $E$ different from $x$.

**Definition 3.4.** A subset of an ordered field $\mathbf{F}$ is called *closed* if it contains all its limit points.

Hence a subset $F$ of $\mathbf{F}$ is closed if it coincides with its closure: $F = \overline{F}$.

The following theorem establishes a close relationship between the concepts of open and closed sets.

**Theorem 3.5.** *A subset of an ordered field* $\mathbf{F}$ *is closed if and only if its complement in* $\mathbf{F}$ *is open.*

*Proof.* Let $E$ be a subset of $\mathbf{F}$. It is easy to see that the result holds if $E = \varnothing$ or $E = \mathbf{F}$. Thus we assume that $E$ is a nonempty proper subset of $\mathbf{F}$.

(Necessity.) Let $E$ be a closed set and $x$ an element of the complement $E^{\mathrm{c}} = \mathbf{F} \setminus E$. Since $x \notin E$, it is not a limit point of $E$. Hence, there is a neighborhood of $x$ which is a subset of $E^{\mathrm{c}}$. It follows that $E^{\mathrm{c}}$ is an open set.

(Sufficiency.) Suppose that the complement $E^{\mathrm{c}}$ of the set $E$ is open and let $x$ be a limit point of $E$. Since every neighborhood of $x$ contains a point of $E$ different from $x$, the point $x$ does not belong to the open set $E^{\mathrm{c}}$. Therefore, $x \in E$, that is, $E$ is a closed set. □

By applying Theorem 3.5 to properties of open sets established in Theorem 3.2, we obtain fundamental properties of closed sets:

**Theorem 3.6.** *The union of every finite family of closed sets is a closed set. The intersection of every family of closed sets is a closed set.*

The following lemma will be needed later in this section.

**Lemma 3.2.** *Let $F$ be a nonempty bounded closed subset of* $\mathbf{R}$. *Then the points* $a = \inf F$ *and* $b = \sup F$ *belong to* $F$.

*Proof.* Since $b$ is the supremum of $F$, every open interval containing $b$ must intersect the set $F$. Hence, $b$ cannot belong to the open set $F^c = \mathbf{F} \setminus F$. It follows that $b \in F$. A similar argument shows that $a \in F$. □

**Definition 3.5.** Let $E$ be a subset of $\mathbf{F}$. A subset $G$ of $E$ is said to be *relatively open* in $E$ if it is an intersection of an open subset of $\mathbf{F}$ with $E$, that is, there is an open set $U \subseteq \mathbf{F}$ such that $G = U \cap E$.

Note that, by Theorem 3.1, a set $G$ is relatively open in $E$ if and only if for every $a \in G$, there is an $\varepsilon$-neighborhood $U_\varepsilon$ of $a$ such that $U_\varepsilon \cap E \subseteq G$.

*Example 3.1.* In $\mathbf{F}$, the half-open interval $[0, 1)$ is a relatively open subset of $[0, 2]$ because

$$[0, 1) = (-\infty, 1) \cap [0, 2].$$

However, $[0, 1)$ is not an open subset of $\mathbf{F}$.

**Definition 3.6.** A subset $E$ of the field $\mathbf{F}$ is said to be *connected* if it is not the union of two disjoint nonempty relatively open subsets of $E$. Otherwise, the set $E$ is said to be *disconnected*.

**Theorem 3.7.** *The intervals of the ordered field* $\mathbf{R}$ *are connected sets. In particular,* $\mathbf{R}$ *is connected.*

*Proof.* Let $I$ be an interval in $\mathbf{R}$. Suppose to the contrary that $I$ is disconnected, that is, there are open sets $U$ and $V$ in $\mathbf{F}$ such that

$$(U \cap I) \cup (V \cap I) = I, \qquad (U \cap I) \cap (V \cap I) = \varnothing,$$

and there exist points $x \in U \cap I$ and $y \in V \cap I$. We may assume that $x < y$.

Because $U$ is an open set, there is an $\varepsilon_1$-neighborhood $(x - \varepsilon_1, x + \varepsilon_1)$ of $x$ which is contained in $U$. Therefore, $[x, x + \varepsilon_1) \subseteq (U \cap I)$. Similarly, there is an $\varepsilon_2$-neighborhood of $y$ such that $(y - \varepsilon_2, y] \subseteq (V \cap I)$ (cf. Figure 3.3).

Let

$$A = \{a \in [x, y] : a \in U\}.$$

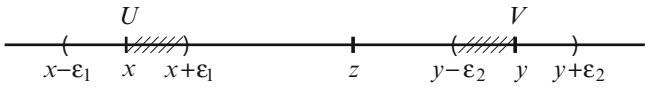



**Fig. 3.3** Proof of Theorem 3.7.

The set $A$ is not empty ($x \in A$) and bounded. Let $z = \sup A$. By the arguments in the preceding paragraphs,

$$x < x + \varepsilon_1 \le z \le y - \varepsilon_2 < y.$$

Moreover, because $z = \sup A$, $(z, y] \subseteq V$.

Since any neighborhood of $z$ contains points in $V$, $z \notin U \cap I$. On the other hand, by the definition of $z$, every neighborhood of $z$ contains points from $U$. (Cf. Theorem 2.1. Note that $z \notin U$.) Hence, $z \notin V \cap I$.

We obtained a contradiction with $(U \cap I) \cup (V \cap I) = I$. □

To establish a connectedness property of an incomplete ordered field, we need the following lemmas.

**Lemma 3.3.** *Let $f : \mathbf{F} \to \mathbf{F}$ be the function defined by*

$$f(x) = px + q, \qquad p > 0,\ x \in \mathbf{F}$$

(*it is a* linear *function on* $\mathbf{F}$). *The function $f$ is a bijection preserving order on* $\mathbf{F}$. *Furthermore, the image $f(G)$ of an open set $G \subseteq \mathbf{F}$ is an open set in* $\mathbf{F}$.

The proof is left as an exercise (cf. Exercise 3.2).

**Lemma 3.4.** *For every two points $a < b$ in an incomplete ordered field* $\mathbf{F}$ *there is a gap $(A, B)$ such that $a \in A$, $b \in B$.*

*Proof.* Since the field $\mathbf{F}$ is incomplete, there is a gap $(C, D)$ in $\mathbf{F}$. Let $c \in C$, $d \in D$ and define $f$ by

$$f(x) = \frac{b-a}{d-c}\,x + \frac{ad-bc}{d-c}, \qquad x \in \mathbf{F}.$$

By Lemma 3.3, the sets $A = f(C)$, $B = f(D)$ define a gap $(A, B)$ in $\mathbf{F}$, because $C \cap D = \varnothing$ and $C \cup D = \mathbf{F}$. It is clear that $a = f(c) \in A$ and $b = f(d) \in B$. □

**Theorem 3.8.** *Every interval in an incomplete ordered field* $\mathbf{F}$ *is disconnected. In particular,* $\mathbf{F}$ *is disconnected.*

*Proof.* Let $I$ be an interval in $\mathbf{F}$ containing points $a < b$. By Lemma 3.4, there is a gap $(A, B)$ in $\mathbf{F}$ such that $a \in A$, $b \in B$. By Lemma 3.1, the sets $A$ and $B$ are open. Since $A \cap B = \varnothing$ and $A \cup B = \mathbf{F}$, we have

$$I = \mathbf{F} \cap I = (A \cup B) \cap I = (A \cap I) \cup (B \cap I)$$

and

$$(A \cap I) \cap (B \cap I) = \varnothing.$$

Therefore, $I$ is disconnected. □

An ordered field $\mathbf{F}$ satisfying the condition of Theorem 3.8 is said to be *totally disconnected*. We summarize the results of Theorems 3.7 and 3.8 as follows.

**Theorem 3.9.** *An ordered field* $\mathbf{F}$ *is either connected* (*and then* $\mathbf{F} = \mathbf{R}$)*, or totally disconnected.*

We conclude this section by introducing an important class of compact sets in an ordered field.

**Definition 3.7.** A nonempty subset $K$ of an ordered field $\mathbf{F}$ is said to be *compact* if every sequence of points in $K$ contains a subsequence converging to a point in $K$.

Compact subsets of the field of real numbers $\mathbf{R}$ are characterized by the following theorem.

**Theorem 3.10 (Heine–Borel Theorem).** *A nonempty subset of the field* $\mathbf{R}$ *is compact if and only if it is closed and bounded.*

*Proof.* (Necessity.) Let $K$ be a compact subset of $\mathbf{R}$ and $x$ a limit point of $K$. For each $n \in \mathbf{N}$ there is a point $x_n \in K$ such that $0 < |x_n - x| < 1/n$. By the Squeeze Theorem (cf. Theorem 1.14), $x_n \to x$. Since $K$ is a compact set, $x \in K$. Hence, $K$ is closed.

Suppose that the compact set $K$ is not bounded above. (The other case is treated similarly.) Then for every $n \in \mathbf{N}$, there is $x_n \in K$ such that $x_n > n$. Because every convergent sequence is bounded, the sequence $(x_n)$ cannot contain a convergent subsequence. It follows that $K$ is a bounded set.

(Sufficiency.) Let $K$ be a nonempty subset of $\mathbf{R}$ which is closed and bounded and $(x_n)$ be a sequence of points in $K$. As $\mathbf{R}$ is a complete ordered field and $(x_n)$ is bounded, it satisfies the Bolzano–Weierstrass Property (cf. Theorem 2.26), that is, $(x_n)$ contains a subsequence converging to a point in $K$ (cf. Exercise 3.4). Hence, $K$ is a compact set. □

A prototypical example of a compact set in $\mathbf{R}$ is the closed interval $[0, 1]$.

The Heine–Borel Theorem does not hold in an incomplete ordered field, as the next theorem shows.

**Theorem 3.11.** *Let* $\mathbf{F}$ *be an incomplete ordered field. There is a bounded closed subset* $E$ *of* $\mathbf{F}$ *that is not compact.*

*Proof.* By Theorem 2.29, an ordered field $\mathbf{F}$ is incomplete if and only if it does not satisfy the Bolzano–Weierstrass Property, that is, there is a bounded sequence $(x_n)$ of points in $\mathbf{F}$ that does not contain a convergent subsequence. Because $(x_n)$ is bounded, there is a closed bounded interval $[a, b]$ that contains the sequence $(x_n)$. It follows that $[a, b]$ is not a compact subset of $\mathbf{F}$. □

A "constructive" proof of Theorem 3.11 is outlined in Exercise 3.9.

By Theorems 3.10 and 3.11, we have the following characterization of the completeness property of an ordered field.

**Theorem 3.12.** *An ordered field* $\mathbf{F}$ *is complete, that is* $\mathbf{F} = \mathbf{R}$*, if and only if every compact subset of* $\mathbf{F}$ *is closed and bounded.*

Compact subsets of the field $\mathbf{R}$ can be characterized by the following, rather odd-sounding property: every open covering of a compact set has a finite subcovering. However, this property is instrumental in some applications. First, we define the notion of an open covering.

**Definition 3.8.** Let $E$ be a subset of an ordered field $\mathbf{F}$. A family $\{U_i\}_{i\in J}$ of open subsets of $\mathbf{F}$ is said to be an *open covering* of $E$ if $E \subseteq \bigcup_{i\in J} U_i$.

If $J' \subseteq J$ and $E \subseteq \bigcup_{i\in J'} U_i$, then the family $\{U_i\}_{i\in J'}$ is called an *open subcovering* of the covering $\{U_i\}_{i\in J}$. A subcovering $\{U_i\}_{i\in J'}$ is said to be *finite* if $J'$ is a finite set.

The following lemma will be needed to prove the last theorem of this section—the Borel–Lebesgue theorem.

**Lemma 3.5.** *Let* $\{U_i\}_{i\in J}$ *be an open covering of a compact set* $K \subseteq \mathbf{R}$*. Then there exists* $\varepsilon > 0$ *such that for every* $x \in K$*,* $(x-\varepsilon, x+\varepsilon) \subseteq U_i$ *for some* $i \in J$*.*

*Proof.* Suppose to the contrary that for every $n \in \mathbf{N}$ there is $x_n \in K$ such that

$$\left(x_n - \frac{1}{n}, x_n + \frac{1}{n}\right) \not\subseteq U_i, \qquad \text{for all } i \in J.$$

Since $K$ is compact, there is a subsequence $(x_{n_k})$ of the sequence $(x_n)$ that converges to a point in $K$, that is, $x_{n_k} \to x \in K$. Since $x \in K$, $x \in U_{i_0}$ for some $i_0 \in J$. Therefore, there is $m \in \mathbf{N}$ such that

$$\left(x - \frac{1}{m}, x + \frac{1}{m}\right) \subseteq U_{i_0}.$$

As $x_{n_k} \to x$, there is $n_0 > 2m$ such that

$$x_{n_0} \in \left(x - \frac{1}{2m}, x + \frac{1}{2m}\right).$$

We have (cf. Exercise 3.5)

$$\begin{aligned}\left(x_{n_0} - \frac{1}{n_0}, x_{n_0} + \frac{1}{n_0}\right) &\subseteq \left(x_{n_0} - \frac{1}{2m}, x_{n_0} + \frac{1}{2m}\right)\\ &\subseteq \left(x - \frac{1}{m}, x + \frac{1}{m}\right) \subseteq U_{i_0},\end{aligned}$$

contradicting our assumption. □

**Theorem 3.13 (Borel–Lebesgue Theorem).** *A subset* $K$ *of the field* $\mathbf{R}$ *is compact if and only if every open covering of* $K$ *contains a finite subcovering.*

*Proof.* (Necessity.) Let $K$ be a compact subset of the field $\mathbf{R}$ and $\{U_i\}_{i\in J}$ an open covering of $K$. By Lemma 3.5, there is $\varepsilon > 0$ such that for every $x \in K$, we have $(x-\varepsilon, x+\varepsilon) \subseteq U_i$ for some $i \in J$. Clearly, it suffices to prove that $K$ can be covered by a finite number of intervals $(x-\varepsilon, x+\varepsilon)$. If $K \subseteq (x_1-\varepsilon, x_1+\varepsilon)$ for some $x_1 \in K$, we are done. Otherwise, let $x_2$ be a point in $K \setminus (x_1-\varepsilon, x_1+\varepsilon)$. If

$$K \subseteq (x_1-\varepsilon, x_1+\varepsilon) \cup (x_2-\varepsilon, x_2+\varepsilon),$$

we are done. If by continuing this process we obtain a finite covering of $K$ by $\varepsilon$-neighborhoods, the proof is over. Otherwise, there is a sequence $(x_n)$ of points in $K$ such that

$$x_{n+1} \notin (x_1-\varepsilon, x_1+\varepsilon) \cup \cdots \cup (x_n-\varepsilon, x_n+\varepsilon)$$

for every $n \in \mathbf{N}$. It is clear that $|x_m - x_n| \geq \varepsilon$ for all $m, n \in \mathbf{N}$. It follows that the sequence $(x_n)$ has no Cauchy subsequences and hence no convergent subsequences. This contradicts our assumption that $K$ is a compact set. Therefore the set $K$ can be covered by a finite number of sets $U_i$.

(Sufficiency.) Let $K$ be a subset of $\mathbf{R}$ such that every open covering of $K$ contains a finite subcovering. Suppose that $(x_n)$ is a sequence of points in $K$ that does not have a convergent subsequence. Then for every point $x \in K$ there is an $\varepsilon_x$-neighborhood of $x$ that contains no points of $(x_n)$ except possibly $x$ itself (cf. Exercise 3.10). The family $\{(x-\varepsilon_x, x+\varepsilon_x)\}_{x\in K}$ is an open covering of $K$ and therefore contains a finite subcovering:

$$K \subseteq (a_1-\varepsilon_1, a_1+\varepsilon_1) \cup \cdots \cup (a_n-\varepsilon_n, a_n+\varepsilon_n)$$

for a finite subset $A = \{a_1, \ldots, a_n\}$ of $K$. By the choice of $\varepsilon_x$-neighborhoods, we have $x_n \in A$ for all $n \in \mathbf{N}$, which contradicts our assumption that $(x_n)$ does not have a convergent subsequence (cf. Exercise 3.8). □

## 3.2 Continuity

In what follows, $E$ is a nonempty subset of an ordered field $\mathbf{F}$.

**Definition 3.9.**

(a) A function $f : E \to \mathbf{F}$ is said to be *continuous at* $a \in E$ if for every $\varepsilon > 0$, there is a $\delta > 0$ such that

$$|x-a| < \delta \quad \text{and} \quad x \in E \qquad \text{imply} \qquad |f(x) - f(a)| < \varepsilon.$$

If $f$ is not continuous at $a \in E$, then it is said to be *discontinuous at* $a$.

(b) $f$ is said to be *continuous on* $E$ if $f$ is continuous at every point in $E$. Otherwise, $f$ is said to be *discontinuous on* $E$.

The following two theorems give equivalent definitions of continuity. The proof of the first theorem is left as an exercise (cf. Exercise 3.15).

**Theorem 3.14.** *A function $f : E \to \mathbf{F}$ is continuous at $a \in E$ if and only if for every $\varepsilon$-neighborhood $U_\varepsilon$ of $f(a)$ there is a $\delta$-neighborhood $U_\delta$ of the point $a$ such that $f(E \cap U_\delta) \subseteq U_\varepsilon$.*

**Theorem 3.15.** *A function $f : E \to \mathbf{F}$ is continuous on $E$ if and only if the inverse image of every open subset of $\mathbf{F}$ is relatively open in $E$.*

*Proof.* (Necessity.) Let $f : E \to \mathbf{F}$ be a continuous function and $U$ an open subset of $\mathbf{F}$. We may suppose that $f^{-1}(U) \neq \varnothing$. For $a \in f^{-1}(U)$, because $U$ is an open set and $f(a) \in U$, there is $\varepsilon > 0$ such that the $\varepsilon$-neighborhood $U_\varepsilon$ of $f(a)$ is contained in $U$. Since $f$ is continuous, there is $\delta > 0$ such that $f(E \cap U_\delta) \subseteq U_\varepsilon \subseteq U$ for the $\delta$-neighborhood $U_\delta$ of $a$ (cf. Theorem 3.14). It follows that $E \cap U_\delta \subseteq f^{-1}(U)$, so $f^{-1}(U)$ is a relatively open subset of $E$.

(Sufficiency.) Suppose that the inverse image of every open subset of $\mathbf{F}$ is relatively open in $E$. Let $a$ be a point in $E$ and $\varepsilon > 0$. Let $V$ be the inverse image of the $\varepsilon$-neighborhood $U_\varepsilon$ of $f(a)$. Since $V$ is a relatively open subset of $E$, there is a $\delta$-neighborhood $U_\delta$ of $a$ such that $E \cap U_\delta \subseteq V = f^{-1}(U_\varepsilon)$, so $f(E \cap U_\delta) \subseteq U_\varepsilon$. By Theorem 3.14, $f$ is continuous at $a \in E$. Since $a$ is an arbitrary point in $E$, $f$ is continuous on $E$. □

We proved that open sets are invariant under inverse images by continuous functions. However, images of open sets are not necessarily open sets.

*Example 3.2.* Let $f : \mathbf{R} \to \mathbf{R}$ be defined by $f(x) = x^2$, $x \in \mathbf{R}$ and $I = (-1, 1)$. Then $f(I) = [0, 1)$, which is not an open subset of $\mathbf{R}$.

The next two theorems show that connected and compact sets are invariant under images by continuous functions.

**Theorem 3.16.** *Let $f : E \to \mathbf{F}$ be a continuous function on a connected set $E \subseteq \mathbf{F}$. Then the image $f(E)$ is a connected subset of $\mathbf{F}$.*

*Proof.* Suppose to the contrary that $f(E)$ is disconnected, that is, there are open subsets $U$ and $V$ of $\mathbf{F}$ such that $f(E) \cap U \neq \varnothing$, $f(E) \cap V \neq \varnothing$, and

$$(f(E) \cap U) \cup (f(E) \cap V) = f(E) \quad \text{and} \quad (f(E) \cap U) \cap (f(E) \cap V) = \varnothing.$$

By Theorem 3.15, the nonempty sets $E \cap f^{-1}(U)$ and $E \cap f^{-1}(V)$ are relatively open in $E$. We have

$$E = f^{-1}[(f(E) \cap U) \cup (f(E) \cap V)] = (E \cap f^{-1}(U)) \cup (E \cap f^{-1}(V))$$

and

$$(E \cap f^{-1}(U)) \cap (E \cap f^{-1}(V)) = f^{-1}[(f(E) \cap U) \cap (f(E) \cap V)] = \varnothing.$$

This contradicts the hypothesis that $E$ is connected. □

To prove the second theorem, we need the following lemma.

**Lemma 3.6.** *If $f : E \to \mathbf{F}$ is continuous at $a \in E$ and $a_n \to a$ in $E$, then $f(a_n) \to f(a)$.*

*Proof.* By Definition 3.9, for $\varepsilon > 0$ there is a $\delta > 0$ such that

$$|x - a| < \delta \quad \text{and} \quad x \in E \qquad \text{imply} \qquad |f(x) - f(a)| < \varepsilon.$$

Since $a_n \to a$, there is $N \in \mathbf{N}$ such that $|a_n - a| < \delta$ for all $n > N$. It follows that $|f(a_n) - f(a)| < \varepsilon$ for all $n > N$, that is, $f(a_n) \to f(a)$. □

**Theorem 3.17.** *Let $f : K \to \mathbf{F}$ be a continuous function on a compact set $K \subseteq \mathbf{F}$. Then the image $f(K)$ is a compact subset of $\mathbf{F}$.*

*Proof.* Let $(a_n)$ be a sequence of points in $f(K)$. Since $K$ is compact, the sequence $(f^{-1}(a_n))$ in $K$ contains a subsequence $(f^{-1}(a_{n_k}))$ that converges to some point $a \in K$. By Lemma 3.6, the subsequence $(a_{n_k}) = (f(f^{-1}(a_{n_k})))$ converges to $f(a) \in f(K)$. Hence, $f(K)$ is a compact set. □

The class of continuous functions $f : E \to \mathbf{F}$ is closed under the usual algebraic operations. For this the reader is referred to Exercises 3.16, 3.17, and 3.18.

*Example 3.3.* Let $f : \mathbf{R} \to \mathbf{R}$ be the function defined by

$$f(x) = \begin{cases} 1, & \text{if } x \text{ is a rational number,} \\ 0, & \text{if } x \text{ is an irrational number.} \end{cases}$$

(Dirichlet's "discontinuous function".) This function is, indeed, discontinuous at every $a \in \mathbf{R}$. If $a \in \mathbf{Q}$, then $f(a) = 1$. By the denseness of irrational numbers in $\mathbf{R}$ (cf. Exercise 2.26), for every $\delta > 0$, there is an irrational number $x$ such that $|x - a| < \delta$. Let $\varepsilon = 1/2$. Because $f(x) = 0$, we have $|f(x) - f(a)| = 1 \geq \varepsilon$. Hence, $f$ is discontinuous at $a$. A similar argument shows that $f$ is discontinuous at every irrational $a \in \mathbf{R}$.

In what follows, we focus on two fundamental theorems (so-called "pillars" of Calculus) about real-valued functions and demonstrate that these results characterize the completeness property of the field $\mathbf{R}$.

**Theorem 3.18 (Intermediate Value Theorem).** *Let $f : [a, b] \to \mathbf{R}$ be a continuous function on a closed interval $[a, b]$. If $f(a) < y < f(b)$ for some $y \in \mathbf{R}$, then there is at least one point $x \in [a, b]$ such that $f(x) = y$.*

*Proof.* Let $A$ be the set defined by

$$A = \{x \in [a, b] : f(x) < y\}.$$

The set $A$ is not empty ($f(a) < y$) and bounded. By the Dedekind Completeness Property (cf. Definition 2.2), the set $A$ has a supremum which we denote by $c$:

$$c = \sup A.$$

Below we prove that $c$ is the required number, that is, $f(c) = y$.

For $n \in \mathbf{N}$, the number $c - \dfrac{1}{n}$ is not an upper bound of $A$, because $c - \dfrac{1}{n} < c$. Therefore, there are $x_n \in A$ such that

$$c - \frac{1}{n} < x_n \le c, \qquad \text{for all } n \in \mathbf{N}.$$

By the Squeeze Theorem (cf, Theorem 1.14), $x_n \to c$. By Lemma 3.6, $f(x_n) \to f(c)$. Since $x_n \in A$, we have $f(x_n) < y$ for all $n \in \mathbf{N}$. By the Comparison Principle (cf. Theorem 1.15), $f(c) \le y$. Because $y < f(b)$, we conclude that $c \ne b$, that is, $c < b$.

By the Archimedean Property (cf. Theorem 2.5) of the complete field $\mathbf{R}$, there is $N \in \mathbf{N}$ such that $N > \dfrac{1}{b - c}$. Therefore,

$$c + \frac{1}{n} < b, \qquad \text{for all } n > N.$$

Since $c = \sup A$, we have $c + \dfrac{1}{n} \notin A$, so $f\left(c + \dfrac{1}{n}\right) \ge y$ for all $n > N$. By Lemma 3.6 and the Comparison Principle,

$$f(c) = \lim f\left(c + \frac{1}{n}\right) \ge y.$$

We already proved that $f(c) \le y$. Hence, $f(c) = y$, which is the desired result. □

The property of a continuous real-valued function on a closed interval established in Theorem 3.18 is known as the Intermediate Value Property.

**Definition 3.10 (Intermediate Value Property).** An ordered field $\mathbf{F}$ satisfies the *Intermediate Value Property* if for every continuous function $f : [a, b] \to \mathbf{F}$ and $f(a) < y < f(b)$ or $f(b) < y < f(a)$, there is $x \in [a, b]$ such that $f(x) = y$.

Theorem 3.18 claims that the complete field $\mathbf{R}$ satisfies the Intermediate Value Property. In fact, the Intermediate Value Property is equivalent to the Dedekind completeness of the field, as the following theorem shows.

**Theorem 3.19.** *An ordered field* $\mathbf{F}$ *is complete if and only if it satisfies the Intermediate Value Property.*

*Proof.* (Necessity.) Theorem 3.18.

(Sufficiency.) Let $\mathbf{F}$ be an incomplete ordered field and $[a, b] \subseteq \mathbf{F}$. It suffices to give an example of a continuous function on $[a, b]$ that does not attain any value in the open interval with the endpoints $f(a)$ and $f(b)$. By Lemma 3.4, there is a gap $(A, B)$ in $\mathbf{F}$ such that $a \in A$ and $b \in B$. Let $f$ be the function on $[a, b]$ defined by

$$f(x) = \begin{cases} 1, & \text{if } x \in [a, b] \cap A, \\ 0, & \text{if } x \in [a, b] \cap B. \end{cases}$$

By Lemma 3.1, the sets $[a, b] \cap A$ and $[a, b] \cap B$ are relatively open in $[a, b]$. Since these sets are disjoint, the function $f$ is continuous on $[a, b]$. Clearly, it does not attain values in $(0, 1)$. Thus an incomplete ordered field does not satisfy the Intermediate Value Property. □

The next theorem is another "pillar" of real analysis. First, we introduce a distinguished property—the Extreme Value Property—of continuous functions on closed intervals.

**Definition 3.11.** An ordered field $\mathbf{F}$ is said to satisfy the *Extreme Value Property* if for every continuous function $f : [a, b] \to \mathbf{F}$, there is $c \in [a, b]$ such that $f(x) \leq f(c)$ for all $x \in [a, b]$.

**Theorem 3.20.** *The field of real numbers* $\mathbf{R}$ *satisfies the Extreme Value Property.*

*Proof.* Let $f : [a, b] \to \mathbf{R}$ be a continuous function. Since the interval $[a, b]$ is a compact set (cf. Theorem 3.10), $A = f([a, b])$ is a compact subset of $\mathbf{R}$ (cf. Theorem 3.17) and therefore is closed and bounded (cf. Theorem 3.10). By Lemma 3.2, $m = \sup A \in A$. Let $c = f^{-1}(m) \in [a, b]$. Clearly, $f(x) \leq f(c) = m$ for all $x \in [a, b]$. □

It is worthwhile noting that the assertion of Theorem 3.20 holds for any compact subset of $\mathbf{R}$.

The next theorem claims that the Extreme Value Property characterizes the complete ordered field $\mathbf{R}$.

**Theorem 3.21.** *An ordered field* $\mathbf{F}$ *is complete if and only if it satisfies the Extreme Value Property.*

*Proof.* (Necessity.) Theorem 3.20.

(Sufficiency.) Let $[a, b]$ be a closed interval in an incomplete ordered field $\mathbf{F}$. It suffices to show that there is a continuous function $f : [a, b] \to \mathbf{F}$ that does not satisfy the Extreme Value Property. By Lemma 3.4, there is a gap $(A, B)$ in $\mathbf{F}$ such that $a \in A$ and $b \in B$. Let $f$ be the function on $[a, b]$ defined by

$$f(x) = \begin{cases} x - a, & \text{if } x \in [a,b] \cap A, \\ 0, & \text{if } x \in [a,b] \cap B. \end{cases}$$

Let $A' = [a,b] \cap A$ and $B' = [a,b] \cap B$. Since $A' \cup B' = [a,b]$ and $A' \cap B' = \varnothing$, the function $f$ is continuous on $[a,b]$ (cf. Lemma 3.1). However, there is no $c \in [a,b]$ such that $f(x) \leq f(c)$ for all $x \in [a,b]$. Indeed, such a $c$ must be in $A'$ because $f$ takes positive values on $[a,b]$. But for every $c \in A'$ there is $c' \in A'$ such that $f(c) < f(c')$. □

A "fixed point" property of functions is important not only in real analysis but also more broadly in mathematics.

**Definition 3.12.** An ordered field $\mathbf{F}$ satisfies the *Fixed Point Property* if for every continuous function $f : [a,b] \to [a,b]$, there is $c \in [a,b]$ such that $f(c) = c$.

**Theorem 3.22.** *The ordered field of real numbers* $\mathbf{R}$ *satisfies the Fixed Point Property.*

*Proof.* Let $f : [a,b] \to [a,b]$ be a continuous function. Clearly, the function

$$g(x) = x - f(x), \qquad \text{for } x \in [a,b],$$

is continuous on $[a,b]$ and

$$g(a) = a - f(a) \leq 0, \quad g(b) = b - f(b) \geq 0.$$

If $g(a) = 0$, then $f(a) = a$ and we are done. Otherwise, $g(a) < 0$. Hence, by the Intermediate Value Theorem (cf. Theorem 3.18), there is $c \in [a,b]$ such that $g(c) = 0$, that is, $f(c) = c$. □

The Fixed Point Property of an ordered field characterizes the complete field $\mathbf{R}$, as the following theorem shows.

**Theorem 3.23.** *An ordered field* $\mathbf{F}$ *is complete if and only if it satisfies the Fixed Point Property.*

*Proof.* (Necessity.) Theorem 3.22.

(Sufficiency.) Let $[a,b]$ be closed interval in an incomplete field $\mathbf{F}$. By Lemma 3.4, there is a gap $(A, B)$ in $\mathbf{F}$ such that $a \in A$, $b \in B$. We define

$$f(x) = \begin{cases} b, & \text{if } x \in A \cap [a,b], \\ a, & \text{if } x \in B \cap [a,b]. \end{cases}$$

Let $A' = A \cap [a,b]$ and $B' = b \cap [a,b]$. Since $A' \cup B' = [a,b]$ and $A' \cap B' = \varnothing$, the function $f$ is continuous on $[a,b]$ (cf. Lemma 3.1). It is not difficult to see that $f$ does not have a fixed point in $[a,b]$. □

We conclude this section by establishing a "continuity property" of the inverse of a strictly increasing function. This property is often introduced for continuous functions. However, this restriction is superfluous, as the next theorem demonstrates.

**Theorem 3.24.** *Let $f : I \to \mathbf{F}$ be a strictly increasing function on the interval $I \subseteq \mathbf{F}$. Then the inverse function $f^{-1} : f(I) \to \mathbf{F}$ is continuous.*

*Proof.* Since $f$ is a strictly increasing function, it is a bijection from $I$ onto $f(I)$. Hence, the inverse function $f^{-1} : f(I) \to \mathbf{F}$ is defined. Moreover, it is not difficult to see that $f^{-1}$ is also a strictly increasing function.

By Theorem 3.15, to prove that $f^{-1}$ is continuous on $f(I)$ it suffices to show that the image under $f$ of every open subset of $I$ is relatively open in $f(I)$. As every relatively open subset of $I$ is a union of relatively open intervals in $I$, we only need to prove that the image under $f$ of a relatively open interval in $I$ is a relatively open subset of $f(I)$.

We prove theorem for $I = [a, b)$. The proofs for other intervals are similar and left as an exercise (cf. Exercise 3.24). There are three kinds of relatively open intervals in $I$. We consider them separately.

Case 1. $(x, y)$, where $x, y \in I$. We show that

$$f((x, y)) = (f(x), f(y)) \cap f(I), \tag{3.1}$$

that is, $f((x, y))$ is relatively open in $f(I)$.

For $p \in f((x, y))$ there is $x < q < y$ such that $p = f(q)$. Since $f$ is strictly increasing, $f(x) < f(q) = p < f(y)$. Hence, $p \in (f(x), f(y))$. Clearly, $p = f(q) \in f(I)$. Therefore,

$$f((x, y)) \subseteq (f(x), f(y)) \cap f(I)$$

On the other hand, for $p \in (f(x), f(y)) \cap f(I)$ there is $q \in I$ such that $p = f(q)$. Since $f(x) < p = f(q) < f(y)$ and the inverse function $f^{-1}$ is strictly increasing, $x < q < y$, that is, $q \in (x, y)$. Hence, $p = f(q) \in f((x, y))$. It follows that

$$f((x, y)) \supseteq (f(x), f(y)) \cap f(I).$$

The last two displayed set inclusions prove the equality in (3.1).

Case 2. $[a, y)$, where $y \in I$. The proof repeats the steps from Case 1.

Case 3. $(x, b)$, where $x \in I$. This interval is a union of open intervals with endpoints in $(x, b) \subseteq I$ (cf. Exercise 3.25). It remains to apply the result of Case 1. □

Clearly, the assertion of Theorem 3.24 also holds for strictly decreasing functions.

## 3.3 Uniform Continuity

The concept of continuity of a function $f : E \to \mathbf{F}$ was introduced "locally" as continuity at every point of the set $E$. Below we introduce a "global" version of the concept.

**Definition 3.13.** A function $f : E \to \mathbf{F}$ is said to be *uniformly continuous on* $E$ if for every $\varepsilon > 0$, there is a $\delta > 0$ such that $|f(x) - f(y)| < \varepsilon$ whenever $|x - y| < \delta$ for $x, y \in E$.

It is easy to see that a uniformly continuous function is continuous (cf. Exercise 3.27). The converse, in general, is not true.

*Example 3.4.* The real-valued function $f(x) = 1/x$ on $(0, 1] \subseteq \mathbf{R}$ is continuous but not uniformly continuous on $(0, 1]$ (cf. Exercise 3.28).

We have, however, the following important theorem.

**Theorem 3.25.** *Let* $[a, b]$ *be a closed interval in* $\mathbf{R}$ *and* $f : [a, b] \to \mathbf{R}$ *a continuous function. Then* $f$ *is uniformly continuous on* $[a, b]$.

*Proof.* Suppose to the contrary that $f$ is not uniformly continuous on $[a, b]$. Then there is $\varepsilon_0 > 0$ such that, for every $n \in \mathbf{N}$ there are $x_n, y_n \in [a, b]$ such that

$$|x_n - y_n| < \frac{1}{n} \qquad \text{and} \qquad |f(x_n) - f(y_n)| \geq \varepsilon_0.$$

By the Bolzano–Weierstrass theorem, there are convergent subsequences $(x_{n_k})$ and $(y_{n_k})$ of the sequences $(x_n)$ and $(y_n)$, respectively. Let $a = \lim x_{n_k}$ and $b = \lim y_{n_k}$. Since $|x_n - y_n| < 1/n$ for all $n \in \mathbf{N}$, $a = b$. As $f$ is a continuous function,

$$\lim \big(f(x_{n_k}) - g(y_{n_k})\big) = f(a) - f(a) = 0,$$

which contradicts our assumption that $|f(x_n) - f(y_n)| \geq \varepsilon_0$ for all $n \in \mathbf{N}$. □

**Theorem 3.26.** *Let* $f : E \to \mathbf{F}$ *be a uniformly continuous function. If* $(x_n)$ *is a Cauchy sequence in* $E$*, then the sequence* $(f(x_n))$ *is Cauchy.*

*Proof.* For $\varepsilon > 0$, in $\mathbf{F}$ let $\delta$ be a positive element of $\mathbf{F}$ such that

$$x, y \in E \quad \text{and} \quad |x - y| < \delta \qquad \text{implies} \qquad |f(x) - f(y)| < \varepsilon.$$

Since $(x_n)$ is Cauchy, there is $N \in \mathbf{N}$ such that $|x_n - x_m| < \delta$ for all $m, n \geq N$. Then $m, n \geq N$ implies $|f(x_n) - f(x_m)| < \varepsilon$. Hence, $(f(x_n))$ is a Cauchy sequence. □

Theorem 3.26 gives a criterion for verifying whether a continuous function is uniformly continuous. For example, the function $f(x) = 1/x$ from Example 3.4 is

not uniformly continuous because the sequence $(1/n)$ in $(0, 1]$ is Cauchy, whereas the sequence $(n)$ is not.

In conclusion of this chapter, we introduce a class of functions that is widely popular in analysis.

**Definition 3.14.** Let $f : E \to \mathbf{F}$ be a function. If there is a constant $C > 0$ in $\mathbf{F}$ such that

$$|f(x) - f(y)| \le C|x - y|, \qquad \text{for all } x, y \in E,$$

then $f$ is said to be a *Lipschitz function* on $E$.

**Theorem 3.27.** *If $f : E \to \mathbf{F}$ is a Lipschitz function, then $f$ is uniformly continuous on $E$.*

*Proof.* Let $f$ be a Lipschitz function on $E$ and $C$ a constant from Definition 3.14. For $\varepsilon > 0$, let $\delta = \varepsilon/C$. If $x, y \in E$ satisfy $|x - y| < \delta$, then

$$|f(x) - f(y) \le C|x - y| < C \cdot \frac{\varepsilon}{C} = \varepsilon, \qquad \text{for all } x, y \in E.$$

Hence, $f$ is uniformly continuous on $E$. □

*Example 3.5.* Let $f(x) = x^2$ on $(-1, 1)$. For $x, y \in (-1, 1)$, we have

$$|x^2 - y^2| = |x + y| \cdot |x - y| \le (|x| + |y|) \cdot |x - y| < 2|x - y|.$$

Therefore, $f$ is Lipschitz on $(-1, 1)$ with $C = 2$. By Theorem 3.27, $f$ is uniformly continuous on $(-1, 1)$. Note that Theorem 3.25 is not applicable in this example.

*Example 3.6.* Let $f(x) = \sqrt{x}$ on $[0, 1]$ (cf. Exercise 3.22). By Theorem 3.25, this function is uniformly continuous. However, $f$ is not Lipschitz. Indeed, suppose, by contradiction, that there is a $C > 0$ such that

$$|\sqrt{x} - \sqrt{y}| \le C|x - y|, \qquad \text{for all } x, y \in [0, 1].$$

For $x = \dfrac{1}{4C^2}$, $y = 0$, we have

$$|\sqrt{x} - \sqrt{y}| = \frac{1}{2C} > C\frac{1}{4C^2} = C|x - y|,$$

a contradiction.

## Notes

A family of subsets of **F** satisfying properties established in Theorem 3.2 is called a *topology* on **F**. Because the family in Theorem 3.2 is defined by means of open intervals in **F**, it is commonly called the *interval topology* on **F**.

By Theorem 3.3, all open sets in **R** are disjoint unions of at most countable many intervals. By Lemma 3.1 and the proof of Theorem 3.4, there are many other open sets in an incomplete ordered field (cf. Exercise 2.30).

The converse of Theorem 3.7 holds: A subset $E$ of **R** is connected if and only if $E$ is an interval (Tao 2009, Theorem 13.4.5). Theorem 3.9 asserts that the connectedness property of an ordered field is equivalent to the completeness property.

Heinrich Eduard Heine (1821–1881) was a German mathematician mostly known for results on special functions and in real analysis. Félix Édouard Justin Émile Borel (1871–1956) was a French mathematician and politician. As a mathematician, he was known for his founding work in the areas of measure theory and probability.

The number $\varepsilon$ in Lemma 3.5 is called the *Lebesgue number* of the open covering $\{U_i\}_{i\in J}$. Henri Léon Lebesgue (1875–1941) was a French mathematician known for his generalization of the Riemann integral.

By the Borel–Lebesgue Theorem (cf. Theorem 3.13), compact subsets of the field **R** are characterized by means of open coverings. This result motivates the definition of a "compact" set in advanced analysis and topology. Namely, a subset of a "topological space" is said to be *compact* if every open covering of this subset has a finite subcovering (Munkres 2000, § 26). Similarly, the assertion of Theorem 3.15 is the definition of continuous function in topology—a function is continuous if the inverse image of every open set in the codomain is an open set in the domain of the function (Munkres 2000, § 18).

Johann Peter Gustav Lejeune Dirichlet (1805–1859) was a German mathematician who made deep contributions to number theory and mathematical analysis.

The assertion of Theorem 3.25 "almost" characterizes the field **R**. Namely, if every continuous function $[a, b] \to \mathbf{F}$ is uniformly continuous, then the field **F** is complete (that is, $\mathbf{F} = \mathbf{R}$), provided that **F** satisfies the "countable cofinality" condition (cf. Definition 4.2 in Chapter 4). For proofs the reader is referred to Teismann (2013).

## Exercises

**3.1.** Prove Theorem 3.1.

**3.2.** Prove Lemma 3.3.

**3.3.** Let $I$ be an interval in **F**. Show that every point in $I$ is its limit point.

**3.4.** Show that a convergent sequence of points in a closed set $F$ converges to a point of $F$.

**3.5.** For $\varepsilon > 0$, let $x$, $y$ be points in $\mathbf{R}$ such that $|x - y| < \varepsilon$. Show that

$$(y - \varepsilon, y + \varepsilon) \subseteq (x - 2\varepsilon, x + 2\varepsilon).$$

**3.6.** Show that the set $\{1/n : n \in \mathbf{N}\}$ is not a compact subset of $\mathbf{R}$.

**3.7.** Give an example of a non-compact subset of $\mathbf{R}$ and an open covering of it that does not satisfy the assertion of Lemma 3.5.

**3.8.** Let $(x_n)$ be a sequence of points in $\mathbf{R}$ such that the set $\{x_n : n \in \mathbf{N}\}$ is finite. Show that $(x_n)$ contains a convergent subsequence.

**3.9.**

(a) Let $\mathbf{F}$ be an Archimedean incomplete ordered field, $(A, B)$ a gap in $\mathbf{F}$, $a_1 \in A$, $b_1 \in B$, and $(a_n)$ the sequence in $A$ obtained by the "bisection method" used in the proof of Theorem 2.4. Show that the set $\{a_n : n \in \mathbf{N}\}$ is closed and bounded, but not compact.
(b) Let $\mathbf{F}$ be a non-Archimedean ordered field. Show that the set of natural numbers $\mathbf{N} \subseteq \mathbf{F}$ is closed and bounded, but not compact.

**3.10.** Show that a point $x \in \mathbf{R}$ is the limit of a subsequence of a sequence $(x_n)$ if an only if every neighborhood of $x$ contains infinitely many terms of $(x_n)$.

**3.11.** Prove that the union of two compact sets is compact.

**3.12.** Prove that the intersection of a family of compact sets is compact.

**3.13.** Let $E$ be a closed subset of a compact set. Prove that $E$ is compact.

**3.14.** Show that a function $f : E \to \mathbf{F}$ is continuous at every isolated point of $E$ (cf. Definition 3.3).

**3.15.** Prove Theorem 3.14.

**3.16.** Let $E$ be a nonempty subset of $\mathbf{F}$ and $f : E \to \mathbf{F}$, $g : E \to \mathbf{F}$ continuous functions. Show that

(a) $f + g$,
(b) $k \cdot f$, $\quad k \in \mathbf{F}$,
(c) $f \cdot g$,
(d) $f/g$, where $g(x) \neq 0$ on $E$, and
(e) $|f|$

are continuous functions on $E$.

**3.17.** Let $E, E' \subseteq \mathbf{F}$ and let $f : E \to \mathbf{F}$ and $g : E' \to \mathbf{F}$ be continuous functions such that $f(E) \subseteq E'$. Show that the composition $g \circ f \to \mathbf{F}$ is a continuous function.

**3.18.** Show that a polynomial function $p : E \to \mathbf{F}$, where

$$p(x) = a_n x^n + a_{n-1}x^{n-1} + \cdots + a_1 x + a_0, \qquad x \in E,\ a_k \in \mathbf{F}\ (0 \le k \le n),$$

and $n \in \mathbf{N} \cup \{0\}$, is continuous on $E$.

**3.19.** Let $E$ be a connected subset of $\mathbf{F}$ and $a < b$ two points in $E$. Show that $[a, b] \subseteq E$.

**3.20.** Prove the following generalization of Theorem 3.18:

Let $f : E \to \mathbf{R}$ be a continuous function on a connected set $E \subseteq \mathbf{R}$. If $a$ and $b$ are any two points in $E$, and $y \in \mathbf{R}$ belongs to the closed interval whose end-points are $f(a)$ and $f(b)$, then there is at least one point $x \in E$ such that $f(x) = y$.

**3.21.**

(a) Let $f : \mathbf{F} \to \mathbf{F}$ be continuous at $a$. Suppose that $f(a) > 0$. Show that there is a $\delta > 0$ such that $f(x) > 0$ for all $x \in (a - \delta, a + \delta)$.

(b) For a continuous function $f : \mathbf{F} \to \mathbf{F}$, show that the set

$$A = \{x \in \mathbf{F} : f(x) = 0\}$$

is closed.

**3.22.** Let $a < b < c$ be points in $\mathbf{F}$, and let $f : [a, b] \to \mathbf{F}$, $g : [b, c] \to \mathbf{F}$ be continuous functions. Define $h$ on $[a, c]$ by

$$h(x) = \begin{cases} f(x), & \text{if } x \in [a, b], \\ g(x), & \text{if } x \in [b, c]. \end{cases}$$

Show that the function $h$ is continuous, provided that $f(b) = g(b)$.

**3.23.** Let $f(x) = x^n$, $x \in [0, \infty) \subseteq \mathbf{R}$, be the $n$th power function. Show that the inverse function $f^{-1}(x) = \sqrt[n]{x}$ exists and is continuous. (Hint: Apply Theorem 3.24.)

**3.24.** Prove Theorem 3.24 for all intervals in $\mathbf{F}$.

**3.25.** Show that every open interval $I$ in $\mathbf{F}$ is a union of open intervals with endpoints in $I$.

**3.26.** Prove that an increasing function on an interval $I \subseteq \mathbf{R}$ is continuous if and only if $f(I)$ is an interval.

**3.27.** Prove that a uniformly continuous function $f : E \to \mathbf{F}$ is continuous at every point in $E$.

**3.28.** Show that the function $f(x) = 1/x$ is not uniformly continuous on $(0, 1] \subseteq \mathbf{R}$.

# Chapter 4
# Differentiation




The notion of the limit of a function is introduced in Section 4.1 where properties of limits are also established. This notion serves a basis for the definition of the derivative in Section 4.2. This section is mainly devoted to the basic results concerning the differentiation of functions.

Section 4.3 is central in this chapter. It focuses on fundamental theorems of the differential calculus. In line with the developments in the previous chapter, it is shown that the statements of these theorems characterize the completeness property of the underlying field.

Convex functions and their differentiability properties are studied in Section 4.4 in the context of the completeness property.

## 4.1 Limits of Functions

In this section, $E$ is a nonempty subset of an ordered field $\mathbf{F}$.

**Definition 4.1.** Let $f : E \to \mathbf{F}$ be a function and $c$ a limit point of the set $E$. We say that a point $L \in \mathbf{F}$ is a *limit of* $f$ *at* $c$ and write

$$f(x) \longrightarrow L \quad \text{as} \quad x \longrightarrow c,$$

if for every $\varepsilon > 0$ there is a $\delta > 0$ such that

$$|f(x) - L| < \varepsilon \quad \text{for all points } x \in E \text{ for which } \quad 0 < |x - c| < \delta.$$

In this case, we also say that $f(x)$ *converges to* $L$ *as* $x$ *approaches* $c$.

For a limit point $c$ of the set $E$ and a positive $\delta$, we define

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$$U_\delta(c) = \{x :\in E : 0 < |x - c| < \delta\} = [(c - \delta, c + \delta) \setminus \{c\}] \cap E$$

and use this notation throughout this chapter. Thus $f(x) \to L$ as $x \to c$ if for every $\varepsilon > 0$ there is a positive $\delta$ such that $L - \varepsilon < f(x) < L + \varepsilon$ for all $x \in U_\delta(c)$.

**Theorem 4.1.** *If a function has a limit, then this limit is unique.*

*Proof.* Suppose to the contrary that $f \to L$ and $f \to M$ as $x \to c$ and $L \neq M$. Then there are $\delta_1 > 0$ and $\delta_2 > 0$ such that

$$|f(x) - L| < \tfrac{1}{2}|L - M| \quad \text{for all } x \in U_{\delta_1}(c)$$

and

$$|f(x) - M| < \tfrac{1}{2}|L - M| \quad \text{for all } x \in U_{\delta_2}(c).$$

Let $\delta = \min\{\delta_1, \delta_2\}$. Then for all $x \in U_\delta(c)$,

$$\begin{aligned} |L - M| &= |(f(x) - M) + (L - f(x)| \le |f(x) - M| + |f(x) - L| \\ &< \tfrac{1}{2}|L - M| + \tfrac{1}{2}|L - M| = |L - M|, \end{aligned}$$

a contradiction. □

If $f(x)$ converges to $L$ as $x$ approaches $c$, we also write

$$\lim_{x \to c} f(x) = L.$$

Note that $c$ in Definition 4.1 need not to be a point of $E$, and even for $c \in E$ one may have $\lim_{x \to c} f(x) \neq f(c)$.

*Example 4.1.*

(a) Let $E = [0, 1) \subseteq \mathbf{F}$. The point 1 is a limit point of $E$ that does not belong to $E$. Let $f(x) = x$ on $E$. Clearly, $\lim_{x \to 1} f(x) = 1$.
(b) Let $f : \mathbf{F} \to \mathbf{F}$ be defined by

$$f(x) = \begin{cases} 1, & \text{if } x = 0, \\ 0, & \text{otherwise.} \end{cases}$$

It is not difficult to see that $\lim_{x \to 0} f(x) = 0 \neq 1$.

The notions of continuity and limit are closely related, as the next theorem asserts.

**Theorem 4.2.** *Let $c \in E$ be a limit point. A function $f : E \to \mathbf{F}$ is continuous at $c$ if and only if*

$$\lim_{x \to c} f(x) = f(c).$$

*Proof.* It suffices to compare Definitions 3.9 and 4.1. We leave the details to the reader. □

The following two theorems are versions of the Squeeze Theorem (cf. Theorem 1.14) and Comparison Principle (cf. Theorem 1.15) for limits of sequences.

**Theorem 4.3.** *Let $f$, $g$, and $h$ be functions on $E$ such that*

$$f(x) \leq g(x) \leq h(x), \qquad \text{for } x \in E,$$

*and let $c$ be a limit point of $E$. If $\lim_{x \to c} f(x) = L$ and $\lim_{x \to c} h(x) = L$, then $\lim_{x \to c} g(x) = L$.*

*Proof.* Let $\varepsilon$ be a positive element of $\mathbf{F}$. There are $\delta_1 > 0$ and $\delta_2 > 0$ such that

$$L - \varepsilon < f(x) < L + \varepsilon, \qquad \text{for all } x \in U_{\delta_1}(c)$$

and

$$L - \varepsilon < h(x) < L + \varepsilon, \qquad \text{for all } x \in U_{\delta_2}(c).$$

Let $\delta = \min\{\delta_1, \delta_2\}$. For every $x \in U_\delta(c)$ we have

$$L - \varepsilon < f(x) \leq g(x) \leq h(x) < L + \varepsilon.$$

It follows that $g(x) \to L$ as $x \to c$. □

**Theorem 4.4.** *Let $f$ and $g$ be functions on $E$ such that*

$$f(x) \leq g(x), \qquad \text{for all } x \in E,$$

*and let $c$ be a limit point of $E$. If $\lim_{x \to c} f(x) = L$ and $\lim_{x \to c} g(x) = M$, then $L \leq M$.*

*Proof.* Suppose to the contrary that $L > M$ and let $\varepsilon = (L - M)/2$. There are $\delta_1 > 0$ and $\delta_2 > 0$ such that

$$f(x) > L - \varepsilon = \frac{L + M}{2}, \qquad \text{for all } x \in U_{\delta_1}(c)$$

and

$$g(x) < M + \varepsilon = \frac{L + M}{2}, \qquad \text{for all } x \in U_{\delta_2}(c).$$

Hence, $f(x) > g(x)$ for $x \in U_{\delta_1}(c) \cap U_{\delta_2}(c) \neq \varnothing$, a contradiction. □

We use the results of the following two lemmas in the proof of Theorem 4.5.

**Lemma 4.1.** *If $f : E \to \mathbf{F}$ has limit at a limit point $c$, then $f$ is bounded in some neighborhood of $c$ in $E$.*

*Proof.* Let $L = \lim_{x\to c} f$. For $\varepsilon = 1$ there is a $\delta > 0$ such that

$$L - 1 < f(x) < L + 1,$$

for $x \in U_\delta(c)$. It is clear that $f$ is bounded on the set $(c - \delta, c + \delta) \cap E$. □

**Lemma 4.2.** *Let $f$ and $g$ be functions from the set $E$ into $\mathbf{F}$. If $f$ is bounded on a neighborhood of a limit point $c$ of the set $E$ and $g(x) \to 0$ as $x \to c$, then $f(x)g(x) \to 0$ as $x \to c$.*

*Proof.* Since $f$ is bounded on a neighborhood of $c$, there is $B > 0$ such that $|f(x)| < B$ on the set $(c - \lambda, c + \lambda) \cap E$ for some $\lambda > 0$. Let $\varepsilon$ be a positive element of $\mathbf{F}$. Since $g(x) \to 0$ as $x \to c$, there is a $\delta > 0$ such that $|g(x)| < \varepsilon/B$ for $x \in U_\delta(c)$. Let $\delta' = \min\{\lambda, \delta\}$. Then

$$|f(x)g(x)| < B \cdot \frac{\varepsilon}{B} = \varepsilon$$

for $x \in U_{\delta'}(c)$. Hence, $f(x)g(x) \to 0$ as $x \to c$. □

As in the case of continuous functions (cf. Exercise 3.16), we have the following "algebraic" properties of limits.

**Theorem 4.5.** *Let $f : E \to \mathbf{F}$ and $g : E \to \mathbf{F}$ be functions and let*

$$\lim_{x\to c} f(x) = L, \qquad \lim_{x\to c} g(x) = M,$$

*where $c$ is a limit point of $E$. Then*

(a) $\lim_{x\to c}(f + g)(x) = L + M$.
(b) $\lim_{x\to c}(kf)(x) = kL$ *for* $k \in \mathbf{F}$.
(c) $\lim_{x\to c} |f(x)| = |L|$.
(d) $\lim_{x\to c}(f \cdot g)(x) = L \cdot M$.
(e) $\lim_{x\to c}(f/g)(x) = L/M$, *provided that* $g(x) \neq 0$ *on* $E$ *and* $M \neq 0$.

*Proof.* Let $\varepsilon > 0$ be an element of the field $\mathbf{F}$.

(a) Since $f(x) \to L$, $g(x) \to M$ at $c$, there are $\delta_1 > 0$ and $\delta_2 > 0$ such that

$$|f(x) - L| < \varepsilon/2, \qquad \text{for } x \in U_{\delta_1}(c),$$

and

$$|g(x) - M| < \varepsilon/2, \qquad \text{for } x \in U_{\delta_2}(c).$$

Let $\delta = \min\{\delta_1, \delta_2\}$. Then

$$\begin{aligned}\big|(f(x)+g(x))-(L+M)\big| &= \big|(f(x)-L)+(g(x)-M)\big| \\ &\le \big|f(x)-L\big| + \big|g(x)-M\big| < \varepsilon,\end{aligned}$$

for all $x \in U_\delta(c)$. Hence, $(f+g) \to L+M$.

(b) The proof is left to the reader (cf. Exercise 4.1).

(c) Since $f(x) \to L$ at $c$, there is $\delta > 0$ such that $|f(x)-L| < \varepsilon$ for all $x \in U_\delta(c)$. We have (cf. Exercise 1.17 (h))

$$\big||f(x)| - |L|\big| \le \big|f(x)-L\big| < \varepsilon$$

for $x \in U_\delta(c)$. Therefore, $|f|(x) \to |L|$ at $c$.

(d) We have

$$\begin{aligned}0 \le \big|f(x)g(x)-LM\big| &= \big|(f(x)g(x)-f(x)M)+(f(x)M-LM)\big| \\ &= \big|f(x)(g(x)-M)+M(f(x)-L)\big| \le |f(x)|\big|g(x)-M\big|+|M|\big|f(x)-L\big|.\end{aligned}$$

By parts (a), (b) and (c) of the proof, Lemmas 4.1 and 4.2, Exercise 4.2, and Theorem 4.3,

$$\lim_{x\to c}(f(x)g(x)) = LM.$$

(e) We may assume that $M > 0$. There is a $\delta > 0$ such that

$$-M/2 < g(x) - M < M/2 \qquad \text{for all } x \in U_\delta(c).$$

Hence, $g(x) > M/2$ on $U_\delta(c)$. Then

$$\begin{aligned}0 \le \left|\frac{f(x)}{g(x)} - \frac{L}{M}\right| = \left|\frac{Mf(x)-Lg(x)}{g(x)M}\right| &< \frac{2}{M^2}\big|Mf(x)-Lg(x)\big| \\ &= \frac{2}{M^2}\big|(Mf(x)-LM)+(LM-Lg(x))\big| \\ &\le \frac{2}{M^2}\Big[M\big|f(x)-L\big| + |L|\big|g(x)-M\big|\Big] \\ &= \frac{2}{M}\big|f(x)-L\big| + \frac{2|L|}{M^2}\big|g(x)-M\big|.\end{aligned}$$

The result follows from parts (a), (b) and (c) of the proof, Exercise 4.2, and Theorem 4.3.

□

The following lemma will be needed later in this chapter (cf. Theorem 4.19).

**Lemma 4.3.** *Let $c$ be a limit point of the set $E$ and $L = \lim_{x\to c} f(x)$. Then*

(a) *If $L > p$ for some $p \in \mathbf{F}$, then there is a $\delta > 0$ such that*

$$f(x) > p, \qquad \text{for all } x \in U_\delta(c).$$

(b) *If $L < p$ for some $p \in \mathbf{F}$, then there is a $\delta > 0$ such that*

$$f(x) < p, \qquad \text{for all } x \in U_\delta(c).$$

*Proof.* We prove only part (a). The proof of part (b) is similar.

For $\varepsilon = L - p > 0$, there is a $\delta > 0$ such that

$$-(L-p) < f(x) - L < L - p, \qquad \text{for all } x \in U_\delta(c).$$

It follows that $p = -(L-p) + L < f(x)$ for all $x \in U_\delta(c)$. □

The following theorem is a sequential characterization of limits.

**Theorem 4.6.** *Let $f : E \to \mathbf{F}$ be a function and $c$ a limit point of the set $E$. Then $f(x) \to L$ as $x \to c$ if and only if $f(x_n) \to L$ for every sequence $(x_n)$ in $E \setminus \{c\}$ such that $x_n \to c$.*

The proof uses an important property in the study of ordered fields that also will be used later in the book.

**Definition 4.2.** An ordered field $\mathbf{F}$ is said to be *countably cofinal* if there exists an unbounded sequence in $\mathbf{F}$, that is, a sequence $(a_n)$ such that for every $x$ in $\mathbf{F}$, $x < |a_n|$ for some $n$ in $\mathbf{N}$.

*Example 4.2.* Since $(n)$ is an unbounded sequence in an Archimedean field, every Archimedean field is countably cofinal.

**Lemma 4.4.** *Let $\mathbf{F}$ be a countably cofinal ordered field. There exists an unbounded strictly increasing sequence of positive elements of $\mathbf{F}$.*

*Proof.* Let $(a_n)$ be an unbounded sequence in $\mathbf{F}$. By removing all zero terms from $(a_n)$ and taking the absolute value of the remaining terms, we obtain an unbounded sequence $(b_n)$ with positive terms. We construct the desired sequence recursively.

Let $k_1 = 1$ and $k_2$ be the first index greater than 1 for which $b_{k_2} > b_{k_1} = b_1$. The number $k_2$ exists because $(b_n)$ is unbounded. Having defined $b_{k_n}$, let $k_{n+1}$ be the first index greater than $k_n$ for which $b_{k_{n+1}} > b_{k_n}$. Again, $k_{n+1}$ exists because $(b_n)$ is unbounded. Clearly,

$$b_{k_1} < b_{k_2} < \cdots < b_{k_n} < \cdots$$

is a strictly increasing sequence of positive elements of $\mathbf{F}$. For every $m \in \mathbf{N}$ there is $k_n$ such that $k_n \leq m < k_{n+1}$. By the above construction, $b_m \leq b_{k_n}$. Since the sequence $(b_n)$ is unbounded, so is the sequence $(b_{k_n})$, which is the desired result. □

**Lemma 4.5.** *Let $(a_n)$ be a sequence in an ordered field $\mathbf{F}$ that is not countably cofinal and let $a_n \to c$. Then $(a_n)$ is "eventually" $c$, that is, there is $N \in \mathbf{N}$ such that $a_n = c$ for all $n > N$.*

*Proof.* Suppose to the contrary that there is a subsequence $(a_{k_n})$ of $(a_n)$ such that $a_{k_n} \neq c$ for all $n$. As $\mathbf{F}$ is not countably cofinal, there exists $m \in \mathbf{F}$ such that $1/|a_{k_n} - c| < m$ for all $n \in \mathbf{N}$. Then $|a_{k_n} - c| > 1/m$, for all $n$, which contradicts the fact that $a_n \to c$. □

We proceed with the proof of Theorem 4.6.

*Proof.* (Necessity.) Since $f \to L$, for every $\varepsilon > 0$ there is a $\delta > 0$ such that $|f(x) - L| < \varepsilon$ for all points $x \in U_\delta(c)$. Let $(x_n)$ be a sequence in $E \setminus \{c\}$ such that $\lim x_n = c$. Then, for some $N \in \mathbf{N}$, $0 < |x_n - c| < \delta$ for all $n > N$. Hence, $|f(x_n) - L| < \varepsilon$ for all $n > N$. It follows that $f(x_n) \to L$.

(Sufficiency.) To this end, suppose that $\lim_{x \to c} f(x) = L$ does not hold. Then there exists some $\varepsilon_0 > 0$ such that for every $\delta > 0$ there is a point $x \in E$, for which $|f(x) - L| \geq \varepsilon_0$ and $0 < |x - c| < \delta$.

First, suppose that $\mathbf{F}$ is countably cofinal. By Lemma 4.4, there is an unbounded strictly increasing sequence $(a_n)$ with positive terms in $\mathbf{F}$. We define $\delta_n = |a_n|^{-1}$ for $n \in \mathbf{N}$. It is clear that $\delta_n \to 0$. Choose $x_n \in E$ such that $0 < |x_n - c| < \delta_n$. By the Squeeze Theorem, $x_n \to c$. However $|f(x_n) - L| \geq \varepsilon_0$ for all $n \in \mathbf{N}$. Thus $x_n \to c$, while the sequence $(f(x_n))$ does not converge to $L$.

Now, suppose that $\mathbf{F}$ is not countably cofinal and let $(x_n)$ be a sequence in $E \setminus \{c\}$. By Lemma 4.5, $(x_n)$ does not converge to $c$. Hence, the sufficiency condition is vacuously true. □

## 4.2 The Derivative

**Definition 4.3.** Let $x$ be a point in an interval $I \subseteq \mathbf{F}$. A function $f : I \to \mathbf{F}$ is said to be *differentiable* at $x$ if the limit

$$\lim_{t \to x} \frac{f(t) - f(x)}{t - x}, \qquad \text{where } t \in I,\ t \neq x, \tag{4.1}$$

exists. The value of this limit is called the *derivative of $f$ at the point $x$*, and is denoted by $f'(x)$. In this way, we associate with the function $f$ a function $f'$ whose domain consists of the points $x \in I$ for which the limit in (4.1) exists. The function $f'$ is called the *derivative of $f$*.

If $f$ is differentiable at every point of $I$, we say that $f$ is *differentiable on $I$*.

Alternative notations for the derivative $f'$ include

$$\frac{dy}{dx}, \quad \frac{df}{dx}, \quad \frac{d}{dx}(f), \quad D_x f.$$

*Example 4.3.*

(a) Let $f(x) = px + q$, $p, q \in \mathbf{F}$, be a "linear function" (cf. Lemma 3.3) on an interval $I$ and let $x \in I$. For $t \neq x$ we have

$$\frac{f(t) - f(x)}{t - x} = \frac{(pt + q) - (px + q)}{t - x} = p.$$

It follows that $f'(x) = p$ on $I$, that is, $f'$ is a constant function on $I$. Specifically, $f'(x) = 1$ on $I$ for the identity function $f(x) = x$, and $f'(x) = 0$ (the zero function on $I$) for a constant function $f(x) = q$.

(b) Let $f(x) = |x|$ for $x \in \mathbf{F}$. The function $f$ is not differentiable at $x = 0$. Indeed, for $x \neq 0$ we have

$$\frac{f(x) - f(0)}{x - 0} = \frac{|x|}{x} = \begin{cases} 1, & \text{if } x > 0, \\ -1, & \text{if } x < 0. \end{cases}$$

Suppose that $\lim_{x \to 0}(|x|/x) = L$ for some $L \in \mathbf{F}$. Then, for $\varepsilon = 1$ there is a $\delta > 0$ such that

$$L - 1 < \frac{|x|}{x} < L + 1$$

provided that $0 < |x| < \delta$. The displayed inequality implies that $L > 0$ for $x > 0$, and $L < 0$ if $x < 0$. The obtained contradiction proves that $f$ is not differentiable at 0.

**Theorem 4.7.** *If a function $f : I \to \mathbf{F}$ is differentiable at $c \in I$, then $f$ is continuous at $c$.*

*Proof.* If $x \neq c$, then, by Theorem 4.5 (d),

$$\begin{aligned} \lim_{x \to c}(f(x) - f(c)) &= \lim_{x \to c}\left(\frac{f(x) - f(c)}{x - c}(x - c)\right) \\ &= \lim_{x \to c}\frac{f(x) - f(c)}{x - c} \cdot \lim_{x \to c}(x - c) = f'(c) \cdot 0 = 0. \end{aligned}$$

By Theorem 4.5 (a), $\lim_{x \to c} f(x) = f(c)$, because $f(x) = f(c) + (f(x) - f(c))$. By Theorem 4.2, $f$ is continuous at $c$. □

Example 4.3 (b) shows that the converse of Theorem 4.7 does not hold—the function $f(x) = |x|$ is continuous at 0, but not differentiable at this point.

The following theorem establishes basic algebraic properties of derivatives.

**Theorem 4.8.** *Let $f : I \to \mathbf{F}$ and $g : I \to \mathbf{F}$ be functions that are differentiable at $c \in I$. Then:*

(a) *The function $f + g$ is differentiable at $c$, and*

$$(f + g)'(c) = f'(c) + g'(c).$$

(b) *For $k \in \mathbf{F}$, the function $kf$ is differentiable at $c$, and*

$$(kf)'(c) = kf'(c).$$

(c) *The function $fg$ is differentiable at $c$, and*

$$(fg)'(c) = f'(c)g(c) + f(c)g'(c).$$

(d) *If $g(c) \neq 0$, then the function $f/g$ is differentiable at $c$, and*

$$\left(\frac{f}{g}\right)'(c) = \frac{f'(c)g(c) - f(c)g'(c)}{(g(c))^2}.$$

*Proof.* We leave proofs of (a) and (b) to the reader (cf. Exercise 4.9).

(c) For $x \neq c$ in $I$,

$$\begin{aligned}\frac{f(x)g(x) - f(c)g(c)}{x - c} &= \frac{[f(x) - f(c)]g(c) + f(x)[g(x) - g(c)]}{x - c} \\ &= \frac{f(x) - f(c)}{x - c}g(c) + f(x)\frac{g(x) - g(c)}{x - c}.\end{aligned}$$

By Theorems 4.5 and 4.2, as $x \to c$, the right-hand side converges to

$$f'(c)g(c) + f(c)g'(c).$$

(d) For $x \neq c$ in $I$,

$$\begin{aligned}\frac{\dfrac{f(x)}{g(x)} - \dfrac{f(c)}{g(c)}}{x - c} &= \frac{f(x)g(c) - f(c)g(x)}{g(x)g(c)(x - c)} \\ &= \frac{f(x)g(c) - f(c)g(c) - f(c)g(x) + f(c)g(c)}{g(x)g(c)(x - c)} \\ &= \frac{\dfrac{f(x) - f(c)}{x - c}g(c) - f(c)\dfrac{g(x) - g(c)}{x - c}}{g(x)g(c)}.\end{aligned}$$

Again, by Theorems 4.5 and 4.2, as $x \to c$, the right-hand side converges to

$$\frac{f'(c)g(c) - f(c)g'(c)}{(g(c))^2}.$$

□

The next theorem is known as the "Chain Rule". It establishes a formula for the derivative of a composite function.

**Theorem 4.9.** *Let $f$ be a function on an interval $I \subseteq \mathbf{F}$, and $g$ a function defined on an interval in $\mathbf{F}$ containing the image $f(I)$. If $f$ is differentiable at $c \in I$ and $g$ is differentiable at $f(c)$, then the composite function $g \circ f$ is differentiable at $c$ and*

$$(g \circ f)'(c) = g'(f(c))\, f'(c).$$

*Proof.* Let $u : I \to \mathbf{F}$ be defined by

$$u(x) = \begin{cases} \dfrac{g(f(x)) - g(f(c))}{f(x) - f(c)}, & \text{if } f(x) \neq f(c), \\ g'(f(c)), & \text{otherwise.} \end{cases}$$

Because $f$ is continuous at $c$ (cf. Theorem 4.2), $f(x) \to f(c)$ as $x \to c$. Therefore, $u(x) \to g'(f(c))$ as $x \to c$. We have, for $x \neq c$,

$$\frac{g(f(x)) - g(f(c))}{x - c} = u(x)\frac{f(x) - f(c)}{x - c}.$$

By taking the limit as $x \to c$, we obtain the desired result. □

Let us recall that, by Theorem 3.24 and Exercise 3.26, the inverse of a strictly increasing continuous function $f : I \to \mathbf{F}$ is a strictly increasing continuous function on the interval $J = f(I)$.

**Theorem 4.10.** *Let $f : I \to \mathbf{F}$ be a strictly increasing continuous function, and let $g$ be its inverse. If $f$ is differentiable at $c \in I$ and $f'(c) \neq 0$, then $g$ is differentiable at the point $d = f(c)$ and*

$$g'(d) = \frac{1}{f'(c)}.$$

*Proof.* For each $y \in J = f(I)$ we have $x = g(y) \in I$ and $y = f(g(y))$. Therefore, for $y \neq d$ (hence, $x \neq c$),

$$\frac{g(y) - g(d)}{y - d} = \frac{1}{\dfrac{f(g(y)) - f(g(d))}{g(y) - g(d)}} = \frac{1}{\dfrac{f(x) - f(c)}{x - c}} \longrightarrow \frac{1}{f'(c)}$$

as $x \to c$. Therefore for $\varepsilon > 0$, there is a $\delta > 0$ such that

$$\left| \frac{g(y) - g(d)}{y - d} - \frac{1}{f'(c)} \right| < \varepsilon,$$

for all $x \in U_\delta(c)$. (Recall that $U_\delta(c) = \{x \in I : 0 < |x - c| < \delta.)$ Since $x = g(y)$ and $c = g(d)$, the continuity of $g$ guarantees the existence of a $\delta'$ such that $0 < |x - c| < \delta$, for all $y \in J$ for which $0 < |y - d| < \delta'$. Consequently,

$$\lim_{y \to d} \frac{g(y) - g(d)}{y - d}$$

exists and is equal to $1/f'(c)$, which is the desired result. □

As the derivative of a differentiable function is a function, one may consider the derivative of the latter provided that it exists.

**Definition 4.4.** If $f : I \to \mathbf{F}$ is differentiable on $I$ and if $f'(x)$ is itself differentiable on $I$, we denote the derivative of $f'(x)$ by $f''(x)$ and call it the *second derivative* of the function $f$.

## 4.3 Main Theorems

**Definition 4.5.** A function $f : [a, b] \to \mathbf{F}$ has a *local maximum* at $c \in [a, b]$ if there is a $\delta > 0$ such that $f(x) \leq f(c)$ for all $x \in (c - \delta, c + \delta) \cap [a, b]$.

It has a *local minimum* at $c \in [a, b]$ if there is a $\delta > 0$ such that $f(x) \geq f(c)$ for all $x \in (c - \delta, c + \delta) \cap [a, b]$.

The following theorem is known as one of Fermat's theorems. We use it below to establish important theorems of Real Analysis.

**Theorem 4.11.** *Let $f : [a, b] \to \mathbf{F}$ be a function that has a local maximum at $c \in (a, b)$. If $f$ is differentiable at $c$, then $f'(c) = 0$.*

*Proof.* According to Definition 4.5, we can choose $\delta > 0$ such that

$$a < c - \delta < c < c + \delta < b.$$

For $c - \delta < x < c$, we have

$$\frac{f(x) - f(c)}{x - c} \geq 0.$$

By Theorem 4.4,

$$f'(c) = \lim_{x \to c} \frac{f(x) - f(c)}{x - c} \geq 0.$$

Similarly, we obtain $f'(c) \leq 0$ by choosing $c < x < c + \delta$. We conclude that $f'(c) = 0$. □

It is not difficult to see that $f : [a, b] \to \mathbf{F}$ has a local minimum at $c \in [a, b]$ if and only if $-f$ has a local maximum at the same point. Therefore the assertion of Theorem 4.11 holds also for local minima.

The following three properties of an ordered field are often referred to as "mean value properties".

**Definition 4.6.**

(a) An ordered field $\mathbf{F}$ satisfies *Rolle's Property* if for every continuous function $f : [a, b] \to \mathbf{F}$, differentiable on $(a, b)$ with

$$f(a) = f(b) = 0,$$

there is a point $c \in (a, b)$ such that $f'(c) = 0$.

(b) An ordered field $\mathbf{F}$ satisfies the *Mean Value Property* if for every continuous function $f : [a, b] \to \mathbf{F}$, differentiable on $(a, b)$, there is a point $c \in (a, b)$ such that

$$f'(c) = \frac{f(b) - f(a)}{b - a}.$$

(c) An ordered field $\mathbf{F}$ satisfies the *Generalized Mean Value Property* if for every pair of continuous functions $f : [a, b] \to \mathbf{F}$, $g : [a, b] \to \mathbf{F}$, both differentiable on $(a, b)$, there is a point $c \in (a, b)$ such that

$$[f(b) - f(a)]g'(c) = [g(b) - g(a)]f'(c).$$

**Theorem 4.12.** *The three properties in Definition 4.6 are equivalent.*

*Proof.* We break the proof into three implications.

(a) ⇒ (c). Let $f$ and $g$ be continuous functions on $[a, b]$, both differentiable on $(a, b)$. We define a function $h : [a, b] \to \mathbf{F}$ by

$$h(x) = [f(b) - f(a)]g(x) - [g(b) - g(a)]f(x) + [f(a)g(b) - f(b)g(a)],$$

for $x \in [a, b]$. By Exercise 3.16, $h$ is continuous on $[a, b]$ and, by Theorem 4.8, it is differentiable on $(a, b)$, with the derivative

$$h'(x) = [f(b) - f(a)]g'(x) - [g(b) - g(a)]f'(x).$$

It is easy to verify that $h(a) = h(b) = 0$. By Rolle's Property (a), there is $c \in (a, b)$ such that $h'(c) = 0$. Therefore,

$$[f(b) - f(a)]g'(c) = [g(b) - g(a)]f'(c).$$

(c) $\Rightarrow$ (b). It suffices to consider $g(x) = x$ on $[a, b]$.
(b) $\Rightarrow$ (a). Obvious. □

**Theorem 4.13.** *The field of real numbers* **R** *satisfies the three mean value properties of Definition* 4.6.

*Proof.* By Theorem 4.12, it suffices to consider the case of Rolle's Property.

Let $f : [a, b] \to \mathbf{R}$ be a function satisfying the conditions of Rolle's Property. Since the derivative of the zero function is the zero function, we suppose that $f$ assumes a nonzero value on $[a, b]$. Moreover, we may assume that this value is a positive real number. (Otherwise, consider the function $-f$.)

By the Extreme Value Theorem (cf. Theorem 3.20), there is a point $c$ in $[a, b]$ such that $f(x) \leq f(c)$ for all $x \in [a, b]$. Since $f$ is a nonzero function satisfying $f(a) = f(b) = 0$, the point $c$ belongs to $(a, b)$. The desired result follows from Theorem 4.11. □

**Theorem 4.14.** *An incomplete ordered field* **F** *does not satisfy Rolle's Property and therefore does not satisfy the other two mean value properties of Definition* 4.6.

*Proof.* Let $[a, b]$ be an interval in an incomplete ordered field **F**. There is a gap $(A, B)$ in **F** such that $a \in A$, $b \in B$ (cf. Lemma 3.4). Define $f : [a, b] \to \mathbf{F}$ by

$$f(x) = \begin{cases} x - a, & \text{if } x \in A \cap [a, b], \\ b - x, & \text{if } x \in B \cap [a, b]. \end{cases}$$

Clearly, $f$ is differentiable on $(a, b)$ (cf. Example 4.3), $f(a) = f(b) = 0$, but for every $x \in (a, b)$, $f'(x) \neq 0$, so Rolle's Property is not satisfied. □

By combining Theorems 4.13 and 4.14 we obtain the following result.

**Theorem 4.15.** *Each of the three mean value properties of Definition* 4.6 *is equivalent to the completeness property of the field* **F**.

**Definition 4.7.** An ordered field **F** satisfies the *Zero Derivative Property* if every differentiable function $f : (a, b) \to \mathbf{F}$ such that $f'(x) = 0$ for all $x \in (a, b)$ is constant on $(a, b)$.

**Theorem 4.16.** *The Mean Value Property implies the Zero Derivative Property.*

*Proof.* Let $f : (a, b) \to \mathbf{F}$ be a function such that $f'(x) = 0$ on the interval $(a, b)$. By the Mean Value Property, for $x_1 < x_2$ in $(a, b)$ there is a point $c \in (x_1, x_2)$ such that

$$\frac{f(x_2) - f(x_1)}{x_2 - x_1} = f'(c) = 0.$$

Hence, $f(x_2) = f(x_1)$. It follows that $f$ is constant on $(a, b)$. □

**Theorem 4.17.** *An ordered field satisfies the Zero Derivative Property if and only if it is complete.*

*Proof.* (Necessity.) Suppose to the contrary that an incomplete field $\mathbf{F}$ satisfies the Zero Derivative Property, that is, every function $f : (a, b) \to \mathbf{F}$ with zero derivative is a constant. There is a gap $(A, B)$ in $\mathbf{F}$ such that $a \in A$, $b \in B$ (cf. Lemma 3.4). Define $f : [a, b] \to \mathbf{F}$ by

$$f(x) = \begin{cases} 1, & \text{if } x \in A \cap [a, b], \\ 0, & \text{if } x \in B \cap [a, b]. \end{cases}$$

It is clear that $f'(x) = 0$ on $(a, b)$. However, $f$ is not a constant function.

(Sufficiency.) By Theorems 4.13 and 4.16, the complete field $\mathbf{R}$ satisfies the Zero Derivative Property. □

**Definition 4.8.** An ordered field $\mathbf{F}$ satisfies the *Monotonicity Property* if every differentiable function $f : (a, b) \to \mathbf{F}$ such that $f'(x) \geq 0$ for all $x$ in $(a, b)$ is increasing.

An equivalent version of the Monotonicity Property is found in Exercise 4.11.

**Theorem 4.18.** *An ordered field satisfies the Monotonicity Property if and only if it is complete.*

*Proof.* (Necessity.) We proceed by contradiction. Suppose that an incomplete field $\mathbf{F}$ satisfies the Monotonicity Property and let $(a, b) \subseteq \mathbf{F}$. There is a gap $(A, B)$ in $\mathbf{F}$ such that $a \in A$, $b \in B$ (cf. Lemma 3.4). Define $f : [a, b] \to \mathbf{F}$ by

$$f(x) = \begin{cases} x - a, & \text{if } x \in A \cap [a, b], \\ x - b, & \text{if } x \in B \cap [a, b]. \end{cases}$$

Clearly, $f'(x) = 1 \geq 0$ on $(a, b)$. For $x_1 \in A \cap (a, b)$ and $x_2 \in B \cap (a, b)$ we have $x_1 < x_2$, whereas $f(x_1) = x_1 - a > x_2 - b = f(x_2)$. Hence, $f$ is not an increasing function.

(Sufficiency.) Let $f : (a, b) \to \mathbf{R}$ be a function with a non-negative derivative. By the Mean Value Property (cf. Theorem 4.13), for $x_1 < x_2$ in $(a, b)$ there is a point $c \in (x_1, x_2)$ such that

$$\frac{f(x_2) - f(x_1)}{x_2 - x_1} = f'(c) \geq 0.$$

It follows that $f(x_1) \leq f(x_2)$, that is, the function $f$ is increasing. □

The last property of an ordered field discussed in this section is also known as the Intermediate Value Property of derivatives (cf. Definition 3.10).

**Definition 4.9.** An ordered field $\mathbf{F}$ satisfies *Darboux's Property* if for every differentiable function $f : [a, b] \to \mathbf{F}$ with $f'(a) \neq f'(b)$ and every point $k \in \mathbf{F}$ that lies between $f'(a)$ and $f'(b)$, there is a point $c \in (a, b)$ such that $f'(c) = k$.

**Theorem 4.19.** *An ordered field is complete if and only if it satisfies Darboux's Property.*

*Proof.* (Necessity.) Let $f$ be a real differentiable function on $[a, b]$. We may assume that $f'(b) < k < f'(a)$. Define $g : [a, b] \to \mathbf{R}$ by $g(x) = f(x) - kx$ for $x \in [a, b]$. Then

$$g'(a) = f'(a) - k > 0 \qquad \text{and} \qquad g'(b) = f'(b) - k < 0.$$

By Lemma 4.3 (cf. proof of Theorem 4.11), there are points $x_1 > a$ and $x_2 < b$ such that $g(x_1) > g(a)$ and $g(x_2) > g(b)$. By the Extreme Value Property (cf. Theorem 3.20), $g$ attains its maximum value at some $c \in (a, b)$. By Theorem 4.11, $g'(c) = 0$, that is, $f'(c) = k$.

(Sufficiency.) Let $\mathbf{F}$ be an incomplete ordered field and $f$ the function from the proof of Theorem 4.14. Clearly, $f'(x) \in \{-1, 1\}$ for all $x \in [a, b]$, contradicting Darboux's Property. □

## 4.4 Convex Functions

The properties of real differentiable functions established in the previous section are used in Calculus to analyze the behavior of such functions on their domains. In this section, we use derivatives to characterize a class of real functions that has important roles in analysis and more broadly in mathematics and its applications.

**Definition 4.10.** A function $f : (a, b) \to \mathbf{F}$ is said to be *convex* if

$$f(\lambda x_1 + (1 - \lambda)x_2) \leq \lambda f(x_1) + (1 - \lambda) f(x_2), \tag{4.2}$$

for all $x_1 < x_2$ in $(a, b)$ and $\lambda \in [0, 1] \subseteq \mathbf{F}$.

For a real function $f$ (cf. Figure 4.1), inequality (4.2) means that each point $R$ on the graph of the function $f$ that lies between two other points $P$ and $Q$ on the graph lies on or below the chord $PQ$ (cf. Exercise 4.13).

Let $x_1 < x_2$ and $x = \lambda x_1 + (1 - \lambda)x_2$. It is not difficult to verify that $x \in [x_1, x_2]$ and

$$\lambda = \frac{x_2 - x}{x_2 - x_1}, \qquad 1 - \lambda = \frac{x - x_1}{x_2 - x_1}.$$

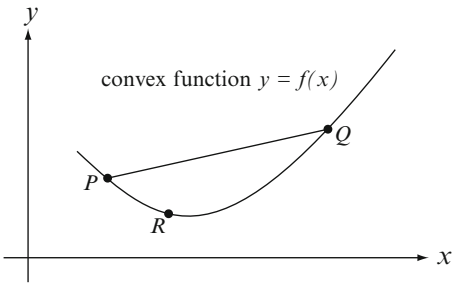



**Fig. 4.1** Real convex function.

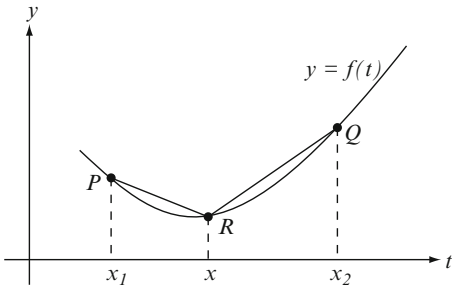



**Fig. 4.2** slope$(PR) \leq$ slope$(RQ)$.

Thus we can rewrite condition (4.2) in the form

$$f(x) \leq \frac{x_2 - x}{x_2 - x_1} f(x_1) + \frac{x - x_1}{x_2 - x_1} f(x_2), \qquad \text{for all } x \in [x_1, x_2], \tag{4.3}$$

where $x_1, x_2 \in (a, b)$. It is an easy algebraic exercise to show that condition (4.3) is equivalent to the following one (cf. Exercise 4.15):

$$\frac{f(x) - f(x_1)}{x - x_1} \leq \frac{f(x_2) - f(x)}{x_2 - x}, \qquad \text{for all } x \in (x_1, x_2), \tag{4.4}$$

where $x_1, x_2 \in (a, b)$. Thus, a function $f : (a, b) \to \mathbf{F}$ is convex if and only if (4.4) holds for every $[x_1, x_2] \subseteq (a, b)$.

The drawing in Figure 4.2 is a geometric illustration of inequality (4.4) for a real function $f$.

**Theorem 4.20.** *If $f : (a, b) \to \mathbf{F}$ is a convex differentiable function, then the derivative $f'(x)$ is an increasing function on $(a, b)$.*

*Proof.* Let $f : (a, b) \to \mathbf{F}$ be a differentiable convex function and let $x_1 < x_2$ be two points in $(a, b)$. From (4.4) we obtain

$$f'(x_1) = \lim_{x \to x_1} \frac{f(x) - f(x_1)}{x - x_1} \leq \frac{f(x_2) - f(x_1)}{x_2 - x_1}$$

and

$$f'(x_2) = \lim_{x \to x_2} \frac{f(x_2 - f(x)}{x_2 - x} \geq \frac{f(x_2) - f(x_1)}{x_2 - x_1},$$

for all $x_1 < x_2$ in $(a, b)$ (cf. Theorem 4.4). Hence, $f'(x_1) \leq f'(x_2)$, that is, the derivative is an increasing function. □

*Example 4.4.* The real function $f(x) = x^2$ is convex on every interval $(a, b)$. Indeed, $f'(x) = 2x$ is clearly an increasing function on $(a, b)$.

The converse of Theorem 4.20 does not hold in an incomplete field, as the following example demonstrates.

*Example 4.5.* Let $(a, b)$ be an interval in an incomplete field $\mathbf{F}$. There is a gap $(A, B)$ in $\mathbf{F}$ such that $a \in A, b \in B$ (cf. Lemma 3.4). Define $f : (a, b) \to \mathbf{F}$ by

$$f(x) = \begin{cases} 1, & \text{if } x \in A \cap (a, b), \\ 0, & \text{if } x \in B \cap (a, b). \end{cases}$$

It is clear that $f'(x) = 0$ is an increasing function on $(a, b)$.

For $x_1 \in A \cap (a, b)$ and $x_2 \in B \cap (a, b)$, let $x$ be a point in $(x_1, x_2) \cap A$. By Exercise 4.14, there is a $\lambda > 0$ such that

$$x = \lambda x_1 + (1 - \lambda)x_2.$$

We have

$$f(x) = 1 > \lambda = \lambda f(x_1) + (1 - \lambda) f(x_2).$$

Hence, $f$ is not a convex function.

However, the converse of Theorem 4.20 does hold for real functions.

**Theorem 4.21.** *A differentiable real function $f : (a, b) \to \mathbf{R}$ is convex if and only if the derivative $f'(x)$ is an increasing function on $(a, b)$.*

*Proof.* (Necessity.) Theorem 4.20.

(Sufficiency.) Suppose that the derivative $f'(x)$ is an increasing function on $(a, b)$ and let $x_1 < x_2$ be two points in $(a, b)$. By the Mean Value Property (cf. Theorem 4.13), for $x \in (x_1, x_2)$ there are points $c_1$ and $c_2$ such that $c_1 \in (x_1, x)$, $c_2 \in (x, x_2)$ and

$$\frac{f(x) - f(x_1)}{x - x_1} = f'(c_1) \quad \text{and} \quad \frac{f(x_2) - f(x)}{x_2 - x} = f'(c_2).$$

Since $f'(c_1) \leq f'(c_2)$, condition (4.4) holds, that is, $f$ is a convex function. □

From Theorem 4.20 and the Monotonicity Property (cf. Theorem 4.18), we obtain the following sufficient condition for convexity of a real function that is useful in applications.

**Theorem 4.22.** *Every twice differentiable real function $f(x)$ on an open interval $(a, b)$ with $f''(x) \geq 0$ for all $x \in (a, b)$ is convex.*

Theorem 4.22 does not hold if the complete field $\mathbf{R}$ is replaced by an incomplete field, as the next theorem claims.

**Theorem 4.23.** *Let $\mathbf{F}$ be an incomplete ordered field. There exists a non-convex twice differentiable function $f : (a, b) \to \mathbf{F}$ with $f''(x) \geq 0$.*

*Proof.* It suffices to consider the function $f$ from Example 4.5. □

By combining Theorems 4.22 and 4.23, we obtain the following characterization of the completeness property.

**Theorem 4.24.** *An ordered field $\mathbf{F}$ is complete if and only if every twice differentiable function $f : (a, b) \to \mathbf{F}$ with $f''(x) \geq 0$ on $(a, b)$ is convex.*

## Notes

In its general form, the cofinality property (cf. Definition 4.2) states that a subset $A$ of a partially ordered set $(X, \leq)$ is *cofinal in* $X$ if for each element $x \in X$ there exists an element $a \in A$ such that $x \leq a$ (Bourbaki 1968; Halmos 1974). The *cofinality* of a partially ordered set $A$ is the least of the cardinalities of the cofinal subsets of $A$. It is known (Schmerl 1985) that for every regular cardinal $\boldsymbol{\kappa}$ there is an ordered field of cofinality $\boldsymbol{\kappa}$. In particular, countable cofinality is $\aleph_0$ cofinality. The notions of a cofinal set and cofinality appear in other branches of mathematics, including algebra (Lang 2002) and general topology (Kelley 1955).

The absolute value function (cf. Example 4.3 (b)) is an example of a continuous function that is not differentiable at a point. There are continuous real functions that are nowhere differentiable (Gelbaum and Olmsted 1964). The *Weierstrass function* is arguably the most known example:

$$f(x) = \sum_{k=0}^{\infty} a^k \cos(b^k \pi x), \qquad x \in \mathbf{R},$$

where $0 < a < 1$, $b$ is a positive odd integer, and $ab > 1 + \frac{3}{2}\pi$.

The theorems of Section 4.3 are important tools in the analysis of real functions. For instance, one can use the Monotonicity Property (cf. Theorem 4.18) to prove Bernoulli's inequality (cf. Exercise 4.12).

Jean-Gaston Darboux (1842–1917) was a French mathematician who made important contributions to geometry and analysis. About twenty objects in mathematics are named in his honor.

It is hard to overestimate the importance of the notion of convexity in mathematics. For instance, the Minkowski's Inequality (cf. Exercise 4.19) is just one of the many inequalities that are essential tools in Real Analysis.

## Exercises

**4.1.** Prove part (b) of Theorem 4.5.

**4.2.** Show that if $\lim_{x\to c} |f(x)| = 0$, then $\lim_{x\to c} f(x) = 0$.

**4.3.** Show that $\lim_{x\to c} f(x) = L$ if and only if $\lim_{x\to 0} f(x + c) = L$.

**4.4.** Show that the results of Exercise 3.16 follows from Theorem 4.5.

**4.5.** Prove that an ordered field is countably cofinal if and only if it contains a convergent sequence which is not eventually constant.

**4.6.** Prove cases (a) and (b) of Theorem 4.8.

**4.7.** We define a real function $f : x \mapsto x^n$ for a natural number $n$ recursively by

(1) $x^0 = 1$,
(2) $x^n = x \cdot x^{n-1}$, for $n \geq 1$.

(a) Show that $(x^n)' = nx^{n-1}$ for $n > 0$.
(b) Let $x^{1/n}$, $n > 1$, be the inverse function of $f$. Show that

$$(x^{1/n})' = \frac{1}{n} x^{1/n-1} \qquad \text{on } (0, \infty).$$

(c) Let $m$ and $n$ be positive integers and $r = m/n$. We define $x^r = (x^m)^{1/n}$. Prove that $(x^r)' = rx^{r-1}$ on $(0, \infty)$.

**4.8.** Find the derivative of the real function

$$f(x) = \sqrt{x + \sqrt{x + \sqrt{x}}}, \qquad x > 0.$$

**4.9.** True or False: If both functions $f$ and $g$ are not differentiable at 0, then the function $f + g$ is not differentiable at 0.

**4.10.** Let $f$ and $g$ be differentiable real-valued functions on $(a, b) \subseteq \mathbf{R}$. Show that $f'(x) = g'(x)$ on $(a, b)$ if and only if $f(x) = g(x) + C$, where $C$ is a constant.

**4.11.** Show that the Monotonicity Property (cf. Definition 4.8) is equivalent to the following condition:

Every differentiable function $f : (a, b) \to \mathbf{F}$ such that $f'(x) \leq 0$ for all $x$ in $(a, b)$ is decreasing.

**4.12.** Prove *Bernoulli's inequality*

$$(1 + x)^r \geq rx + 1,$$

where $r \geq 1$ is a rational number and $x \geq -1$ is a real number.

**4.13.** Prove that a function $f : (a, b) \to \mathbf{R}$ is convex if and only if for every three consecutive points $P$, $R$, $Q$ on the graph of $f$, the point $R$ lies below or on the chord $PQ$ (cf. Figure 4.1).

**4.14.** Let $x_1 < x_2$ be points in $\mathbf{F}$. Show that for $x \in [x_1, x_2]$ there is a unique $\lambda \in [0, 1]$ such that $x = \lambda x_1 + (1 - \lambda)x_2$.

**4.15.** Show that the inequalities in (4.3) and (4.4) are equivalent.

**4.16.** For $x_1, \ldots, x_n \in (a, b) \subseteq \mathbf{R}$, the number

$$x = \lambda_1 x_1 + \cdots + \lambda_n x_n$$

is said to be a *convex combination* of the numbers $x_1, \ldots, x_n$, provided that $0 \leq \lambda_k \leq 1$ for $1 \leq k \leq n$ and $\lambda_1 + \cdots + \lambda_n = 1$. Show that

$$\min\{x_1, \ldots, x_n\} \leq x \leq \max\{x_1, \ldots, x_n\}.$$

Accordingly, $x \in (a, b)$.

**4.17.** Prove that a function $f : (a, b) \to \mathbf{R}$ is convex if and only if

$$f\left(\sum_{k=1}^{n} \lambda_k x_k\right) \leq \sum_{k=1}^{n} \lambda_k f(x_k),$$

for all $x_1, \ldots, x_n \in (a, b)$ and $0 \leq \lambda_k \leq 1$ $(1 \leq k \leq n)$ such that $\sum_{k=1}^{n} \lambda_k = 1$ (cf. Exercise 4.16).

**4.18.** Show that the *power function* $f(x) = x^p$, where $p \geq 1$ is a rational number, is convex over every interval $(a, b)$ $(a \geq 0)$.

**4.19. Minkowski's Inequality.** Prove that for arbitrary real numbers $x_1, \ldots, x_n$, $y_1, \ldots, y_n$ and a rational number $p \geq 1$,

$$\left[\sum_{k=1}^{n} |x_k + y_k|^p\right]^{1/p} \leq \left[\sum_{k=1}^{n} |x_k|^p\right]^{1/p} + \left[\sum_{k=1}^{n} |y_k|^p\right]^{1/p}.$$

(Hint: Use the results of Exercises 4.17 and 4.18.)

# Chapter 5
# Integration




Integration in ordered fields is more variegated than in the field of real numbers. Notions that are equivalent in **R** may lead to different sets of integrable functions. In this chapter, the Riemann and Darboux integrable functions and the respective integrals are introduced in the context of a general ordered field.

Partitions of closed bounded intervals are key elements of any elementary integration theory. The notion of a partition is introduced for an arbitrary ordered field **F** in Section 5.1, where some features of this concept specific to ordered fields are also discussed.

The definitions of the Riemann integral and Riemann integrability in Section 5.2 is taken word for word from Calculus. In Sections 5.2 and 5.3, properties of Riemann integrable functions are investigated in real and general ordered fields. It is shown that Riemann integrability of continuous functions is equivalent to the completeness property of the underlying field.

Step functions on an ordered field are introduced in Section 5.4 where their Riemann integrability is proved and a formula for the Riemann integral of a step function is established. Step functions are building blocks of the Darboux integrability.

The notions of Darboux integrability and the Darboux integral are introduced in Section 5.5, where the relations between these two concepts are investigated. Standard properties of Darboux integrable functions are presented in Section 5.6

## 5.1 Partitions and Gaps

Let $[a, b]$ be an interval in an ordered field **F**. A finite subset $P$ of $[a, b]$ containing the end points $a$ and $b$ is called a *partition* of $[a, b]$. Since $P$ is a finite set, its elements can be enumerated and ordered as follows (cf. Exercise 5.1)

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$$a = x_0 < x_1 < \cdots < x_{n-1} < x_n = b,$$

where $n+1$ is the cardinality of the set $P$. The partition $P$ divides the interval $[a, b]$ into $n$ closed subintervals

$$[x_0, x_1], [x_1, x_2], \ldots, [x_{n-1}, x_n].$$

Clearly, $[a, b] = [x_0, x_1] \cup [x_1, x_2] \cup \cdots \cup [x_{n-1}, x_n]$.

Let $\triangle_k = x_k - x_{k-1}$ for $k = 1, \ldots, n$. The *norm* $\|P\|$ of the partition $P$ is the maximum of the set $\{\triangle_k\}_{1 \le k \le n}$ (cf. Exercise 5.2), that is,

$$\|P\| = \max_{1 \le k \le n} \triangle_k.$$

A *refinement* of a partition $P$ of the interval $[a, b]$ is a partition $Q$ of the same interval such that $P \subset Q$. Note that $P$ is a refinement of itself. It is not difficult to see that subintervals of a refinement $Q$ of the partition $P$ are subintervals of the subintervals of $P$ and that $\|Q\| \le \|P\|$ (cf. Exercise 5.3).

Let $P_1$ and $P_2$ be partitions of $[a, b]$. Clearly, the union $Q = P_1 \cup P_2$ is a refinement (called the *common refinement*) of both $P_1$ and $P_2$. Hence, $\|Q\| \le \min\{\|P_1\|, \|P_2\|\}$.

Recall (cf. Theorem 2.17) that an Archimedean field $\mathbf{F}$ is a subfield of the field $\mathbf{R}$ of real numbers. Hence the elements of $\mathbf{F}$ can be regarded as real numbers. In this context, the concepts introduced in the previous paragraphs are familiar notions from Calculus. In particular, the norm of a partition of the interval $[a, b]$ can be made as small as we wish. Indeed, let $P_n$ be a partition $\{x_0, x_1, \ldots, x_n\}$ of $[a, b]$ into $n$ equal subintervals, that is,

$$\triangle_k = \frac{b-a}{n}, \qquad 1 \le k \le n.$$

Since $\mathbf{F}$ is Archimedean, for every $\varepsilon > 0$ there is $n \in \mathbf{N}$ such that

$$\|P_n\| = \frac{b-a}{n} < \varepsilon$$

(choose $n > (b-a)/\varepsilon$). Hence, $\inf\{\|P\| : P \text{ is a partition of } [a, b]\} = 0$.

The notion of a partition is less intuitive if the field $\mathbf{F}$ is not Archimedean. We will show below that the norms of partitions of a closed bounded interval in a non-Archimedean field $\mathbf{F}$ are bounded below by a positive element of $\mathbf{F}$. To this end, we introduce the following definitions.

**Definition 5.1.** Let $\mathbf{F}$ be an ordered field. We define

(a) $\mathcal{I}(\mathbf{F}) = \{x \in \mathbf{F} : |x| < 1/n, \text{ for all } n \in \mathbf{N}\}$;
(b) $\mathcal{F}(\mathbf{F}) = \{x \in \mathbf{F} : |x| < n, \text{ for some } n \in \mathbf{N}\}$;
(c) $\mathcal{L}(\mathbf{F}) = \{x \in \mathbf{F} : |x| > n, \text{ for all } n \in \mathbf{N}\}$.

The elements in $\mathcal{I}(\mathbf{F})$, $\mathcal{F}(\mathbf{F})$, and $\mathcal{L}(\mathbf{F})$ are called *infinitesimals*, *finite*, and *infinitely large*, respectively.

It is not difficult to verify that $\mathcal{I}(\mathbf{F}) \subseteq \mathcal{F}(\mathbf{F})$, $\mathbf{F} = \mathcal{F}(\mathbf{F}) \cup \mathcal{L}(\mathbf{F})$, and $\mathcal{F}(\mathbf{F}) \cap \mathcal{L}(\mathbf{F}) = \varnothing$ (cf. Exercise 5.4). The field $\mathbf{F}$ is Archimedean if and only if $\mathbf{F} = \mathcal{F}(\mathbf{F})$ (equivalently, $\mathcal{L}(\mathbf{F}) = \varnothing$). Note, that $0 \in \mathcal{I}(\mathbf{F})$, so $\mathcal{I}(\mathbf{F}) \neq \varnothing$.

Recall that a gap in an ordered field $\mathbf{F}$ is a cut without a cut point (cf. Definition 2.11).

**Definition 5.2.** Let $(A, B)$ be a gap in the field $\mathbf{F}$. This gap is said to be *irregular* if there is a positive $\delta$ in $\mathbf{F}$ such that $b - a \geq \delta$ for every $a \in A$ and $b \in B$. Otherwise, the gap $(A, B)$ is said to be *regular*.

**Theorem 5.1.** *For every positive $\delta$ in $\mathbf{F}$, where $\mathbf{F}$ is a non-Archimedean ordered field, there is a gap $(A, B)$ in $\mathbf{F}$ such that*

$$b - a \geq \delta, \qquad \text{for all } a \in A,\ b \in B.$$

*Proof.* It suffices to consider only infinitely large positive elements $\delta$, because then the conclusion of the theorem clearly holds for any finite positive $\delta$. So, let $\delta$ be a positive element of $\mathcal{L}(\mathbf{F})$. We define

$$A_\delta = \bigcup_{n \in \mathbf{N}} (-\infty, n\delta), \qquad B_\delta = \mathbf{F} \setminus A_\delta.$$

Because $\delta$ is infinitely large, $n\delta < \delta^2$ for all $n \in \mathbf{N}$. Hence, $\delta^2 \in B_\delta$, so the set $B_\delta$ is not empty. It is not difficult to see that $(A_\delta, B_\delta)$ is a gap. For an $a \in A_\delta$, there is $n \in \mathbf{N}$ such that $a < n\delta$. If $b \in B_\delta$, then $b > n\delta$ for all $n \in \mathbf{N}$. Therefore,

$$a < n\delta < (n+1)\delta < b.$$

Hence, $b - a > (n+1)\delta - n\delta = \delta$. □

The next result is a version of Lemma 3.4 for non-Archimedean fields.

**Lemma 5.1.** *For every two points $a < b$ in a non-Archimedean ordered field $\mathbf{F}$ there is an irregular gap $(A, B)$ such that $a \in A$, $b \in B$.*

*Proof.* Let $(A_\delta, B_\delta)$ be the gap in $\mathbf{F}$ defined in the proof of Theorem 5.1. For $c \in A_\delta$ and $d \in B_\delta$, define $f$ by

$$f(x) = \frac{b-a}{d-c}\, x + \frac{ad - bc}{d-c}, \qquad x \in \mathbf{F}.$$

By Lemma 3.4, the sets $A = f(A_\delta)$, $B = f(B_\delta)$ define a gap $(A, B)$ in $\mathbf{F}$. It is clear that $a = f(c) \in A$ and $b = f(d) \in B$.

Since the function $f$ is a bijection, for $a_1 \in A$ and $b_1 \in B$ there are unique $a_2 \in A_\delta$ and $b_2 \in B_\delta$ such that $a_1 = f(a_2)$ and $b_1 = f(b_2)$. Then

$$b_1 - a_1 = f(b_2) - f(a_2) = \frac{b-a}{d-c}(b_2 - a_2) \geq \frac{b-a}{d-c}\,\delta.$$

It follows that the gap $(A, B)$ is irregular. □

Let $[a, b]$ be an interval in a non-Archimedean ordered field $\mathbf{F}$ and

$$P = \{x_0, x_1, \ldots, x_n\}$$

a partition of $[a, b]$. By Lemma 5.1, there is an irregular gap $(A, B)$ in $\mathbf{F}$ such that $a \in A$ and $b \in B$. Because the elements of $P$ are ordered, there is an index $k$, $1 \leq k \leq n$, such that $x_{k-1} \in A$ and $x_k \in B$. Since $(A, B)$ is irregular, there is a $\delta > 0$ such that $\triangle_k = x_k - x_{k-1} > \delta$. Hence, $\|P\| > \delta$, so the norms of all partitions of $[a, b]$ are bounded below by a positive element of the field $\mathbf{F}$. The following example shows that the norm $\|P\|$ can be bounded below even by an infinitely large positive element of a non-Archimedean ordered field $\mathbf{F}$.

*Example 5.1.* Let $\mathbf{F}$ be a non-Archimedean ordered field and $\mathbf{F}^+$ the set of positive elements of $\mathbf{F}$. We define

$$B = \mathcal{L}(\mathbf{F}) \cap \mathbf{F}^+ \quad \text{and} \quad A = \mathbf{F} \setminus B.$$

It is clear that $(A, B)$ is a gap in $\mathbf{F}$. Let $[a, b]$ be a bounded closed interval in $\mathbf{F}$ with $a \in A$ and $b \in B$, and let $P = \{x_0, x_1 \ldots, x_n\}$ be a partition of $[a, b]$. As $x_0 \in A$ and $x_n \in B$, there is an index $k$, $1 \leq k \leq n$, such that $x_{k-1} \in A$ and $x_k \in B$. Since $\triangle_k = x_k - x_{k-1}$ is an infinitely large element of $\mathbf{F}$ (cf. Exercise 5.5), the norm $\|P\|$ is also an infinitely large positive element of the field $\mathbf{F}$.

The last theorem of this section characterizes incomplete ordered fields (that is, ordered fields different from $\mathbf{R}$) with regular gaps.

**Theorem 5.2.** *An incomplete ordered field is Archimedean if and only if every gap in the field is regular.*

*Proof.* (Necessity.) Suppose that an incomplete ordered field $\mathbf{F}$ is Archimedean. Let $\delta$ be a positive element of $\mathbf{F}$ and $(A, B)$ a gap in $\mathbf{F}$. Let $a_1 \in A$ and $b_1 \in B$. Define recursively sequences $(a_n)$ and $(b_n)$ as follows:

(1) $\dfrac{a_n + b_n}{2} \in A$. Set $a_{n+1} = \dfrac{a_n + b_n}{2}$ and $b_{n+1} = b_n$.

(2) $\dfrac{a_n + b_n}{2} \in B$. Set $a_{n+1} = a_n$ and $b_{n+1} = \dfrac{a_n + b_n}{2}$.

It is clear from (1) and (2) that $(a_n)$ and $(b_n)$ belong to the sets $A$ and $B$, respectively. Moreover,

$$0 < b_{n+1} - a_{n+1} = \frac{b_n - a_n}{2} = \cdots = \frac{b_1 - a_1}{2^n} < \frac{b_1 - a_1}{n}.$$

As $\mathbf{F}$ is Archimedean, there is $n \in \mathbf{N}$ such that $(b_1 - a_1)/n < \delta$, so $b_{n+1} - a_{n+1} < \delta$. Hence, $(A, B)$ is a regular gap.

(Sufficiency.) Follows immediately from Theorem 5.1. □

## 5.2 The Riemann Integral

**Definition 5.3.** Let $f : [a, b] \to \mathbf{F}$ be a function, $P = \{x_0, \ldots, x_n\}$ a partition of $[a, b]$, and $\{c_k\}_{1 \le k \le n}$ a set of elements of $\mathbf{F}$ such that $c_k \in [x_{k-1}, x_k]$ for each $k = 1, \ldots, n$. The sum

$$S_P = \sum_{k=1}^{n} f(c_k) \triangle_k$$

is called a *Riemann sum* for $f$ on $[a, b]$. (Note that the Riemann sum $S_P$ also depends on the choice of elements $c_k$'s.)

A number $J$ is said to be the *Riemann integral* of $f$ if for every $\varepsilon > 0$ there is a $\delta > 0$ such that for every partition $P = \{x_0, \ldots, x_n\}$ of $[a, b]$ with $\|P\| < \delta$ and any choice of points $c_k \in [x_{k-1}, x_k]$, $k = 1, \ldots, n$, we have

$$\left| \sum_{k=1}^{n} f(c_k) \triangle_k - J \right| < \varepsilon. \tag{5.1}$$

In this case, we say that $f$ is *Riemann integrable* and write $J = \int_a^b f$.

It can be readily verified (cf. Exercise 5.6) that any constant function on a closed bounded interval $[a, b] \subseteq \mathbf{F}$ is Riemann integrable. Thus, the class of Riemann integrable functions on $[a, b]$ is not empty. More examples of Riemann integrable functions are found in Section 5.4.

Any Riemann integrable function is bounded. Indeed, suppose that $f : [a, b] \to \mathbf{F}$ is unbounded and let $\sum_{k=1}^{n} f(c_k) \triangle_k$ be a Riemann sum for some partition $P$. Since $f$ is unbounded on $[a, b]$, it is unbounded on some $[x_{k-1}, x_k]$. By choosing $c_k \in [x_{k-1}, x_k]$ we can make $f(c_k) \triangle_k$, and therefore the entire Riemann sum $\sum_{k=1}^{n} f(c_k) \triangle_k$ as large (in absolute value) as we wish. Hence, (5.1) does not hold for any partition $P$ and $J \in \mathbf{F}$.

Accordingly, in the rest of this chapter we consider only bounded functions $f : [a, b] \to \mathbf{F}$ and denote the set of bounded functions on $[a, b]$ by $B[a, b]$. It is clear that $B[a, b]$ is a vector space over the field $\mathbf{F}$ (cf. Exercise 5.7).

As in many instances in the previous chapters, the complete ordered field $\mathbf{R}$ differs drastically from incomplete fields in the theory of the integral. For this

reason, we begin our exposition by presenting elements of the theory for real functions.

Unfortunately, Definition 5.3 is not practical for developing integrability criteria, that is, it is inconvenient to use it to decide whether a given real function is integrable or not. To overcome this difficulty, we introduce (following Darboux) the lower and upper sums.

Let $f : [a, b] \to \mathbf{R}$ be a function and $P = \{x_0, \ldots, x_n\}$ a partition of $[a, b]$. Define the numbers $m_k$ and $M_k$ for $1 \leq k \leq n$ by

$$m_k = \inf_{x \in [x_{k-1}, x_k]} f(x), \qquad M_k = \sup_{x \in [x_{k-1}, x_k]} f(x).$$

Note that these numbers exist because the real function $f$ is assumed to be bounded and $\mathbf{R}$ is a complete ordered field.

**Definition 5.4.** The numbers

$$L_P = \sum_{k=1}^{n} m_k \, \triangle_k \qquad \text{and} \qquad U_P = \sum_{k=1}^{n} m_k \triangle_k$$

are called the *lower* and *upper* sums corresponding to the partition $P$, respectively.

The drawing in Figure 5.1 is a geometric interpretation of the upper sum $U_P$. The "area" under the graph of the function $f$ on the interval $[a, b]$ is approximated from above by rectangles with dotted vertical sides.

Clearly,

$$m_k \leq f(c_k) \leq M_k$$

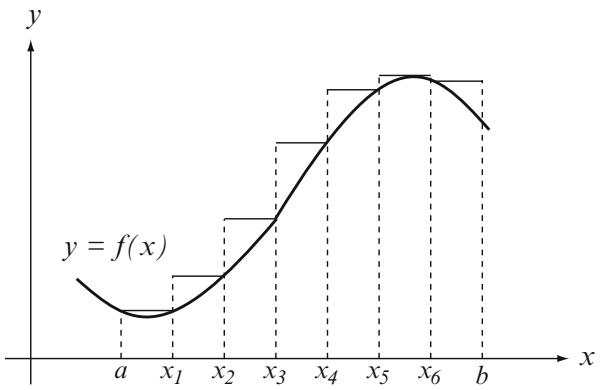



**Fig. 5.1** A geometric illustration of the upper sum $U_P$.

for any choice of $c_k \in [x_{k-1}, x_k]$, $1 \le k \le n$. By multiplying each term of the above inequality by $\triangle_k$ and taking the sum from $k = 1$ to $k = n$, we obtain

$$L_P \le S_P \le U_P, \tag{5.2}$$

that is, for a partition $P$, every Riemann sum lies between the lower sum and upper sum.

For a given partition $P$, the numbers $L_P$ and $U_P$ are constants, whereas the Riemann sum $S_P$ is a variable depending on the choice of numbers $c_k$'s. However, we can choose each $f(c_k)$ as close to $M_k$ as we wish. It is not difficult to see that we can select these numbers to make $S_P$ as close to $U_P$ as we wish. A similar argument show that we can make $S_P$ as close to $L_P$ as we wish. Thus, for a given partition $P$ the sums $L_P$ and $U_P$ are the infimum and supremum, respectively, of the set of Riemann sums $\{S_P\}$ (cf. Exercise 5.8).

We need the next two lemmas to establish an integrability criterion for real functions on closed bounded intervals.

**Lemma 5.2.** *If a partition $Q$ is a refinement of the partition $P$, that is, $Q \supseteq P$, then*

$$L_Q \ge L_P \qquad \text{and} \qquad U_Q \le U_P.$$

*Proof.* Let $P = \{x_0, \ldots, x_n\}$. It is clear that it suffices to consider the case when $Q = P \cup \{x'\}$ where $x_{k-1} < x' < x_k$ for some $1 \le k \le n$. We define

$$M'_k = \sup_{x \in [x_{k-1}, x']} f(x), \qquad M''_k = \sup_{x \in [x', x_k]} f(x).$$

Clearly,

$$M'_k(x' - x_{k-1}) \le M_k(x' - x_{k-1})$$

and

$$M''_k(x_k - x') \le M_k(x_k - x').$$

By adding the above inequalities, we obtain

$$M'_k(x' - x_{k-1}) + M''_k(x_k - x') \le M_k(x_k - x_{k-1}).$$

The sum $U_Q$ is obtained from the sum $U_P$ by replacing the term $M_k \triangle_k$ by the sum $M'_k(x' - x_{k-1}) + M''_k(x_k - x')$. Hence the last displayed inequality proves that $U_Q \le U_P$.

The case of the lower sums is treated similarly. □

**Lemma 5.3.** *Any lower sum is not greater than every upper sum. That is, for any two partitions $P'$ and $P''$, $L_{P'} \le U_{P''}$.*

*Proof.* The partition $P = P' \cup P''$ is a refinement of the partitions $P'$ an $P''$. By Lemma 5.2, $L_{P'} \le L_P$ and $U_P \le U_{P''}$. Since $L_P \le U_P$, we have $L_{P'} \le U_{P''}$. □

By Lemma 5.3, the set $\{L_P\}$ of all lower sums is bounded above and the set $\{U_P\}$ is bounded below. This observation justifies the following definition.

**Definition 5.5.** For a bounded real function $f : [a, b] \to \mathbf{R}$, the numbers

$$J_* = \sup\{L_P\} \qquad \text{and} \qquad J^* = \inf\{U_P\},$$

where the supremum and the infimum are taken over all partitions of $[a, b]$, are called the *lower* and *upper* integrals of $f$, respectively.

Note that $J_* \le J^*$ (cf. Exercise 2.6).

**Theorem 5.3.** *A real function $f : [a, b] \to \mathbf{R}$ is Riemann integrable if and only if for every $\varepsilon > 0$ there is a $\delta > 0$ such that*

$$|U_P - L_P| < \varepsilon, \tag{5.3}$$

*for every partition $P$ of $[a, b]$ with $\|P\| < \delta$.*

*Proof.* (Necessity.) Suppose that $f$ is Riemann integrable and let $\varepsilon > 0$. Then there is a real number $J$ such that there is a $\delta > 0$ such that for every partition $P$ with $\|P\| < \delta$ and any choice of $c_k \in [x_{k-1}, x_k]$, $1 \le k \le n$, we have

$$|S_P - J| < \frac{\varepsilon}{4}. \tag{5.4}$$

Since $L_P = \inf\{S_P\}$ and $U_P = \sup\{S_P\}$ (cf. Exercise 5.8), there are sums $S'_P$ and $S''_P$ such that

$$|S'_P - L_P| < \frac{\varepsilon}{4} \qquad \text{and} \qquad |S''_P - U_P| < \frac{\varepsilon}{4}.$$

By (5.4),

$$|S''_P - S'_P| = |(S''_P - J) + (J - S'_P| \le |S''_P - J| + |S'_P - J| < \frac{\varepsilon}{4} + \frac{\varepsilon}{4} = \frac{\varepsilon}{2}.$$

Consequently,

$$\begin{aligned} |U_P - L_P| &= |U_P - S''_P + S''_P - S'_P + S'_P - L_P| \\ &\le |U_P - S''_P| + |S''_P - S'_P| + |S'_P - L_P| \end{aligned}$$

$$< \frac{\varepsilon}{4} + \frac{\varepsilon}{2} + \frac{\varepsilon}{4} = \varepsilon,$$

which is the desired result.

(Sufficiency.) For every partition $P$ we have

$$L_P \le J_* \le J^* \le U_P.$$

By (5.3), for $\varepsilon > 0$ there is a $\delta > 0$ such that $|U_P - L_P| < \varepsilon$ for a partition $P$ with $\|P\| < \delta$. Then

$$|J^* - J_*| \le |U_P - L_P| < \varepsilon.$$

Since $\varepsilon$ is an arbitrary positive real number, $J_* = J^*$. Let $J = J_* = J^*$. Clearly,

$$L_P \le J \le U_P$$

for any partition $P$. By (5.2), for every Riemann sum $S_P$,

$$L_P \le S_P \le U_P.$$

By (5.3) again, for $\varepsilon > 0$ there is a $\delta > 0$ such that $|U_P - L_P| < \varepsilon$ for a partition $P$ with $\|P\| < \delta$. As

$$|S_P - J| \le |U_P - L_P| < \varepsilon$$

and $\varepsilon$ is an arbitrary positive real number, the function $f$ is Riemann integrable. □

**Theorem 5.4.** *Every continuous real function on a closed bounded interval is Riemann integrable.*

*Proof.* Let $f : [a, b] \to \mathbf{R}$ be a continuous function and $\varepsilon > 0$. Because $f$ is uniformly continuous on $[a, b]$ (cf. Theorem 3.25), there exists $\delta > 0$ such that

$$|f(x) - f(y)| < \frac{\varepsilon}{b-a} \quad \text{provided that} \quad |x - y| < \delta. \tag{5.5}$$

Let $P$ be a partition of $[a, b]$ with $\|P\| < \delta$. By the Extreme Value Property (cf. Theorem 3.20), for every $1 \le k \le n$, there are $x_k'$ and $x_k''$ in $[x_{k-1}, x_k]$ such that $m_k = f(x_k')$ and $M_k = f(x_k'')$. For every $1 \le k \le n$, we have $\triangle_k < \delta$, because $\|P\| < \delta$. By (5.5), $M_k - m_k < \varepsilon/(b-a)$. We have

$$U_P - L_P = \sum_{k=1}^{n} (M_k - m_k)\triangle_k < \frac{\varepsilon}{b-a} \sum_{k=1}^{n} \triangle_k = \varepsilon.$$

By Theorem 5.3, the function $f$ is Riemann integrable. □

Another example of a Riemann integrable real function on a closed bounded interval is a monotone function (cf. Exercise 5.9).

A standard example of a real function that is not Riemann integrable is the Dirichlet function (cf. Example 3.3).

*Example 5.2.* Let $f : [a, b] \to \mathbf{R}$ be defined by

$$f(x) = \begin{cases} 1, & \text{if } x \in [a, b] \cap \mathbf{Q}, \\ 0, & \text{if } x \in [a, b] \setminus \mathbf{Q}. \end{cases}$$

Let $P = \{x_0, \ldots, x_n\}$ be a partition of $[a, b]$. We have $S_P = 1$, by choosing all $c_k$'s in $Q$, and $S_P = 0$ by choosing all $c_k$'s in $[a, b] \setminus \mathbf{Q}$. By (5.1), the function $f$ is not Riemann integrable.

The last theorem of this section claims that the property of Riemann integrability of continuous functions (cf. Theorem 5.4) is equivalent to the completeness property of the underlying field.

**Theorem 5.5.** *An ordered field* $\mathbf{F}$ *is complete if and only if every continuous function* $f : [a, b] \to \mathbf{F}$ *is Riemann integrable over* $[a, b]$.

*Proof.* (Necessity.) Theorem 5.4.

(Sufficiency.) We follow the steps of the proof of Theorem 3.19. Let $\mathbf{F}$ be an incomplete ordered field and $[a, b] \subseteq \mathbf{F}$. It suffices to construct an example of a continuous function on $[a, b]$ that is not Riemann integrable. By Lemma 3.4, there is a gap $(A, B)$ in $\mathbf{F}$ such that $a \in A$ and $b \in B$. Let $f$ be the function on $[a, b]$ defined by

$$f(x) = \begin{cases} 1, & \text{if } x \in [a, b] \cap A, \\ 0, & \text{if } x \in [a, b] \cap B. \end{cases}$$

By Lemma 3.1, the sets $[a, b] \cap A$ and $[a, b] \cap B$ are relatively open in $[a, b]$. Since these sets are disjoint, the function $f$ is continuous on $[a, b]$.

Suppose that $f$ is Riemann integrable on $[a, b]$ and let $J = \int_a^b f$. Then for every $\varepsilon > 0$ there is a $\delta > 0$ such that for every partition $P = \{x_0, x_1, \ldots, x_n\}$ with $\|P\| < \delta$ and any choice of $c_k$'s in $[x_{k-1}, x_k]$ we have (5.1).

For a partition $P$, there is a $k$ such that $x_{k-1} \in A$ and $x_k \in B$. For every choice of $c_k$'s, we have

$$\sum_{i=1}^{n} f(c_i)\triangle_i = x_{k-1} - a + (x_k - x_{k-1})\lambda,$$

where $\lambda \in \{0, 1\}$. For $\lambda = 0$, by (5.1),

$$(J + a) - \varepsilon < x_{k-1} < (J + a) + \varepsilon.$$

Similarly, for $\lambda = 1$,

$$(J + a) - \varepsilon < x_k < (J + a) + \varepsilon.$$

Therefore, every $\varepsilon$-neighborhood of $J + a$ contains points from the sets $A$ and $B$, implying that $J + a$ is a cut point for $(A, B)$ (cf. Exercise 2.29). This contradicts our assumption that $(A, B)$ is a gap. □

## 5.3 Properties of the Riemann Integral

The following theorem establishes the *linearity* property of the Riemann integral.

**Theorem 5.6.** *Let $f$ and $g$ be Riemann integrable functions on the interval $[a, b] \subseteq \mathbf{F}$ and $k \in \mathbf{F}$ be a constant. Then the functions $kf$ and $f + g$ are Riemann integrable and*

$$\int_a^b kf = k \int_a^b f \qquad \text{and} \qquad \int_a^b (f + g) = \int_a^b f + \int_a^b g.$$

*Proof.* The first property obviously holds for $k = 0$. Let $k \neq 0$, $J = \int_a^b f$, and $\varepsilon > 0$. There is a positive $\delta$ such that for every partition $P = \{x_0, \ldots, x_n\}$ of $[a, b]$ with $\|P\| < \delta$ and any choice of $c_k \in [x_{k-1}, x_k]$, $1 \leq k \leq n$,

$$\left| \sum_{k=1}^{n} f(c_k)\, \triangle_k - J \right| < \frac{\varepsilon}{|k|}.$$

Hence,

$$\left| \sum_{k=1}^{n} kf(c_k)\, \triangle_k - kJ \right| < \varepsilon,$$

so $\int_a^b kf = k \int_a^b f$.

To prove the second property, let $J' = \int_a^b f$, $J'' = \int_a^b g$, and $\varepsilon > 0$. There is a positive $\delta$ such that for every partition $P = \{x_0, \ldots, x_n\}$ of $[a, b]$ with $\|P\| < \delta$ and any choice of $c_k \in [x_{k-1}, x_k]$, $1 \leq k \leq n$,

$$\left| \sum_{k=1}^{n} f(c_k)\, \triangle_k - J' \right| < \frac{\varepsilon}{2} \qquad \text{and} \qquad \left| \sum_{k=1}^{n} g(c_k)\, \triangle_k - J'' \right| < \frac{\varepsilon}{2}.$$

Therefore,

$$\left|\sum_{k=1}^{n}(f(c_k)+g(c_k))\,\triangle_k - (J'+J'')\right|$$

$$= \left|\left[\sum_{k=1}^{n} f(c_k)\,\triangle_k - J'\right] + \left[\sum_{k=1}^{n} g(c_k)\,\triangle_k - J''\right]\right|$$

$$\leq \left|\sum_{k=1}^{n} f(c_k)\,\triangle_k - J'\right| + \left|\sum_{k=1}^{n} g(c_k)\,\triangle_k - J''\right| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon.$$

It follows that $\int_a^b (f+g) = \int_a^b f + \int_a^b g$. □

The linearity property of Theorem 5.6 shows that the set of Riemann integrable functions on the interval $[a, b]$ is a vector space over the field $\mathbf{F}$.

The next result is known as the *positivity property* of the Riemann integral.

**Theorem 5.7.** *If a Riemann integrable function $f : [a, b] \to \mathbf{F}$ is a non-negative on $[a, b]$, then $\int_a^b f \geq 0$.*

*Proof.* Let $J = \int_a^b f$. Suppose to the contrary that $J < 0$. For every $\varepsilon > 0$ there is a $\delta > 0$ such that for every partition $P = \{x_0, \ldots, x_n\}$ of $[a, b]$ with $\|P\| < \delta$ and any choice of $c_k \in [x_{k-1}, x_k]$, we have (5.1). Then

$$\sum_{k=1}^{n} f(c_k)\,\triangle_k + (-J) < \varepsilon,$$

where $\sum_{k=1}^{n} f(c_k)\triangle_k \geq 0$ and $-J > 0$. We reach a contradiction by choosing $\varepsilon < -J$. □

The *monotonicity property* of the Riemann integral follows immediately from the positivity property (cf. Exercise 5.10).

**Theorem 5.8.** *Let $f$ and $g$ be Riemann integrable functions on the interval $[a, b]$. If $g(x) \leq f(x)$ for all $x \in [a, b]$, then $\int_a^b g \leq \int_a^b f$.*

Let $f$ and $g$ be functions on $[a, b]$. We write $g \leq f$ if $g(x) \leq f(x)$ for all $x \in [a, b]$ and use this convention throughout the book.

Then the monotonicity property can be symbolically expressed as

$$g \leq f \quad \Longrightarrow \quad \int_a^b g \leq \int_a^b f.$$

The last theorem in this section establishes the *additivity* property of the Riemann integral over intervals.

**Theorem 5.9.** *A function $f : [a, b] \to \mathbf{F}$ which is Riemann integrable on the intervals $[a, c]$ and $[c, b]$, is Riemann integrable on $[a, b]$ and*

$$\int_a^b f = \int_a^c f + \int_c^b f.$$

*Proof.* Denote $J' = \int_a^c f$ and $J'' = \int_c^b f$. Let $\varepsilon > 0$. There is a $\delta' > 0$ such that for every partition $P'$ of $[a, c]$ with $\|P'\| < \delta'$, we have

$$|S_{P'} - J'| < \varepsilon/3.$$

Similarly, there is a $\delta'' > 0$ such that for every partition $P''$ of $[c, b]$ with $\|P''\| < \delta'$, we have

$$|S_{P''} - J''| < \varepsilon/3.$$

As $f$ is bounded, there is $M \in \mathbf{F}$ such that $|f(x)| < M$ over $[a, b]$.

We define $\delta = \min\left\{\delta', \delta'', \varepsilon/(6M), (b-a)/3\right\}$. Let $P = \{x_0, \ldots, x_n\}$ be a partition of $[a, b]$ with $\|P\| < \delta$. Note that $n \geq 3$. We consider two cases.

(1) $c \in P$. Let $P' = P \cap [a, c]$ and $P'' = P \cap [c, b]$ be partitions of $[a, c]$ and $[c, b]$, respectively. Clearly, $\|P'\| < \delta'$, $\|P''\| < \delta''$, and $S_P = S_{P'} + S_{P''}$. Then

$$|S_P - (J' + J'')| = |(S_{P'} - J') + (S_{P''} - J''| \leq |S_{P'} - J'| + |S_{P''} - J''| < \frac{\varepsilon}{3} + \frac{\varepsilon}{3} < \varepsilon.$$

(2) $x_{m-1} < c < x_m$ for some $m$, $1 \leq m \leq n$. Consider partitions

$$P' = \{x_0, \ldots, x_{m-1}, c\} \qquad \text{and} \qquad P'' = \{c, x_m, \ldots, x_n\}$$

of the intervals $[a, c]$ and $[c, b]$, respectively. Clearly, $\|P'\| < \delta'$ and $\|P''\| < \delta''$. Let $c_k \in [x_{k-1}, x_k]$ for $k = 1, \ldots, n$. Without loss of generality, we may assume that $c_m \in [x_{m-1}, c]$. Let $c'_m \in [c, x_m]$ and consider the Riemann sums

$$S_{P'} = \begin{cases} \sum_{i=1}^{m-1} f(c_i)\,\triangle_i + f(c_m)(c - x_{m-1}), & \text{if } m > 1, \\ f(c_m)(c - x_{m-1}), & \text{if } m = 1, \end{cases}$$

$$S_{P''} = \begin{cases} f(c'_m)(x_m - c) + \sum_{i=m+1}^{n} f(c_i)\triangle_i, & \text{if } m < n, \\ f(c'_m)(x_m - c), & \text{if } m = n, \end{cases}$$

and

$$S_P = \sum_{i=1}^{n} f(c_i)\,\triangle_i\,.$$

A straightforward calculation shows that

$$S_P = \sum_{i=1}^{n} f(c_i)\triangle_i = S_{P'} + S_{P''} + (f(c_m) - f(c'_m))(x_m - c).$$

Therefore,

$$\begin{aligned} |S_P - (J' + J'')| &= |(S_{P'} - J') + (S_{P''} - J'') + (f(c_m) - f(c'_m))(x_m - c)| \\ &\le |S_{P'} - J'| + |S_{P''} - J''| + |(f(c_m) - f(c'_m))(x_m - c)| \\ &< \frac{\varepsilon}{3} + \frac{\varepsilon}{3} + 2M\delta < \frac{\varepsilon}{3} + \frac{\varepsilon}{3} + \frac{\varepsilon}{3} = \varepsilon. \end{aligned}$$

Since the cases (1) and (2) exhaust all possibilities, $f$ is Riemann integrable on $[a, b]$ and $\int_a^b f = \int_a^c f + \int_c^b f$. □

## 5.4 Step Functions

For the present, in the case of an incomplete ordered field, only constant functions are known to be Riemann integrable, whereas the class of real Riemann integrable functions includes continuous and monotone functions (cf. Theorem 5.4 and Exercise 5.9).

In this section we introduce a class of Riemann integrable functions that is pivotal in the development of an alternative integration theory in the last sections of this chapter.

**Definition 5.6.** A function $s : [a, b] \to \mathbf{F}$ is said to be a *step function* if there is a partition $P = \{x_0, \ldots, x_n\}$ such that the values $s(x)$ for $x \in (x_{k-1}, x_k)$ are equal to a constant $s_k \in \mathbf{F}$, $k = 1, 2, \ldots, n$ (cf. Figure 5.2). Any partition that satisfies this condition is called a *step partition* for $s$.

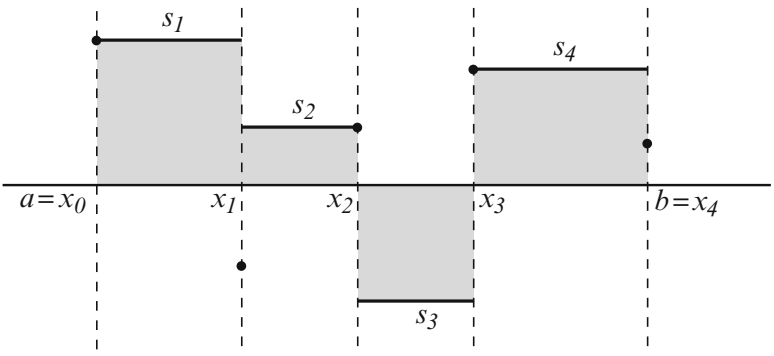



**Fig. 5.2** An example of a step function.

*Example 5.3.* Let $s$ be a constant function on $[a, b]$. It is clear that any partition of $[a, b]$ is a step partition for $s$, so $s$ is a (trivial) example of a step function. In general, let $s$ be a step function on $[a, b]$ and $P$ a step partition for $s$. Then any refinement $Q \supseteq P$ is a step partition for $s$ (cf. Exercise 5.12).

The set of step functions on an interval is closed under the standard algebraic operations as the next theorem asserts. (The proof is left to the reader (cf. Exercise 5.13)).

**Theorem 5.10.** *Let $s$ and $t$ be step function on the interval $[a, b]$. The following functions are step functions on $[a, b]$:*

(a) *$ks$, for every $k \in \mathbf{F}$.*
(b) *$s + t$.*
(c) *$s \cdot t$.*
(d) *$1/s$, provided that $s$ does not vanish on $[a, b]$.*

*In view of parts (a) and (b), the set of step functions on $[a, b]$ is a vector space over the field $\mathbf{F}$.*

In Calculus, the integral $\int_a^b f$ is defined informally as the signed area of the region in the plane that is bounded by the graph of the function $f$, the $x$-axis and the vertical lines $x = a$ and $x = b$. The next theorem justifies this informal definition in the case of step functions (cf. Figure 5.2).

**Theorem 5.11.** *Let $s : [a, b] \to \mathbf{F}$ be a step function, $P = \{x_0, \ldots, x_n\}$ a step partition for $s$, and $s_k = s(x)$ for $x \in (x_{k-1}, x_k)$, $1 \le k \le n$. The function $s$ is Riemann integrable and*

$$\int_a^b s = \sum_{k=1}^{n} s_k \, \triangle_k \, . \tag{5.6}$$

*Proof.* The proof is by induction on $n$.

The base case, $n = 1$. Note that $s$ is a bounded function, so $|s(x)| < M$ for some positive $M \in \mathbf{F}$. Let $\varepsilon > 0$ and define $\delta = \min\{\varepsilon/(4M), (b - a)/2\}$.

Let $J = s_1(b - a)$, $Q = \{y_0, \ldots, y_m\}$ a partition of $[a, b]$ with $\|Q\| < \delta$, and $c_i \in [y_{i-1}, y_i]$ for $i = 1, \ldots, m$. Note that $m > 2$. We have

$$\begin{aligned} \sum_{i=1}^{m} s(c_i)\triangle_i &= s(c_1)\,\triangle_1 + \sum_{i=2}^{m-1} s(c_i)\,\triangle_i + s(c_m)\triangle_m \\ &= s(c_1)\,\triangle_1 + s(c_m)\,\triangle_m + s_1(y_{m-1} - y_1). \end{aligned}$$

Then

$$\begin{aligned}
&\left|\sum_{i=1}^{m} s(c_i)\,\triangle_i - J\right| = \\
&= \left|s(c_1)\,\triangle_1 + s(c_m)\,\triangle_m + (s_1 y_{m-1} - s_1 y_1) - (s_1 y_m - s_1 y_0)\right| \\
&= \left|s(c_1)\,\triangle_1 + s(c_m)\,\triangle_m - s_1\,\triangle_m - s_1\,\triangle_1\right| \\
&= \left|(s(c_1) - s_1)\,\triangle_1 + (s(c_m) - s_1)\,\triangle_m\right| \\
&\le \left|(s(c_1) - s_1)\right|\,\triangle_1 + \left|(s(c_m) - s_1)\right|\triangle_m < 2M\delta + 2M\delta = 4M\delta \le \varepsilon.
\end{aligned}$$

Hence, $J = \int_a^b s$, so (5.6) holds for $n = 1$.

Suppose that (5.6) holds for some $n \ge 1$ and let $P = \{x_0, \ldots, x_n, x_{n+1}\}$ be a step partition for a step function $s$. By the hypothesis, the function $s$ is Riemann integrable on the interval $[a, x_n]$ and

$$\int_a^{x_n} s = \sum_{k=1}^{n} s_k\,\triangle_k\,.$$

By the base step, $s$ is Riemann integrable on $[x_n, b]$ with $\int_{x_n}^b s = s_{n+1}\triangle_{n+1}$. The claim of the theorem follows from Theorem 5.9. □

**Corollary 5.1.** *The sum in the right-hand side of* (5.6) *does not depend on the choice of a step partition for the step function s.*

Note that since step functions are Riemann integrable, their integrals have the properties established in Section 5.3.

## 5.5 The Darboux Integral

We denote the set of step functions on $[a, b] \subseteq \mathbf{F}$ by $S[a, b]$. Let $f : [a, b] \to \mathbf{F}$ be a bounded function. The upper and lower sets of step functions (relative to $f$) are defined by

$$\mathcal{U}_f = \{s \in S[a, b] : s \ge f\} \qquad \text{and} \qquad \mathcal{L}_f = \{s \in S[a, b] : s \le f\},$$

respectively. These sets are not empty because $f$ is assumed to be bounded. Note that $g \le h$ for any $g \in \mathcal{L}_f$ and $h \in \mathcal{U}_f$.

The following definition is motivated by the Riemann integrability criterion (cf. Theorem 5.3).

**Definition 5.7.** A bounded function $f : [a, b] \to \mathbf{F}$ is said to be *Darboux integrable* if for every $\varepsilon > 0$ there exist step functions $g \in \mathcal{L}_f$ and $h \in \mathcal{U}_f$ such that $\int_a^b h - \int_a^b g < \varepsilon$.

The set of Darboux integrable functions on $[a, b]$ will be denoted by $I[a, b]$. Note that Definition 5.7 does not say anything about the integral of the function $f$. For this, we have another definition.

**Definition 5.8.** A bounded function $f : [a, b] \to \mathbf{F}$ is said to have a *Darboux integral* if there is a unique element $J \in \mathbf{F}$ such that

$$\int_a^b g \le J \le \int_a^b h,$$

for all $g \in \mathcal{L}_f$, $h \in \mathcal{U}_f$. In this case, we set $\int_a^b f = J$.

The set of functions on $[a, b]$ that have a Darboux integral will be denoted by $D[a, b]$. It is not difficult to show that $S[a, b] \subseteq D[a, b]$ (cf. Exercise 5.14).

**Theorem 5.12.** *If a function $f : [a, b] \to \mathbf{F}$ has a Darboux integral, then it is Darboux integrable, that is, $D[a, b] \subseteq I[a, b]$.*

*Proof.* Let $J = \int_a^b f$. Suppose to the contrary that $f$ is not Darboux integrable. Then there exists $\varepsilon > 0$ such that

$$\int_a^b h - \int_a^b g \ge \varepsilon, \tag{5.7}$$

for all $g \in \mathcal{L}_f$, $h \in \mathcal{U}_f$. There is $g \in \mathcal{L}_f$ such that $\int_a^b g > J - \varepsilon/2$. Indeed, otherwise,

$$\int_a^b g \le J - \frac{\varepsilon}{2} < J \le \int_a^b h,$$

for all $g \in \mathcal{L}_f$, $h \in \mathcal{U}_f$, which contradicts the uniqueness of $J$ (cf. Definition 5.8). Similarly, there is $h \in \mathcal{U}_f$ such that $\int_a^b h < J + \varepsilon/2$. Hence, $\int_a^b h - \int_a^b g < (J + \varepsilon/2) - (J - \varepsilon/2) = \varepsilon$, contradicting (5.7). □

In summary, we have the following chain of inclusions:

$$S[a, b] \subseteq D[a, b] \subseteq I[a, b] \subseteq B[a, b]. \tag{5.8}$$

It is not difficult to show (cf. Exercise 5.15) that Darboux integrability and the Darboux and Riemann integrals coincide if $\mathbf{F} = \mathbf{R}$, the field of real numbers. In this case, $D[a, b] = I[a, b]$ in (5.8), whereas the remaining two inclusions are proper.

In the rest of this section, we study relations between the sets $D[a, b]$ and $I[a, b]$ for different incomplete ordered fields.

First, we prove the following lemma (cf. Lemma 5.1).

**Lemma 5.4.** *If a field $\mathbf{F}$ has a regular gap, then for every $a < b$ there is a regular gap $(A, B)$ in $\mathbf{F}$ such that $a \in A$, $b \in B$.*

*Proof.* Let $(C, D)$ be a regular gap in $\mathbf{F}$. Since $(C, D)$ is a regular gap, there are $c \in C$ and $d \in D$ such that $d - c < b - a$. Define a function $f : \mathbf{F} \to \mathbf{F}$ by

$$f(x) = x + (a - c),$$

and let $A = f(C)$, $B = f(D)$. It is not difficult to see that $(A, B)$ is a gap. Let $\varepsilon > 0$ and $x \in C$, $y \in D$ such that $y - x < \varepsilon$. Then $f(x) \in A$, $f(y) \in B$, and $f(y) - f(x) = y - x < \varepsilon$. Therefore, $(A, B)$ is a regular gap. □

**Theorem 5.13.** *If an ordered field* $\mathbf{F}$ *has a regular gap, then for every interval* $[a, b] \subseteq \mathbf{F}$ *there is a function that is Darboux integrable, but does not have a Darboux integral.*

*In other words, the set* $D[a, b]$ *is a proper subset of the set* $I[a, b]$.

*Proof.* By Lemma 5.4, we may assume that there is a regular gap $(A, B)$ in $\mathbf{F}$ such that $a \in A$, $b \in B$. We define

$$f(x) = \begin{cases} 1, & \text{if } x \in [a, b] \cap A, \\ 0, & \text{if } x \in [a, b] \cap B. \end{cases} \tag{5.9}$$

(We used this function in the proofs of Theorems 3.19 and 5.5. Specifically, it is shown in the proof of Theorem 5.5 that the continuous function $f$ in (5.9) is not Riemann integrable.)

For $r \in (a, b)$, we define a step function $f_r(x)$ by

$$f_r(x) = \begin{cases} 1, & \text{if } x \in [a, r], \\ 0, & \text{if } x \in (r, b]. \end{cases}$$

It is clear that $f_r \in \mathcal{L}_f$ if $r \in (a, b) \cap A$, and $f_r \in \mathcal{U}_f$ if $r \in (a, b) \cap B$. Also, $\int_a^b f_r = r - a$.

Since $(A, B)$ is a regular gap, for every $\varepsilon > 0$ there are $c \in A$ and $d \in B$ such that $d - c < \varepsilon$. Without loss of generality, we may assume that $c \in (a, b) \cap A$ and $d \in (a, b) \cap B$. Then

$$\int_a^b f_d - \int_a^b f_c = (d - a) - (c - a) = d - c < \varepsilon.$$

Hence, $f$ is Darboux integrable.

Suppose to the contrary that the Darboux integral $J = \int_a^b f$ exists. Then

$$c - a = \int_a^b f_c \le J \le \int_a^b f_d = d - a,$$

for all $c \in (a, b) \cap A$, $d \in (a, b) \cap B$. It follows that

$$c \le a + J \le d,$$

for all $c \in A$, $d \in B$. Hence, $a + J$ is a cut point for $(A, B)$, which contradicts our assumption that $(A, B)$ is a gap. □

The next theorem gives a characterization of the completeness property for Archimedean fields.

**Theorem 5.14.** *Let* $\mathbf{F}$ *be an Archimedean ordered field. Then* $\mathbf{F} = \mathbf{R}$ *if and only if* $D[a, b] = I[a, b]$ *for every* $[a, b] \subseteq \mathbf{F}$.

*Proof.* (Necessity.) Exercise 5.15.

(Sufficiency.) Suppose to the contrary that $D[a, b] = I[a, b]$, but $\mathbf{F} \neq \mathbf{R}$. As $\mathbf{F}$ is an Archimedean incomplete field, there is a regular gap $(A, B)$ in $\mathbf{F}$ (cf. Theorem 5.2). By Theorem 5.13, $D[a, b]$ is a proper subset of $I[a, b]$, a contradiction. Hence, $\mathbf{F} = \mathbf{R}$. □

Since $D[a, b] \subseteq I[a, b]$, we can reformulate the last theorem as follows.

**Theorem 5.15.** *An Archimedean ordered field is complete if and only if every Darboux integrable function has a Darboux integral.*

It remains to consider the case of ordered fields that do not have regular gaps.

**Theorem 5.16.** *Let* $\mathbf{F}$ *be an incomplete ordered field such that every gap in* $\mathbf{F}$ *is irregular. Then every Darboux integrable function has a Darboux integral.*

*In other words,* $D[a, b] = I[a, b]$ *for every interval* $[a, b] \subseteq \mathbf{F}$.

*Proof.* Let $f : [a, b] \to \mathbf{F}$ be a Darboux integrable function. We define subsets $A$ and $B$ of $\mathbf{F}$ by $A = \bigcup_{g \in \mathcal{L}_f} \left(-\infty, \int_a^b g\right]$ and $B = \bigcup_{h \in \mathcal{U}_f} \left[\int_a^b h, \infty\right)$. There are three possible cases:

(1) $A \cup B \neq \mathbf{F}$. Let $x$ be an element of $\mathbf{F}$ that does not belong to $A \cup B$. Then $x > \int_a^b g$ for all $g \in \mathcal{L}_f$ and $x < \int_a^b h$ for all $h \in \mathcal{U}_f$, that is,

$$\int_a^b g < x < \int_a^b h,$$

for all $g \in \mathcal{L}_f$, $H \in \mathcal{U}_f$. Since $f$ is Darboux integrable, for every $\varepsilon > 0$ there are $g \in \mathcal{L}_f$ and $h \in \mathcal{U}_f$ such that $\int_a^b h - \int_a^b g < \varepsilon$. Therefore, $x$ is a unique element of $\mathbf{F}$ satisfying the displayed inequalities. It follows that $x$ is the Darboux integral of $f$, that is, $x = \int_a^b f$.

(2) $A \cap B \neq \varnothing$. Let $x \in A \cap B$. Then there are $g_0 \in \mathcal{L}_f$ and $h_0 \in \mathcal{U}_f$ such that $x \le \int_a^b g_0$ and $x \ge \int_a^b h_0$. We have

$$\int_a^b g \le \int_a^b h_0 \le x \le \int_a^b g_0 \le \int_a^b h,$$

for all $g \in \mathcal{L}_f$, $h \in \mathcal{U}_f$. As in the previous case, $x = \int_a^b f$.

(3) $A \cup B = \mathbf{F}$, $A \cap B = \varnothing$. Clearly, $(A, B)$ is a cut. Suppose that this cut is a gap. Since $f$ is Darboux integrable, for every $\varepsilon > 0$ there are $g \in \mathcal{L}_f$ and $h \in \mathcal{U}_f$ such that $\int_a^b h - \int_a^b g < \varepsilon$. Since $\int_a^b g \in A$ and $\int_a^b h \in B$, the gap $(A, B)$ is regular. This contradicts our assumption that all gaps in $\mathbf{F}$ are irregular. Therefore, the cut $(A, B)$ has a (unique) cut point, say, $J$. It is clear that

$$\int_a^b g \le J \le \int_a^b h,$$

for all $g \in \mathcal{L}_f$, $h \in \mathcal{U}_f$. Hence, $J$ is the Darboux integral of $f$, that is, $J = \int_a^b f$. □

The next theorem follows immediately from Theorems 5.13 and 5.16.

**Theorem 5.17.** *Let* $\mathbf{F}$ *be an incomplete ordered field. Then* $D[a, b] = I[a, b]$ *if and only if all gaps in* $\mathbf{F}$ *are irregular.*

We conclude this section with an example of a continuous function that is not Darboux integrable.

*Example 5.4.* Let $f$ be defined as in (5.9) for an irregular gap $(A, B)$, that is, there is a $\delta > 0$ such that $y - x \ge \delta$ for all $x \in A$, $y \in B$. The function $f$ is not Darboux integrable on $[a, b]$ and hence does not have a Darboux integral.

Indeed, let $g \in \mathcal{L}_f$ and $h \in \mathcal{U}_f$. Without loss of generality, we may assume that $g$ and $f$ have the same step partition $P = \{x_0, \ldots, x_n\}$ with $n \ge 3$ (cf. Example 5.3). Let $g_i = g(x)$ and $h_i = h(x)$ for $x \in (x_{i-1}, x_i)$ $(1 \le i \le n)$.

There is a unique $k \in \{1, \ldots, n\}$ such that $x_{k-1} \in A$ and $x_k \in B$. Then $\triangle_k = x_k - x_{k-1} \ge \delta$. Clearly, $g_i \le h_i$ for $i \ne k$ and $h_k - g_k \ge 1$. We have

$$\int_a^b h - \int_a^b g = \sum_{i=1}^{k-1} (h_i - g_j)\, \triangle_i + (h_k - g_k)\, \triangle_k + \sum_{i=k}^{n} (h_i - g_j) \triangle_i \ge \delta.$$

It follows that $f$ is not Darboux integrable.

## 5.6 Properties of Darboux Integrable Functions

Recall that step functions are Riemann integrable (cf. Theorem 5.11). Below we use some of the properties established in Section 5.3.

We begin with the linearity property for Darboux integrable functions.

**Theorem 5.18.** *Let* $f$ *and* $f'$ *be Darboux integrable functions on the interval* $[a, b] \subseteq \mathbf{F}$ *and* $k \in \mathbf{F}$. *Then the functions* $kf$ *and* $f + f'$ *are Darboux integrable.*

*Proof.* First, consider $kf$. The case $k = 0$ is trivial.

Suppose that $k > 0$ and let $\varepsilon > 0$. There are $g \in \mathcal{L}_f$ and $h \in \mathcal{U}_f$ such that $\int_a^b h - \int_a^b g < \varepsilon/k$. Hence, $\int_a^b kh - \int_a^b kg < \varepsilon$. It is clear that $kg \in \mathcal{L}_{kf}$ and $kh \in \mathcal{U}_{kf}$. It follows that $kf$ is a Darboux integrable function.

Now suppose that $k < 0$ and again let $\varepsilon > 0$. There are $g \in \mathcal{L}_f$ and $h \in \mathcal{U}_f$ such that $\int_a^b h - \int_a^b g < \varepsilon/|k|$. Since $|k| = -k$, we have

$$\int_a^b k(-h) - \int_a^b k(-g) < \varepsilon,$$

or, equivalently,

$$\int_a^b kg - \int_a^b kh < \varepsilon. \tag{5.10}$$

Further, since $k < 0$,

$$g \le f \le h \qquad \text{implies} \qquad kh \le kf \le kg,$$

so $kh \in \mathcal{L}_{kf}$ and $kg \in \mathcal{U}_{kf}$. By (5.10), the function $kf$ is Darboux integrable.

Let $f$ and $f'$ be Darboux integrable functions. Then for $\varepsilon > 0$ there are functions $g \in \mathcal{L}_f$, $h \in \mathcal{U}_f$, $g' \in \mathcal{L}_{f'}$, and $h' \in \mathcal{U}_{f'}$ such that

$$\int_a^b h - \int_a^b g < \frac{\varepsilon}{2} \qquad \text{and} \qquad \int_a^b h' - \int_a^b g' < \frac{\varepsilon}{2}.$$

Note that $g + g' \le f + f' \le h + h'$, so $g + g' \in \mathcal{L}_{f+f'}$ and $h + h' \in \mathcal{U}_{f+f'}$. By adding the last two displayed inequalities, we obtain

$$\int_a^b (h + h') - \int_a^b (g + g') < \varepsilon.$$

It follows that $f + f'$ is a Darboux integrable function. □

By Theorem 5.18, the set $I[a, b]$ is a vector space over the field $\mathbf{F}$ under operations of addition and multiplication by a constant.

We define $x^+ = \max\{x, 0\}$ and $x^- = \min\{x, 0\}$ for $x \in \mathbf{F}$.

**Theorem 5.19.** *Let $f : [a, b] \to \mathbf{F}$ be a Darboux integrable function. Then the functions $f^+$, $f^-$, and $|f|$ are Darboux integrable on $[a, b]$.*

*Proof.* First, we prove the claim for $f^+$, leaving the case of $f^-$ to the reader (cf. Exercise 5.16). Let $\varepsilon > 0$ and choose $g \in \mathcal{L}_f$, $h \in \mathcal{U}_f$ such that

$$\int_a^b h - \int_a^b g < \varepsilon.$$

By Exercise 5.17 (a), $g \le f \le h$ implies

$$g^+ \le f^+ \le h^+ \qquad \text{and} \qquad g - g^+ \le h - h^+.$$

From the last inequality we have $h^+ - g^+ \le h - g$. Hence,

$$\int_a^b h^+ - \int_a^b g^+ \le \int_a^b h - \int_a^b g < \varepsilon.$$

It follows that $f^+$ is a Darboux integrable function.

To prove that the function $|f|$ is Darboux integrable, it suffices to note that $|f| = f^+ - f^-$ (cf. Exercise 5.17 b) and Theorem 5.18). □

We conclude this section by proving that the product of Darboux integrable functions is integrable. For this, we first prove a weaker statement.

**Lemma 5.5.** *If a function $f : [a, b] \to \mathbf{F}$ is Darboux integrable, then so is the function $f^2$.*

*Proof.* Since $f^2 = |f|^2$ and $|f|$ is Darboux integrable (cf. Theorem 5.19), we may assume that $f \ge 0$ on $[a, b]$. Since $f$ is bounded, there is $M > 0$ such that $f(x) < M$ for all $x \in [a, b]$. Let $\varepsilon > 0$. As $f$ is Darboux integrable, there are $g \in \mathcal{L}_f$ and $h \in \mathcal{U}_f$ such that

$$\int_a^b (h - g) = \int_a^b h - \int_a^b g < \frac{\varepsilon}{2M}.$$

Define step functions $s$ and $t$ by

$$s(x) = \max\{g(x), 0\} \qquad \text{and} \qquad t(x) = \min\{h(x), M\}, \qquad x \in [a, b].$$

It is not difficult to verify that

$$g \le s \le f \le t \le h$$

and

$$0 \le s \le f \le t \le M$$

on $[a, b]$. It follows that $s^2 \le f^2 \le t^2$, that is, $s^2 \in \mathcal{L}_{f^2}$ and $t^2 \in \mathcal{U}_{f^2}$. We have

$$t^2 - s^2 = (t + s)(t - s) \le 2M(t - s) \le 2M(h - g).$$

Therefore,

$$\int_a^b t^2 - \int_a^b s^2 = \int_a^b (t^2 - s^2) \leq 2M \int_a^b (h - g) < 2M \cdot \frac{\varepsilon}{2M} = \varepsilon.$$

The result follows. □

**Theorem 5.20.** *If two functions are Darboux integrable, then their product is Darboux integrable.*

*Proof.* Let $f_1$ and $f_2$ be Darboux integrable functions on the interval $[a, b]$. We have

$$f_1 f_2 = \frac{1}{4}\Big[(f_1 + f_2)^2 - (f_1 - f_2)^2\Big].$$

The result follows from Lemma 5.5 and Theorem 5.18. □

## Notes

The properties of Riemann integrable functions over a general ordered field established in Section 5.3 do not include some properties known in the case of the real Riemann integrable functions. This is due to the absence of the Riemann criterion (cf. Theorem 5.3) for arbitrary ordered fields. This deficiency of the Riemann integral is partly corrected by bringing to the table Darboux integrable functions in Section 5.5 and establishing their properties in Section 5.6.

Theorem 5.5 gives another characterization of the completeness property of the underlying ordered field in addition to characterizations established in the previous chapters. One more characterization for an Archimedean field is found in Theorem 5.15.

The notions of Darboux integrable functions and the Darboux integral were introduced in (Olmsted 1973), where a different terminology is used. We adopt these terms from (Teismann 2013).

Ordered fields that do not have regular gaps (cf. Theorem 5.16) are known as *Scott complete* fields (Teismann 2013). Scott (1969) calls an ordered field *complete* if it is not properly contained in another ordered field in which the given field is dense. Scott complete fields are called *quasi-complete* in (Olmsted 1973).

For the treatment of the Fundamental Theorem of Calculus in relation to the completeness property the reader is referred to (Deveau and Tiesmann 2015).

## Exercises

**5.1.** Prove that any finite subset $A$ of an ordered field containing more than one element can be enumerated in such a way that $A = \{a_1, \ldots, a_n\}$ where

$$a_1 < a_2 < \cdots < a_n.$$

**5.2.** Prove that every finite nonempty subset of an ordered field contains a maximum element, that is, an element that is greater than all other elements of the set.

**5.3.** Let $Q$ be a refinement of the partition $P$ of the interval $[a, b]$. Show that subintervals of $Q$ are subintervals of subintervals of $P$ and $\|Q\| \leq \|P\|$.

**5.4.** Prove that (cf. Definition 5.1)

(a) $\mathcal{I}(\mathbf{F}) \subseteq \mathcal{F}(\mathbf{F})$.
(b) $\mathbf{F} = \mathcal{F}(\mathbf{F}) \cup \mathcal{L}(\mathbf{F})$.
(c) $\mathcal{F}(\mathbf{F}) \cap \mathcal{L}(\mathbf{F}) = \varnothing$.

**5.5.** Prove that the sets $\mathcal{I}(\mathbf{F})$ and $\mathcal{L}(\mathbf{F}) \cap \mathbf{F}^+$ are closed under addition. Conclude that $x - y$ is an infinitely large element of $\mathbf{F}$ for all $x \in B$, $y \in A$, where the sets $A$ and $B$ are defined in Example 5.1.

**5.6.** Show that a constant function $f(x) = k$ is Riemann integrable on $[a, b] \subseteq \mathbf{F}$ and $\int_a^b k = k(b - a)$.

**5.7.** Let $f$, $g$ be bounded functions on $[a, b]$ and $k \in \mathbf{F}$. Show that $kf$ and $f + g$ are bounded functions on $[a, b]$.

**5.8.** Let $P$ be a partition of $[a, b] \subseteq \mathbf{R}$. Prove that

$$U_P = \sup\{S_P\}, \quad L_P = \inf\{S_P\},$$

where the operations sup and inf are taken over the set

$$\{(c_1, \ldots, c_n) : c_k \in [x_{k-1}, x_k],\ 1 \leq k \leq n\}.$$

**5.9.** Prove that every monotone real function on a closed bounded interval is Riemann integrable.

**5.10.** Prove Theorem 5.8.

**5.11.** Prove Theorem 5.9.

**5.12.** Show that any refinement of a step partition for a step function is also a step partition for that function.

**5.13.** Prove Theorem 5.10. (Hint: the union of two partitions is their common refinement).

**5.14.** Prove that every step function on the interval $[a, b]$ has a Darboux integral which is equal to its Riemann integral.

**5.15.** Show that a real function on $[a, b]$ is Darboux integrable if and only if it is Riemann integrable. Also, show that the Darboux and Riemann integrals of this function are equal.

**5.16.** Prove Theorem 5.19 for $f^-$.

**5.17.**

(a) Prove that functions $x^+$ and $x - x^+$ are increasing.
(b) Show that $|x| = x^+ - x^-$.

**5.18.** Prove that a function $f : [a, b] \to \mathbf{F}$ is Darboux integrable if and only if its restrictions to intervals $[a, c]$ and $[c, b]$, where $c \in (a, b)$, are Darboux integrable on these intervals (cf. Theorem 5.9).

# Chapter 6
# Infinite Series




The concept of a series with terms in an ordered field and its convergence are introduced in Section 6.1. It is shown there that in the case of "large" ordered fields convergent series are "finite", that is, they terminate with zeros. Also, a sufficiency condition for convergence is established for series with terms in a non-Archimedean, Cauchy complete field.

A standard theorem in Real Analysis states that a real series with non-negative terms converges if and only if the sequence of its partial sums is bounded. It is shown in Section 6.2 that this theorem characterizes the completeness property of the field **R**. The Comparison Test property is also covered in this section.

Another standard theorem in Real Analysis is the Alternating Series Test. In Section 6.3, this result is established for every Cauchy complete ordered field.

Various relations between convergence and absolute convergence of series with terms in an ordered field are discussed in Section 6.4. In particular, it is shown that an Archimedean ordered field is complete if and only if absolute convergence of a series implies its convergence.

Finally, in Section 6.5, it is shown that the validity of the Ratio Test for series with terms in an ordered field is equivalent to the completeness property of the underlying field.

## 6.1 Introduction

**Definition 6.1.** If $(a_n)$ is a sequence in an ordered field **F**, then the sequence $(s_n)$, where

$$s_n = a_1 + a_2 + \cdots + a_n = \sum_{k=1}^{n} a_k,$$

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is called a *series with terms* $a_k$ and denoted by

$$a_1 + a_2 + \cdots + a_n + \cdots, \qquad \text{or by} \qquad \sum_{k=1}^{\infty} a_k.$$

The terms of the sequence $(s_n)$ are called *partial sums* of the series $\sum_{k=1}^{\infty} a_k$. For $m \in \mathbf{N}$, the series

$$a_{m+1} + a_{m+2} + \cdots = \sum_{k=1}^{\infty} a_{m+k}$$

is said to be a *remainder* of the series $\sum_{k=1}^{\infty} a_k$.

If the sequence $(s_n)$ converges with $\lim s_n = S$, we say that the series $\sum_{k=1}^{\infty} a_k$ is *convergent*, call $S$ the *sum* of the series, and write $\sum_{k=1}^{\infty} a_k = S$. Otherwise, the series $\sum_{k=1}^{\infty} a_k$ is said to be *divergent*.

Note that the sequence $(a_{m+1}, a_{m+2}, \ldots)$ defining the remainder $\sum_{k=1}^{\infty} a_{m+k}$ is a tail of the sequence $(a_n)$ (cf. Definition 1.15).

**Theorem 6.1.** *A series is convergent if and only if a remainder of the series converges.*

*Proof.* Let $\sum_{k=1}^{\infty} a_k$ be a series in $\mathbf{F}$ and $m \in \mathbf{N}$. For $n > m$ we have

$$s_n = \underbrace{(a_1 + \cdots + a_m)}_{s_m} + \underbrace{(a_{m+1} + \cdots + a_n)}_{s'_n},$$

where $s'_n$ is the $(n-m)$-th partial sum of the reminder, that is,

$$s_n = s_m + s'_n. \tag{6.1}$$

It is clear that the sequence $(s_n)$ converges if and only if so does the sequence $(s'_n)$, which is the desired result. □

From (6.1) we immediately obtain the following theorem.

**Theorem 6.2.** *If the series* $\sum_{k=1}^{\infty} a_k$ *converges, then*

$$S = s_m + S'_m,$$

*for every* $m \in \mathbf{N}$*, where* $S$ *is the sum of the series,* $s_m$ *is its partial sum, and* $S'_m$ *is the sum of the remainder. Symbolically,*

$$\sum_{k=1}^{\infty} a_k = \sum_{k=1}^{m} a_k + \sum_{k=1}^{\infty} a_{m+k}.$$

If a series $\sum_{k=1}^{\infty} a_k$ converges, then the sequence $(s_n)$ of its partial sums is Cauchy (cf. Theorem 1.16). Hence the following result (cf. Exercise 6.1).

**Theorem 6.3.** *If a series $\sum_{k=1}^{\infty} a_k$ is convergent, then for every $\varepsilon > 0$ there is $N \in \mathbf{N}$ such that*

$$|a_m + \cdots + a_n| < \varepsilon,$$

*for all $n \geq m > N$.*

This theorem immediately yields a necessary condition for convergence of a series (cf. Exercise 6.1).

**Corollary 6.1.** *If the series $\sum_{k=1}^{\infty} a_k$ converges, then $a_n \to 0$.*

Note that the necessary condition of Corollary 6.1 is not sufficient in a general ordered field, as the following two examples illustrate.

*Example 6.1.* The series with real terms

$$1 + \frac{1}{2} + \cdots + \frac{1}{n} + \cdots = \sum_{k=1}^{\infty} \frac{1}{k}$$

is called the *harmonic series*. Clearly, $1/n \to 0$ in $\mathbf{R}$. However, the sequence $(s_n)$ of partial sums of this series diverges. Indeed, we have

$$s_{2^{m+1}} - s_{2^m} = \frac{1}{2^m + 1} + \frac{1}{2^m + 2} + \cdots + \frac{1}{2^{m+1}} \geq (2^{m+1} - 2^m)\frac{1}{2^{m+1}} = \frac{1}{2}.$$

Hence, the sequence $(s_n)$ is not Cauchy and therefore diverges.

*Example 6.2.* Let $\mathbf{F}$ be an ordered field that is not Cauchy complete and let $(x_n)$ be a divergent Cauchy sequence in $\mathbf{F}$. We define

$$a_1 = x_1, \quad a_n = x_n - x_{n-1}, \quad \text{for } n > 1.$$

Since $(x_n)$ is Cauchy, $a_n = x_n - x_{n-1} \to 0$. As

$$s_n = \sum_{k=1}^{n} a_n = x_1 + (x_2 - x_1) + \cdots + (x_n - x_{n-1}) = x_n,$$

the series $\sum_{k=1}^{\infty} a_k$ does not converge in $\mathbf{F}$.

Two basic algebraic operations on convergent series are the subject of the next theorem. The proof is left to the reader (cf. Exercise 6.2).

**Theorem 6.4.** *Let $\sum_{k=1}^{\infty} a_k$ and $\sum_{k=1}^{\infty} b_k$ be convergent series and $c \in \mathbf{F}$. Then the series $\sum_{k=1}^{\infty}(a_k + b_k)$ and $\sum_{k=1}^{\infty}(ca_k)$ are convergent, and*

$$\sum_{k=1}^{\infty}(a_k + b_k) = \sum_{k=1}^{\infty} a_k + \sum_{k=1}^{\infty} b_k \qquad \text{and} \qquad \sum_{k=1}^{\infty}(ca_k) = c\sum_{k=1}^{\infty} a_k.$$

Are there convergent series in every ordered field? The following example gives an affirmative answer to this question.

*Example 6.3.* Let $(a_n)$ be a nonzero sequence in an ordered field $\mathbf{F}$ with a zero tail, that is,

$$(a_n) = (a_1, \ldots, a_m, 0, 0, \ldots),$$

for some $m \in \mathbf{N}$. Then for $k \geq m$ the partial sums $s_k$ are equal to:

$$s_m = s_{m+1} = s_{m+2} = \cdots,$$

that is, the sequence $(s_n)$ is eventually constant (cf. Lemma 4.5) and therefore convergent. Hence, the series $\sum_{k=1}^{\infty} a_k$ is convergent with the sum

$$\sum_{k=1}^{m} a_k.$$

In this case, we say that the series $\sum_{k=1}^{\infty} a_k$ is *finite* and write

$$\sum_{k=1}^{\infty} a_k = \sum_{k=1}^{m} a_k.$$

Although they seem to provide a trivial example of a convergent series, the finite series are the only convergent series in some (very large) ordered fields.

**Theorem 6.5.** *Let $\mathbf{F}$ be an ordered field that is not countably cofinal. Then a series $\sum_{k=1}^{\infty} a_k$ converges in $\mathbf{F}$ if and only if it is finite.*

*Proof.* Clearly, it suffices to show that every convergent series $\sum_{k=1}^{\infty} a_k$ in $\mathbf{F}$ is finite. Since the sequence of partial sums $(s_n)$ converges in $\mathbf{F}$, it is eventually constant (cf. Lemma 4.5), that is, there is $m \in \mathbf{N}$ such that

$$s_m = s_{m+1} = s_{m+2} = \cdots.$$

As $a_n = s_n - s_{n-1}$ for all $n > 1$, we have $a_n = 0$ for all $n > m$. Hence, the series under consideration is finite. □

According to Examples 6.1 and 6.2, the condition $a_n \to 0$ is not sufficient for the convergence of the series $\sum_{k=1}^{\infty} a_k$ in the cases of the field of real numbers $\mathbf{R}$ and ordered fields that are not Cauchy complete. In fact, this condition is sufficient for all other ordered fields, that is, Cauchy complete non-Archimedean fields. To prove it, we need a property of countably cofinal fields.

**Lemma 6.1.** *Let* $\mathbf{F}$ *be a countably cofinal non-Archimedean field. There exists sequence of infinitesimals* $(\lambda_n) \subset \mathbf{F}$ *such that* $\lambda_n \to 0$ *and* $\lambda_{n+1}/\lambda_n$ *is an infinitesimal for all* $n \in \mathbf{N}$.

*Proof.* By Lemma 4.4, there is an unbounded strictly increasing sequence $(a_n)$ of positive elements of $\mathbf{F}$, that is,

$$0 < a_1 < a_2 < \cdots < a_n < \cdots .$$

Since the field $\mathbf{F}$ is non-Archimedean, we may assume that all $a_n$'s are infinitely large. We define

$$x_1 = a_1, \quad \text{and} \quad x_n = a_1 a_2 \cdots a_n, \quad \text{for } n = 2, 3, \ldots .$$

The sequence $(x_n)$ is positive, strictly increasing and unbounded. Furthermore, $x_{n+1}/x_n = a_n$ is infinitely large.

Clearly, the sequence $\lambda_n = x_n^{-1}$, $n \in \mathbf{N}$, satisfies the requirements of the lemma (cf. Exercise 6.6). □

**Theorem 6.6.** *Let* $\mathbf{F}$ *be an ordered field that is Cauchy complete and non-Archimedean. If a sequence* $(a_n) \subset \mathbf{F}$ *converges to zero, then the series* $\sum_{k=1}^{\infty} a_k$ *converges in* $\mathbf{F}$.

*Proof.* We consider two mutually exclusive cases.

Case (1). The field $\mathbf{F}$ is countably cofinal. Let $(\lambda_n)$ be the sequence provided by Lemma 6.1. For $k \in \mathbf{N}$, we can choose $N_k$ such that $|a_n| \leq \lambda_{k+1}$ for all $n \geq N_k$. Then for all $n < m$ greater than $N_k$ we have

$$|a_n + \cdots + a_m| \leq |a_n| + \cdots + |a_m| \leq (m - n + 1)\lambda_{k+1} < \lambda_k,$$

because $\lambda_k/\lambda_{k+1}$ is infinitely large. Inasmuch as $\lambda_n \to 0$, the sequence $(\sum_{k=1}^{n} a_k)$ is Cauchy and therefore converges.

Case (2). The field $\mathbf{F}$ is not countably cofinal. By Lemma 4.5, the sequence $(a_n)$ has a zero tail. Hence, the series $\sum_{k=1}^{\infty} a_k$ is finite and therefore convergent (cf. Example 6.3). □

We conclude this section with two examples of convergent series in the field of real numbers $\mathbf{R}$.

*Example 6.4.* The series $\sum_{k=1}^{\infty} ar^{k-1}$, where $|r| < 1$, is called a *geometric series*. Since

$$\sum_{k=1}^{n} ar^{k-1} = \frac{a(1-r^n)}{1-r} = \frac{a}{1-r} + \frac{a}{1-r} r^n$$

and $r^n \to 0$ (cf. Exercise 6.4), we have

$$\sum_{k=1}^{\infty} ar^{k-1} = \frac{a}{1-r}.$$

*Example 6.5.* We show that every real number $x \in [0, 1)$ admits a *binary representation* as an infinite series:

$$x = \sum_{k=1}^{\infty} \frac{c_k}{2^k}, \tag{6.2}$$

where $c_k \in \{0, 1\}$ for $k = 1, 2, \ldots$. To this end, we prove by mathematical induction that there is a sequence $(c_n)$ such that $c_n \in \{0, 1\}$ and

$$0 \le x - \sum_{k=1}^{n} \frac{c_k}{2^k} < \frac{1}{2^n} \tag{6.3}$$

for all $n \in \mathbf{N}$.

(1) The base case $n = 1$. If $x < 1/2$, then (6.3) holds for $c_1 = 0$. Otherwise, (6.3) holds for $c_1 = 1$, because $x < 1$.
(2) Assume that (6.3) holds for some natural number $m > 1$. If

$$x - \sum_{k=1}^{m} \frac{c_k}{2^k} < \frac{1}{2^{m+1}},$$

we set $c_{m+1} = 0$. Then, clearly, (6.3) holds for $n = m + 1$. If

$$x - \sum_{k=1}^{m} \frac{c_k}{2^k} \ge \frac{1}{2^{m+1}},$$

we set $c_{m+1} = 1$. Then

$$0 \le x - \sum_{k=1}^{m+1} \frac{c_k}{2^k} = x - \sum_{k=1}^{m} \frac{c_k}{2^k} - \frac{1}{2^{m+1}} < \frac{1}{2^m} - \frac{1}{2^{m+1}} = \frac{1}{2^{m+1}}.$$

Hence, (6.3) holds for $n = m + 1$.

It follows that (6.3) holds for all $n \in \mathbf{N}$. Since the sequence $(1/2^n)$ converges to zero (cf. Exercise 6.4), (6.3) implies (6.2).

## 6.2 Series with Non-negative Terms

We begin with a standard result of Real Analysis.

**Theorem 6.7.** *If $(a_n)$ is a sequence of non-negative real numbers, then the series $\sum_{k=1}^{\infty} a_k$ converges if and only if the sequence of partial sums $(\sum_{k=1}^{n} a_k)$ is bounded.*

*Proof.* (Necessity.) A convergent sequence of partial sums $(\sum_{k=1}^{n} a_k)$ is bounded (cf. Corollary 1.1).

(Sufficiency.) Clearly, the sequence $(\sum_{k=1}^{n} a_k)$ is increasing. By Theorem 2.22 (Bounded Monotone Convergence Property of **R**), this sequence converges. Therefore, the series $\sum_{k=1}^{\infty} a_k$ converges. □

The following example shows that if the underlying field is not complete, then there are series with bounded partial sums that do not converge.

*Example 6.6.*

(1) Let **F** be an incomplete Archimedean field, that is, a proper subset of the field **R**. Let $\sum_{k=1}^{\infty} c_k/2^k$ be a binary representation of a real number in $[0, 1) \cap (\mathbf{R} \setminus \mathbf{F})$ (cf. Example 6.5). We have (cf. Example 6.4)

$$\sum_{k=1}^{n} \frac{c_k}{2^k} \le \sum_{k=1}^{n} \frac{1}{2^k} < \sum_{k=1}^{\infty} \frac{1}{2^k} = 1.$$

Hence, the partial sums of the series $\sum_{k=1}^{\infty} c_k/2^k$ are bounded, whereas the series is not convergent in **F**.

(2) Suppose that **F** is not Archimedean. The series $\sum_{k=1}^{\infty} 1$ ($a_k = 1$ for all $k \in \mathbf{N}$) is divergent (cf. Exercise 6.9), whereas the partial sums $\sum_{k=1}^{n} 1 = n$ are bounded by any infinitely large element of **F**.

Theorem 6.7 and Example 6.6 yield the following characterization of the field **R**.

**Theorem 6.8.** *An ordered field is complete if and only if every series with non-negative terms and bounded partial sums converges.*

The Comparison Test is an important theorem in Real Analysis. We introduce this test as a property of a general ordered field.

**Definition 6.2.** An ordered field satisfies the *Comparison Test* property if for every pair of series $\sum_{k=1}^{\infty} a_k$ and $\sum_{k=1}^{\infty} b_k$ such that

$$0 \le a_k \le b_k, \qquad k = 1, 2, \ldots,$$

convergence of the series $\sum_{k=1}^{\infty} b_k$ implies convergence of the series $\sum_{k=1}^{\infty} a_k$.

**Theorem 6.9.** *Every Cauchy complete field satisfies the Comparison Test property.*

*Proof.* Let $\mathbf{F}$ be a Cauchy complete ordered field and let $\sum_{k=1}^{\infty} a_k$, $\sum_{k=1}^{\infty} b_k$ be two series such that

$$0 \le a_k \le b_k, \qquad k = 1, 2, \ldots,$$

and the series $\sum_{k=1}^{\infty} b_k$ converges. We consider two possible cases:

Case (1). $\mathbf{F} = \mathbf{R}$. Since the series $\sum_{k=1}^{\infty} b_k$ converges, the sequence $(\sum_{k=1}^{n} b_k)$ is bounded. Hence, the sequence $(\sum_{k=1}^{n} a_k)$ is bounded. By Theorem 6.7, the series $\sum_{k=1}^{\infty} a_k$ converges.

Case (2). $\mathbf{F} \neq \mathbf{R}$, that is, $\mathbf{F}$ is Cauchy complete and non-Archimedean. By Corollary 6.1, $b_n \to 0$. By the Squeeze Theorem (cf. Theorem 1.14), $a_n \to 0$. By Theorem 6.6, the series $\sum_{k=1}^{\infty} a_k$ converges. □

*Example 6.7.* The series $\sum_{k=1}^{\infty} c_k/2^k$ and $\sum_{k=1}^{\infty} 1/2^k$ of Example 6.6 satisfy the conditions

$$0 \le \frac{c_k}{2^k} \le \frac{1}{2^k}, \qquad k = 1, 2, \ldots.$$

However, the first series diverges, whereas the second is a convergent geometric series. Therefore, the Comparison Test property does not hold in an incomplete Archimedean field.

From this example and Theorem 6.9 we obtain the following characterization of the completeness property for Archimedean fields.

**Theorem 6.10.** *An Archimedean ordered field is complete if and only if it satisfies the Comparison Test property.*

## 6.3 Alternating Series

**Definition 6.3.** An *alternating* series is a series of the form

$$\sum_{k=1}^{\infty} (-1)^{k-1} a_k = a_1 - a_2 + \cdots + (-1)^{n-1} a_n + \cdots$$

where $0 < a_{n+1} < a_n$, $n = 1, 2, \ldots$, and $a_n \to 0$.

**Lemma 6.2.** *The sequence of partial sums $(s_n)$ of an alternating series is Cauchy.*

*Proof.* For $n > m$ we have

$$\begin{aligned} |s_n - s_m| &= \left| \sum_{k=1}^{n} (-1)^{k-1} a_k - \sum_{k=1}^{m} (-1)^{k-1} a_k \right| \\ &= \left| (-1)^m a_{m+1} + \cdots + (-1)^{n-1} a_n \right| \\ &= \left| a_{m+1} - a_{m+2} + \cdots + (-1)^{n-m-1} a_n \right|. \end{aligned}$$

If $n - m$ is an odd integer, then

$$|s_n - s_m| = a_{m+1} - (a_{m+2} - a_{m+3}) - \cdots - (a_{n-1} - a_n) \leq a_{m+1}.$$

Otherwise,

$$|s_n - s_m| = a_{m+1} - (a_{m+2} - a_{m+3}) - \cdots - (a_{n-2} - a_{n-1}) - a_n < a_{m+1}.$$

In both cases we have

$$|s_n - s_m| \leq a_{m+1}.$$

As $a_n \to 0$, the sequence $(s_n)$ is Cauchy. □

The next theorem follows immediately from Lemma 6.2.

**Theorem 6.11.** *An alternating series with terms in a Cauchy complete field is convergent.*

In Real Analysis, Theorem 6.11 is known as the Alternating Series Test. The next theorem gives an estimate for a remainder of a convergent alternating series (cf. Theorem 6.2).

**Theorem 6.12.** *Let $\sum_{k=1}^{\infty} (-1)^{k-1} a_k$ be a convergent alternating series in an ordered field with the sum $S$. Then for every $n \in \mathbf{N}$,*

$$0 < \left| S - \sum_{k=1}^{n} (-1)^{k-1} a_k \right| < a_{n+1}.$$

*Proof.* Since $s_{2m} < S < s_{2m+1}$ (cf. Exercise 6.8), we have

$$0 < |S - s_{2m}| < s_{2m+1} - s_{2m} = a_{2m+1}.$$

Similarly, $s_{2m} < S < s_{2m-1}$ implies

$$0 < |S - s_{2m-1}| < s_{2m-1} - s_{2m} = a_{2m}.$$

The result follows. □

*Example 6.8.* By Theorem 6.11, the alternating series

$$1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \cdots + \frac{(-1)^{n-1}}{n} + \cdots = \sum_{k=1}^{\infty} \frac{(-1)^{k-1}}{k}$$

converges in the field of real numbers $\mathbf{R}$.

The next example shows that an alternating series may be divergent if the underlying field is not Cauchy complete.

*Example 6.9.* We show that the alternating series

$$\sum_{k=1}^{\infty} \frac{(-1)^{k-1}}{k!} = 1 - \frac{1}{2!} + \frac{1}{3!} - \cdots + \frac{(-1)^{n-1}}{n!} + \cdots$$

does not converge in $\mathbf{Q}$.

Suppose that this series converges to a rational number $r$, that is,

$$\sum_{k=1}^{\infty} \frac{(-1)^{k-1}}{k!} = r.$$

Note that $r > 0$ (cf. Exercise 6.8). By Theorem 6.12,

$$0 < \left| r - \sum_{k=1}^{n} \frac{(-1)^{k-1}}{k!} \right| < \frac{1}{(n+1)!}.$$

Let $r = \dfrac{p}{q} = \dfrac{p(q-1)!}{q!}$, $p, q \in \mathbf{N}$, and $n = q$. From the above inequalities, we obtain

$$0 < \left| \frac{p(q-1)!}{q!} - \sum_{k=1}^{q} \frac{(-1)^{k-1}}{k!} \right| < \frac{1}{(q+1)!},$$

which is equivalent to

$$0 < \left| p(q-1)! - \sum_{k=1}^{q} \frac{(-1)^{k-1} q!}{k!} \right| < \frac{1}{q+1}.$$

These inequalities do not hold because the middle number is an integer. This contradiction shows that the series under consideration does not converge in the field $\mathbf{Q}$. (It converges to $e^{-1}$ in $\mathbf{R}$ (cf. Exercise 6.13).)

## 6.4 Absolute Convergence

**Definition 6.4.** A series $\sum_{k=1}^{\infty} a_k$ with terms in an ordered field $\mathbf{F}$ is said to be *absolutely convergent* if the series $\sum_{k=1}^{\infty} |a_k|$ converges in $\mathbf{F}$.

A series $\sum_{k=1}^{\infty} a_k$ is said to be *conditionally convergent* if it converges, but not absolutely.

*Example 6.10.* The alternating series

$$1 - \frac{1}{2} + \frac{1}{3} - \cdots \frac{(-1)^{n-1}}{n} + \cdots$$

converges conditionally in $\mathbf{R}$ because it converges (cf. Example 6.8), whereas the harmonic series

$$1 + \frac{1}{2} + \frac{1}{3} + \cdots \frac{1}{n} + \cdots$$

diverges (cf. Example 6.1).

It is clear that for series with non-negative terms there is no difference between ordinary convergence and absolute convergence.

Here is another theorem in Real Analysis that, in fact, holds in every Cauchy complete ordered field.

**Theorem 6.13.** *If a series $\sum_{k=1}^{\infty} a_k$ in a Cauchy complete field $\mathbf{F}$ converges absolutely, then it converges.*

*Proof.* Since the series $\sum_{k=1}^{\infty} |a_k|$ converges, for every $\varepsilon > 0$ there is $N \in \mathbf{N}$ such that

$$|a_m| + \cdots + |a_n| < \varepsilon,$$

for all $n \geq m > N$ (cf. Theorem 6.3). Therefore, since

$$|a_m + \cdots + a_n| < |a_m| + \cdots + |a_n|,$$

the sequence of partial sums of the series $\sum_{k=1}^{\infty} a_k$ is Cauchy. Finally, as the field $\mathbf{F}$ is Cauchy complete, the series $\sum_{k=1}^{\infty} a_k$ converges. □

The converse of Theorem 6.13 does not hold in general, as Example 6.10 demonstrates. However, we have the following result.

**Theorem 6.14.** *In a non-Archimedean Cauchy complete field, every convergent series $\sum_{k=1}^{\infty} a_k$ converges absolutely.*

*Proof.* By Corollary 6.1, $a_n \to 0$. Hence, $|a_n| \to 0$. By Theorem 6.6, the series $\sum_{k=1}^{\infty} |a_k|$ converges in $\mathbf{F}$. □

Theorems 6.13 and 6.14 can be summarized as follows. In the real field **R** every absolutely convergent series converges, but in **R** there is a convergent series that is not absolutely convergent. In a Cauchy complete non-Archimedean field a series converges if and only if it converges absolutely.

The next theorem addresses the relation between convergence and absolute convergence in fields that are not Cauchy complete.

**Theorem 6.15.** *Let* **F** *be an incomplete Archimedean field, that is,* **F** *is a proper subfield of* **R**. *There is an absolutely convergent series in* **F** *that is not convergent.*

*Proof.* Let $x \in \mathbf{R} \setminus \mathbf{F}$. Clearly, we may assume that $0 < x < 1$. In the field of real numbers **R**, the irrational number $x$ admits a binary representation by a convergent series

$$x = \sum_{k=1}^{\infty} \frac{c_k}{2^k}, \tag{6.4}$$

where all $c_k \in \{0, 1\}$ (cf. Example 6.2). For every maximal set of consecutive zeros in (6.4), we apply the following identity (cf. Exercise 6.5):

$$\frac{0}{2^m} + \cdots + \frac{0}{2^{n-1}} + \frac{1}{2^n} = \frac{1}{2^m} - \cdots - \frac{1}{2^{n-1}} - \frac{1}{2^n} \tag{6.5}$$

to obtain another representation of $x$ in **R** by a convergent series

$$x = \sum_{k=1}^{\infty} \frac{d_k}{2^k}, \tag{6.6}$$

where $d_k \in \{-1, 1\}$ (cf. Exercise 6.12). Clearly, the series in (6.6) is not convergent in **F**. On the other hand this series converges absolutely in **F** because

$$\sum_{k=1}^{\infty} \frac{|d_k|}{2^k} = \sum_{k=1}^{\infty} \frac{1}{2^k} = 1 \in \mathbf{F}$$

(cf. Example 6.4). □

The following characterization of the field **R** is an immediate consequence of the preceding theorem.

**Theorem 6.16.** *An Archimedean field* **F** *is complete if and only if every absolutely convergent series in* **F** *converges.*

## 6.5 The Ratio Test

The power of the Comparison Test property is illustrated by the following theorem, which is called the *Ratio Test* for absolute convergence.

**Theorem 6.17.** *Let $(a_n)$ be a sequence of nonzero real numbers such that*

$$\lim \frac{|a_{n+1}|}{|a_n|} = r < 1,$$

*Then the series $\sum_{k=1}^{\infty} a_k$ converges absolutely.*

*Proof.* For $0 < \varepsilon < 1 - r$ there is $N \in \mathbf{N}$ such that

$$\left| \frac{|a_{n+1}|}{|a_n|} - r \right| < \varepsilon, \qquad \text{for all } n > N.$$

Hence,

$$\frac{|a_{n+1}|}{|a_n|} < r + \varepsilon, \qquad \text{for all } n > N.$$

Let $r_1 = r + \varepsilon$. Clearly, $0 < r_1 < 1$. By mathematical induction, we easily obtain

$$|a_n| \leq r_1 |a_{n-1}| \leq \cdots \leq r_1^{n-N-1} |a_{N+1}|,$$

for $n > N$. We compare the remainder

$$|a_{N+1}| + |a_{N+2}| + \cdots + |a_n| + \cdots$$

of the series $\sum_{k=1}^{\infty} |a_k|$ with the convergent geometric series (cf. Example 6.4)

$$|a_{N+1}| + |a_{N+1}| r_1 + \cdots + |a_{N+1}| r_1^{n-N-1} \cdots .$$

By the Comparison Test property (cf. Theorem 6.9), the remainder

$$\sum_{k=N+1}^{\infty} |a_k|$$

converges. By Theorem 6.1, the series $\sum_{k=1}^{\infty} |a_k|$ converges. □

**Theorem 6.18.** *An ordered field* **F** *is complete if and only if for every series $\sum_{k=1}^{\infty} a_k$ with nonzero terms in* **F***, the condition*

$$\lim \frac{|a_{n+1}|}{|a_n|} = r < 1,$$

*implies absolute convergence of the series* $\sum_{k=1}^{\infty} a_k$ *in* $\mathbf{F}$.

*Proof.* (Necessity.) Theorem 6.17.

(Sufficiency.) Suppose that the claim of the theorem holds for an incomplete ordered field $\mathbf{F}$.

First, we show that $\mathbf{F}$ is Archimedean. Suppose to the contrary that there is an infinitesimal element $\varepsilon > 0$ in $\mathbf{F}$. The series $\sum_{k=1}^{\infty}(1/2)^k$ satisfies the conditions of the theorem. Therefore, it converges in $\mathbf{F}$. However, the sequence $(\sum_{k=1}^{n}(1/2)^k)$ of partial sums is not Cauchy, because for all $m < n$, we have

$$\frac{1}{2^m} + \cdots + \frac{1}{2^n} > \varepsilon.$$

Since every convergent series is Cauchy, $\mathbf{F}$ is an Archimedean ordered field.

Recall that an Archimedean field is a subfield of the field of real numbers. By our assumption, $\mathbf{F}$ is a proper subfield of $\mathbf{R}$. Let $x \in \mathbf{R} \setminus \mathbf{F}$ be an irrational number in the interval $(0, 1)$ and let

$$x = \sum_{k=1}^{\infty} \frac{d_k}{2^k}. \tag{6.7}$$

where $d_k \in \{-1, 1\}$, as in (6.6). Clearly, the series in (6.7) satisfies the conditions of the theorem and hence converges in $\mathbf{F}$. We obtained a contradiction, because $x \notin \mathbf{F}$. □

## Notes

Case (2) of the proof of Theorem 6.6 does not use the Cauchy completeness property of the field $\mathbf{F}$. This comes as no surprise, because it is not difficult to show that every ordered field that is not countably cofinal is Cauchy complete (cf. Exercise 6.7).

Theorem 6.6 states that the condition $a_n \to 0$ is sufficient for the convergence of the series $\sum_{k=1}^{\infty} a_k$ with terms in a Cauchy complete ordered field that is different from the field of real numbers $\mathbf{R}$. In this connection, note that the Alternating Series Test (Theorem 6.11) holds for *every* Cauchy complete field, including $\mathbf{R}$.

The assumption that the field $\mathbf{F}$ in Theorem 6.15 is Archimedean is superfluous. In fact, the following two statements hold in every Cauchy incomplete field (Clark and Diepeveen 2014):

(a) the field admits an absolutely convergent series that is not convergent;
(b) the field admits a convergent series that is not absolutely convergent.

## Exercises

**6.1.** Prove Theorem 6.3 and Corollary 6.1.

**6.2.** Prove Theorem 6.4.

**6.3.** Prove that for all $n > 1$,

$$\frac{1}{n+1} + \frac{1}{n+2} + \cdots + \frac{1}{2n} > \frac{1}{2}.$$

**6.4.**

(1) Prove by mathematical induction the *Bernoulli inequality*

$$(1+x)^n \geq 1 + nx,$$

where $x > -1$ is a real number and $n \in \mathbf{N}$.

(2) Use the Bernoulli inequality to show that $r^n \to 0$, if $-1 < r < 1$.

**6.5.** Prove that the sum of geometric series formula (cf. Example 6.4)

$$\sum_{k=1}^{\infty} ar^{k-1} = \frac{a}{1-r}, \qquad \text{where } |r| < 1,$$

holds in every Archimedean field.

**6.6.** Prove that a nonzero element $x$ of a non-Archimedean field is infinitesimal if and only if the element $1/x$ is infinitely large.

**6.7.** Prove that a Cauchy sequence in an ordered field that is not countably cofinal is eventually constant (cf. Example 6.2). Conclude that an ordered field that is not countably cofinal is Cauchy complete.

**6.8.** Let $(s_n)$ be the sequence of partial sums of a convergent alternating series with the sum $S$. Prove that the sequence $(s_{2n})$ is strictly increasing and the sequence $(s_{2n-1})$ is strictly decreasing. Conclude that

$$s_{2n} < S < s_{2m-1}, \qquad \text{for all } m, n \in \mathbf{N}.$$

**6.9.** Prove that the series

$$1 + 1 + \cdots + 1 + \cdots$$

diverges in any ordered field.

**6.10.** Show that if a convergent series has only a finite number of negative terms, then it is absolutely convergent.

**6.11.** Prove identity (6.5).

**6.12.** Prove that the series in (6.6) converges to the same real number as the series in (6.4).

**6.13.** Prove that for every real number $x$ the series

$$\sum_{k=0}^{\infty} \frac{x^k}{k!}$$

converges. (By definition, $0! = 1$, $1! = 1$, and $(n+1)! = n!(n+1)$ for $n = 1, 2, 3, \ldots,$. Also, $x^0 = 1$ for all $x \in \mathbf{R}$.) The sum of this series is called the (real) *exponential function* and denoted by $e^x$ (cf. Example 6.9).

**6.14.**

(a) Prove that the series

$$\sum_{k=0}^{\infty} (-1)^k \frac{x^{2k+1}}{(2k+1)!}$$

converges for every $x \in \mathbf{R}$. The sum of this series is called the *sine function* and is denoted by $\sin x$.

(b) Prove that the series

$$\sum_{k=0}^{\infty} (-1)^k \frac{x^{2k}}{(2k)!}$$

converges for every $x \in \mathbf{R}$. The sum of this series is called the *cosine function* and is denoted by $\cos x$.

# Appendix A
# Natural Numbers and Integers

Algebraic structures of natural numbers, integers, and rings are introduced and a summary of results about these structures is given.

## A.1 Natural Numbers

**Definition A.1.** Let $N$ be a nonempty set, $s$ a function $s : N \to N$, and 1 an element of $N$. The triple $\langle N, s, 1\rangle$ is called an *algebraic structure*. The set $N$ and the element 1 are called the *underlying set* and the *distinguished element* of the algebraic structure $\langle N, s, 1\rangle$, respectively.

An algebraic structure $\langle N, s, 1\rangle$ is called a *Peano system* if the following conditions are satisfied:

**P1.** $s(a) \neq 1$ for all $a \in N$.
**P2.** $s(a) = s(b)$ implies $a = b$ for all $a, b \in N$.
**P3.** (Axiom of Induction) If $M$ is a subset of $N$ such that

(1) $1 \in M$, and
(2) $a \in M$ implies $s(a) \in M$,

then $M = N$.

If $\langle N, s, 1\rangle$ is a Peano system, then $s$ is called the *successor function*. Conditions **P1**–**P3** are called *Peano Axioms*.

Informally, the natural numbers are "constructed" by starting with the number 1 and taking its successors, 2, 3, and so on. Here, a successor of a natural number $n$ is the "next" natural number $n + 1$. Definition A.1 formalizes the intuitive concepts of the number 1 and the successor function.

Since the natural numbers are necessary for the development of other number systems, we a adopt a simple assumption.

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**Axiom A.1.** *There exists a Peano system.*

We proceed with defining operations of addition and multiplication on the underlying set of a Peano system $\langle N, s, 1\rangle$, and formulating their properties.

**Theorem A.1.** *There exists a unique binary operation $+$ on $N$ such that*

$$a + 1 = s(a), \qquad \text{for all } a \in N, \tag{A.1}$$

*and*

$$a + s(b) = s(a + b), \qquad \text{for all } a, b \in N. \tag{A.2}$$

The unique binary operation $+$, the existence of which is asserted by Theorem A.1, is called the operation of *addition* (on the underlying set $N$). The element $a + b$ of $N$ obtained by addition of $b$ to $a$ is called the *sum* of elements $a$ and $b$. The elements $a$ and $b$ in the sum $a + b$ are called *summands*.

The following are standard algebraic properties of the addition operation:

**Theorem A.2.**

(i) **Associative Property of Addition.**

$$(a + b) + c = a + (b + c),$$

*for all $a, b, c \in N$.*

(ii) **Commutative Property of Addition.**

$$a + b = b + a,$$

*for all $a, b \in N$.*

(iii) **Trichotomy Property of Addition.**

*For any two elements $a$ and $b$ of the set $N$, one and only one of the following statements holds:*

(1) $a = b$.
(2) *There is $m \in N$ such that $a = b + m$.*
(3) *There is $n \in N$ such that $b = a + n$.*

Now we introduce the multiplication operation on the underlying set of $\langle N, s, 1\rangle$ state its properties.

**Theorem A.3.** *There exists a unique binary operation $\cdot$ on $N$ such that*

$$a \cdot 1 = a, \qquad \text{for all } a \in N,$$

*and*

$$a \cdot s(b) = (a \cdot b) + a, \qquad \textit{for all } a, b \in N.$$

The unique binary operation $\cdot$, the existence of which is asserted in Theorem A.3, is called the operation of *multiplication*. The symbol $\cdot$ is read *times*, and the element $a \cdot b$ of $N$ is called the *product* of $a$ and $b$. We often write $ab$ for $a \cdot b$ and omit parentheses following the standard order of operations rules known from elementary algebra.

**Theorem A.4.**

(i) **Right Distributive Property.**

$$(a + b)c = ac + bc,$$

*for all* $a, b, c \in N$.

(ii) **Commutative Property of Multiplication.**

$$ab = ba,$$

*for all* $a, b) \in N$.

(iii) **Associative Property of Multiplication.**

$$(ab)c = a(bc),$$

*for all* $a, b, c \in N$.

Note that the Left Distributive Property

$$c(a + b) = ca + cb,$$

for all $a, b, c \in N$, follows immediately from the first two claims of Theorem A.4.

Intuitively, the natural numbers are ordered. We give a formal definition of various order relations on the underlying set $N$ of a Peano system $\langle N, s, 1\rangle$.

**Definition A.2.** For any $a, b \in N$, we put:

$$\begin{aligned} a < b, &\quad \text{if there is } n \in N \text{ such that } b = a + n. \\ a \leq b, &\quad \text{if } a < b \text{ or } a = b. \\ a > b, &\quad \text{if } b < a. \\ a \geq b, &\quad \text{if } b \leq a. \end{aligned}$$

The binary relations $<$, $\leq$, $>$, and $\geq$ are called *order relations* on $N$.

**Theorem A.5.**

(i) **Trichotomy Property of the Order Relation** $<$.
*For any* $a, b \in N$, *exactly one of the three cases*

$$a < b, \qquad a = b, \qquad b < a$$

*holds.*

(ii) **Transitive Property of the Order Relation** $<$.
*For any* $a, b, c \in N$,

$$\text{if} \quad a < b \quad \text{and} \quad b < c, \quad \text{then } a < c.$$

(iii) *For any* $a, b, c \in N$, *if* $a < b$, *then*

$$a + c < b + c \qquad \text{and} \qquad ac < bc.$$

**Definition A.3.** Let $M$ be a nonempty subset of $N$. An element $a \in M$ is called a *least* element of $M$ if $a \leq b$ for all $b \in M$. Similarly, $a \in M$ is a *greatest* element of $M$ if $a \geq b$ for all $b \in M$.

**Theorem A.6 (Well-Ordering Principle).** *Any nonempty subset of* $N$ *has a least element.*

Axiom A.1 states that there is at least one Peano system. Theorem A.7 below claims that this system is unique up to isomorphism.

**Definition A.4.** Two algebraic structures $\langle N, s, 1\rangle$ and $\langle N', s', 1'\rangle$ are said to be *isomorphic* if there is a bijection, that is, a one-to-one and onto function $f : N \to N'$, such that

$$\begin{aligned} &f(1) = 1', \\ &f(s(a)) = s'(f(a)), \quad \text{for all } a \in N. \end{aligned}$$

The function $f$ is called an *isomorphism* of $\langle N, s, 1\rangle$ onto $\langle N', s', 1'\rangle$.

**Theorem A.7.** *Any two Peano systems* $\langle N, s, 1\rangle$ *and* $\langle N', s', 1'\rangle$ *are isomorphic.*

In conclusion of this section, we present a set-theoretic model of the Peano system.

For every set $x$ we define the *successor* $s(x)$ of $x$ to be the set

$$x \cup \{x\}. \tag{A.3}$$

and introduce an axiom which is another form of Axiom A.1.

**Axiom A.2.** *There exists a set that contains the empty set* $\varnothing$ *and the successor of each of its elements.*

We say that a set $A$ is *inductive* if $\varnothing \in A$ and if $x \cup \{x\} \in A$ whenever $x \in A$. Thus, Axiom A.2 says that there exists an inductive set $A$. It is not difficult to show that the intersection of any family of inductive sets is itself an inductive set. Specifically, the intersection of all inductive sets included in the inductive set $A$ is an inductive set. We denote this set by $\mathbf{N}$ and use this notation throughout the book. The set $\mathbf{N}$ is the minimum inductive set.

By definition, a *natural number* is an element of the set $\mathbf{N}$. It is clear that sets $1 = \varnothing$ and

$$2 = s(1) = 1 \cup \{1\} = \varnothing \cup \{1\} = \{1\},$$
$$3 = s(2) = 2 \cup \{2\} = \{1\} \cup \{2\} = \{1, 2\},$$
$$4 = s(3) = 3 \cup \{3\} = \{1, 2\} \cup \{3\} = \{1, 2, 3\},$$

and so on, are natural numbers. The notation

$$\mathbf{N} = \{1, 2, 3, \ldots\}$$

is used very often. In this notation, the ellipsis ... stands for "and so on" in our informal definition. Note that according to our definition, natural numbers are sets.

**Theorem A.8.** *The triple* $\langle \mathbf{N}, s, 1 \rangle$*, where the function* $s$ *is given by* (A.3)*, is a Peano system.*

## A.2 Integers

Informally, integers are differences of natural numbers. According to the Trichotomy properties (cf. Theorems A.2(iii) and A.5(i)), the equation

$$m + x = n, \qquad m, n \in \mathbf{N}$$

has a solution in the set $\mathbf{N}$ if and only if $m < n$. This solution $x$ is the difference $n - m$ of the numbers $m$ and $n$. In general, we may have

$$n - m = q - p$$

for distinct pairs $(m, n)$ and $(p, q)$ of natural numbers. This equation can be written in the equivalent form

$$m + q = p + n,$$

which involves only natural numbers and the addition operation on $\mathbf{N}$, the concepts that were defined in Section A.1.

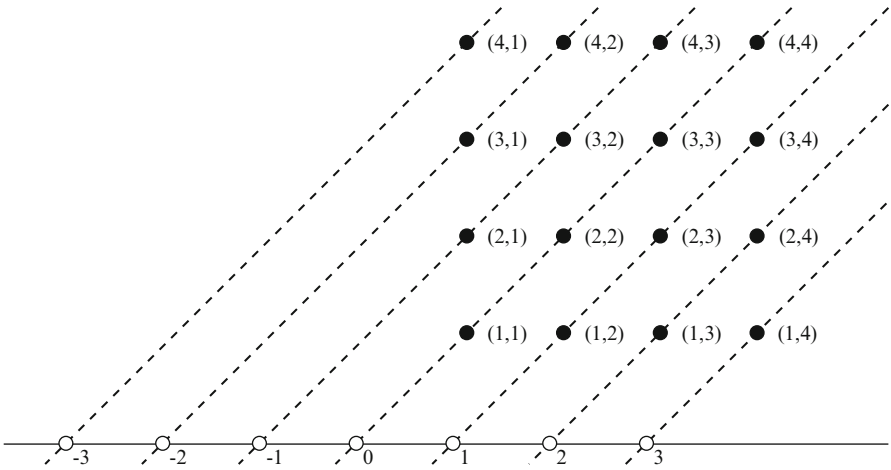



**Fig. A.1** Geometric construction of integers.

Now we proceed with a formal development of the theory of integers.

**Theorem A.9.** *The binary relation* $\sim$ *on the set* $\mathbf{N} \times \mathbf{N}$ *defined by*

$$(m, n) \sim (p, q) \quad \textit{if and only if} \quad m + q = n + p,$$

*for all* $m, n, p, q \in \mathbf{N}$*, is an equivalence relation.*

We denote by $[m, n]$ the equivalence class of the pair $(m, n)$ with respect to $\sim$:

$$[m, n] = \{(p, q) \in \mathbf{N} \times \mathbf{N} : (p, q) \sim (m, n)\}.$$

**Definition A.5.** The equivalence classes of the relation $\sim$ are called *integers*. The set of all integers is denoted by $\mathbf{Z}$.

A fragment of the set $\mathbf{N} \times \mathbf{N}$ is depicted in Figure A.1, where solid dots represent ordered pairs of natural numbers. Equivalence classes of the relation $\sim$ consist of ordered pairs on the dashed lines. These classes are distinguished by hollow dots on the horizontal solid line. The drawing illustrates the connection between the formal definition of integers and the traditional representation of the integers on the number line.

The drawing in Figure A.1 suggests that natural numbers can be associated with some equivalence classes of the relation $\sim$. Let $\langle N', s', 1' \rangle$ be an algebraic structure with

$$N' = \{[1, n + 1] : n \in \mathbf{N}\}, \quad s' : [1, n + 1] \mapsto [1, n + 2], \quad \text{and} \quad 1' = [1, 2].$$

It can be shown that the algebraic structure $\langle N', s', 1' \rangle$ is a Peano system and therefore is isomorphic to the Peano system $\langle \mathbf{N}, s, 1 \rangle$ (cf. Theorem A.7). For instance, the classes $[1, 2]$, $[1, 3]$, and $[1, 4]$ can be identified with the natural numbers 1, 2, and 3, respectively (cf. Figure A.1). In general, we identify the natural number $n$ with the integer $[1, n + 1]$ by means of the function $\psi : \mathbf{N} \to \mathbf{Z}$ defined by

$$\psi(n) = [1, n + 1], \qquad \text{for all } n \in \mathbf{N}. \tag{A.4}$$

In this book, we regard the set $\mathbf{N}$ of natural numbers as a subset of the set $\mathbf{Z}$ of integers.

Our immediate goal is to expand the operations of addition and multiplication and the order relations from the set $\mathbf{N}$ to the set of integers $\mathbf{Z}$.

Let $a = [m, n]$ and $b = [p, q]$ be integers. The operation of addition $+$ on $\mathbf{Z}$ is defined by

$$a + b = [m + p, n + q].$$

It is readily seen that the sum of two integers does not depend on the ordered pairs of natural numbers defining these integers (cf. Exercise A.5). Moreover, we have

$$\psi(m + n) = \psi(m) + \psi(n),$$

that is, the addition operation on integers coincides with the addition operation on natural numbers when the summands are natural numbers.

The integers of the form $[1, n + 1]$ (natural numbers) are also called *positive integers*. We denote $0 = [1, 1]$ and call this equivalence class the *zero element* of $\mathbf{Z}$. The integer $[n + 1, 1]$ is denoted by $-n$ and the integers in this form are called *negative integers* (cf. Figure A.1). We also say that $-n$ is the *negative* of the integer $n$.

The following theorem summarizes some properties of addition of integers.

**Theorem A.10.**

(i) **Commutative Property of Addition.** *For any $a, b \in \mathbf{Z}$ we have*

$$a + b = b + a.$$

(ii) **Associative Property of Addition.** *If $a$, $b$, and $c$ are integers, then*

$$a + (b + c) = (a + b) + c.$$

(iii) **Cancellation Property of Addition.** *If $a + c = b + c$ for some integers $a$, $b$, and $c$, then $a = b$.*

(iv) *For any integer $a$, $a + 0 = a$.*

(v) *Let $a$ and $b$ be integers. The equation $a + x = b$ has a unique integer solution $x$.*

By Theorem A.10(v), for every integer $a$ there is a unique integer $b$ such that $a + b = 0$. This integer $b$ is called the *additive inverse* of $a$ and is denoted by $-a$. For a positive $a$, its additive inverse is the negative $-a$ of $a$, so the two notations agree.

Let $a = [m, n]$ and $b = [p, q]$ be integers. The *product* $a \cdot b$ is defined by

$$a \cdot b = [mq + np, nq + mp].$$

We usually write the product $a \cdot b$ as $ab$. As in the case of addition, the product of two integers does not depend on ordered pairs of natural numbers defining these integers (cf. Exercise A.6). Moreover,

$$\psi(m \cdot n) = \psi(m) \cdot \psi(n),$$

that is, the multiplication operation on the integers is an extension of the multiplication on the natural numbers.

The next theorems summarize a number of properties of the multiplication operation on integers.

**Theorem A.11.** *For all $a, b, c \in \mathbf{Z}$:*

(i) **Commutative Property.** $ab = ba$.
(ii) **Associative Property.** $a(bc) = (ab)c$.
(iii) **Distributive Property.** $a(b + c) = ab + ac$.

The integer $1 = [1, 2]$ is called the *identity element* of $\mathbf{Z}$.

**Theorem A.12.** *For any integer $a$, we have $a \cdot 1 = a$.*

The result of the next theorem is known as the "zero-factor property" in algebra.

**Theorem A.13.** *In $\mathbf{Z}$, $a \cdot b = 0$ if and only if $a = 0$ or $b = 0$.*

We extend the definitions of the order relation on $\mathbf{N}$ (cf. Definition A.2) to the set $\mathbf{Z}$ as follows:

**Definition A.6.** For any $a, b \in \mathbf{Z}$, we put:

$$\begin{aligned} a < b, &\quad \text{if there is } n \in \mathbf{N} \text{ such that } b = a + n. \\ a \leq b, &\quad \text{if } a < b \text{ or } a = b. \\ a > b, &\quad \text{if } b < a. \\ a \geq b, &\quad \text{if } b \leq a. \end{aligned}$$

The binary relations $<$, $\leq$, $>$, and $\geq$ are called *order relations* on $\mathbf{Z}$.

Some properties of the order relations are found in the next theorems.

**Theorem A.14.**

(i) **Trichotomy Property of Order Relation** $<$. *For any* $a, b \in \mathbf{Z}$*, exactly one of the following holds:*

$$a < b, \qquad a = b, \qquad b < a.$$

(ii) **Transitivity Property of Order Relation** $<$. *For any* $a, b, c \in \mathbf{Z}$*,*

$$a < b \quad \textit{and} \quad b < c \qquad \textit{implies} \qquad a < c.$$

**Definition A.7.** A binary relation $<$ on a nonempty set (not necessarily $\mathbf{Z}$) that satisfies the Trichotomy and Transitive Properties is called a *linear order* (the terms *simple order* and *total order* are also used).

By Theorem A.14, the relation $<$ on $\mathbf{Z}$ is a linear order. The name "linear order" is motivated by the plot of integers on the number line such as the one shown at the bottom of Figure A.1.

**Theorem A.15.**

(i) *An integer* $a$ *is positive if and only if* $a > 0$.
(ii) *An integer* $a$ *is negative if and only if* $a < 0$.

**Theorem A.16.**

(i) $a < b$ *if and only if* $a + c < b + c$*, for all* $a, b, c \in \mathbf{Z}$.
(ii) $a < b$ *if and only if* $a \cdot c < b \cdot c$*, for all* $a, b \in \mathbf{Z}$ *and* $c > 0$ *in* $\mathbf{Z}$.

Note that the function $\psi$ (cf. (A.4)) preserves the order relation on $\mathbf{N}$:

$$\text{If } m < n, \text{ then } \psi(m) < \psi(n).$$

## A.3 Rings

**Definition A.8.** Let $R$ be a nonempty set endowed with binary operations $+$ and $\cdot$ such that for all elements $a, b, c \in R$:

**R1** $a + b = b + a$.
**R2** $a + (b + c) = (a + b) + c$.
**R3** There is $x \in R$ such that $a + x = b$.
**R4** $a \cdot (b \cdot c) = (a \cdot b) \cdot c$.
**R5** $a \cdot (b + c) = a \cdot b + a \cdot c$ and $(b + c) \cdot a = b \cdot a + c \cdot a$.

An algebraic structure $\langle R, +, \cdot \rangle$ is called a *ring* if the conditions **R1**–**R5** are satisfied. The operations $+$ and $\cdot$ are called *addition* and *multiplication*, respectively.

The element $a+b$ is called the *sum* of $a$ and $b$. The element $a \cdot b$ is called the *product* of $a$ and $b$ and is often denoted by $ab$.

Conditions **R1** and **R2** are called the Commutative and Associative Properties of Addition, respectively. Condition **R4** is called the Associative Property of Multiplication and conditions **R5** are called the Left and Right Distributive Properties, respectively.

An element 0 of a ring $R$ is called a *zero element* of $R$ if

$$a + 0 = a, \qquad \text{for every } a \in R.$$

**Theorem A.17.** *In any ring $R$ there exists a unique zero element.*

Let $a$ be an element of a ring $R$. An element $-a$ of $R$ such that

$$a + (-a) = 0$$

is called an *additive inverse* of $a$.

**Theorem A.18.** *For every element $a$ of a ring $R$ there exists a unique additive inverse $-a$.*

It is clear that the set of integers $\mathbf{Z}$ with the addition and multiplication operations defined in Section A.2 is a ring.

An element 1 of a ring $R$ is an *identity element* (also called *unity*) of $R$ if

$$a \cdot 1 = 1 \cdot a = a, \qquad \text{for all } a \in R$$

(cf. Exercise A.11). Note that there are rings without an identity element—see Exercise A.9 (c).

A ring is said to be *commutative* if the multiplication operation is commutative, that is,

$$a \cdot b = b \cdot a, \qquad \text{for all } a, b \in R.$$

Clearly, in a commutative ring one needs only one of the Distributive Properties **R5**. The set of integers $\mathbf{Z}$ with the usual operations of addition and multiplication is a commutative ring with a unity.

Let $R$ be a ring. A nonzero element $a$ of $R$ is said to be a *zero divisor* if there is $a$ nonzero element $b$ of $R$ such that $ab = 0$.

**Definition A.9.** A commutative ring with the identity element $1 \neq 0$ is called an *integral domain* if it has no zero divisors.

It is clear that in an integral domain

$$ab = 0 \qquad \text{implies} \qquad a = 0 \quad \text{or} \quad b = 0$$

(cf. Theorem A.13).

The prototypical example of an integral domain is the ring $\mathbf{Z}$ of all integers. Another example is the ring $\mathbf{Z}_2$ from Exercise A.9 (b).

**Definition A.10.** An integral domain $\langle D, +, \times, 0, 1\rangle$ with a linear order $<$ on $D$ (cf. Definition A.7) is said to be an *ordered integral domain* if

(i) $a < b$ if and only if $a + c < b + c$, for all $a, b, c \in D$, and
(ii) $a < b$ if and only if $a \cdot c < b \cdot c$, for all $a, b \in D$ and $c > 0$ in $D$.

By Theorem A.16, the ring $\mathbf{Z}$ is an ordered integral domain with its usual order.

We conclude this section with important examples of rings.

*Example A.1.* Let $R$ be a ring. We denote by $R[x]$ the set of formal symbols

$$R[x] = \{a_n x^n + a_{n-1}x^{n-1} + \cdots + a_1 x + a_0 : a_k \in R,\ 0 \le k \le n,$$
$$n \text{ is a nonnegative integer}\}.$$

The elements of the set $R[x]$ are called (formal) *polynomials in* $x$. Two polynomials

$$a_n x^n + a_{n-1}x^{n-1} + \cdots + a_1 x + a_0$$

and

$$b_m x^m + b_{m-1}x^{m-1} + \cdots + b_1 x + b_0$$

are said to be equal if and only if $m = n$ and $b_k = a_k$, for all $0 \le k \le n$.

Let

$$f = a_n x^n + a_{n-1}x^{n-1} + \cdots + a_1 x + a_0$$

and

$$g = b_m x^m + b_{m-1}x^{m-1} + \cdots + b_1 x + b_0$$

be polynomials in $x$. The addition $+$ and multiplication $\cdot$ operations on $R[x]$ are defined as follows:

$$f + g = (a_p + b_p)x^p + (a_{p-1} + b_{p-1})x^{p-1} + \cdots + (a_1 + b_1)x + (a_0 + b_0),$$

where $p$ is the maximum of $n$ and $m$, $a_k = 0$ for $k > n$, and $b_k = 0$ for $k > m$, and

$$f \cdot g = c_{n+m}x^{n+m} + c_{n+m-1}x^{n+m-1} + \cdots + c_1 x + c_0,$$

with

$$c_k = a_k b_0 + a_{k-1}b_1 + \cdots + a_1 b_{k-1} + a_0 b_k,$$

for $0 \le k \le n + m$, where again $a_k = 0$ for $k > n$ and $b_k = 0$ for $k > m$.

The zero element of $R[x]$ is the *zero polynomial*, that is, the unique polynomial with $n = 0$ and $a_0 = 0$.

It can be shown that $R[x]$ is a ring (cf. Exercise A.13). This ring is called the *polynomial ring* over $R$ in the indeterminate $x$. If $R$ is an integral domain, then also $R[x]$ is an integral domain (cf. Exercise A.13). In particular, $\mathbf{Z}[x]$ is an integral domain. Let

$$f = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0$$

and

$$g = b_m x^m + b_{m-1} x^{m-1} + \cdots + b_1 x + b_0$$

be elements of $\mathbf{Z}[x]$. We define

$$f < g \qquad \text{if and only if} \qquad a_p < b_p,$$

where $p$ is the maximum of $n$ and $m$, $a_k = 0$ for $k > n$, and $b_k = 0$ for $k > m$. In words, $f < g$ if and only if "the leading coefficient of $f$ is smaller than the leading coefficient of $g$". This convention makes $\mathbf{Z}[x]$ an ordered integral domain (cf. Exercise A.14). Note that $f > 0$ if and only if the leading coefficient of $f$ is positive.

## Notes

For comprehensive treatments of number system and underlying algebraic structures the reader is referred to Cohen and Enrlich (1963), Mendelson (2008), and Ovchinnikov (2015).

## Exercises

In Exercises $\langle N, s, 1\rangle$ is a Peano system.

**A.1.** Show that the function $s$ is a bijection from $N$ onto $N \setminus \{1\}$. Conclude that the set $N$ is infinite.

**A.2.** Show that for every $a \in N$, $s(a) \neq a$.

**A.3.** Show that for all $a, b \in N$, $a + b \neq b$. (Hint: Fix $a \in N$ and define $M = \{b \in N : a + b \neq b\}$, then use the Axiom of Induction.)

**A.4.**

(a) Prove that if a nonempty subset of $N$ has a least (greatest) element, then this element is unique.
(b) Show that 1 is the least element of $N$ and there is no greatest element in $N$.

**A.5.** Show that for any $(m', n') \in [m, n]$ and $(p', q') \in [p, q]$ we have

$$[m' + p', n' + q'] = [m + p, n + q].$$

**A.6.** Let $a = [m, n]$, $b = [p, q]$ be integers. Show that, if $(m', n') \sim (m, n)$ and $(p', q') \sim (p, q)$, then

$$[m', n'] \cdot [p', q'] = [m, n] \cdot [p, q].$$

**A.7.** Prove Theorem A.17.

**A.8.** Prove Theorem A.18.

**A.9.** Verify that the following algebraic structures are indeed rings:

(a) Let $R$ be a singleton. Because a ring must have a zero element, necessarily $R = \{0\}$. The addition and multiplication operations are uniquely defined and trivial. The ring $R$ itself is called *trivial*.
(b) The set $R = \{0, 1\}$ with the operations $+$ and $\cdot$ defined by

| $+$ | 0 | 1 |
|---|---|---|
| 0 | 0 | 1 |
| 1 | 1 | 0 |

and

| $\cdot$ | 0 | 1 |
|---|---|---|
| 0 | 0 | 0 |
| 1 | 0 | 1 |

is a ring. This ring is denoted by $\mathbf{Z}_2$.
(c) Let $R$ be the set of all even integers, that is,

$$R = 2\mathbf{Z} = \{\ldots, -4, -2, 0, 2, 4, \ldots\}.$$

This set is a ring with the operations inherited from $\mathbf{Z}$.
(d) Let $R = 2^X$, the set of all subsets of a set $X$. We define the *symmetric difference* of two sets $A, B \in 2^X$ by

$$A \triangle B = (A \setminus B) \cup (B \setminus A).$$

The set $R$ endowed with the operations $\triangle$ (addition) and $\cap$ (multiplication) is a ring. The empty set $\varnothing$ is the zero element of this ring. We also have $-A = A$ for any element $A$ of the ring $R$.

(e) Let $X$ be a nonempty set and let $F$ be the set of all functions $f : X \to \mathbf{Z}$. We define

$$(f + g)(x) = f(x) + g(x), \qquad (f \cdot g)(x) = f(x)g(x), \qquad \text{for } x \in X.$$

The algebraic structure $\langle F, +, \cdot \rangle$ is a ring. The zero function, $0(x) = 0$ for all $x \in X$, is the zero element of this ring.

**A.10.** Prove the Cancellation Property for rings (cf. Theorem A.10(iii)): Let $a$, $b$, and $c$ be elements of a ring. Then

$$a + c = b + c \qquad \text{implies} \qquad a = b.$$

**A.11.** Prove that if a ring $R$ has an identity element, then this element is unique.

**A.12.** Let $a$, $b$, and $c$ be elements of an integral domain. Show that if $ab = ac$ and $a \neq 0$, then $b = c$.

**A.13.** Prove that $R[x]$ is a ring (cf. Definition A.10). Show that if $R$ is an integral domain, then so is $R[x]$.

**A.14.** Show that the ring $\mathbf{Z}[x]$ (cf. Definition A.10) is an ordered integral domain.

# Appendix B
# Dedekind's Construction of Real Numbers

There are essentially two ways of constructing real numbers. In Chapter 2, we used Cantor's method of Cauchy sequences. In Section B.1 of this Appendix, we outline the construction based on completion by cuts. This construction was introduced by Dedekind. Theorem B.1 claims that Dedekind's construction produces a complete ordered field, that is, the same field $\mathbf{R}$ that was constructed in Chapter 2. We omit the long and tedious proof of this theorem. It can be found in many monographs and textbooks (cf., for instance, Rudin 1976, Appendix to Chapter 1).

The idea of using cuts to construct real numbers can be traced back to Euclid's *Elements*. In Section B.2, we collect some relevant citations from *Elements* and pertinent historical literature.

## B.1 Dedekind Cuts

**Definition B.1.** A *Dedekind cut* in the ordered field of rational numbers $\mathbf{Q}$ is a nonempty proper subset $\alpha$ of $\mathbf{Q}$ (so $\alpha \neq \varnothing$ and $\alpha \neq \mathbf{Q}$) such that

(a) If $q \in \alpha$ and $p < q$, then $p \in \alpha$.
(b) The set $\alpha$ does not have a greatest element.

The set of all Dedekind cuts in $\mathbf{Q}$ is denoted by $\mathbf{R}$ (Theorem B.1 below justifies this notation).

A Dedekind cut $\alpha$ in $\mathbf{Q}$ defines a cut $(A, B)$ (cf. Definition 2.11) in $\mathbf{Q}$ with $A = \alpha$ and $B = \mathbf{Q} \setminus \alpha$. Note that a cut point for $(A, B)$ (if there is one) belongs to $B$ (cf. Definition B.1 (b)).

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*Example B.1.* The sets

$$\{p \in \mathbf{Q} : p < 7\} \qquad \text{and} \qquad \{p \in \mathbf{Q} : p < 0 \text{ or } p^2 < 2\} \quad \text{(cf. Example 2.2)}$$

are Dedekind cuts, whereas the set $\{p \in \mathbf{Q} : p \leq 7\}$ is not.

We introduce a structure of an ordered field on $\mathbf{R}$ as follows.

First, the zero and identity elements are defined by

$$0 = \{p \in \mathbf{Q} : p < 0\} \qquad \text{and} \qquad 1 = \{p \in \mathbf{Q} : p < 1\},$$

respectively. (Note that we use the same symbols 0 and 1 for both $\mathbf{Q}$ and $\mathbf{R}$.)

Second, we define the relation $<$ on $\mathbf{R}$ by

$$\alpha < \beta \qquad \text{if and only if} \qquad \alpha \text{ is a proper subset of } \beta.$$

We write $\alpha > \beta$, if $\beta < \alpha$. If $\alpha > 0$, then we say that the cut $\alpha$ is *positive*. Similarly, a cut $\alpha$ is *negative* if $\alpha < 0$. The *negation* (*additive inverse*) of a cut $\alpha$ is given by

$$-\alpha = \{p \in \mathbf{Q} : p < -q \text{ for some choice of } q \in \mathbf{Q} \text{ with } q \notin \alpha\}.$$

Note that $-\alpha = 0$ if $\alpha = 0$, $-\alpha < 0$ if $\alpha > 0$, and $-\alpha > 0$ if $\alpha < 0$.

Next, the operation of addition on $\mathbf{R}$ is defined by

$$\alpha + \beta = \{r \in \mathbf{Q} : r = p + q,\ p \in \alpha,\ q \in \beta\}.$$

The definition of the multiplication operation on $\mathbf{R}$ is more involved. We set $\alpha \cdot \beta = 0$ if $\alpha = 0$ or $\beta = 0$, and define $\alpha \cdot \beta$ for positive cuts $\alpha$ and $\beta$ by

$$\alpha \cdot \beta = \{p \in \mathbf{Q} : p < qr \text{ for some choice of } q \in \alpha, r \in \beta \text{ with } q > 0, r > 0.\}$$

Furthermore, by definition,

$$\alpha \cdot \beta = \begin{cases} -((-\alpha) \cdot \beta), & \text{if } \alpha < 0, \beta > 0, \\ -(\alpha \cdot (-\beta)), & \text{if } \alpha > 0.\ \beta < 0, \\ (-\alpha) \cdot (-\beta), & \text{if } \alpha < 0, \beta < 0. \end{cases}$$

The following theorem is the main result about the algebraic structure $\langle \mathbf{R}, 0, 1, +, \cdot, < \rangle$ introduced above.

**Theorem B.1.** *The algebraic structure* $\langle \mathbf{R}, 0, 1, +, \cdot, < \rangle$ *is a complete ordered field isomorphic to the field of real numbers.*

## B.2 Historical Notes

Definition 4 from Euclid's *Elements*, book V (Fitzpatrick 2008, p.130):

> 4. (Those) magnitudes are said to have a ratio with respect to one another which, being multiplied, are capable of exceeding one another.

Fitzpatrick comments (Fitzpatrick 2008, p.131):

> In other words, $\alpha$ has a ratio with respect to $\beta$ if $m\alpha > \beta$ and $n\beta > \alpha$, for some [natural numbers] $m$ and $n$.

In fact, this is the Archimedean property of the field of real numbers $\mathbf{R}$.

Definition 5 from *Elements*, book V (Fitzpatrick 2008, p.130):

> 5. Magnitudes are said to be in the same ratio, the first to the second, and the third to the fourth, when equal multiples of the first and the third either both exceed, are both equal to, or are both less than, equal multiples of the second and the fourth, respectively, being taken in corresponding order, according to any kind of multiplication whatever.

Fitzpatrick comments (Fitzpatrick 2008, p.131):

> In other words, $\alpha : \beta :: \gamma : \delta$ if and only if $m\alpha > n\beta$ whenever $m\gamma > n\delta$, and $m\alpha = n\beta$ whenever $m\gamma = n\delta$, and $m\alpha < n\beta$ whenever $m\gamma < n\delta$, for all $m$ and $n$. This definition is the kernel of Eudoxus' theory of proportion, and is valid even if $\alpha$, $\beta$, etc., are irrational.

One can say that this is a definition of equal reals in terms of Dedekind cuts.

In his essay "The Nature and Meaning of Numbers" (Dedekind 1863, pp. 39–40), Richard Dedekind writes:

> ... if ... one regards the irrational number as the ratio of two measurable quantities, then ... this manner [by means of cuts] of determining it already set forth in the clearest possible way in the celebrated definition which Euclid gives of the equality of two ratios (*Elements*, book V, definition 5).

Morris Kline (Kline 1972, p.982) writes:

> Euclid in Book V of the *Elements* had treated incommensurable ratios of magnitudes and had defined the equality and inequality of such ratios. His definition of equality ... amounted to dividing the rational numbers $m/n$ into two classes, those for which $m/n$ is less than incommensurable ratio $a/b$ of magnitudes $a$ and $b$ and those for which $m/n$ is greater.
>
> ...
>
> Actually Dedekind did make use of Euclid's work and acknowledged this debt (cf. Dedekind 1863).

# Appendix C
# A Panorama of Ordered Fields

Properties of ordered fields were used throughout the book in relation to theorems of elementary Real Analysis. The diagram in Section C.1 provides a visualization of the relationships between major classes of ordered fields.

## C.1 Main Classes of Ordered Fields

The rectangular box in Figure C.1 represents the class of all ordered fields. To explain the relationships indicated in the diagram, we first make several observations.

(1) There is only one ordered field that is Archimedean and Cauchy complete. This is the field of real numbers $\mathbf{R}$.
(2) The field $\mathbf{Q}$ of rational numbers is Archimedean, but not Cauchy Complete.
(3) The NCC class of fields consists of all ordered fields that are not countably cofinal. Every element of this class is a Cauchy complete ordered field (cf. Exercise 6.7). The fields in NCC are really large—for every regular cardinal $\kappa$ there is an ordered field of cofinality $\kappa$.
(4) The field of Laurent series $\mathbf{R}((x))$ is countably cofinal (so it does not belong to the NCC class), non-Archimedean, and Cauchy complete (cf. Section C.2).
(5) The field $\mathbf{R}(x)$ of rational functions over the field $\mathbf{R}$ is not Archimedean and not Cauchy complete (cf. Section C.3).

## C.2 The Field $\mathbf{R}((x))$

A formal Laurent series is a formal expression

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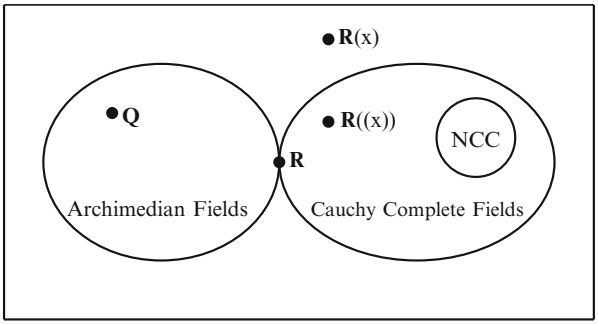



**Fig. C.1** A diagram of classes of ordered fields.

$$\sum_{k=m}^{\infty} a_k x^k = a_m x^m + a_{m+1} x^{m+1} + \cdots$$

where coefficients $a_k$ are real numbers and $m$ is an integer. The set of all Laurent series is a field denoted by $\mathbf{R}((x))$. In what follows we denote elements of this field by lower Greek letters.

It is convenient to represent elements of $\mathbf{R}((x))$ as formal series extended in both directions:

$$\alpha = \sum_{k=-\infty}^{\infty} a_k x^k, \qquad \text{where } a_k = 0 \text{ for all } k \text{ less than some integer } m.$$

If $a_k = 0$ for all $k \in \mathbf{Z}$, then $\alpha = 0$ in $\mathbf{R}((x))$. We also write

$$\alpha = a_m x^m + a_{m+1} x^{m+1} + \cdots$$

if $a_m$ is the first nonzero coefficient in $\alpha$. An element $\alpha$ of $\mathbf{R}((x))$ is positive if the "leading" coefficient $a_m$ in $\alpha$ is positive. The order on $\mathbf{R}((x))$ is defined by

$$\beta < \alpha \qquad \text{if and only if} \qquad \alpha - \beta > 0.$$

The field $\mathbf{R}((x))$ endowed with this order is an ordered field.

Let $0 < \varepsilon = a_m x^m + a_{m+1} x^{m+1} + \cdots$. For every $n > \max\{m, 0\}$, $x^n < \varepsilon$ because the leading coefficient of $\varepsilon - x^n$ is $a_m > 0$.

Below we use the notation $\varepsilon_n = x^n$ for $n = 1, 2, \ldots$. It is clear that $\varepsilon_n > 0$ for all $n \in \mathbf{N}$,

$$\varepsilon_1 > \varepsilon_2 > \cdots > \varepsilon_n > \cdots,$$

and $\varepsilon_n \to 0$. Hence, the sequence $(\varepsilon_n^{-1})$ is unbounded. It follows that the field $\mathbf{R}((x))$ is countably cofinal (cf. Definition 4.2). This field is not Archimedean because $nx < 1$ for all $n \in \mathbf{N}$.

We use the following observations in the proof of Cauchy completeness of the field $\mathbf{R}((x))$ (Theorem C.1 below).

Let $\alpha = \sum_{k=-\infty}^{\infty} a_k x^k$. It is not difficult to see that $|\alpha| < \varepsilon_p$ if $a_k = 0$ for all $k \le p$. Let

$$\alpha = \sum_{k=-\infty}^{\infty} a_k x^k, \qquad \text{where } a_k = 0 \text{ for } k < m$$

and

$$\beta = \sum_{k=-\infty}^{\infty} b_k x^k, \qquad \text{where } b_k = 0 \text{ for } k < m'.$$

We have

$$\alpha - \beta = \sum_{k=-\infty}^{\infty} (a_k - b_k) x^k, \qquad \text{where } a_k = b_k = 0 \text{ for all } k < \min\{m, m'\}.$$

Then

$$|\alpha - \beta| < \varepsilon_p, \tag{C.1}$$

if $a_k = b_k$ for all $k \le p$.

**Theorem C.1.** *The field $\mathbf{R}((x))$ is Cauchy complete.*

*Proof.* Let $(\alpha_n)$ be a Cauchy sequence in $\mathbf{R}((x))$ and $p \in \mathbf{N}$. Then there is $N_p \in \mathbf{N}$ such that

$$|\alpha_n - \alpha_{n'}| < \varepsilon_p, \qquad \text{for } n, n' > N_p.$$

Let

$$\alpha_n = \sum_{k=-\infty}^{\infty} a_{nk} x^k \quad \text{and} \quad \alpha_{n'} = \sum_{k=-\infty}^{\infty} a_{n'k} x^k.$$

By (C.1), $a_{nk} = a_{n'k}$ for $k \le p$. Thus we can write $\alpha_n$ and $a_{n'}$ as

$$\alpha_n = \sum_{k=-\infty}^{p} c_k x^k + \sum_{k=p+1}^{\infty} a_{nk} x^k \quad \text{and} \quad \alpha_{n'} = \sum_{k=-\infty}^{p} c_k x^k + \sum_{k=p+1}^{\infty} a_{n'k} x^k,$$

respectively, where $c_k = a_{nk} = a_{n'k}$ for all $k \leq p$. It follows that coefficients $c_k$'s do not depend on the choice of $n, n' > N_p$.

By applying the previous argument successively for $p = 1, 2, \ldots$, we obtain a formal Laurent series

$$\gamma = \sum_{k=-\infty}^{\infty} c_k x^k.$$

Let $\varepsilon > 0$ and $p \in \mathbf{N}$ such that $\varepsilon_p < \varepsilon$. For all $n > N_p$ we have

$$|\gamma - \alpha_n| = \left| \sum_{k=p+1}^{\infty} (c_k - a_{nk}) x^k \right| < \varepsilon_p < \varepsilon.$$

Hence, $\alpha_n \to \gamma \in \mathbf{R}((x))$, that is, the field of formal Lauren series is Cauchy complete. □

## C.3 The Field R(*x*)

A real rational function is a formal expression $f/g$, where $f$ and $g$ are formal polynomials with real coefficients and $g \neq 0$. The set of all real rational functions is a field denoted by $\mathbf{R}(x)$.

Since every polynomial is an element of $\mathbf{R}((x))$ and $\mathbf{R}((x))$ is a field, $\mathbf{R}(x)$ is a subfield of $\mathbf{R}((x))$.

An important remark is in order. In this section, we consider $\mathbf{R}(x)$ as an *ordered* subfield of $\mathbf{R}((x))$ with the order induced by the order on $\mathbf{R}((x))$. In this order, a polynomial is positive if the sign of its *trailing* term (that is, the coefficient of the nonzero term of lowest degree) is positive, whereas in Example A.1 a polynomial is positive if the sign of the *leading* term is positive.

**Theorem C.2.** *The field* $\mathbf{R}(x)$ *is a proper subfield of the field* $\mathbf{R}((x))$.

*Proof.* It suffices to show that

$$\alpha(x) = 1 + x + x^4 + \cdots + x^{k^2} + \cdots,$$

which is an element of $\mathbf{R}((x))$, cannot be written as $f/g$, where $f$ and $g$ are polynomials. Suppose to the contrary that $\alpha = f/g$, that is, $g \cdot \alpha = f$. Let $g(x) = a_0 + a_1 x + \cdots + a_d x^d$. Then

$$(a_0 + a_1 x + \cdots + a_d x^d)(1 + x + x^4 + \cdots + x^{k^2} + \cdots) = f(x).$$

For $m > d$, the coefficient of $x^d \cdot x^{m^2} = x^{d+m^2}$ is $a_d$. Indeed, suppose that for some $0 \le k < d$,

$$x^{d+m^2} = x^k \cdot x^{n^2} = x^{k+n^2},$$

so $d + m^2 = k + n^2$. Clearly, $n > m$. We have

$$d \ge d - k = n^2 - m^2 \ge (m+1)^2 - m^2 = 2m + 1 > 2d + 1,$$

a contradiction. Thus the product $g \cdot \alpha$ has infinitely many nonzero terms $a_d x^{m^2}$, $m > d$, and hence is not a polynomial. □

**Theorem C.3.** *The field* $\mathbf{R}(x)$ *is a dense subfield of the field* $\mathbf{R}((x))$.

*Proof.* It suffices to show that for every $\alpha \in \mathbf{R}((x)) \setminus \mathbf{R}(x)$ and $\varepsilon_p$ there is $\beta \in \mathbf{R}(x)$ such that $|\alpha - \beta| < \varepsilon_p$.

Let $\alpha = a_m x^m + a_{m+1} x^{m+1} + \cdots$ and $n = \max\{m, p\}$. Clearly, there are infinitely many nonzero coefficients in $\alpha$. (Otherwise, $\alpha \in \mathbf{R}(x)$.) Without loss of generality, we may assume that $a_{n+1} \ne 0$. For

$$\beta = a_m x^m + \cdots + a_n x^n \in \mathbf{R}[x] \subseteq \mathbf{R}(x)$$

we have

$$|\alpha - \beta| = \left|a_{n+1} x^{n+1} + \cdots\right| = \begin{cases} a_{n+1} x^{n+1} + \cdots, & \text{if } a_{n+1} > 0, \\ -a_{n+1} x^{n+1} - \cdots, & \text{if } a_{n+1} < 0. \end{cases}$$

In both cases, $|\alpha - \beta| < \varepsilon_n \le \varepsilon_p$. □

As $\mathbf{R}(x)$ is a proper dense subfield of $\mathbf{R}((x))$ and $\mathbf{R}((x))$ is Cauchy complete, we conclude that the field $\mathbf{R}(x)$ is not Cauchy complete (cf. Theorem 2.12 in Chapter 2). Since $nx < 1$ for all $n \in \mathbf{N}$, the field $\mathbf{R}(x)$ is not Archimedean (cf. Figure C.1).

## Notes

In algebra, for every ordered integral domain $D$ (including the field $\mathbf{R}$), the ordered field $D(x)$ is the quotient field of the polynomial ring $D[x]$ (cf. Examples A.1, 1.2, and 1.3). Similarly, the ordered field $D((x))$ is the quotient field of the ring $D[\![x]\!]$ of formal power series with coefficient in $D$. It can be shown that this field is isomorphic to the field of formal Laurent series introduced in Section C.2 (cf. Mac Lane and Birkhoff 1999, Ch. VIII, §9).

Theorems in Appendix C are known results in the area of ordered fields (cf. Bourbaki 2003, Ch. IV). Because elementary proofs of these theorems are difficult to access, these theorems are proved in Sections C.2 and C.3.

Note that it is shown in the proof of Theorem C.3 that the ring $\mathbf{R}[x]$ is dense in the field $\mathbf{R}((x))$, assuming that the order on $\mathbf{R}[x]$ is induced by the order on $\mathbf{R}((x))$.

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